SLIM3 CONTROLLER

PRO L

MAINTENANCE DRAWINGS

Document No: V090i-C1E-X03

Edition No.: C

MAKINO

MAKINO ASIA PTE LTD

Drawing Number - V090i-C1E-X03

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CIRCUIT DRAWING	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No.
				--

HISTORY OF MODIFICATIONS

EDITION NO.	DATE	DRAWING UPDATE RELEASE NO.	CONTENT	DRAWING NO.	REMARKS	PREPARED BY
A	14-08-2006	MV09-0206	FIRST RELEASE	V09OI-C1E-X01	——	TAI KENG SONG
B	25-08-2008	MV09-0428	SECOND RELEASE	V09OI-C1E-X02	——	TAI KENG SONG
C	25-11-2010	MV09-0758	THIRD RELEASE	V09OI-C1E-X03	——	PRAVEEN B

Revisions			Date	Name
	④			
	③	APPROVED	25-NOV-2010	GOH WILLIAM
	②	CHECKED	25-NOV-2010	S MAHESH
	①	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
HISTORY OF MODIFICATIONS	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. A01

DRAWING CROSS REFERENCE:

1.FORMAT:

C01/C05

- ZONE NO.
- PAGE NO.

2.ZONE NO. REFERENCE

C 05

- VERTICAL (DWG SHEET)
- HORIZONTAL (DWG SHEET)

3.PAGE NO. REFERENCE

C 01

- PAGE NO. UNDER PARTICULAR GROUP
- GROUP NO.
  - A-HISTORY OF MODIFICATIONS
  - B-CONTENTS
  - C-CIRCUIT DRAWINGS
  - D-TERMINAL DRAWINGS
  - E-CONNECTOR TABLE
  - F-DI/DO ADDRESS TABLE
  - G-MACHINE INTERCONNECTION
  - H-LAYOUT,PARTS LIST

4.DRAWING REFERENCE:

V090i-C1E-XAA

- RUNNING NO.
- X=STD;
- ENGLISH
- 1 = STD M/C and APC M/C;
- CIRCUIT DRAWING
- CONTROLLER TYPE
- SLIM M/C
- VERTICAL M/C

COMMENT FIELD

Revisions			Date	Name
	4			
	3	APPROVED	25-NOV-2010	GOH WILLIAM
	2	CHECKED	25-NOV-2010	S MAHESH
	1	DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
GENERAL	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. B01

Table of Contents

PAGE NO.	PAGE DESCRIPTION
--	CIRCUIT DRAWING
A01	HISTORY OF MODIFICATIONS
B01	GENERAL
B02	Table of contents: (-- - C22)
B03	Table of contents: (C23 - C50)
B04	Table of contents: (C51 - C80)
B05	Table of contents: (C81 - F06)
B06	Table of contents: (F07 - H09)
C01	INPUT POWER SOURCE
C02	EXPORT TRANSFORMER (OPT)
C03	POWER SOURCE
C04	CONTROL POWER SOURCE
C05	POWER SUPPLY MODULE
C06	SPINDLE AMPLIFIER MODULE (15K)
C07	HIGH TORQUE SPINDLE (8K)
C08	SPINDLE AMPLIFIER MODULE (HSK-A40)
C09	SERVO AMPLIFIER CONNECTION (STD)
C10	SERVO AMPLIFIER CONNECTION (OPT)
C11	SERVO AMPLIFIER CONNECTION (OPT)
C12	ATC BETA AMPLIFIER MODULE (OPT)
C13	FEEDBACK SCALE (OPT)
C14	X AND Y SERVO AMPLIFIER MODULE
C15	Z AND C SERVO AMPLIFIER MODULE
C16	A SERVO AMPLIFIER MODULE (OPT)
C17	B SERVO AMPLIFIER MODULE (OPT)
C18	B AND Z SERVO AMPLIFIER MODULE (OPT)
C19	Z AXIS MOTOR BRAKE
C20	CONTROLLER UNIT
C21	I/O LINK CONNECTION (STD M/C)
C22	I/O LINK CONNECTION (APC M/C)

Revisions			Date	Name
	④			
	③	APPROVED	25-NOV-2010	GOH WILLIAM
	②	CHECKED	25-NOV-2010	S MAHESH
	①	DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
Table of contents: (-- - C22)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. B02

Table of Contents

PAGE NO.	PAGE DESCRIPTION
C23	COOLANT PUMP
C24	THRU SPDL COOLANT 1.5MPa (OPT)
C25	THRU SPDL COOLANT SIGNAL 1.5MPa (OPT)
C26	THRU SPDL COOLANT SIGNAL (OPT)
C27	COOLANT LEVEL SENSOR (OPT)
C28	THRU SPDL COOLANT 3MPa (OPT)
C29	THRU SPDL COOLANT CNTRL 3MPa (OPT)
C30	THRU SPDL COOLANT SIGNAL 3MPa (OPT)
C31	LIFT UP CHIP CONVEYOR (OPT)
C32	CONTROL VOLTAGE SOURCE
C33	NC POWER ON CONTROL
C34	EMERGENCY CIRCUIT
C35	CB OVERLOAD CONTROL
C36	SPINDLE SWITCHING CONTROL
C37	SPINDLE TEMPERATURE CONTROLLER
C38	HSK TOOL SEAT CHECK (HSK-A50)
C39	TOOL CLAMP/UNCLAMP
C40	HSK TOOL SEAT CHECK (HSK-A40)
C41	POWER PACK CONTROL
C42	LS REFERENCE AND OVERTRAVEL
C43	ATC MAGAZINE AIR
C44	COOLANT (OPT)
C45	MACHINE LIGHT
C46	NOZZLE COOLANT
C47	ADDITIONAL A AXIS (OPT)
C48	ADDITIONAL B AXIS (OPT)
C49	PRAGATI INDEX TABLE (OPT)
C49-1	PRAGATI INDEX TBL - PAL 1(OPT)
C49-2	PRAGATI INDEX TBL - PAL 2(OPT)
C50	AWM-MARPOSS (OPT)

Revisions		Date	Name
4			
3	APPROVED	25-NOV-2010	GOH WILLIAM
2	CHECKED	25-NOV-2010	S MAHESH
1	DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
Table of contents: (C23 - C50)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. B03

Table of Contents

PAGE NO.	PAGE DESCRIPTION
C51	ATLM-RENISHAW TOUCH PROBE (OPT)
C52	AWM-RENISHAW MP10 (OPT)
C53	ATLM-BLUM Z-NANO TYPE (OPT)
C54	ATLM-BLUM Z-NANO TYPE (OPT)
C55	S/G DOOR INTERLOCK
C56	AIR BLOW (OPT)
C57	MAIN OPERATION PANEL(24V FOR I/O MODULE)
C58	MODE SELECTION, FLEXIBLE SWITCHES
C59	MODE SELECTION, FLEXIBLE SWITCHES
C60	MODE SELECTION, FLEXIBLE SWITCHES
C61	MODE SELECTION, FLEXIBLE SWITCHES
C62	MANUAL PULSE GENERATOR
C63	PORTABLE MANUAL PULSE GENERATOR (OPT)
C64	SIGNAL LIGHT/PATROL LIGHT
C65	APC PALLET OPERATOR PANEL (APC M/C)
C66	APC PALLET CONTROL (APC M/C)
C67	ROBOT DR LEFT OR RIGHT (OPT)
C68	ROBOT DR LEFT & RIGHT (OPT)
C69	AUTO DOOR WITH PUSHBUTTON (OPT)
C70	TPC JR (WITHOUT OPT PCB)
C71	TPC JR (WITH OPT PCB)
C72	USER MISCELLANEOUS CONTROL (OPT)
C73	USER MISCELLANEOUS CONTROL (OPT)
C74	CUSTOMISED CONTROL (OPT)
C75	AIRCON (OPT)
C76	AIR DRYER (OPT)
C77	TOOL BROKEN SENSOR (OPT)
C78	AUTO GREASE (OPT)
C79	HYDRAULIC OPT (AMF)
C80	HYDRAULIC PRESSURE SW SOL (AMF)

Revisions		Date	Name
④			
③	APPROVED	25-NOV-2010	GOH WILLIAM
②	CHECKED	25-NOV-2010	S MAHESH
①	DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
Table of contents: (C51 - C80)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. B04

Table of Contents

PAGE NO.	PAGE DESCRIPTION
C81	HYDRAULIC OPT (YUKEN)
C82	HYDRAULIC PRESSURE SW, SOL(YUKEN)
C83	HYDRAULIC OPT (HYDAC)
C84	HYDRAULIC PR SW, SOL(1-2) STK (HYDAC)
C85	HYDRAULIC PR SW, SOL(3-4) STK (HYDAC)
C86	AIR SEAT CHECK SENSOR (OPT)
C87	HYDRAULIC DISPLAY (OPT)
C88	HYDRAULIC DISPLAY-2ND SET UN/CLAMP (OPT)
C89	TWO HAND PUSHBUTTON (OPT)
C90	OIL MIST COLLECTOR (OPT)
C91	SAFETY LIGHT CURTAIN (OPT)
D01	TERMINAL BLOCK - MTC
E01	CONNECTOR TABLE - RELAY PCB
E02	CONNECTOR TABLE - RELAY PCB
E03	CONNECTOR TABLE - OPT PCB
E04	CONNECTOR TABLE - OPT PCB
E05	CONNECTOR TABLE - OPT PCB
E06	CONNECTOR TABLE - MTC
E07	CONNECTOR TABLE - MTC
E08	CONNECTOR TABLE - OPR. PANEL
E09	CONNECTOR TABLE - OPR. PANEL
E10	CONNECTOR TABLE - COOLANT TERMINAL BOX
E11	CONNECTOR TABLE - APC PCB
E12	CONNECTOR TABLE - INTERFACE CONNECTORS
F01	DI/DO ADDRESS TABLE (MACHINE I/O)
F02	DI/DO ADDRESS TABLE (MACHINE I/O)
F03	DI/DO ADDRESS TABLE (MACHINE I/O)
F04	DI/DO ADDRESS TABLE (MACHINE I/O)
F05	DI/DO ADDRESS TABLE (MACHINE I/O)
F06	DI/DO ADDRESS TABLE (MACHINE I/O)

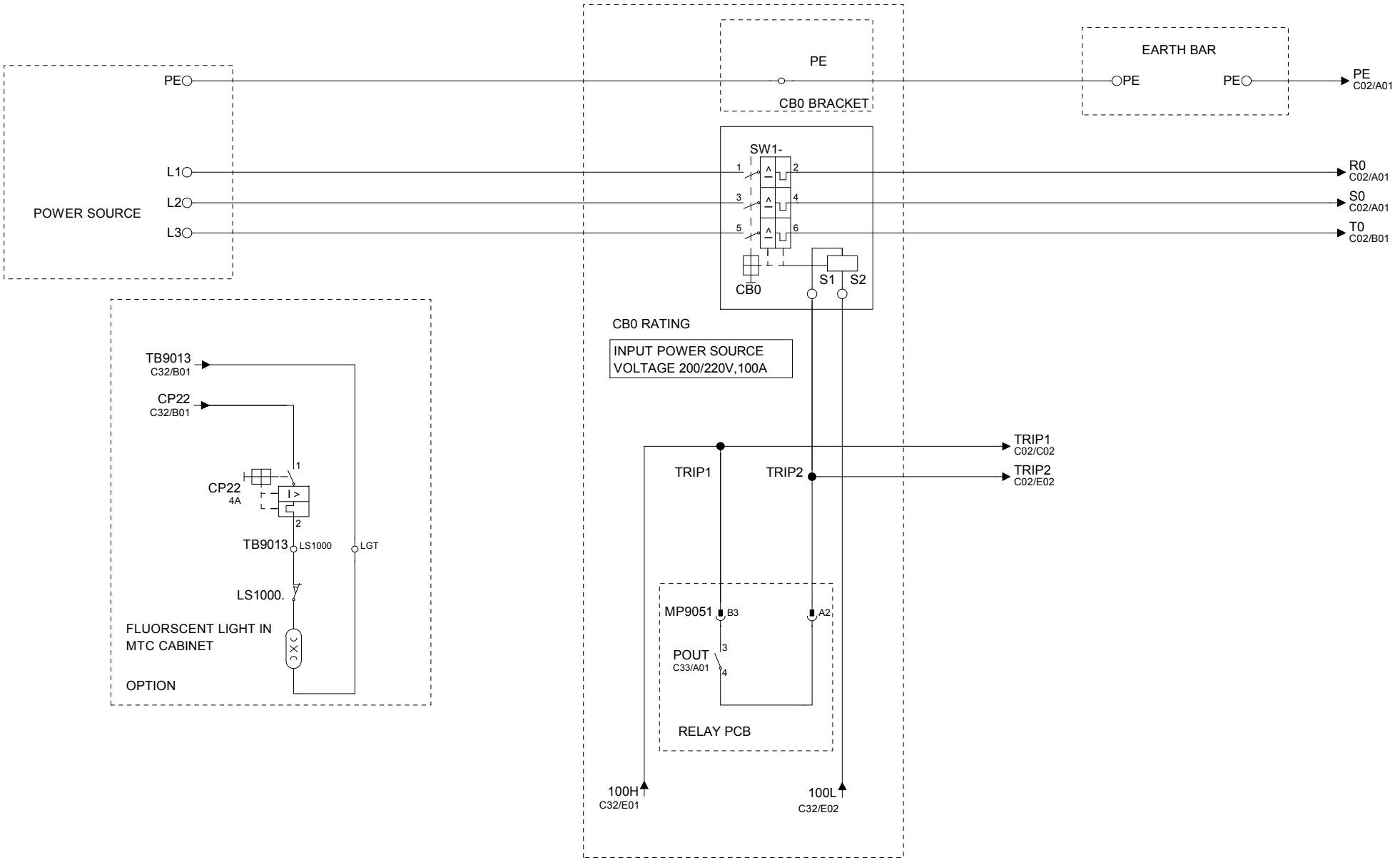
Table of Contents

PAGE NO.	PAGE DESCRIPTION
F07	DI/DO ADDRESS TABLE (MACHINE I/O)
F08	DI/DO ADDRESS TABLE (OPR. PANEL I/O)
F09	DI/DO ADDRESS TABLE (OPR. PANEL I/O)
G01	MACHINE INTERCONNECTION (STD & APC M/C)
H01	MTC LAYOUT, PARTS
H02	MTC LAYOUT, PARTS
H03	MTC LAYOUT, PARTS
H04	OPERATOR PANEL LAYOUT, PARTS
H05	RELAY PCB LAYOUT, PARTS
H06	OPERATOR PCB LAYOUT, PARTS
H07	OPTION PCB LAYOUT, PARTS
H08	HYDRAULIC PCB LAYOUT, PARTS
H09	APC PCB LAYOUT, PARTS

Revisions		Date	Name
4			
3	APPROVED	25-NOV-2010	GOH WILLIAM
2	CHECKED	25-NOV-2010	S MAHESH
1	DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
Table of contents: (F07 - H09)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. B06

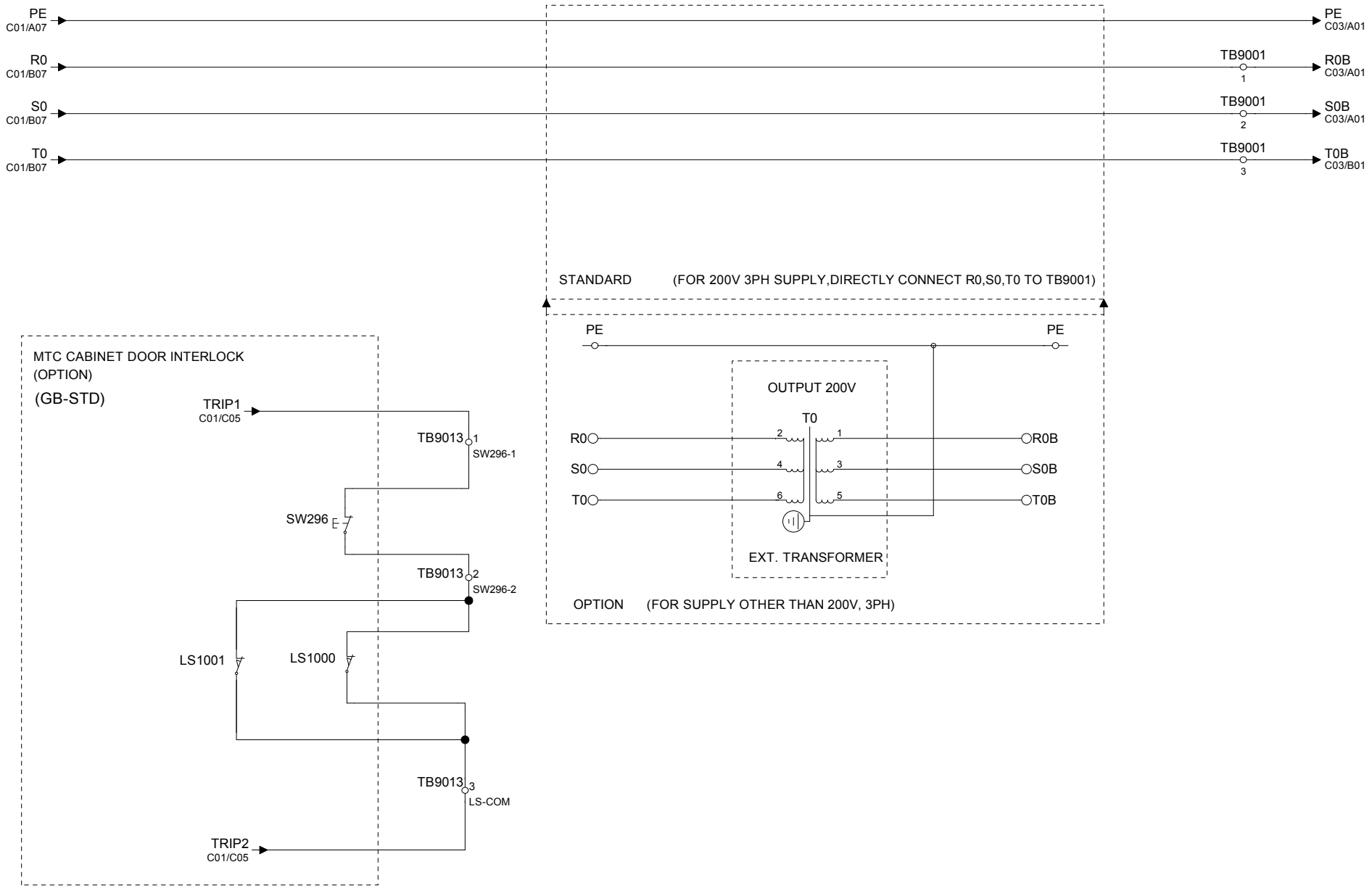


3PH AC SOURCE	MTC LIGHT CONTROL	CB0 POWER, TRIP CONTROL	

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
INPUT POWER SOURCE	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C01



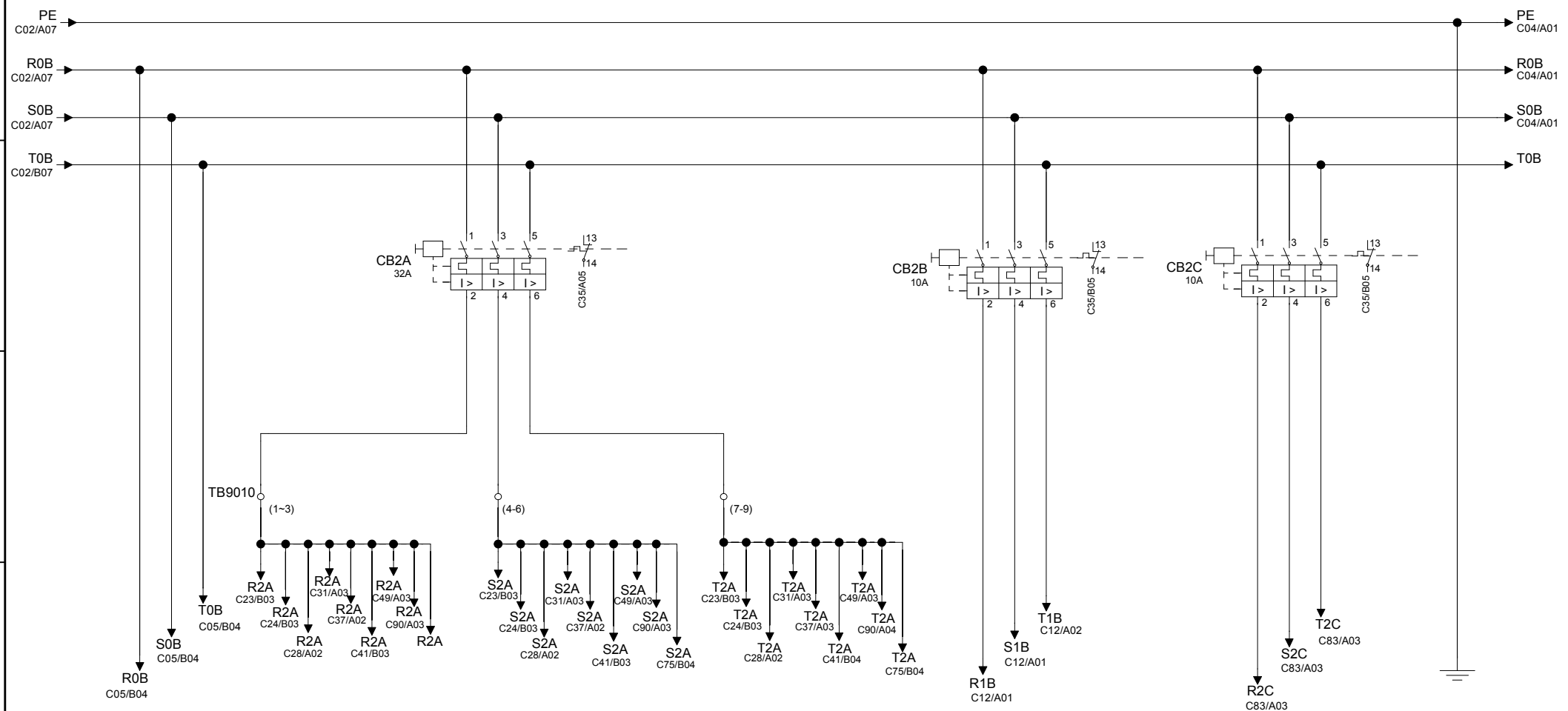
MTC DOOR INTERLOCK

EXPORT TRANSFORMER POWER

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
EXPORT TRANSFORMER (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C02



C23-B03	COOLANT PUMP
C24-B03	THRU SP/TL COOLANT 1.5MPa (OPT)
C28-A02	THRU SP/TL COOLANT 3MPa (OPT)
C31-A03	LIFT UP CHIP CONVEYOR (OPT)
C37-A02	SPINDLE TEMPERATURE CONTROLLER
C41-B03	POWER PACK CONTROL
C49-A03	PRAGATI INDEX TABLE (OPT)
C75-B04	AIRCON (OPT)
C90-A03	OIL MIST COLLECTOR (OPT)

POWER SUPPLY MODULE
PSM26i

ACR2

FIXTURE HYD CONTROLS

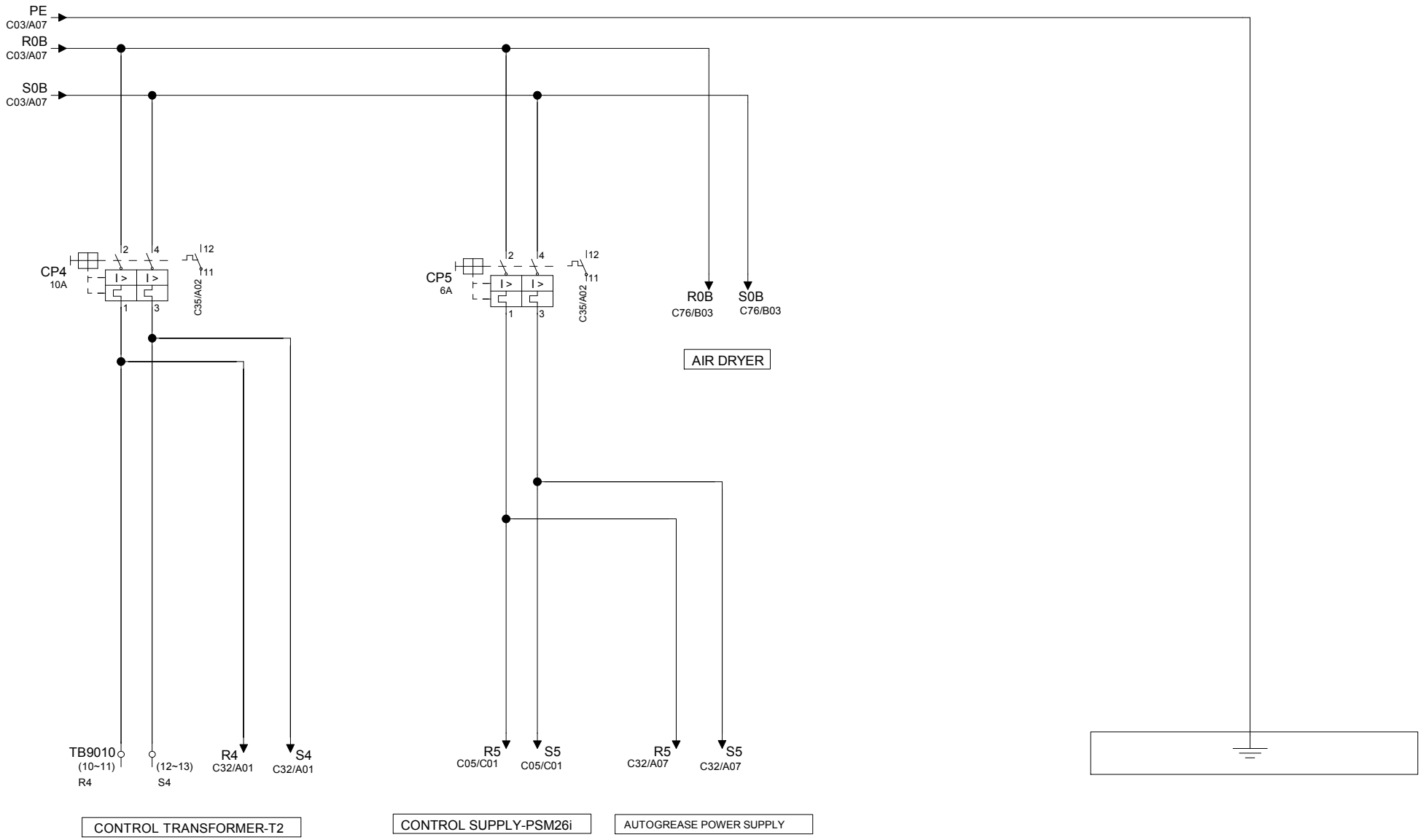
MAIN POWER DISTRIBUTION

Revisions			Date	Name
	④			
	③	APPROVED	25-NOV-2010	GOH WILLIAM
	②	CHECKED	25-NOV-2010	S MAHESH
	①	DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.
POWER SOURCE	NC TYPE	FS0i-MC	V090i-C1E-X03

GROUP =
SHEET No. C03

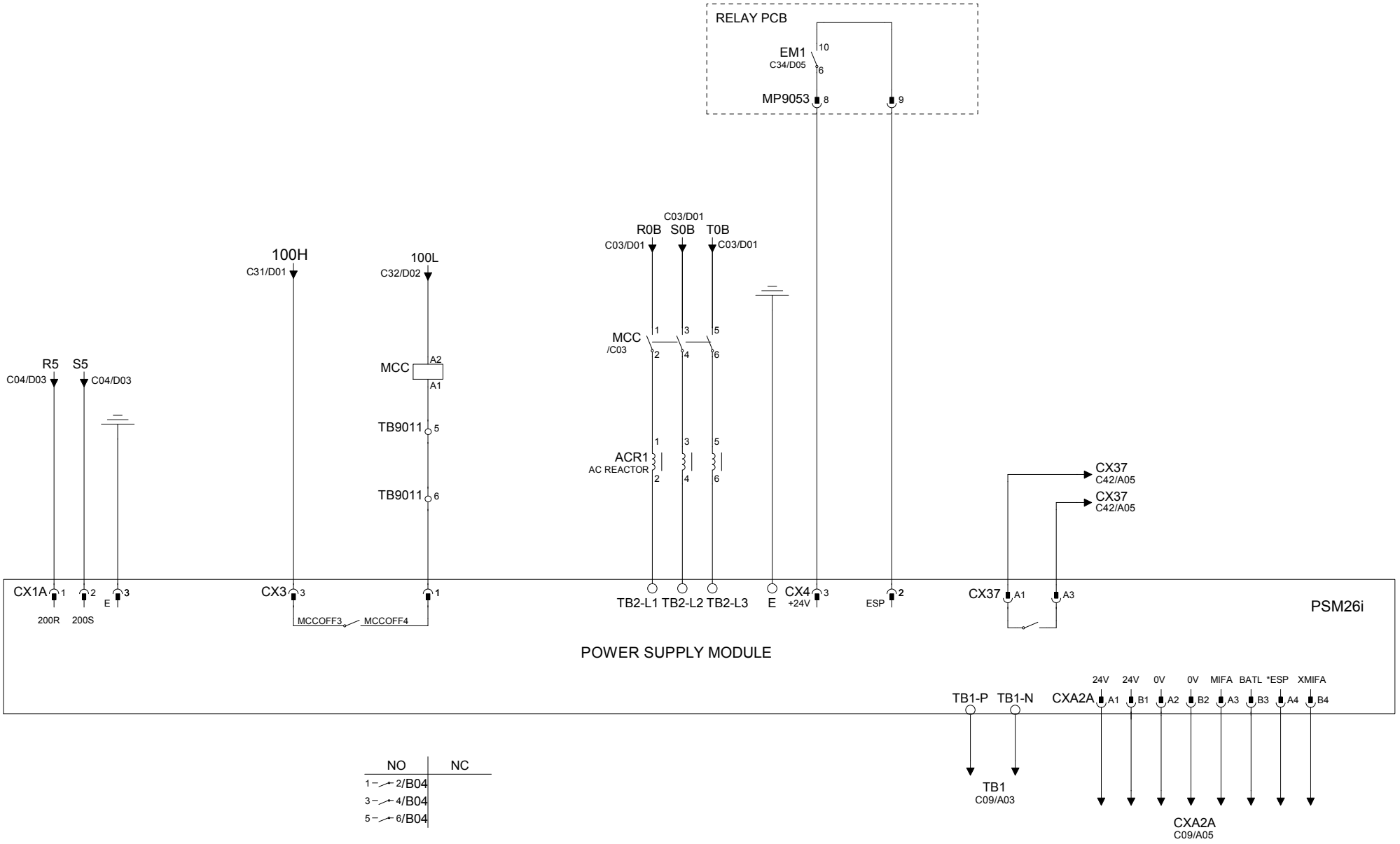


CONTROL POWER DISTRIBUTION

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONTROL POWER SOURCE	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C04



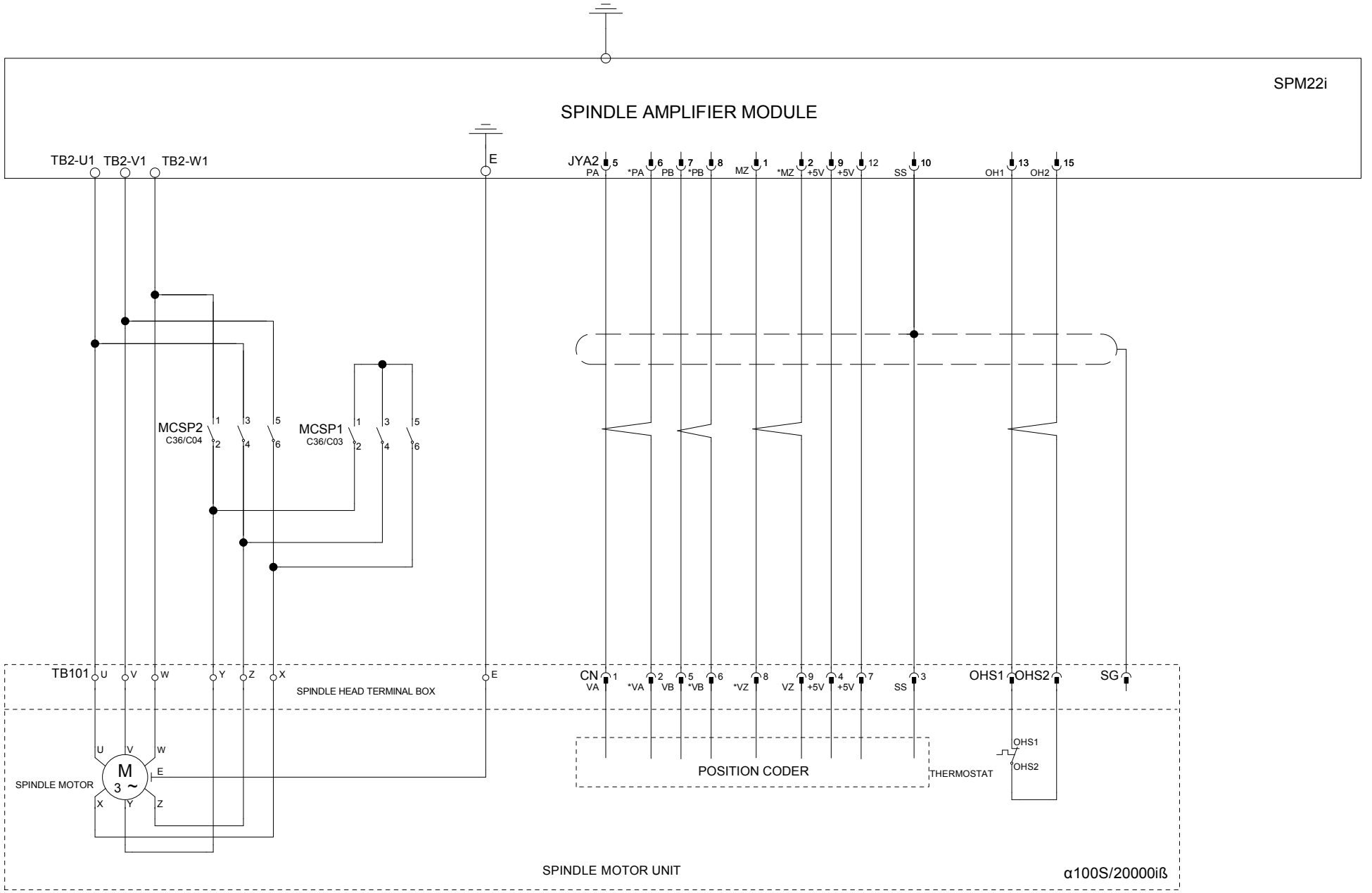
NOTE: COMMON FOR HSK A50 AND HSK A40

CONTROL SUPPLY	DRIVE CONTROL	POWER SUPPLY	EMERGENCY INTERLOCK	DC BUS	INTERFACE SIGNALS

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
POWER SUPPLY MODULE	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C05

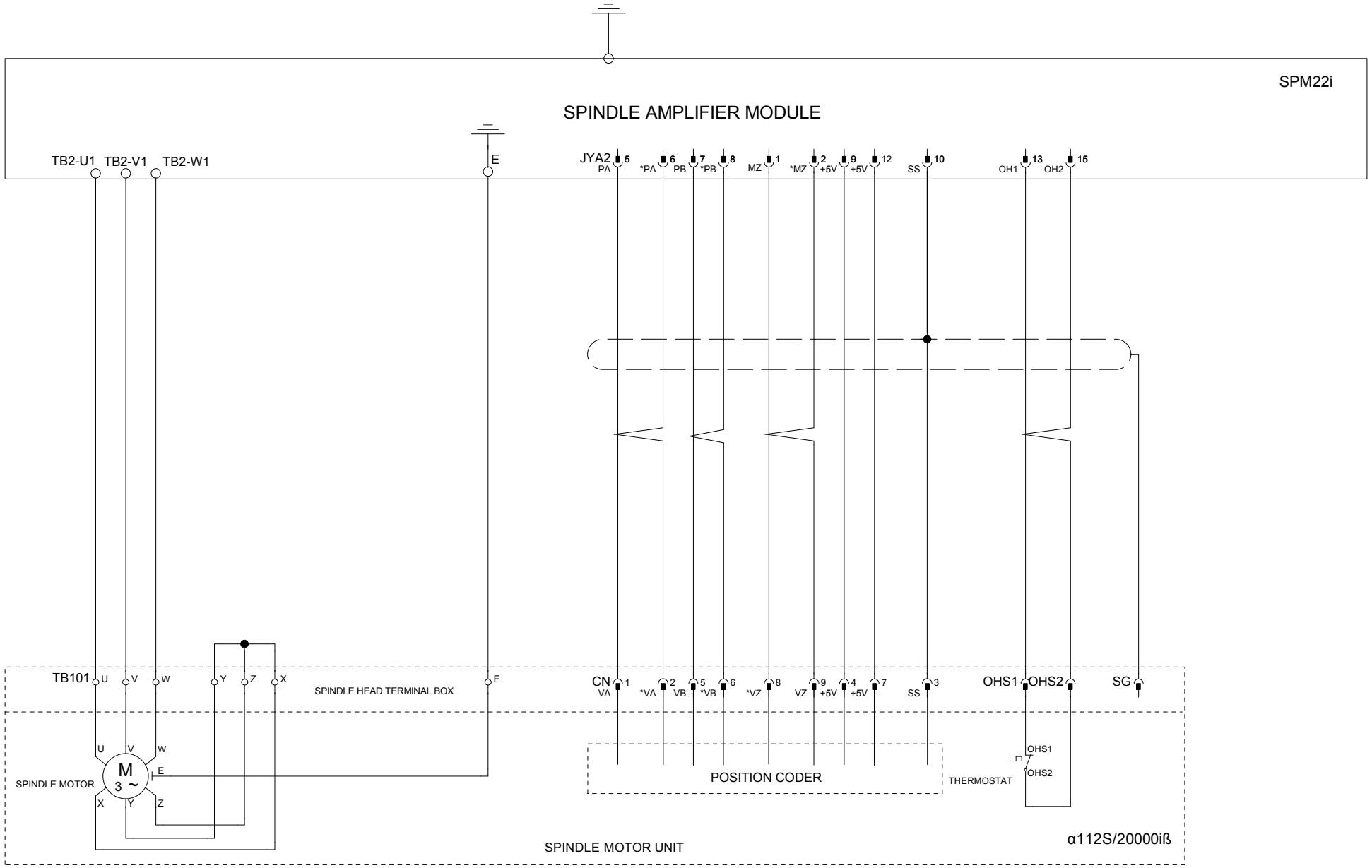


SPINDLE MOTOR POWER - 15000RPM	SPINDLE POSITION SIGNAL	OVER HEAT CONTROL

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
SPINDLE AMPLIFIER MODULE (15K)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C06

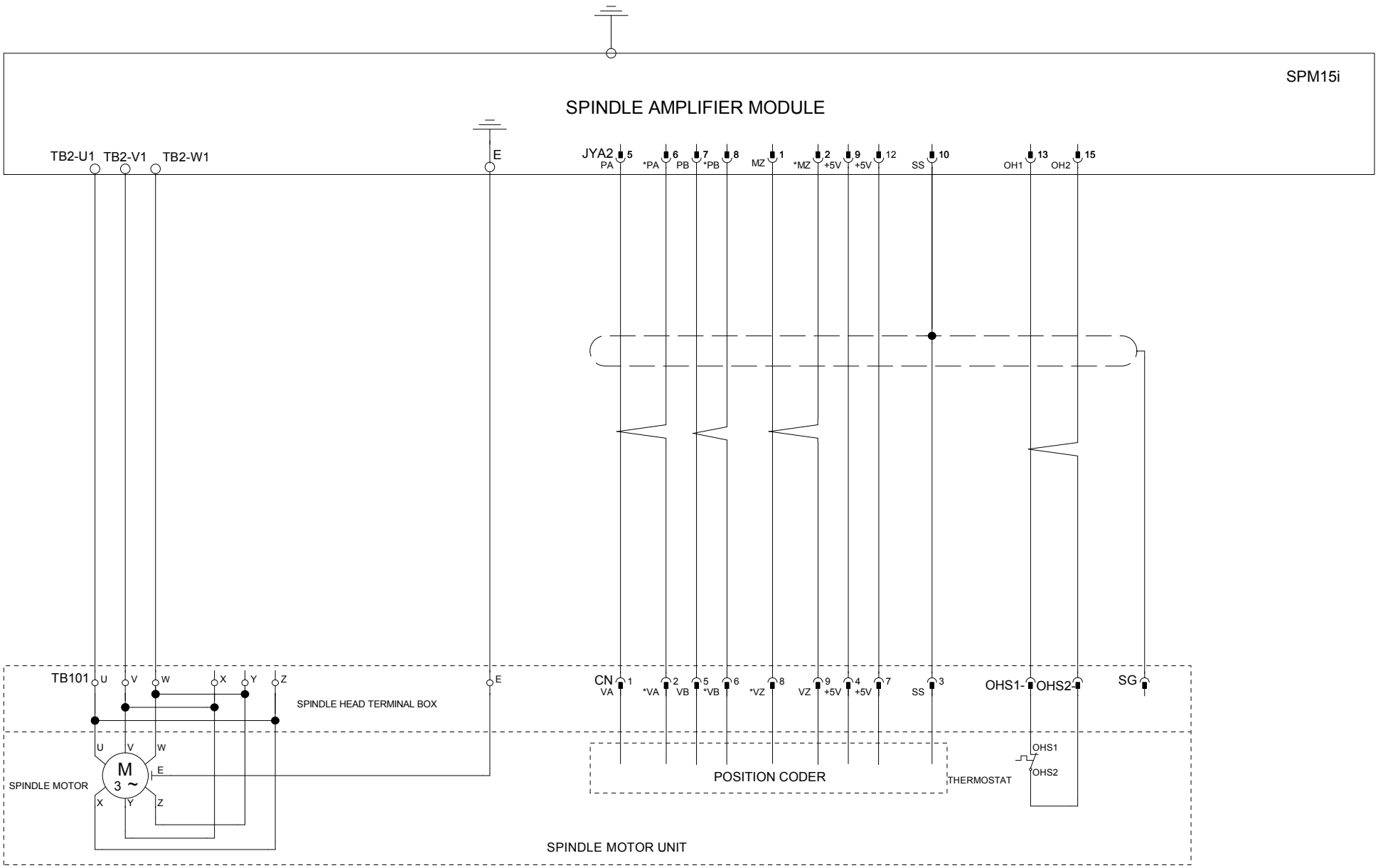


SPINDLE MOTOR POWER - 8000RPM	SPINDLE POSITION SIGNAL	OVER HEAT CONTROL

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
HIGH TORQUE SPINDLE (8K)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C07

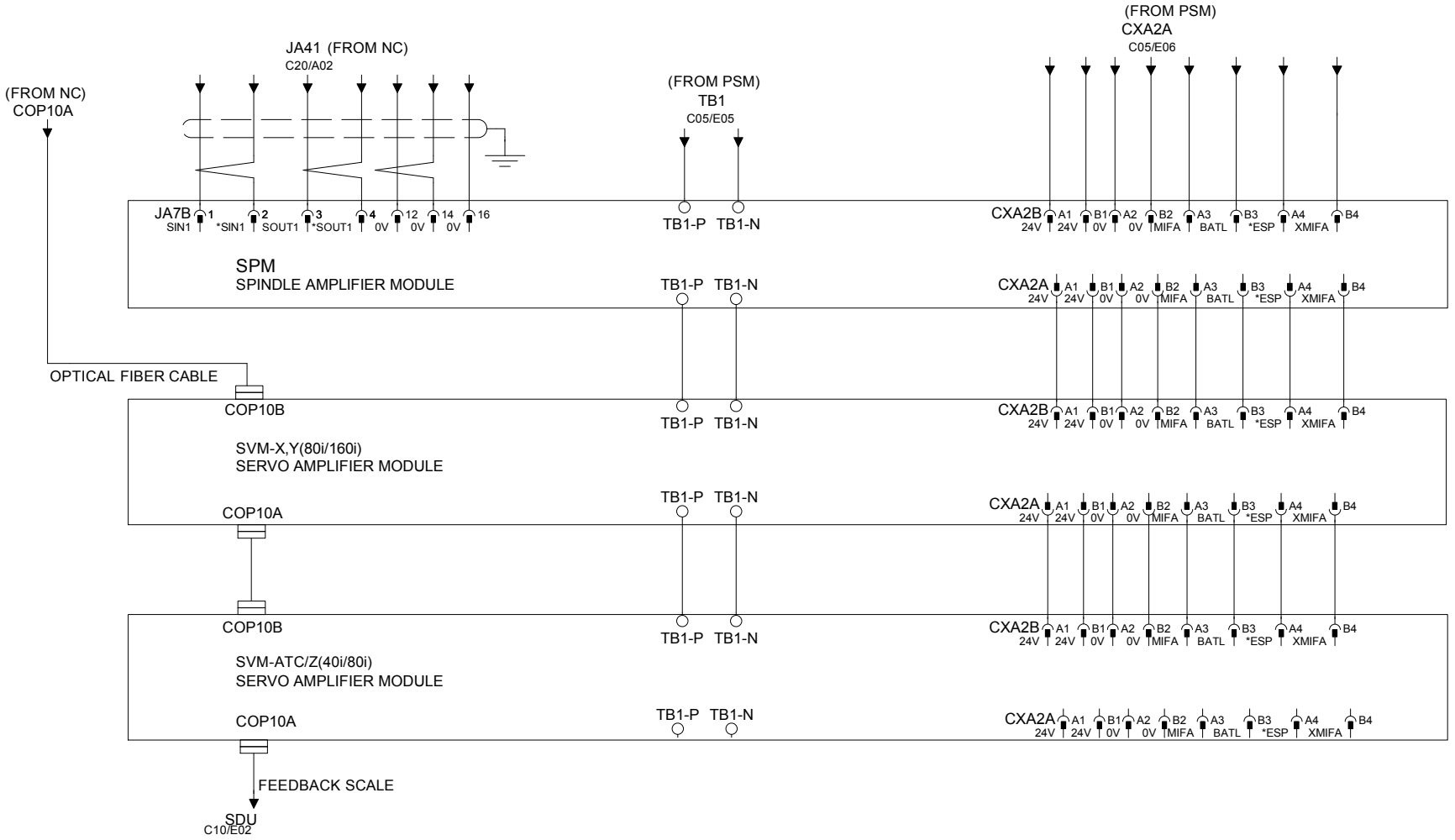


SPINDLE MOTOR POWER - 15000RPM	SPINDLE POSITION SIGNAL	OVER HEAT CONTROL

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
SPINDLE AMPLIFIER MODULE (HSK-A40)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C08

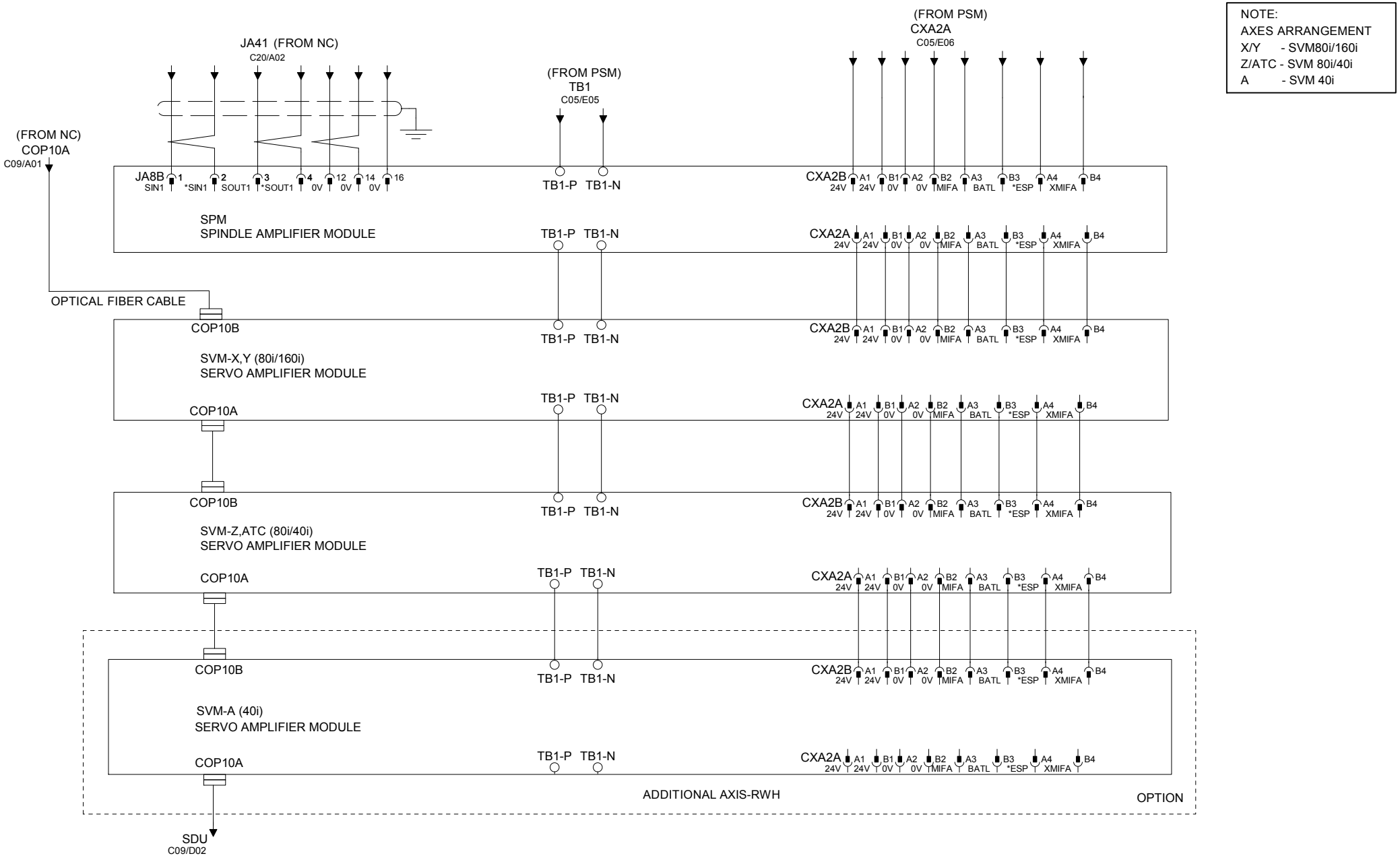


SERVO AMPLIFIER INTERCONNECTIONS - STANDARD

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
SERVO AMPLIFIER CONNECTION (STD)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C09

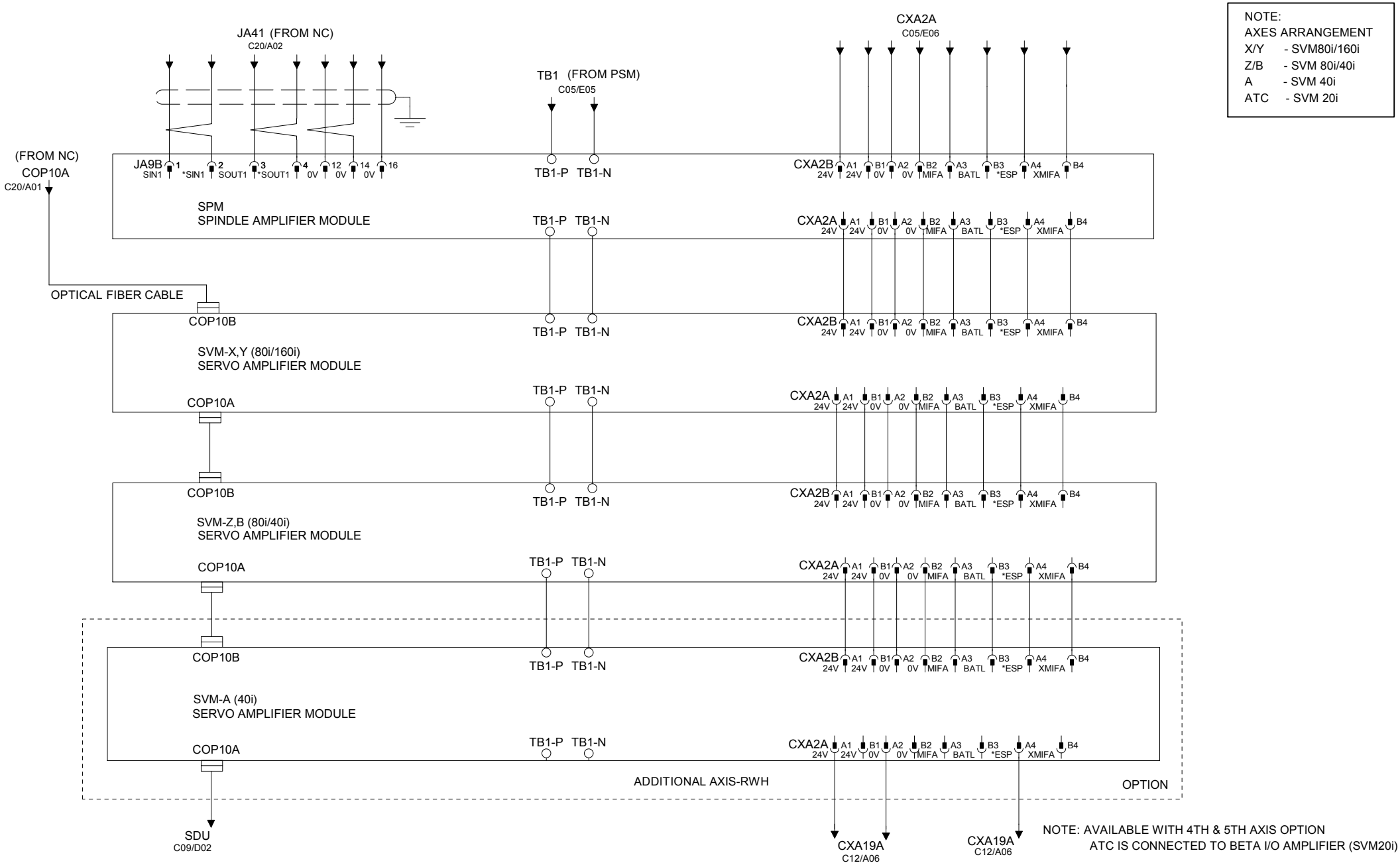


SERVO AMPLIFIER INTERCONNECTIONS - OPTION

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
SERVO AMPLIFIER CONNECTION (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C10

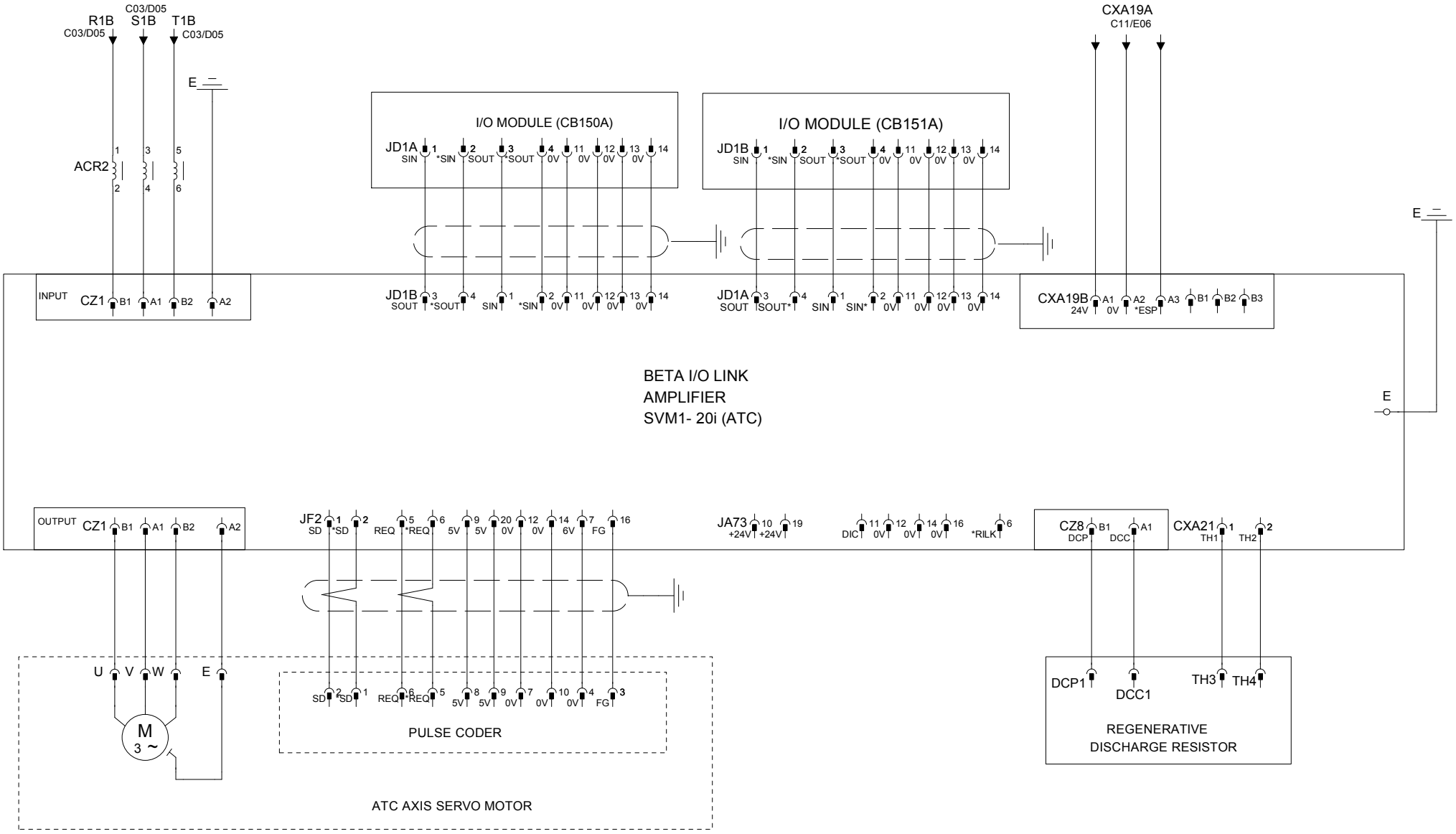


SERVO AMPLIFIER INTERCONNECTIONS - OPTION

Revisions			Date	Name
	4			
	3	APPROVED	25-NOV-2010	GOH WILLIAM
	2	CHECKED	25-NOV-2010	S MAHESH
	1	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
SERVO AMPLIFIER CONNECTION (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C11



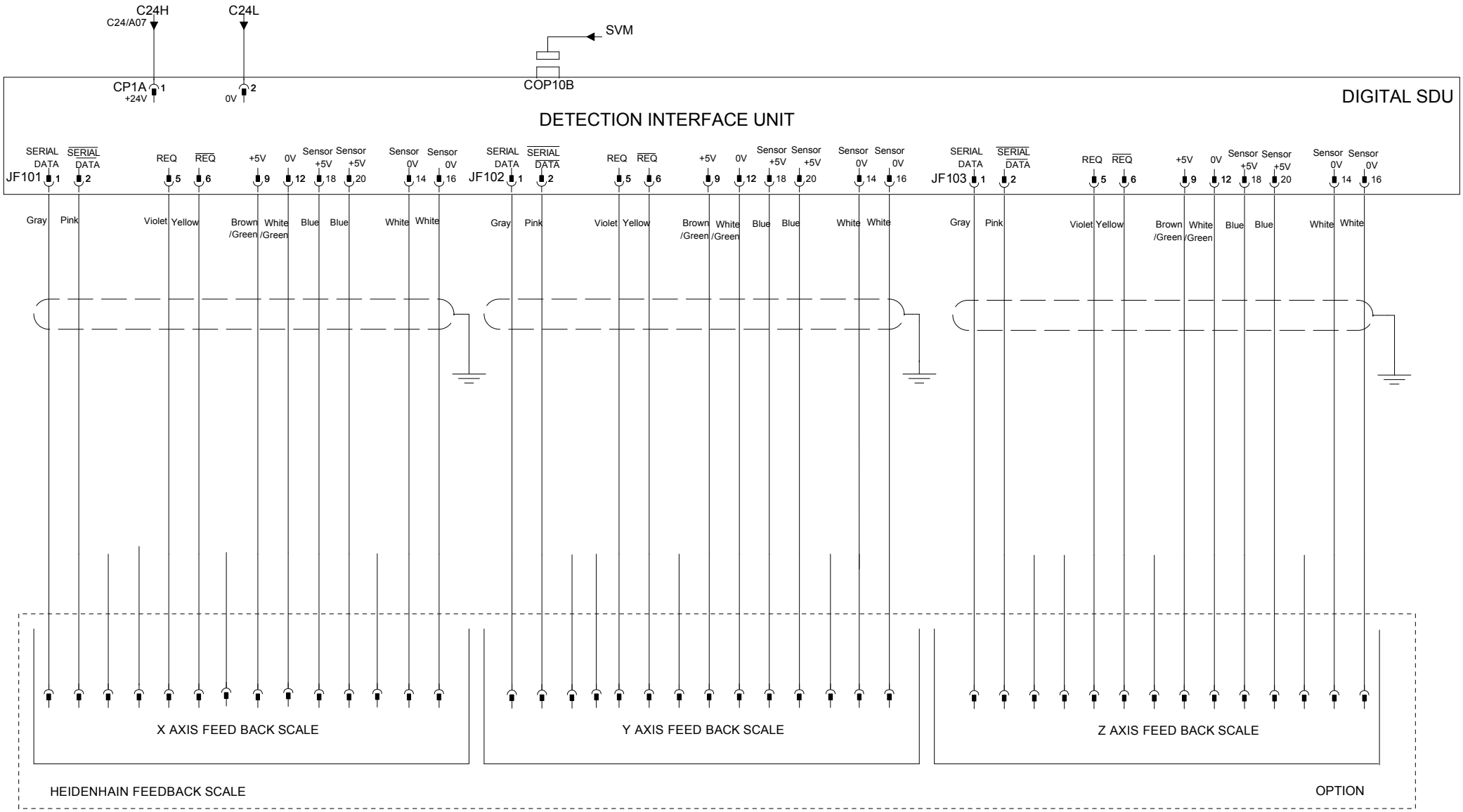
NOTE
1) VALID FOR M/Cs WITH BETA AMP FOR ATC

ATC AXIS MOTOR POWER	ATC AXIS MOTOR SIGNAL	

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
ATC BETA AMPLIFIER MODULE (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C12



FEED BACK SCALE CONNECTION

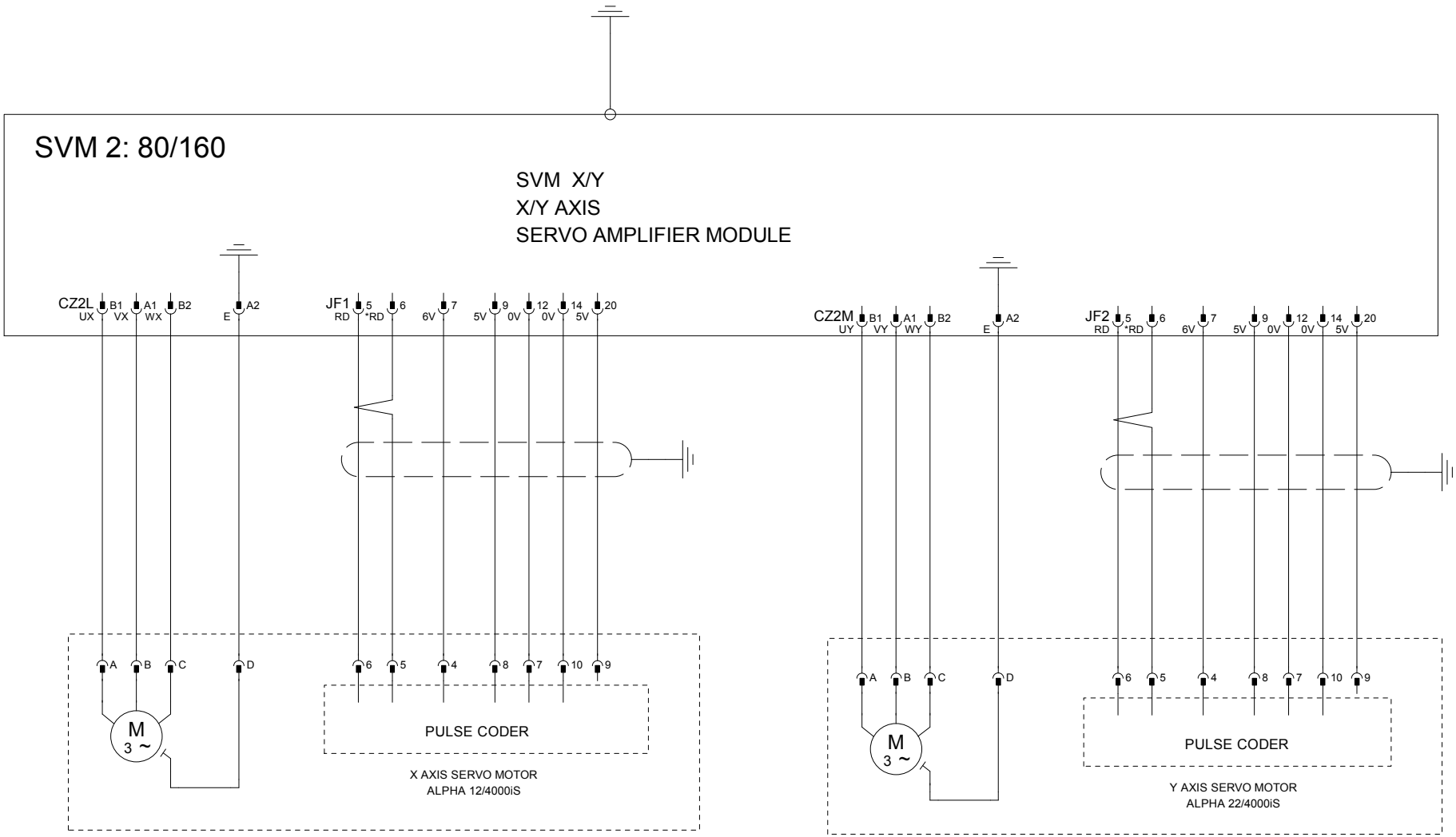
Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
FEEDBACK SCALE (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C13

SVM 2: 80/160

SVM X/Y
X/Y AXIS
SERVO AMPLIFIER MODULE



X AXIS MOTOR POWER	X AXIS MOTOR SIGNAL	Y AXIS MOTOR POWER	Y AXIS MOTOR SIGNAL

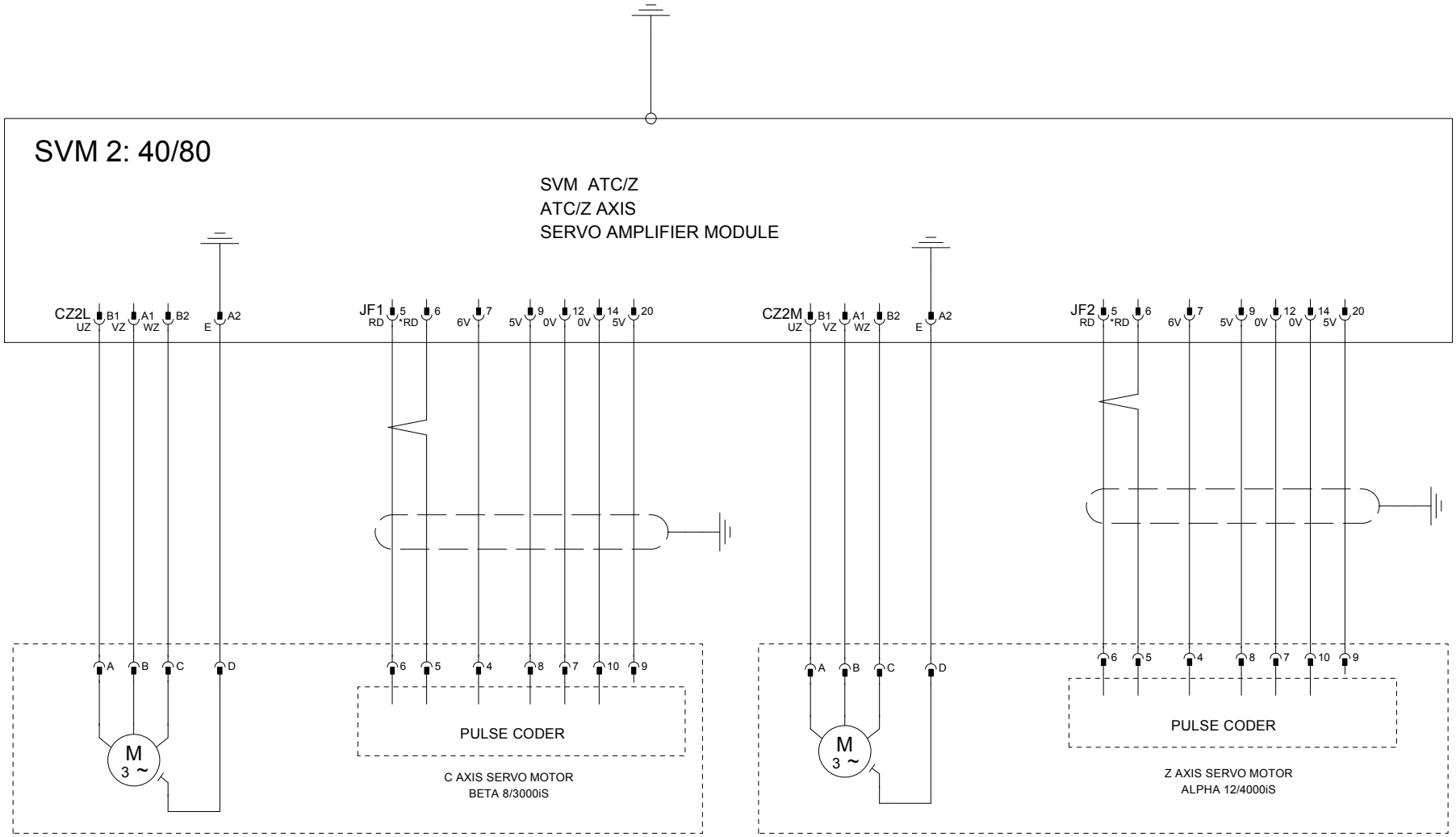
Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
X AND Y SERVO AMPLIFIER MODULE	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C14

SVM 2: 40/80

SVM ATC/Z
ATC/Z AXIS
SERVO AMPLIFIER MODULE

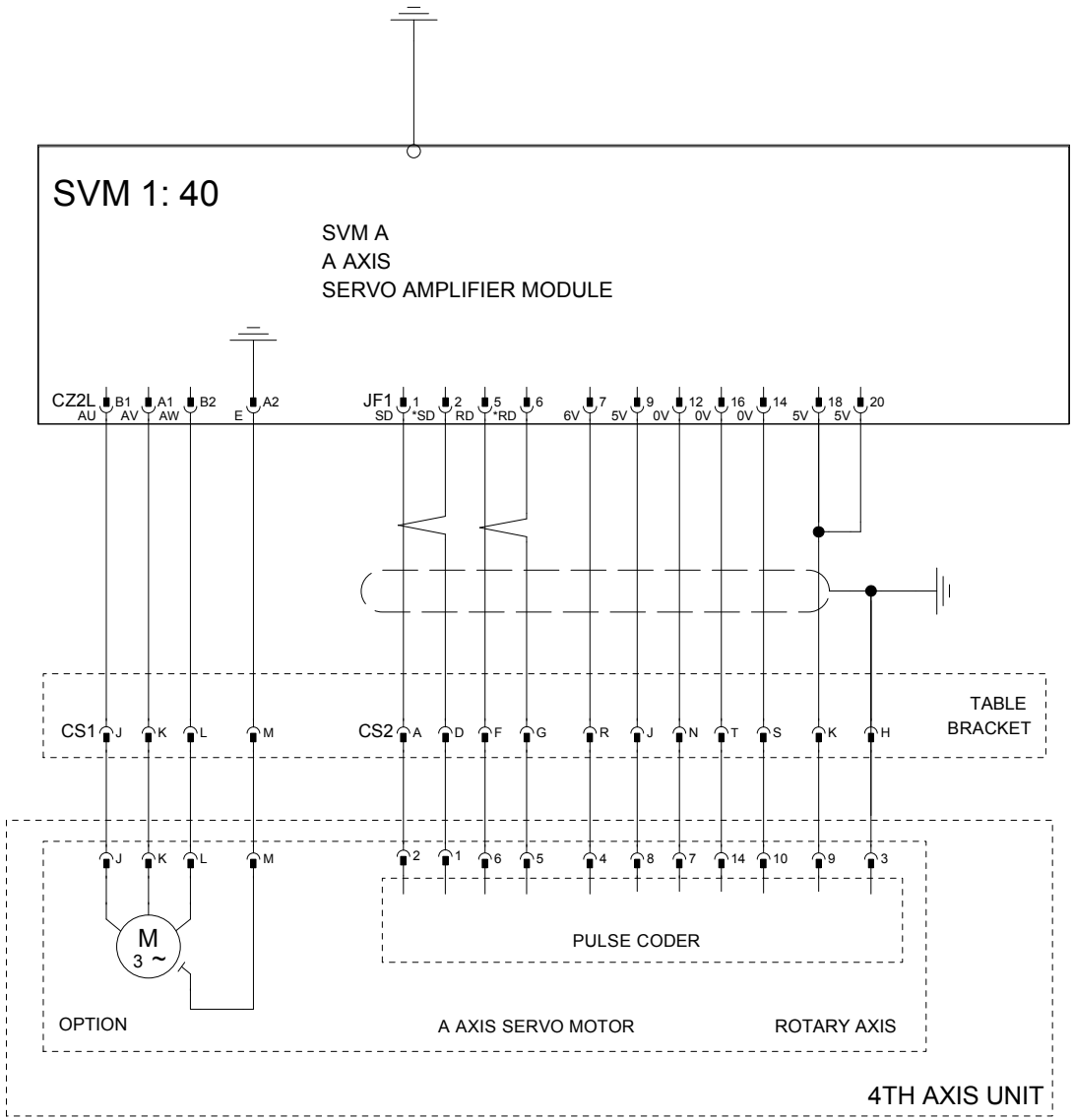


ATC AXIS MOTOR POWER	ATC AXIS MOTOR SIGNAL	Z AXIS MOTOR POWER	Z AXIS MOTOR SIGNAL

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
Z AND C SERVO AMPLIFIER MODULE	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C15

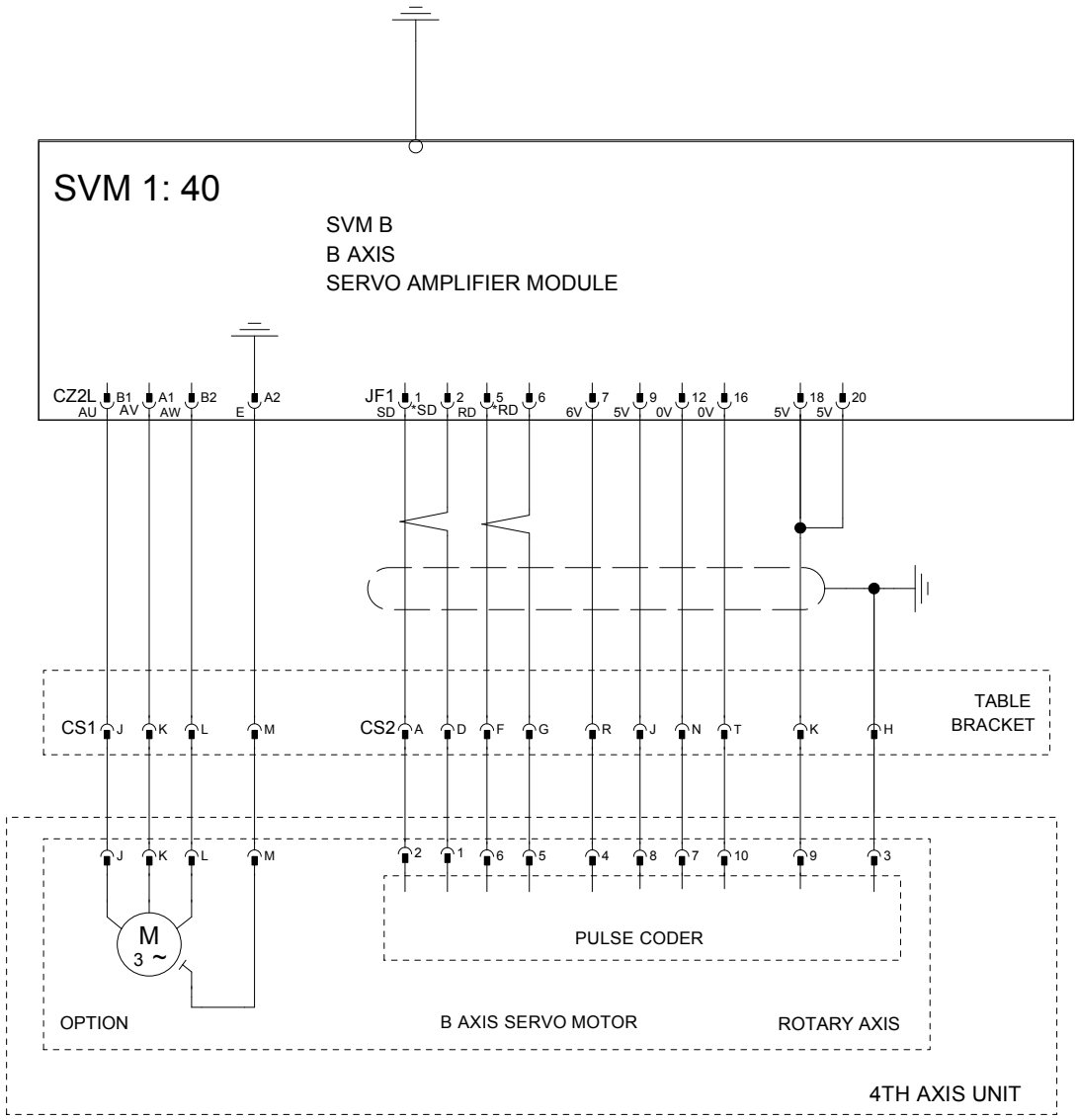


A AXIS MOTOR POWER | A AXIS MOTOR SIGNAL

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
A SERVO AMPLIFIER MODULE (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C16



B AXIS MOTOR POWER | B AXIS MOTOR SIGNAL

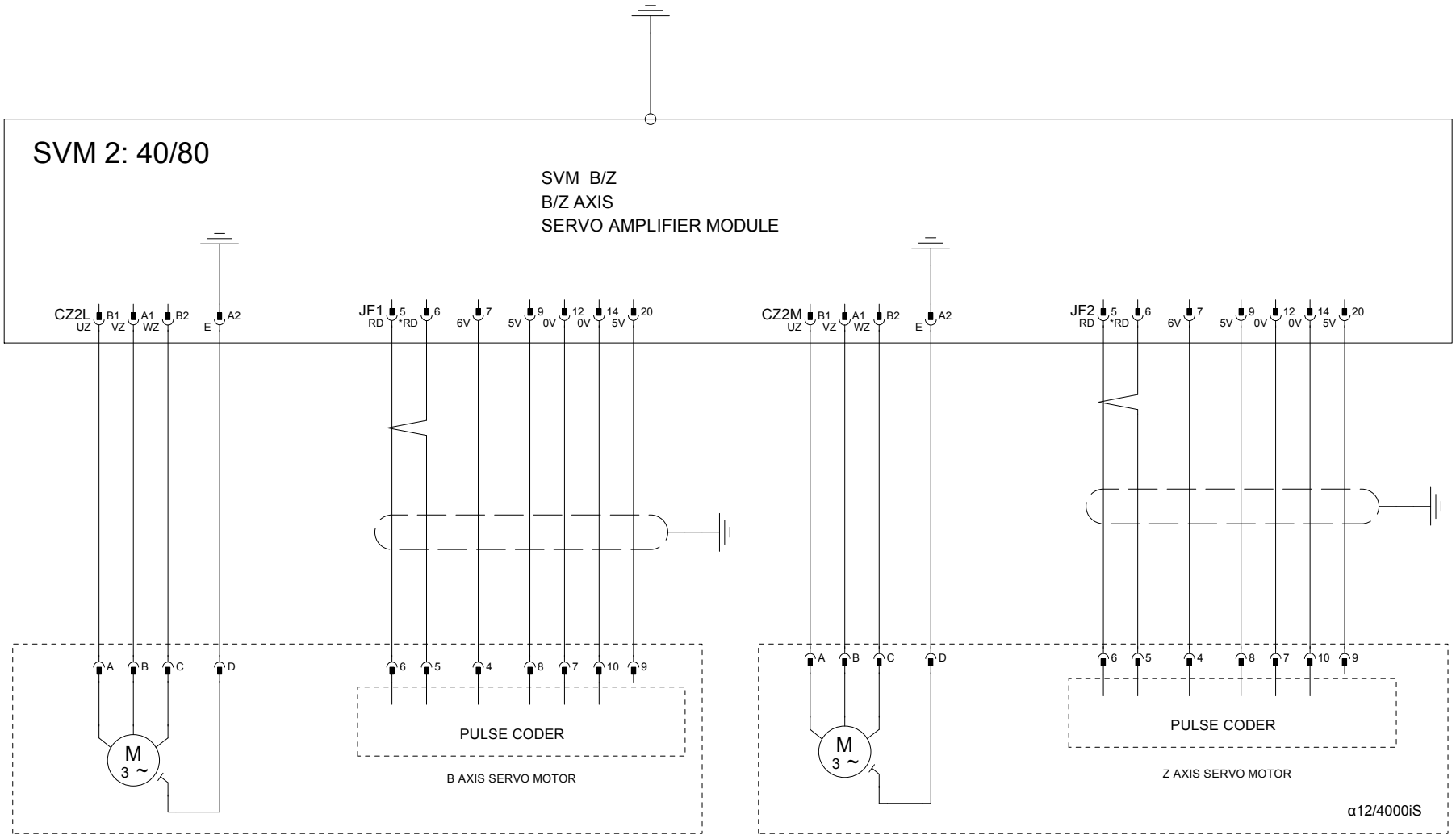
Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
B SERVO AMPLIFIER MODULE (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C17

SVM 2: 40/80

SVM B/Z
B/Z AXIS
SERVO AMPLIFIER MODULE

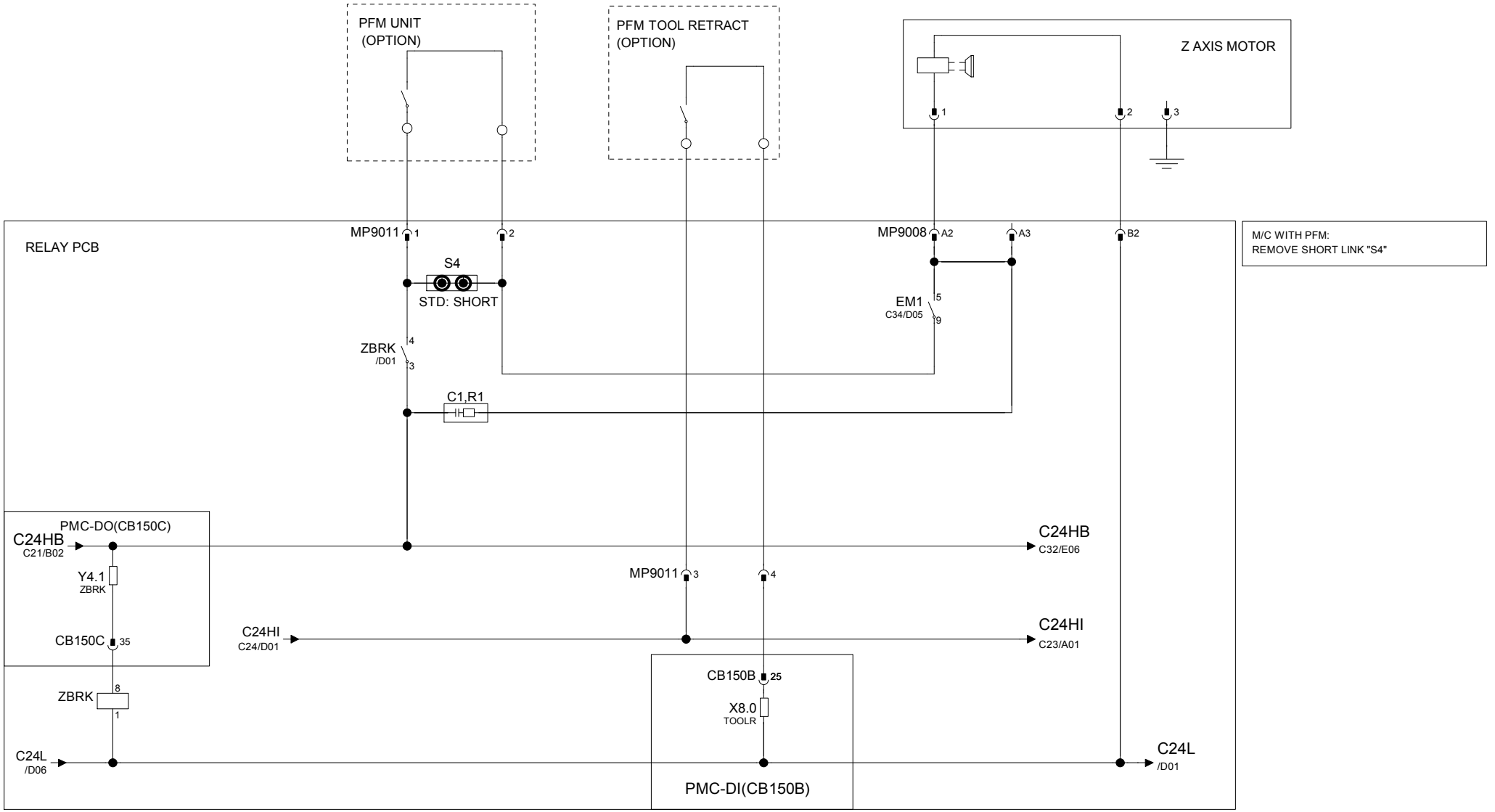


B AXIS MOTOR POWER	B AXIS MOTOR SIGNAL	Z AXIS MOTOR POWER	Z AXIS MOTOR SIGNAL

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
B AND Z SERVO AMPLIFIER MODULE (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C18

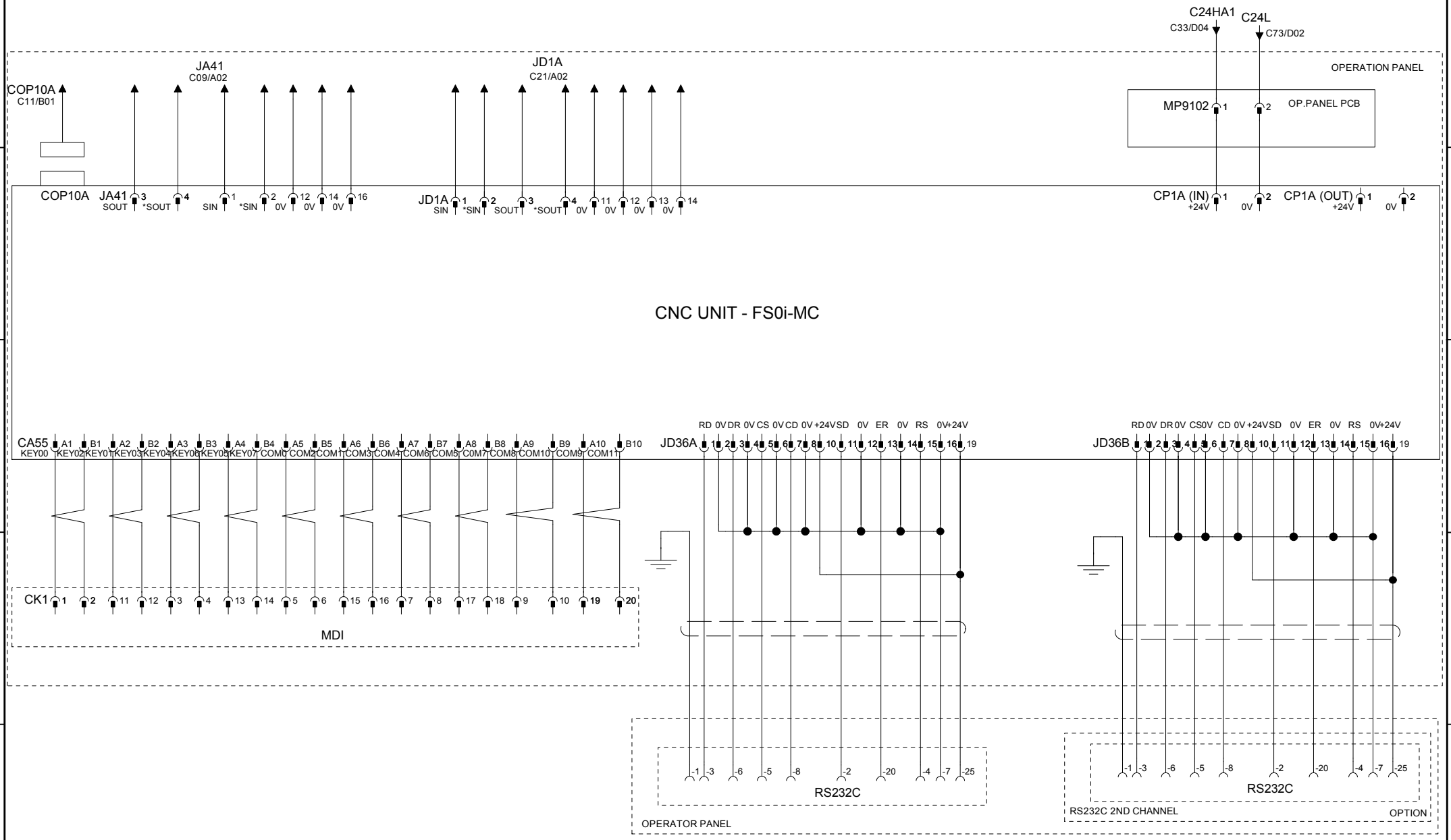


SERVO READY	Z-AXIS MOTOR BRAKE CONTROL	PFM TOOL RETRACT	Z AXIS MOTOR BRAKE

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
Z AXIS MOTOR BRAKE	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C19

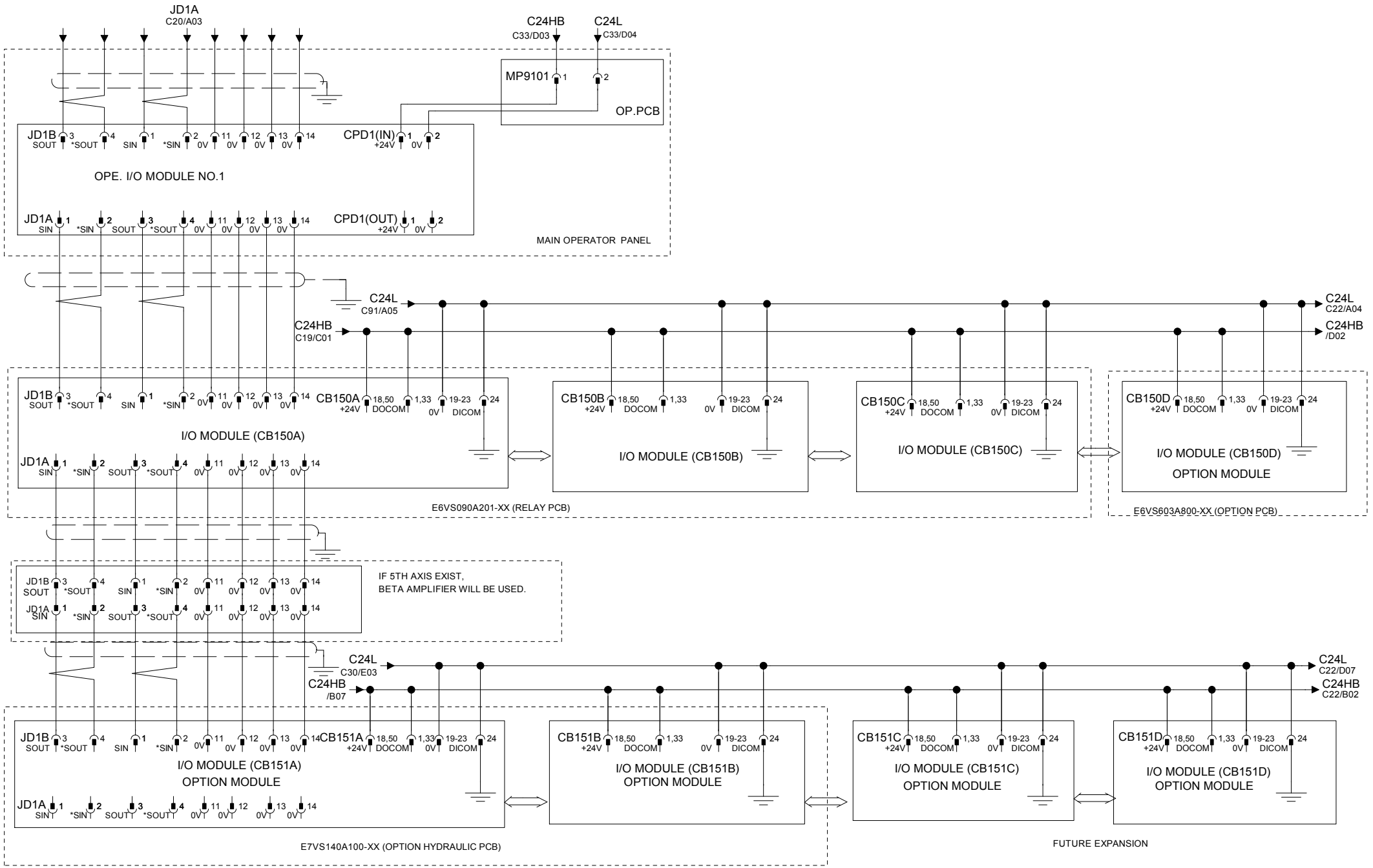


FS0i-MC CNC CONTROLLER

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONTROLLER UNIT	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C20

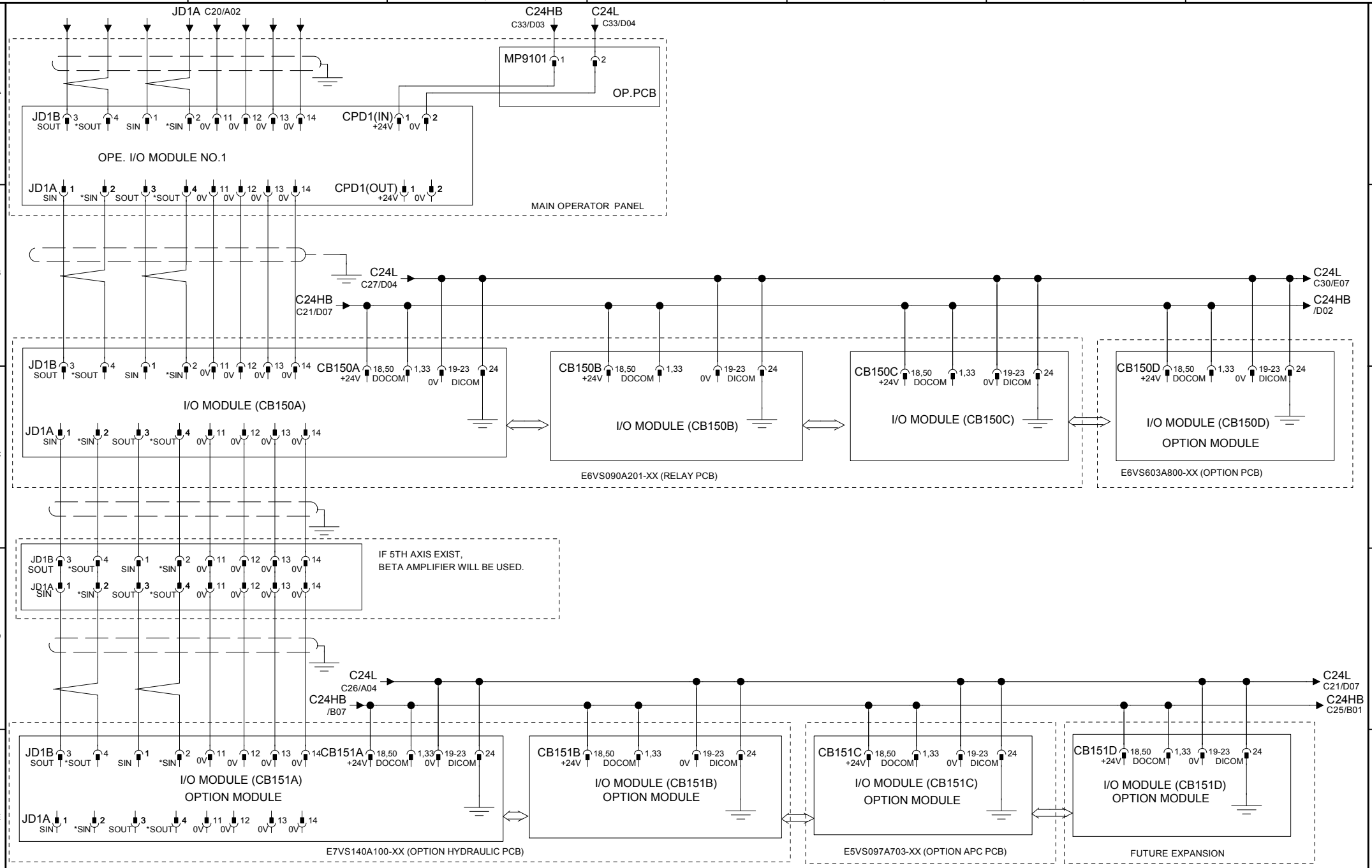


I/O MODULE POWER,LINK CONNECTION

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
I/O LINK CONNECTION (STD M/C)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C21

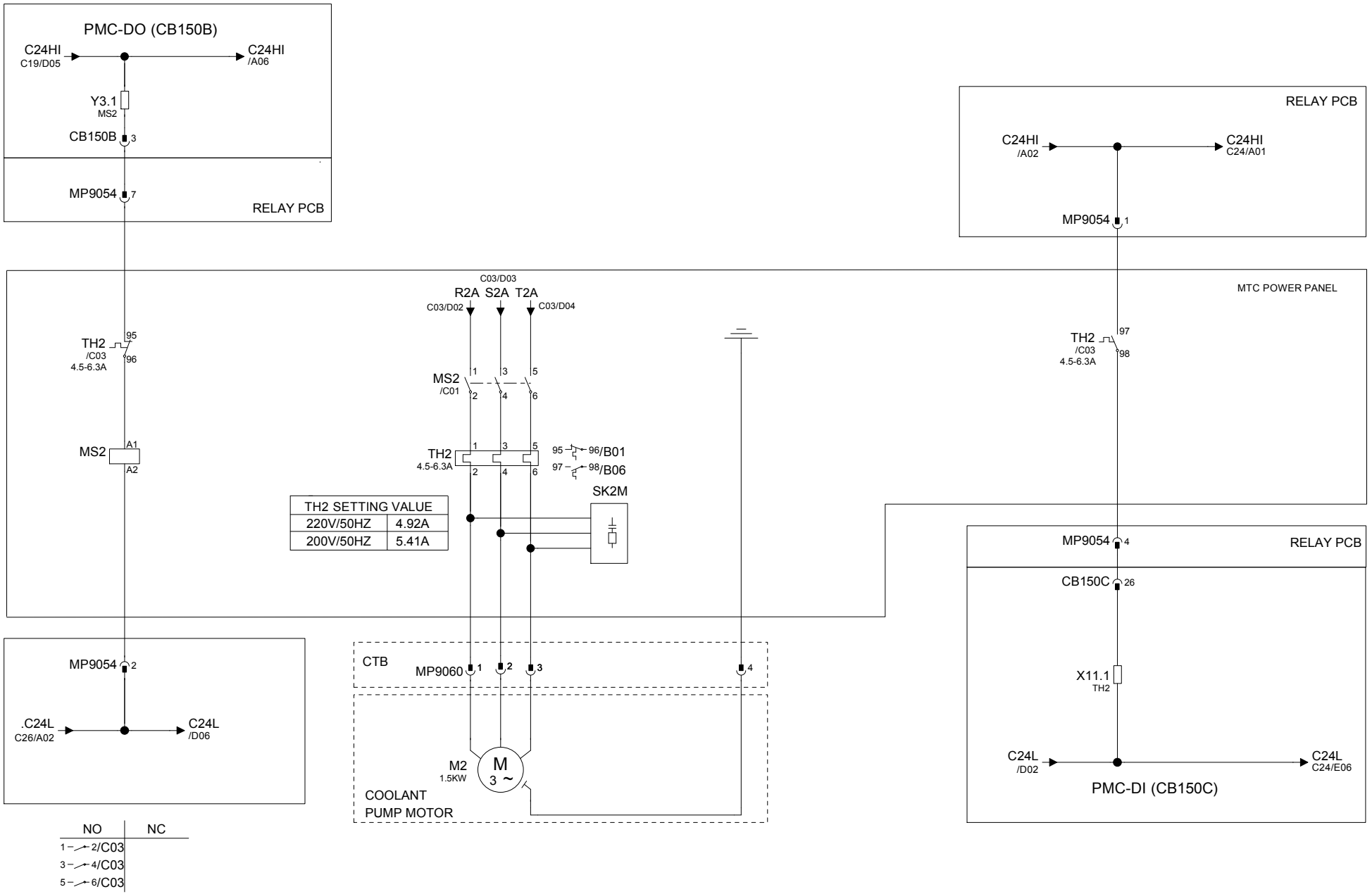


I/O MODULE POWER,LINK CONNECTION

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
I/O LINK CONNECTION (APC M/C)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C22

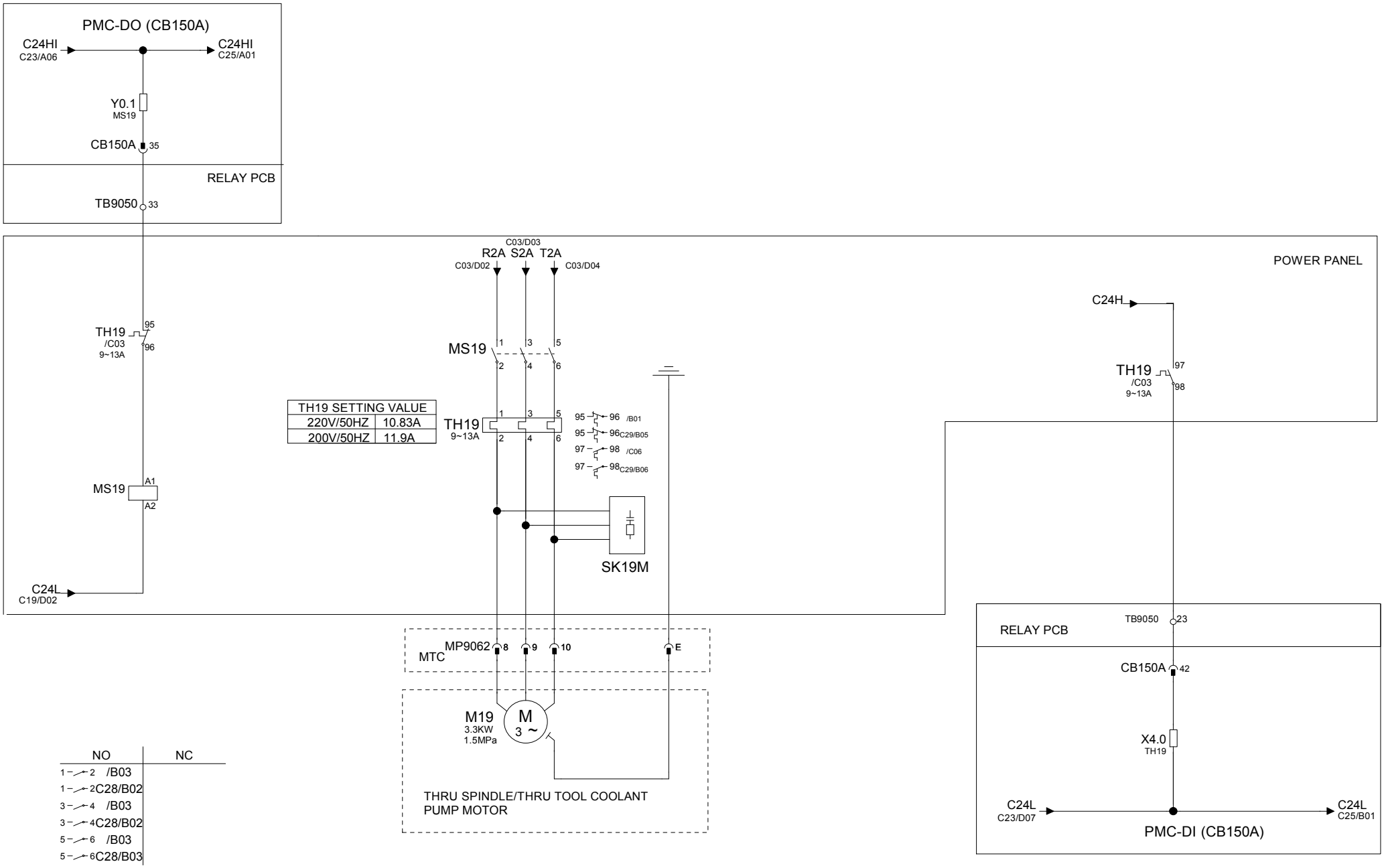


COOLANT CONTROL | COOLANT PUMP POWER | COOLANT THERMAL ALARM

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
COOLANT PUMP	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C23

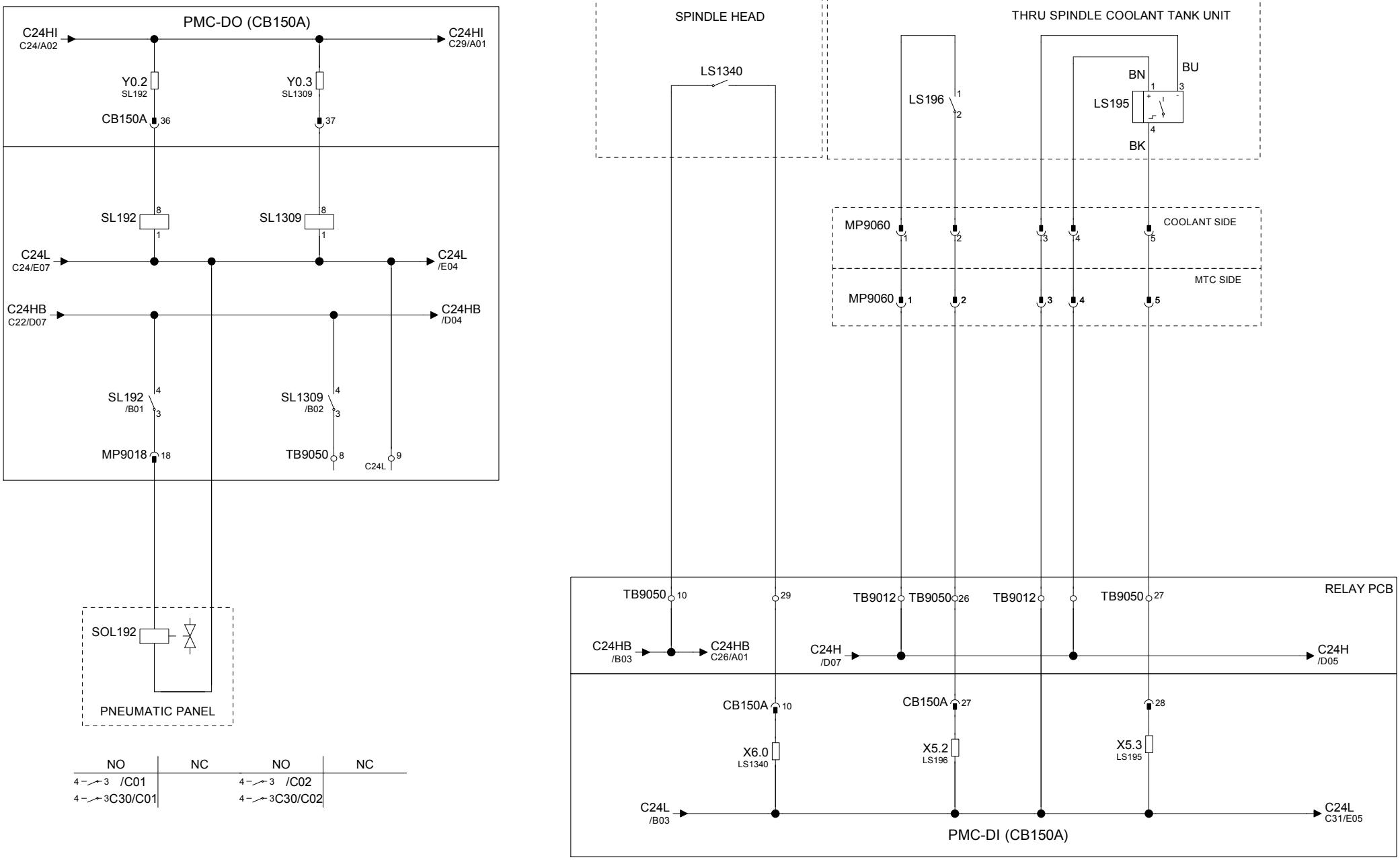


THRU SPINDLE COOLANT CONTROL	THRU SPINDLE COOLANT PUMP POWER 1.5 Mpa	THRU SPINDLE COOLANT THERMAL ALARM

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
THRU SPDL COOLANT 1.5MPa (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C24

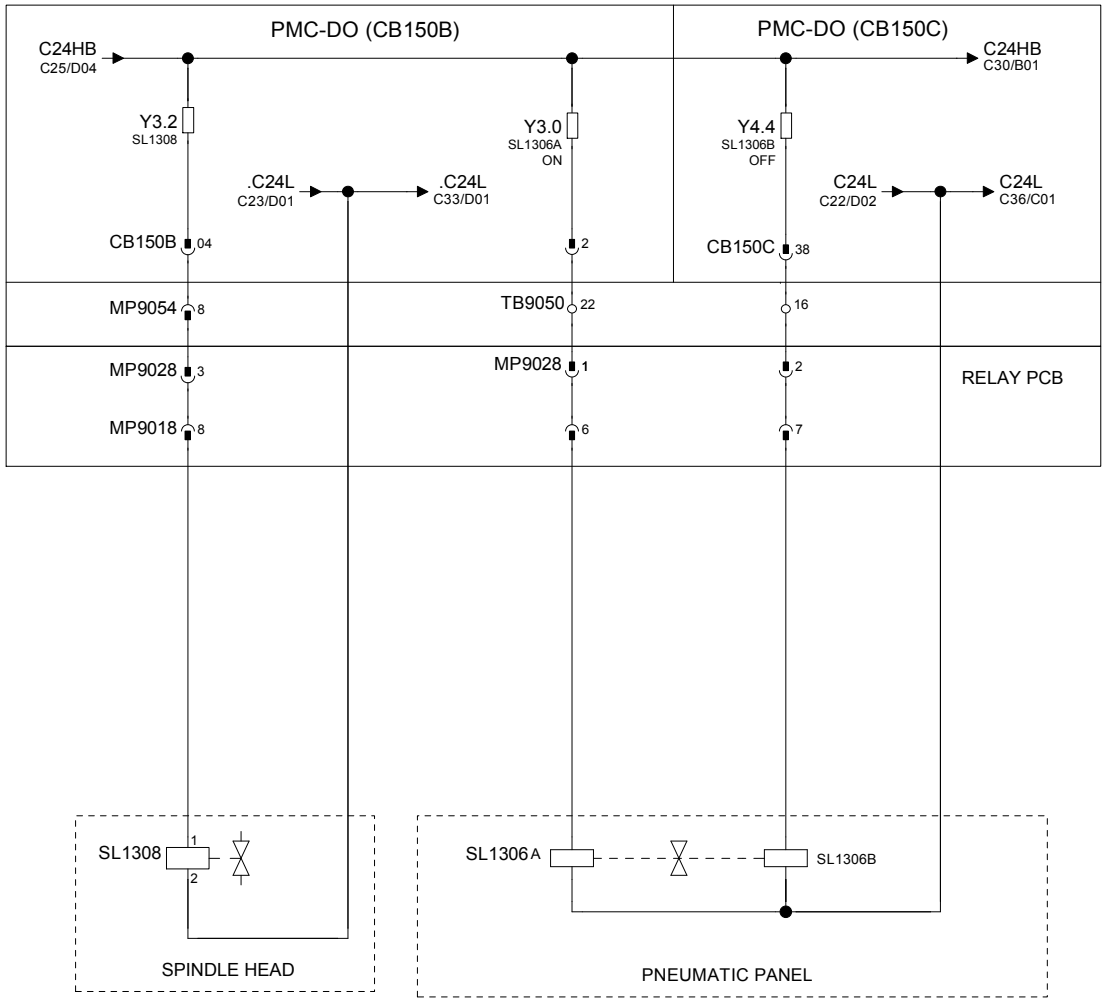


THRU SPINDLE AIR	ROTARY UNION WEAR & TEAR	TSC COOLANT LEVEL LOW	TSC FLOW SWITCH

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
THRU SPDL COOLANT SIGNAL 1.5MPa (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C25

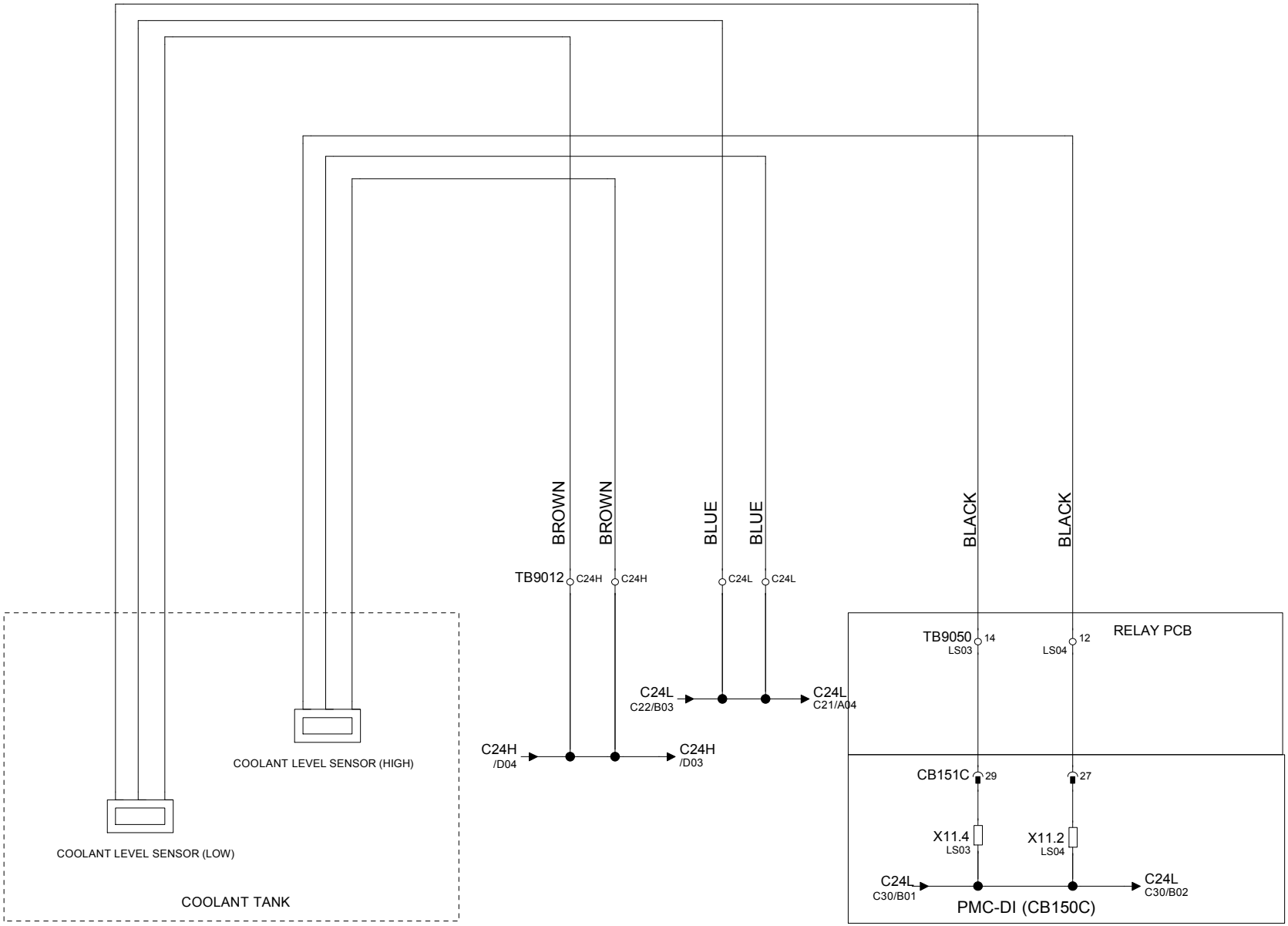


ROTARY UNION BACKUP AIR	COOLANT DRAWBACK ON	COOLANT DRAWBACK OFF

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
THRU SPDL COOLANT SIGNAL (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C26



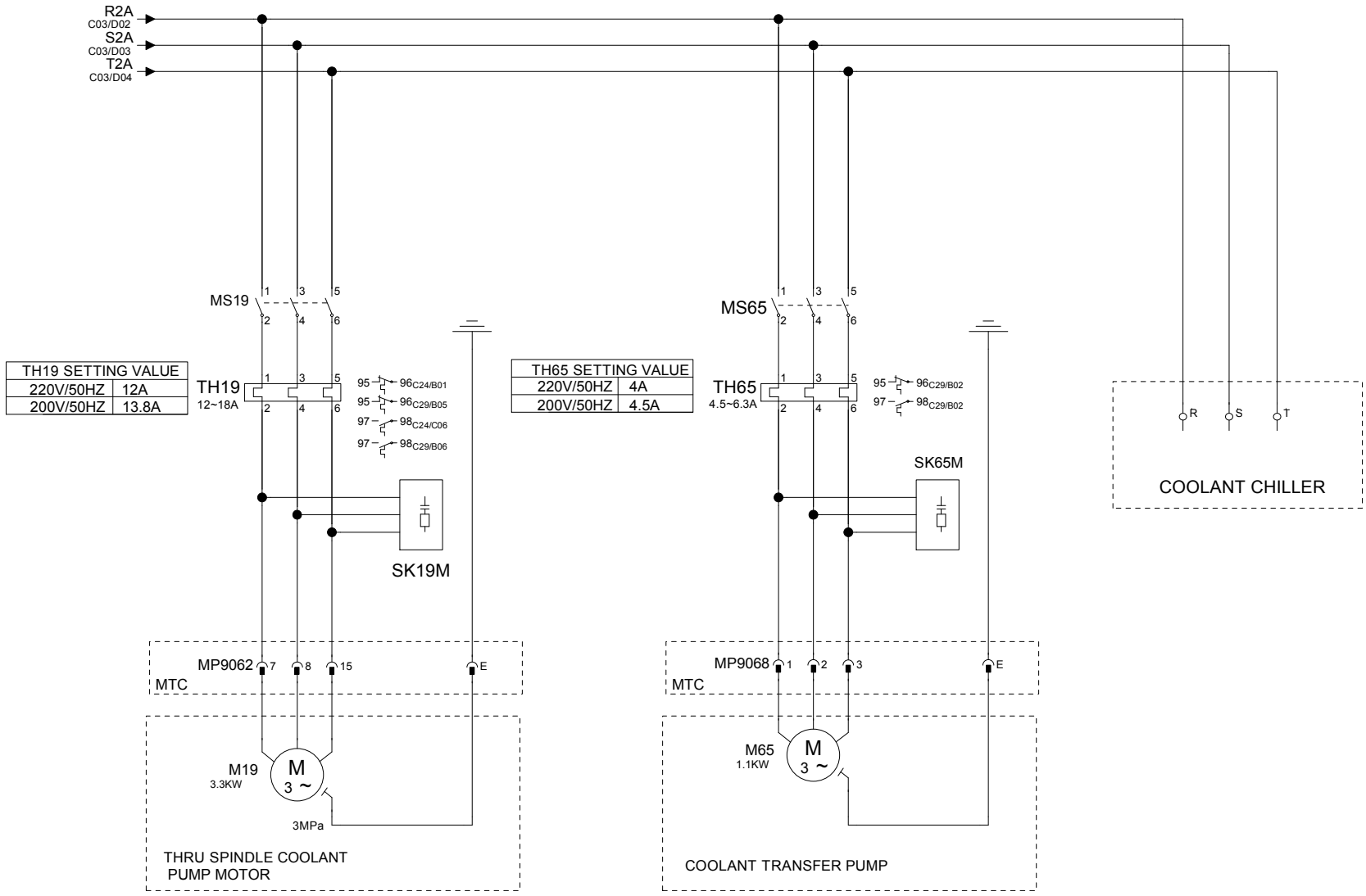
NOTE: HIGH & LOW LEVEL SENSOR IS AVAILABLE AS OPTION

COOLANT LEVEL SENSOR
LOW HIGH

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
COOLANT LEVEL SENSOR (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C27

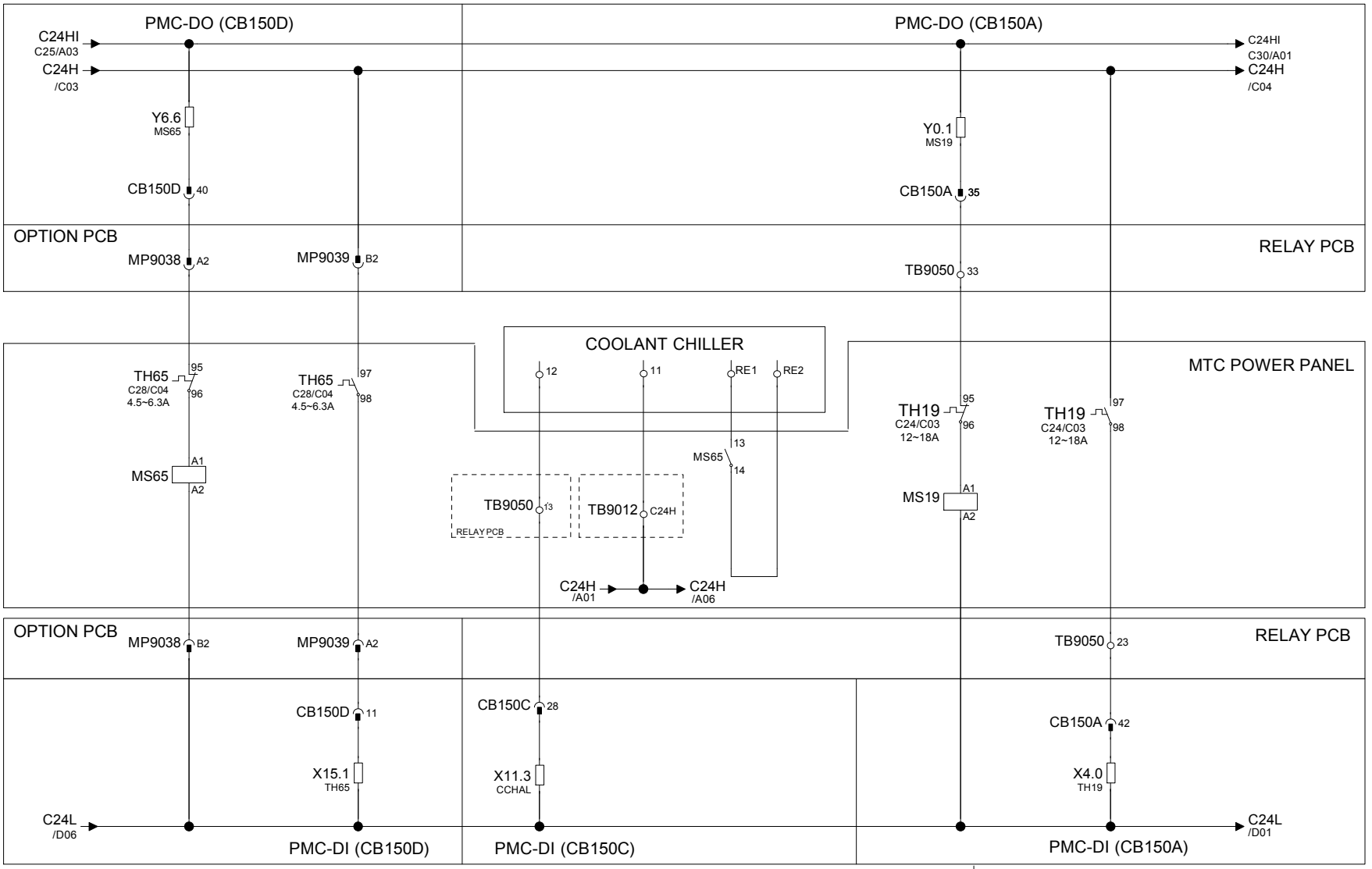


THRU SP COOLANT PUMP		COOLANT TRANSFER PUMP		COOLANT CHILLER

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
THRU SPDL COOLANT 3MPa (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C28

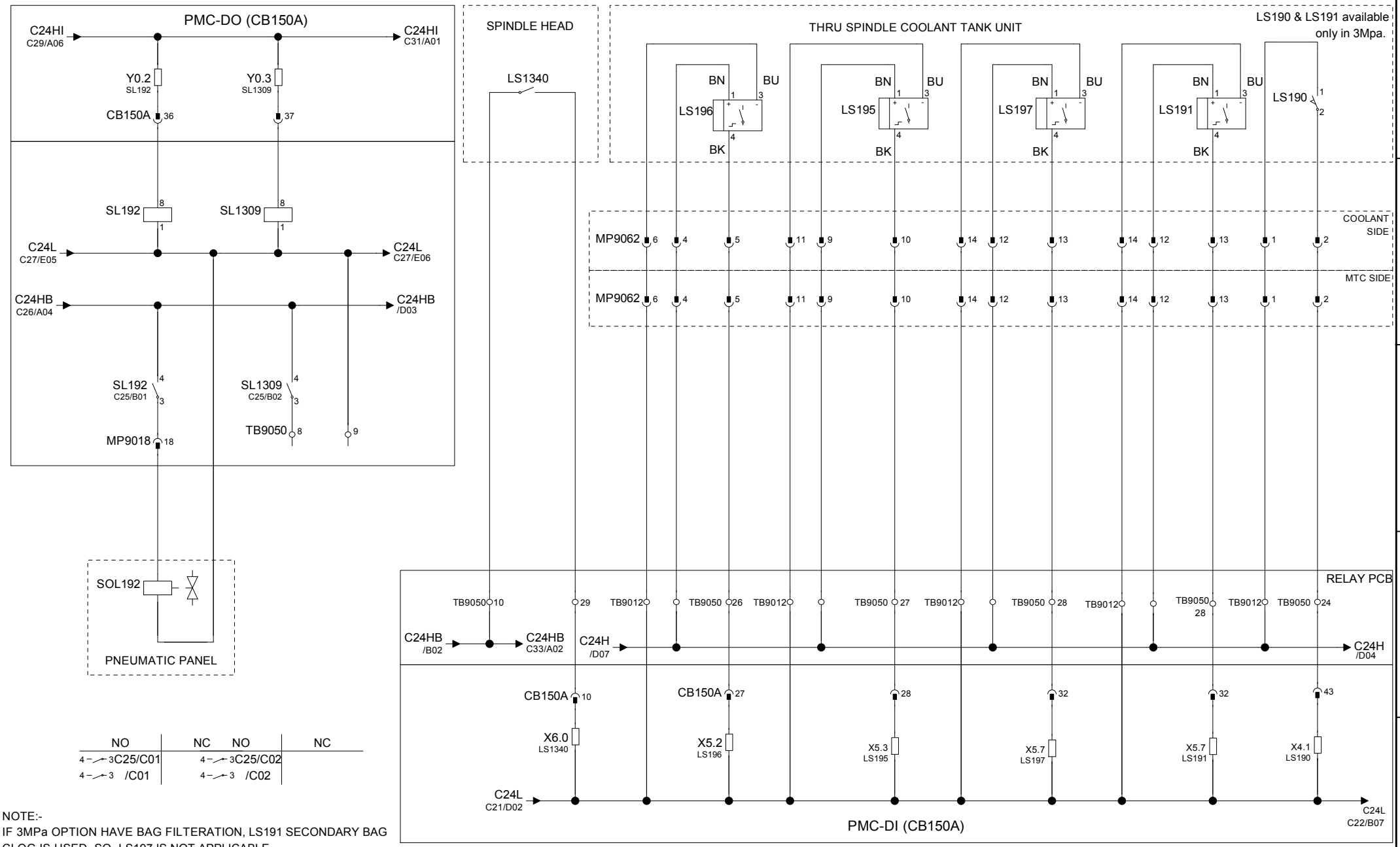


THRU SP COOLANT CONTROL	THRU SP COOLANT THERMAL TRIP	COOLANT CHILLER ALARM	COOLANT CHILLER CONTROL	COOLANT TRANSFER PUMP CONTROL	COOLANT TRANSFER PUMP THERMAL TRIP

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
THRU SPDL COOLANT CNTRL 3MPa (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C29



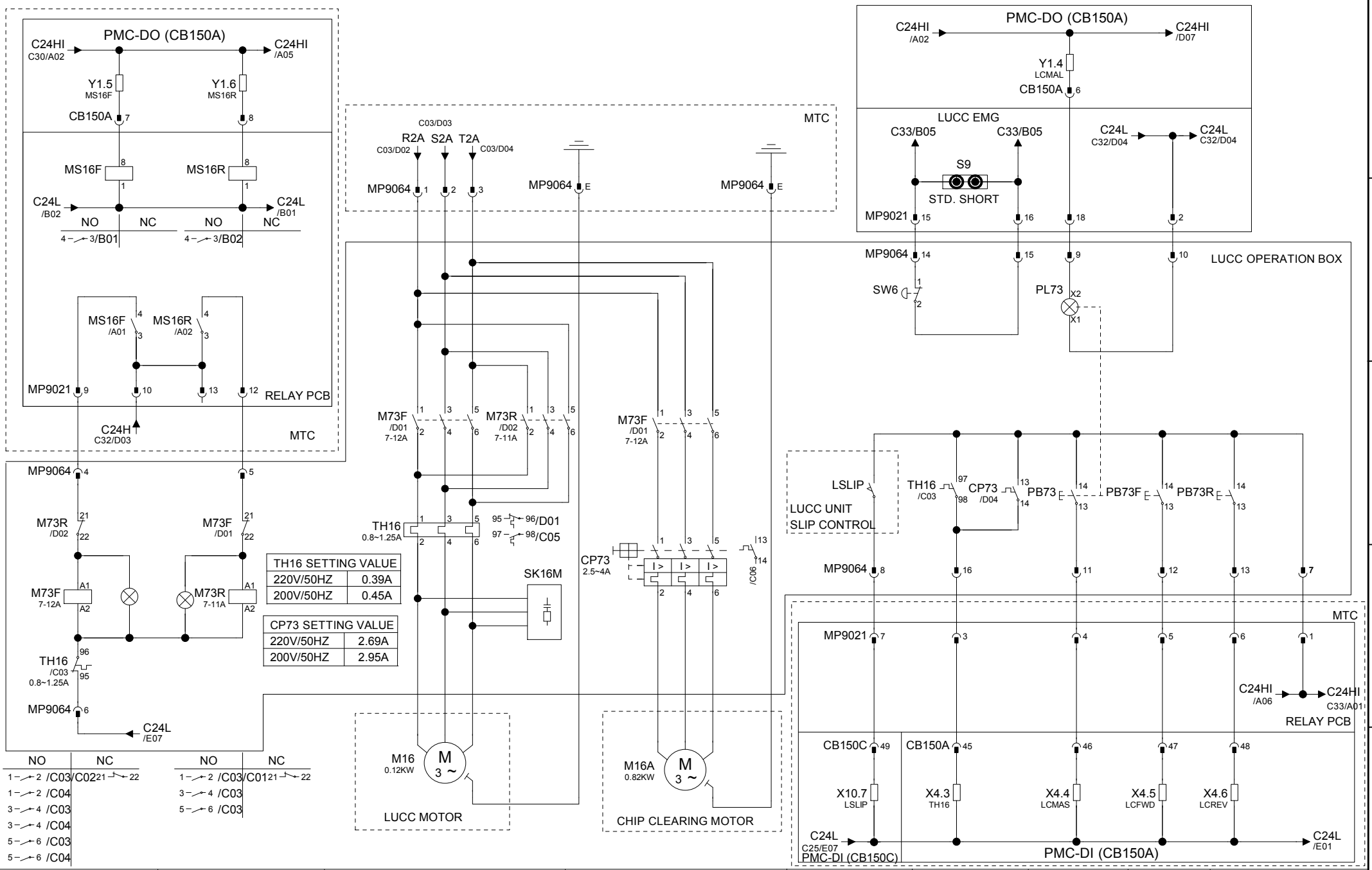
NOTE:-
IF 3MPa OPTION HAVE BAG FILTERATION, LS191 SECONDARY BAG CLOG IS USED. SO, LS197 IS NOT APPLICABLE.

THRU SPINDLE AIR		ROTARY UNION WEAR & TEAR	TSC COOLANT LEVEL LOW	TSC FLOW SWITCH	TSC COOLANT LEVEL HIGH	SECONDARY BAG FILTER CLOG	TSC SUCTION FILTER CLOG

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
THRU SPDL COOLANT SIGNAL 3MPa (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C30

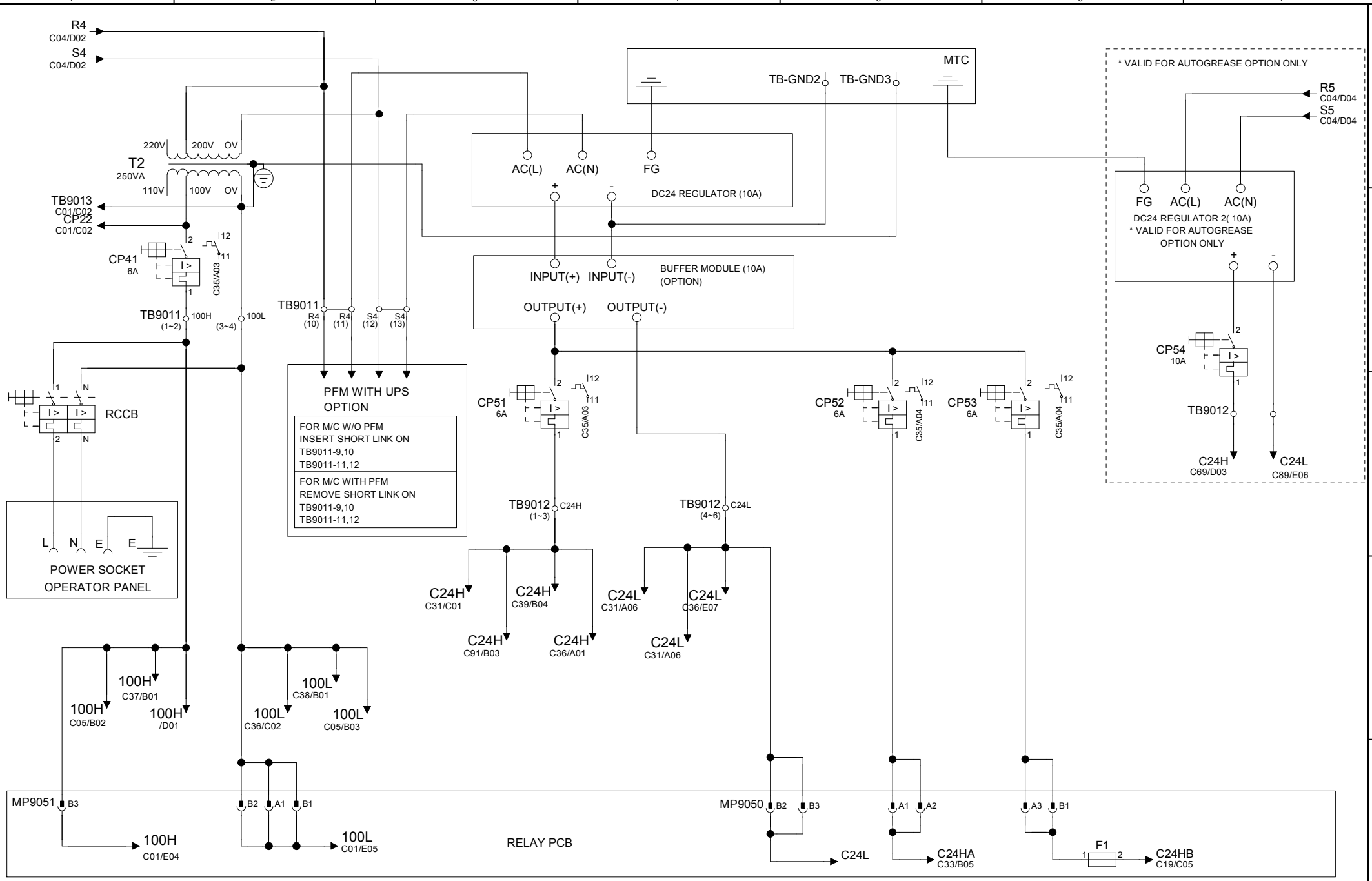


LIFT UP CHIP CONVEYOR FORWARD CONTROL	LIFT UP CHIP CONVEYOR REVERSE CONTROL	LIFT UP CHIP CONVEYOR MOTOR POWER	BACK WASH MOTOR POWER	LUCC SLIP CONTROL (OPTION)	LUCC THERMAL /COOLANT ALARM	LUCC MANUAL CONTROL	LUCC FORWARD	LUCC REVERSE

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
LIFT UP CHIP CONVEYOR (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C31

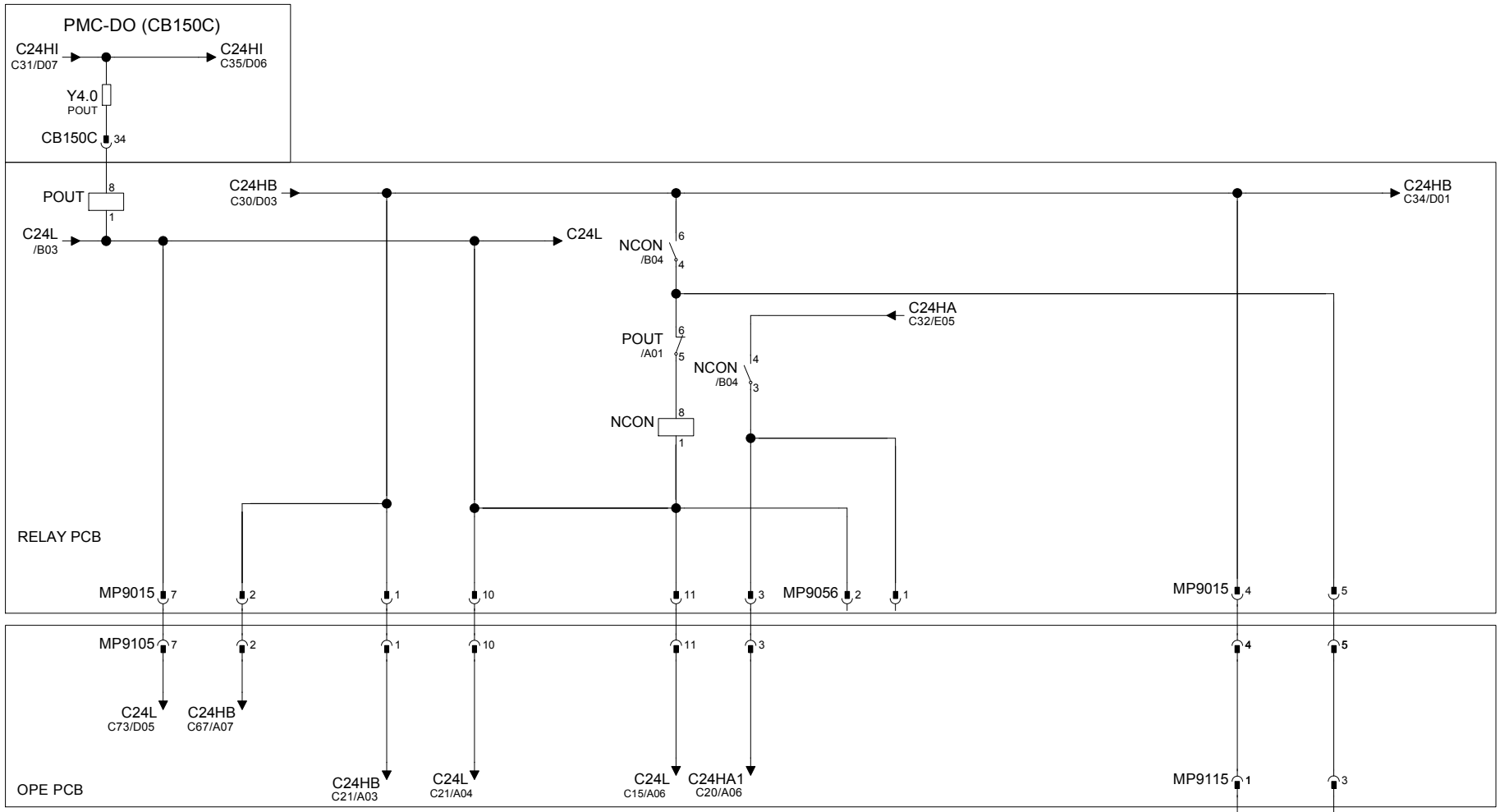


110V AC POWER SOCKET | 100V AC SOURCE, DISTRIBUTION | 24VDC SOURCE, DISTRIBUTION

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONTROL VOLTAGE SOURCE	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C32



NO	NC
3 – 4 C01/D04	/B04 6 – 5

NO	NC
4 – 3/B04	
6 – 4/B04	

PS1 IS A TWO CONTACTS SWITCH
NEXT AUXILLARY CONTACT IS LOCATED
AT PAGE C34/A01

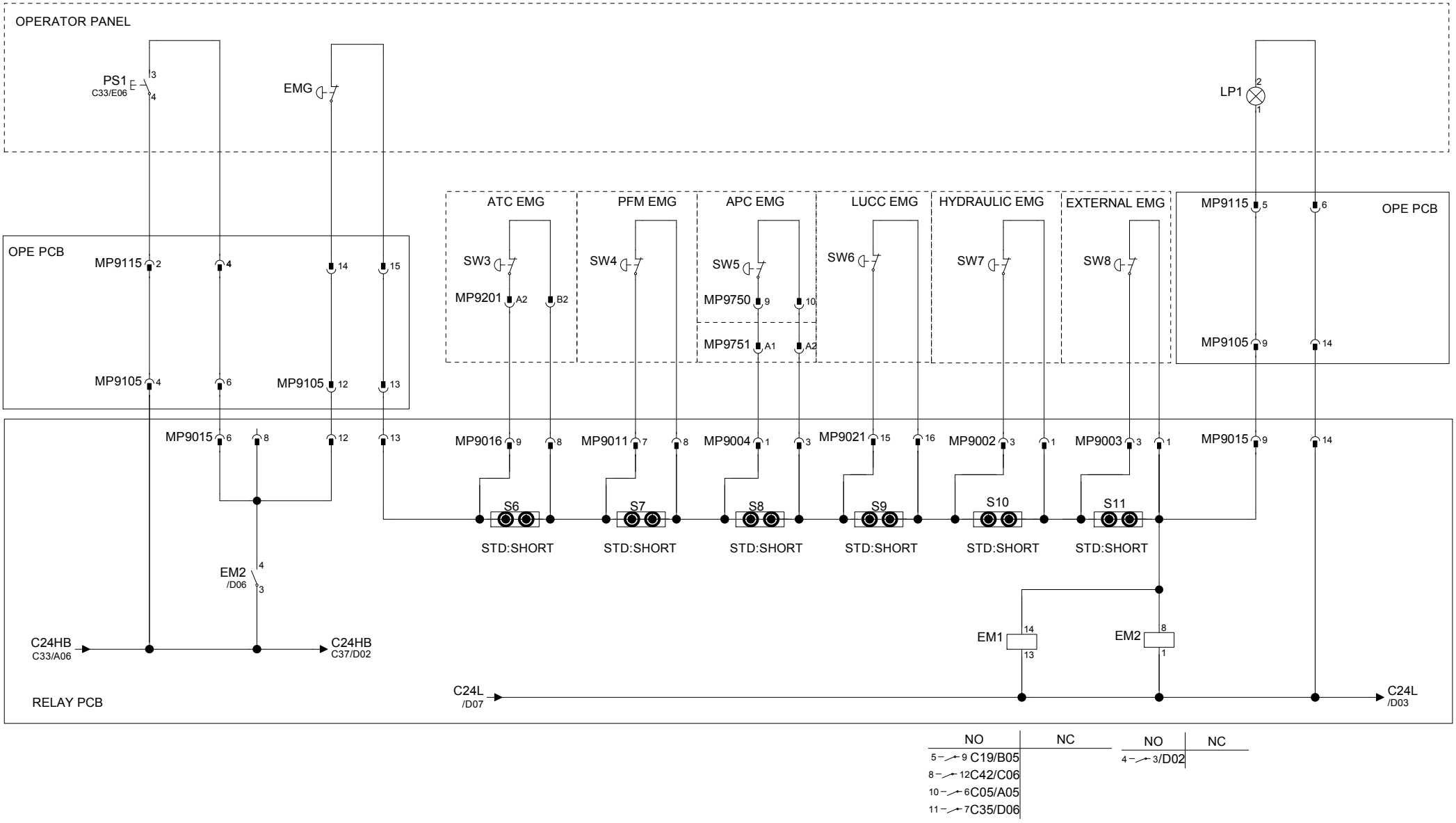
AUTO POWER OFF	OP. I/O MODULE POWER	NC POWER		NC ON CONTROL	

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
NC POWER ON CONTROL	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C33

PS1 IS A 2 CONTACTS SWITCH
NEXT AUXILLARY CONTACT IS
LOCATED AT PAGE C33/E06

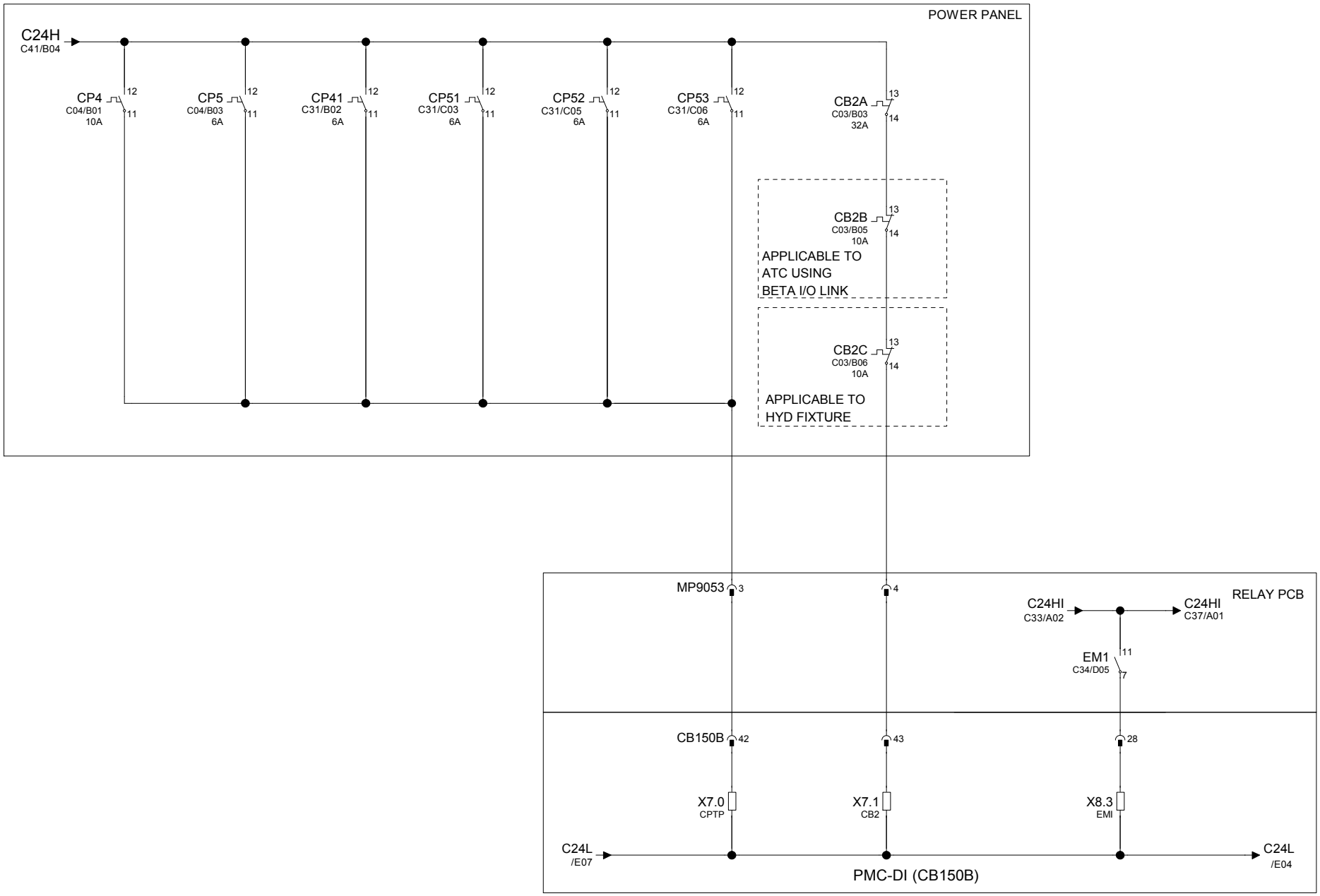


NC ON CONTROL	EMERGENCY SWITCH	EMERGENCY STOP CONTROL	MACHINE ON SIGNAL

Revisions		Date	Name
4			
3	APPROVED	25-NOV-2010	GOH WILLIAM
2	CHECKED	25-NOV-2010	S MAHESH
1	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
EMERGENCY CIRCUIT	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C34

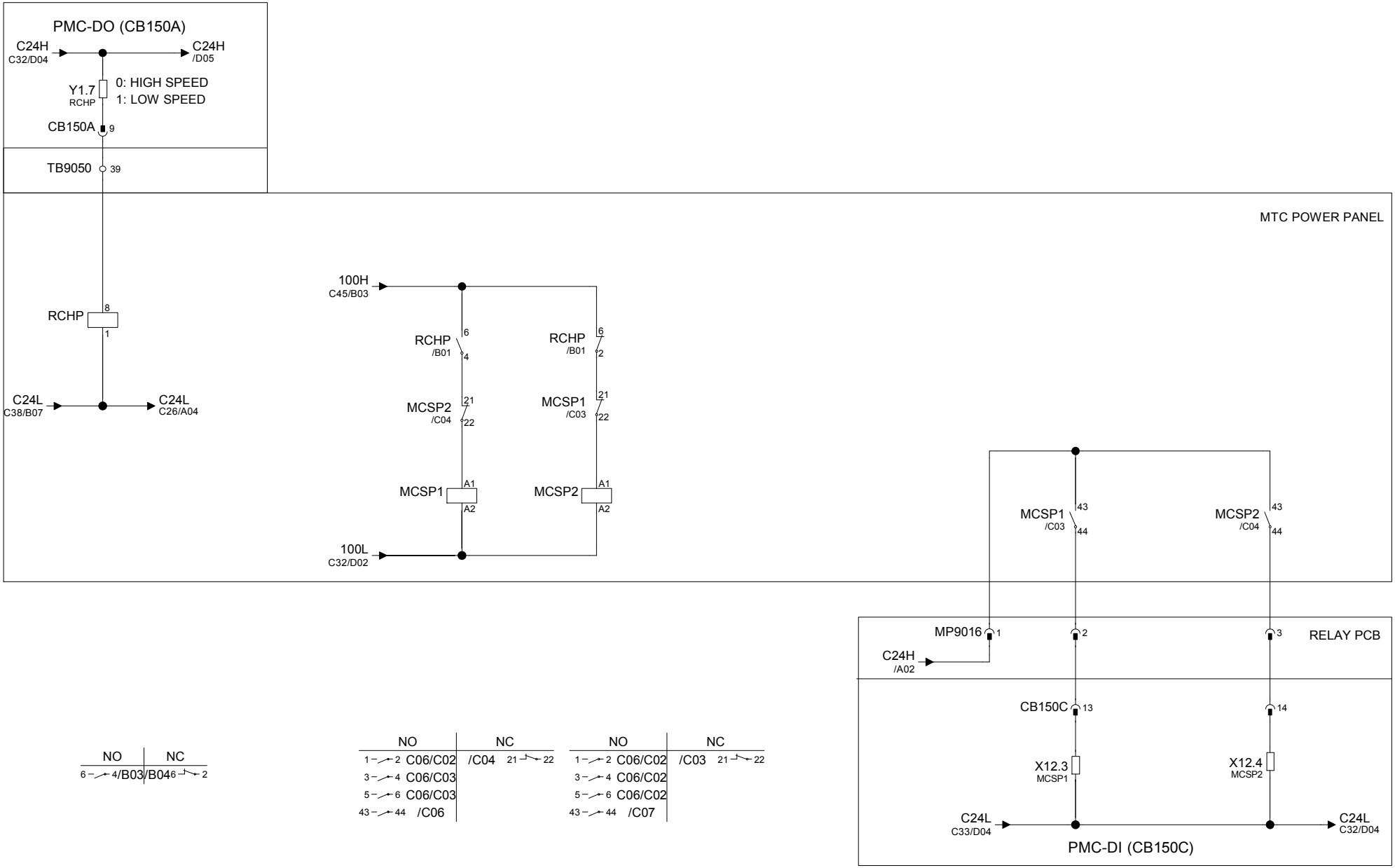


INSIDE CABINET PROTECTION TRIP	ASSIST EQUIPMENT POWER LINE BREAKER	EMERGENCY STOP

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CB OVERLOAD CONTROL	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C35

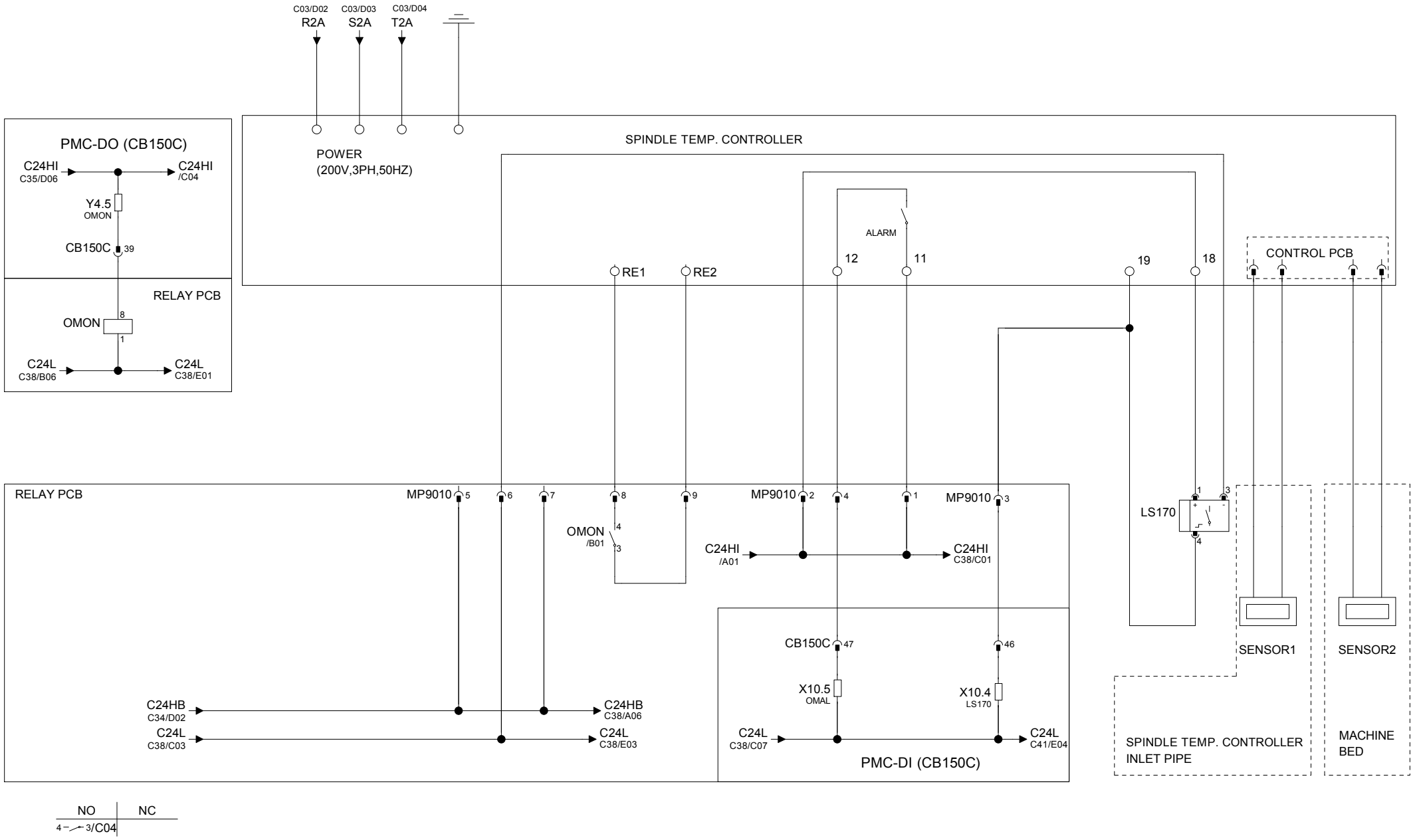


SPINDLE SPEED CONTROL	LOW SPEED SELECTION	HIGH SPEED SELECTION		SPINDLE LOW SPEED CONFIRM	SPINDLE HI SPEED CONFIRM	

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
SPINDLE SWITCHING CONTROL	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C36

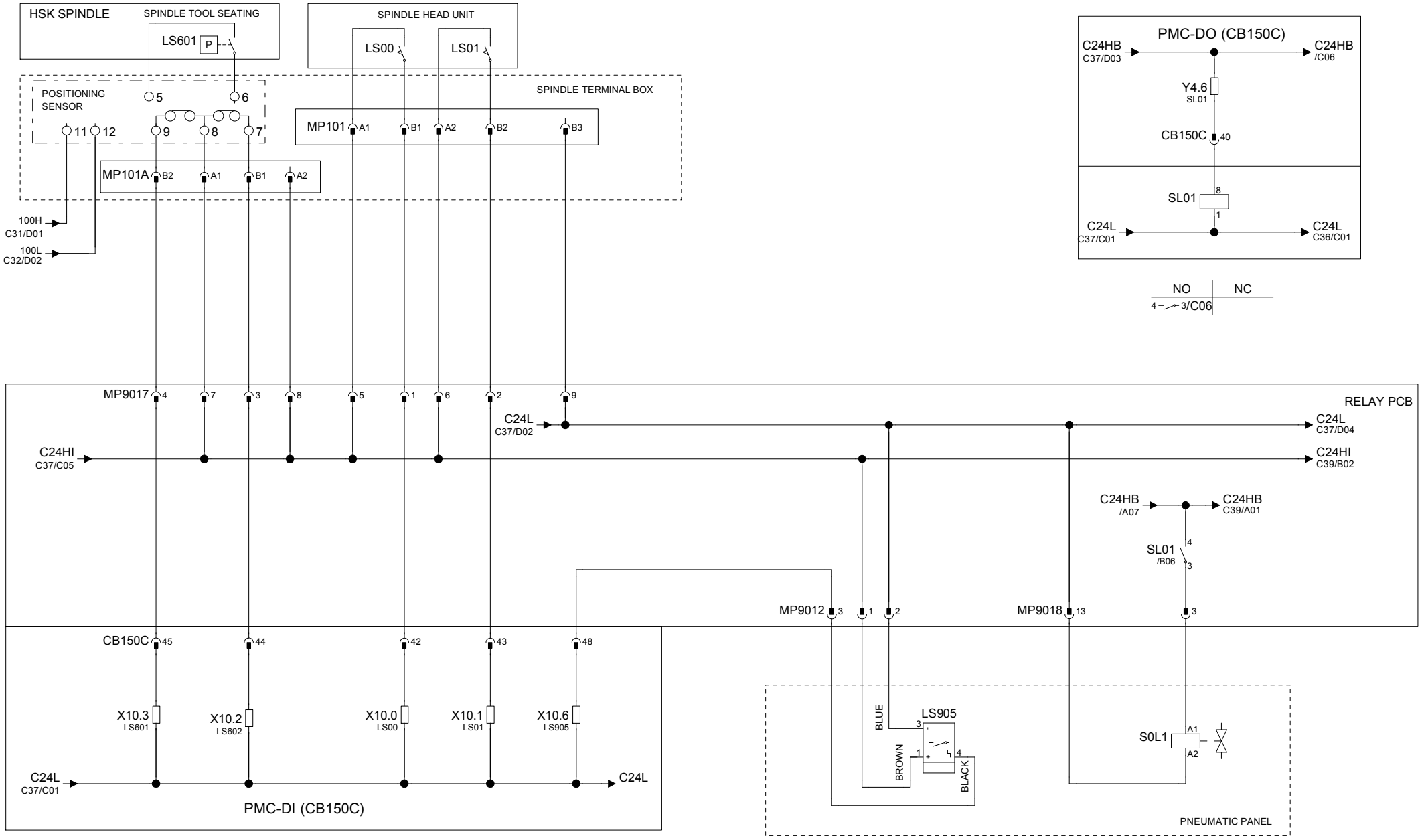


OILMATIC ON CONTROL	INCOMING POWER	SPINDLE TEMP CONTROLLER ALARM	SPINDLE OIL FLOW CONTROL
		OIL TEMP SENSOR	AMBIENT TEMP SENSOR

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
SPINDLE TEMPERATURE CONTROLLER	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C37

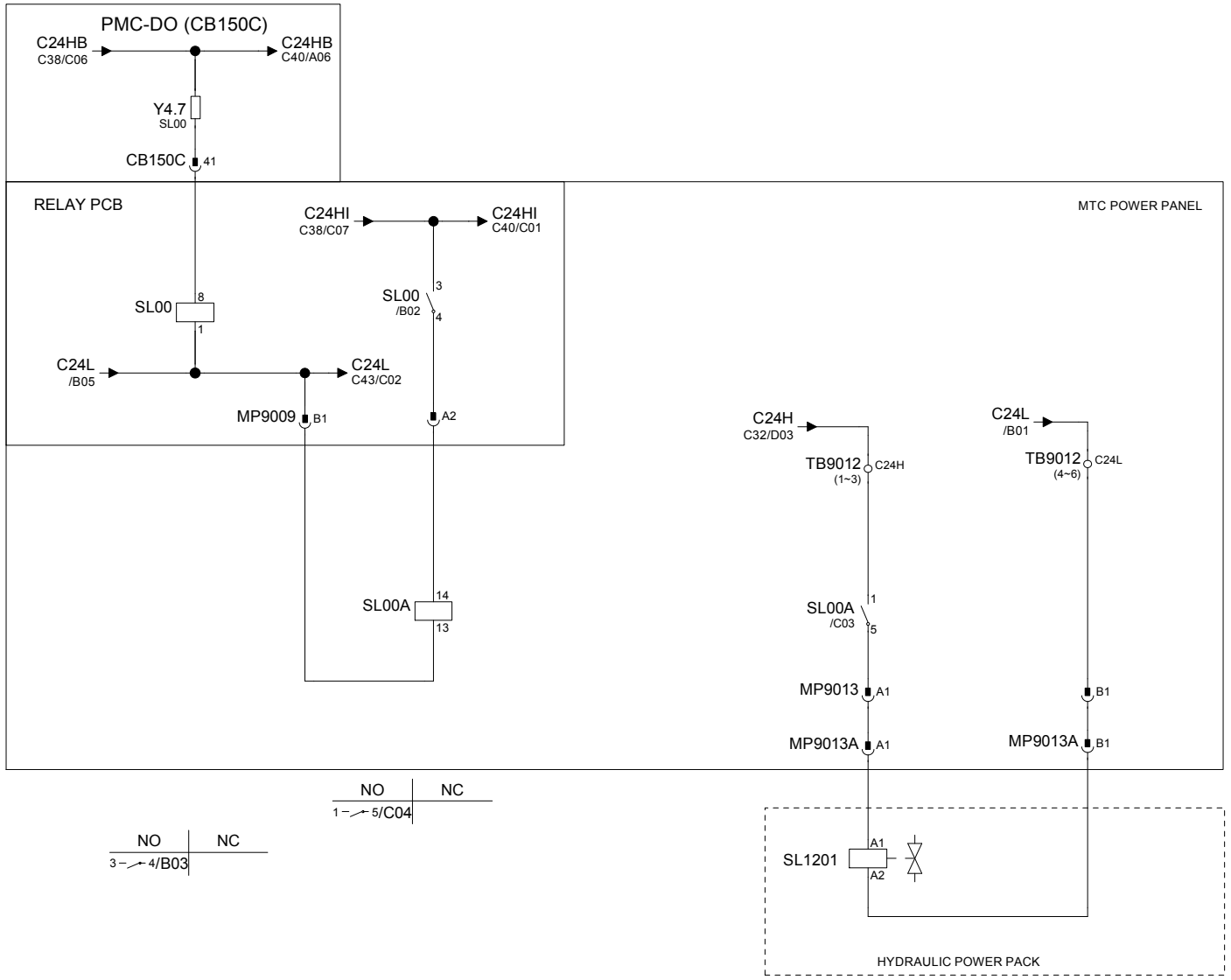


SP.TOOL (HSK TYPE ONLY) SEAT CHECK	SP. TOOL(HSK TYPE ONLY) EXIST CHECK	TOOL CLAMP CONFIRM	TOOL UNCLAMP CONFIRM		AIR PRESSURE LS	SP.TAPER CLEANING AIR CONTROL	

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
HSK TOOL SEAT CHECK (HSK-A50)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C38



TOOL UNCLAMP CONTROL

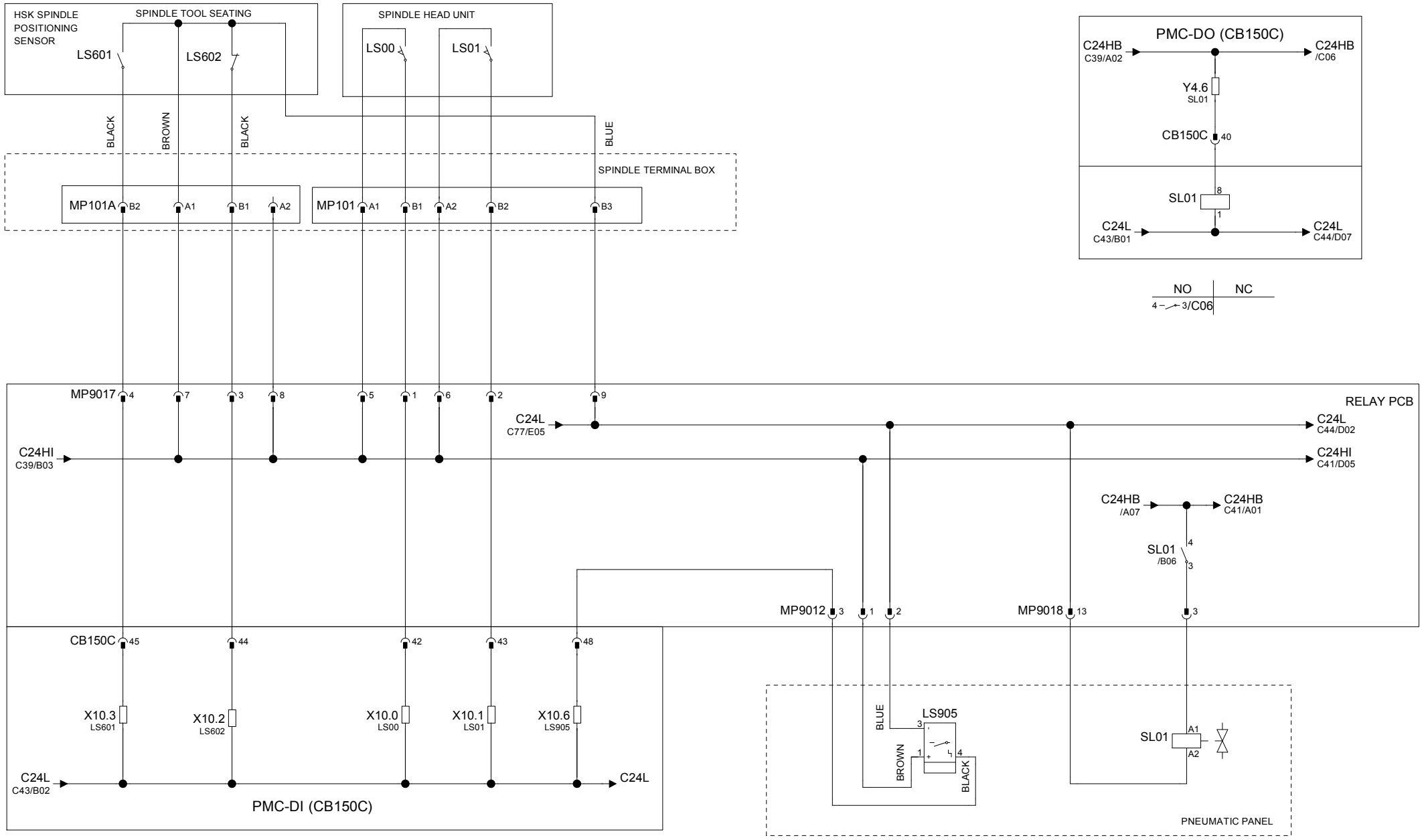
TOOL UNCLAMP SOLENOID

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO

MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
TOOL CLAMP/UNCLAMP	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C39

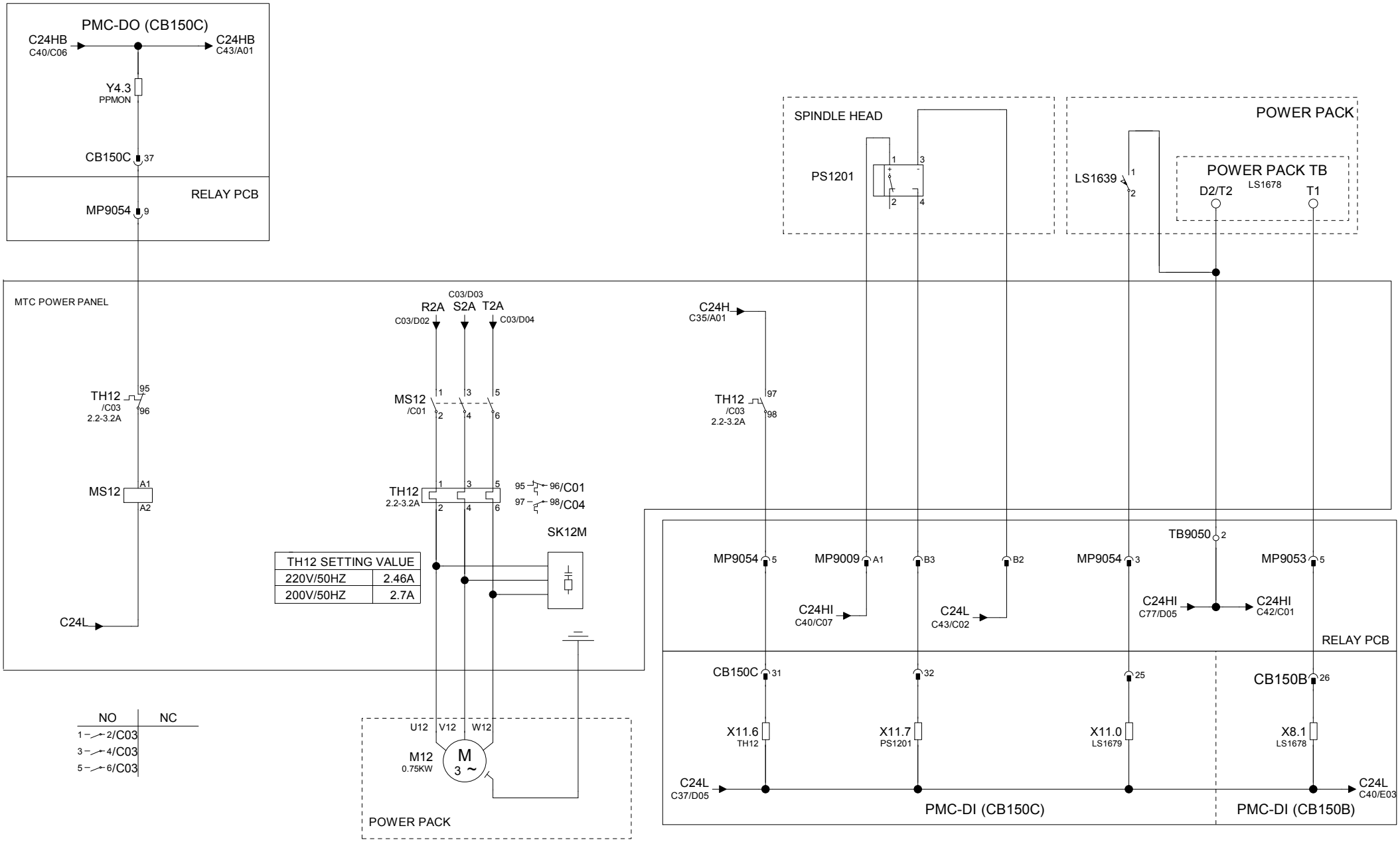


SP.TOOL (HSK TYPE ONLY) SEAT CHECK	SP. TOOL(HSK TYPE ONLY) EXIST CHECK	TOOL CLAMP CONFIRM	TOOL UNCLAMP CONFIRM	AIR PRESSURE SWITCH		AIR PRESSURE LS	SP.TAPER CLEANING AIR CONTROL	

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
HSK TOOL SEAT CHECK (HSK-A40)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C40

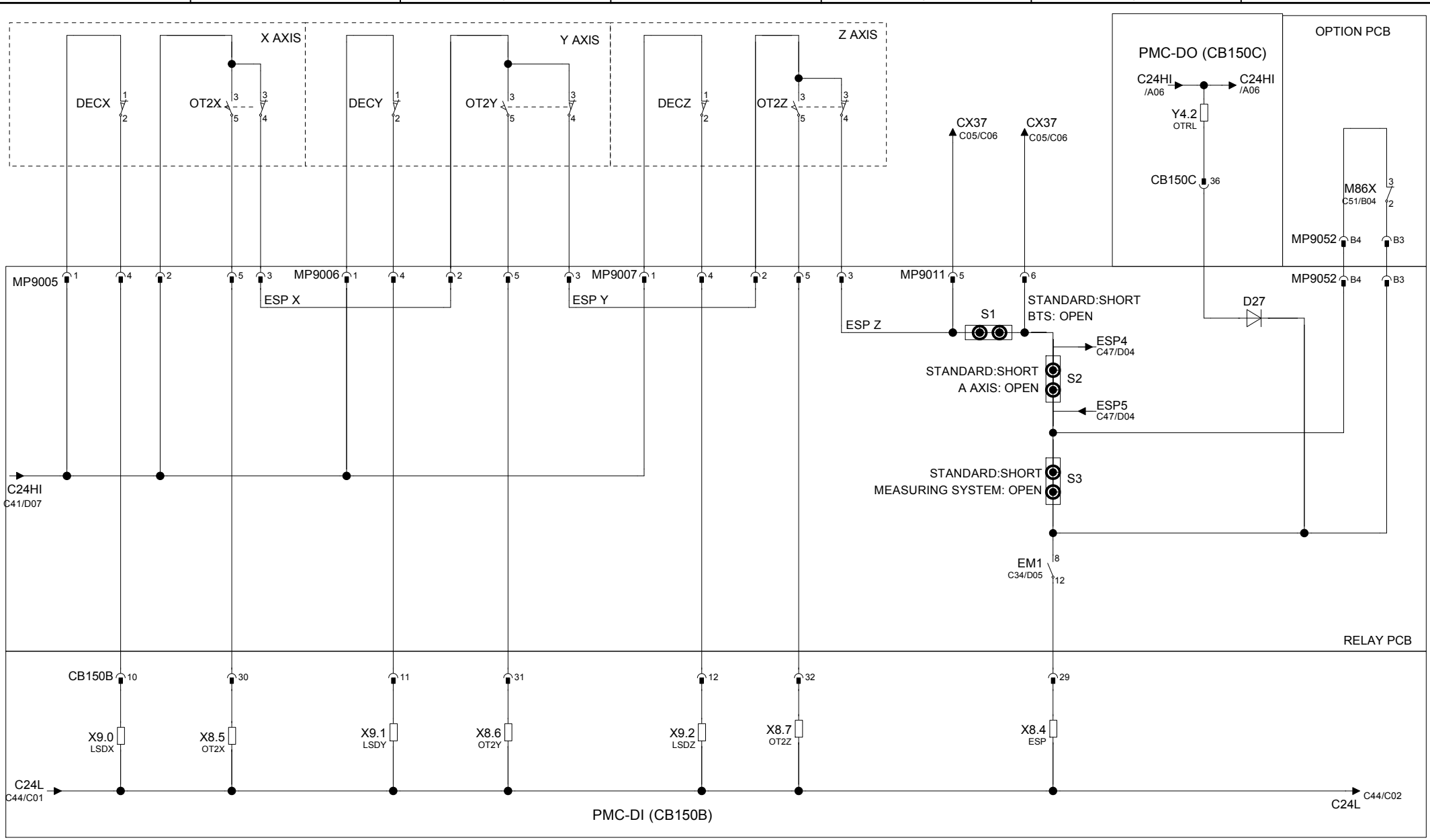


POWER PACK MOTOR ON	POWER PACK PUMP	POWER PACK ALARM	POWER PACK PRESSURE SW	POWER PACK OIL LEVEL LOW	POWER PACK TEMP. ALARM

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
POWER PACK CONTROL	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C41

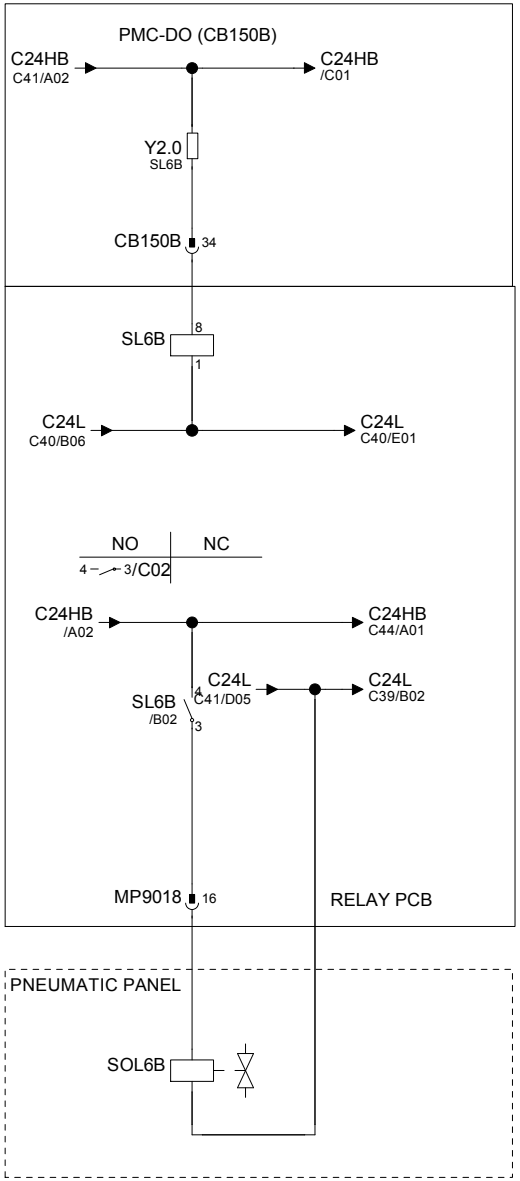


REFERENCE RETURN SLOWDOWN - X AXIS	OVERTRAVEL X-AXIS	REFERENCE RETURN SLOWDOWN - Y AXIS	OVERTRAVEL Y-AXIS	REFERENCE RETURN SLOWDOWN - Z AXIS	OVERTRAVEL Z-AXIS	EMERGENCY STOP SIGNAL	OT RELEASE

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
LS REFERENCE AND OVERTRAVEL	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C42

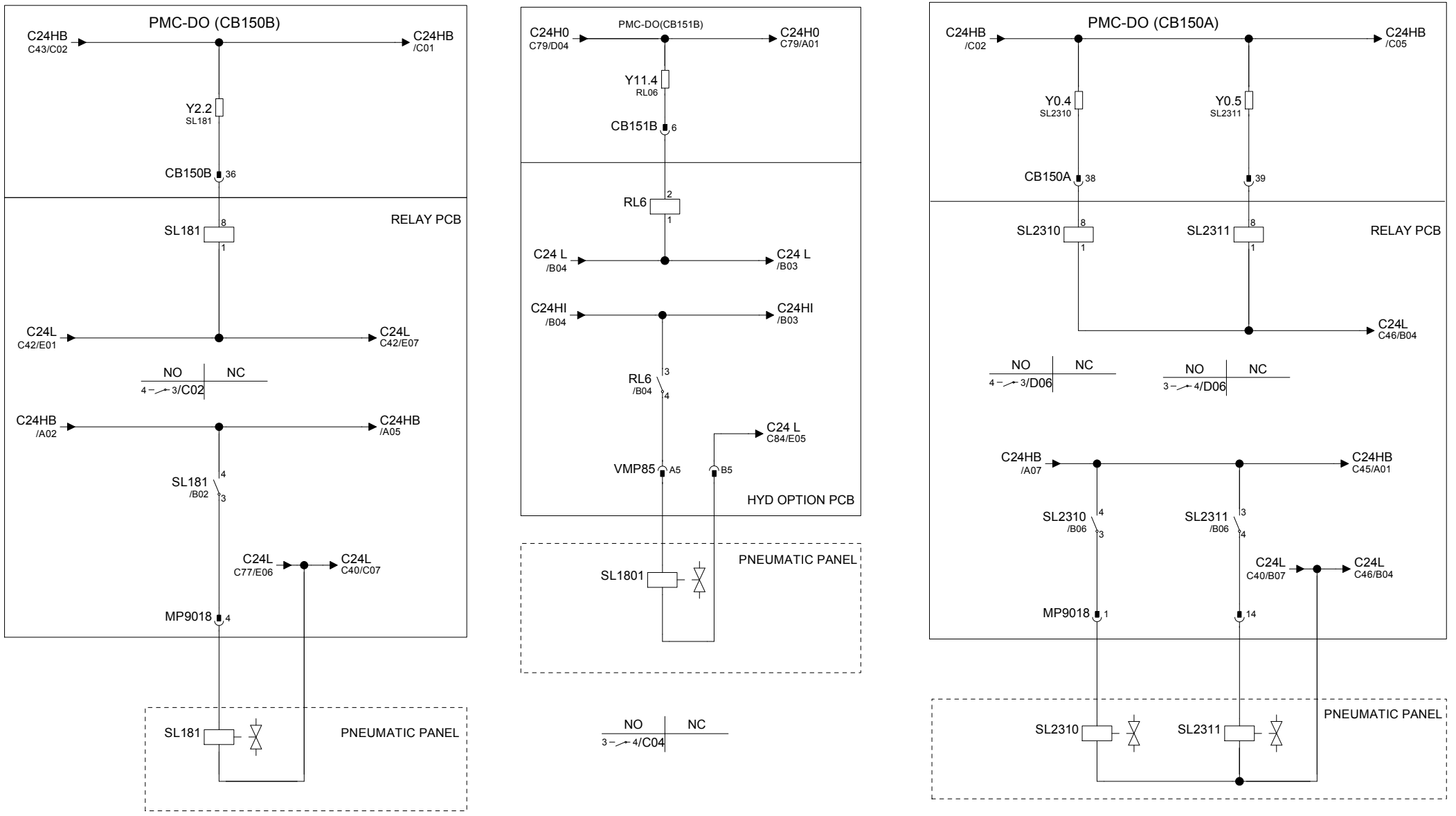


ATC MAGAZINE AIR

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
ATC MAGAZINE AIR	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C43

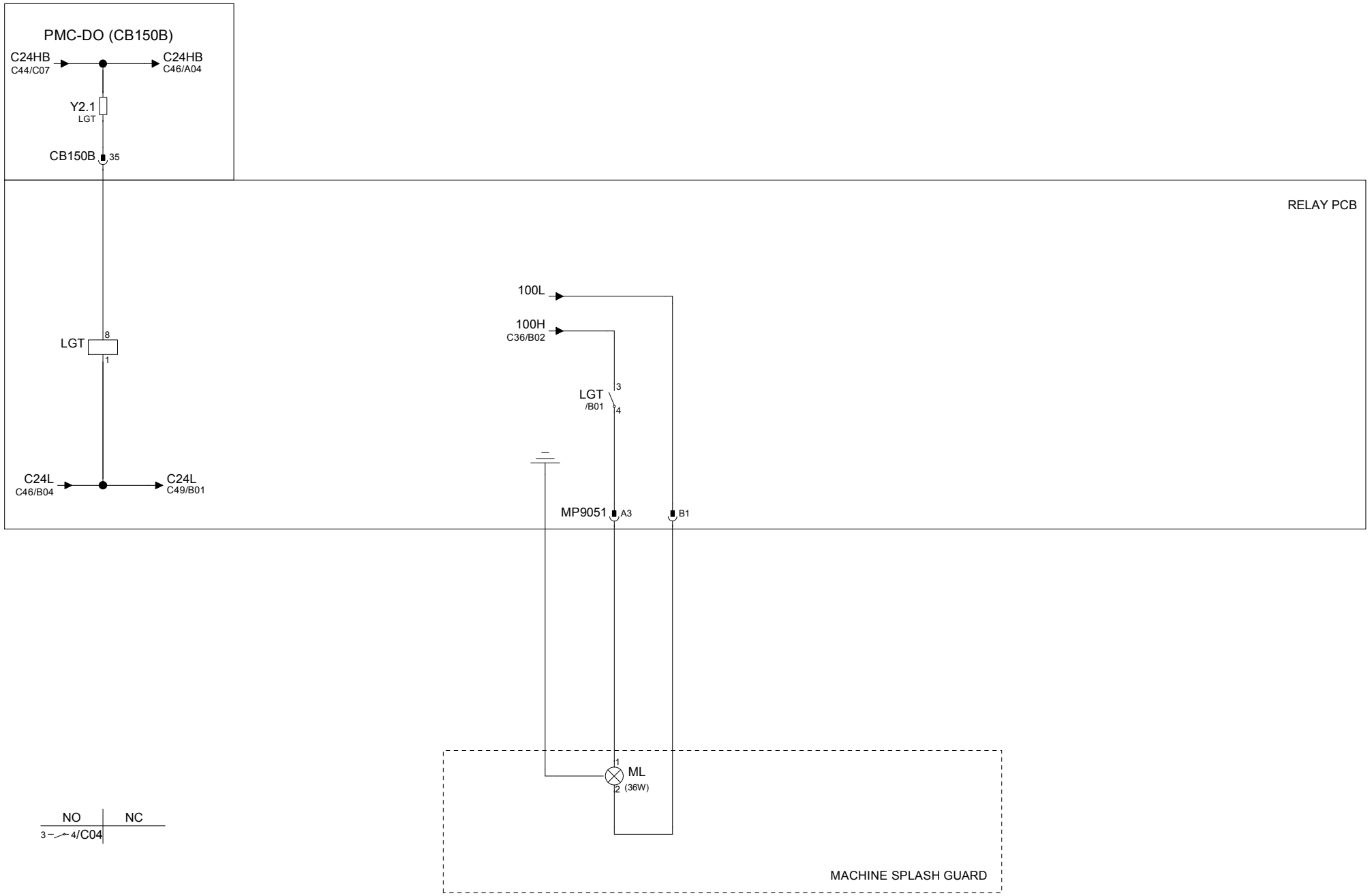


SHOWER COOLANT (OPTION)	FIXTURE COOLANT (OPTION)	SPINDLE AIRSEAL	NOZZLE COOLANT AIR PURGE

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
COOLANT (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C44



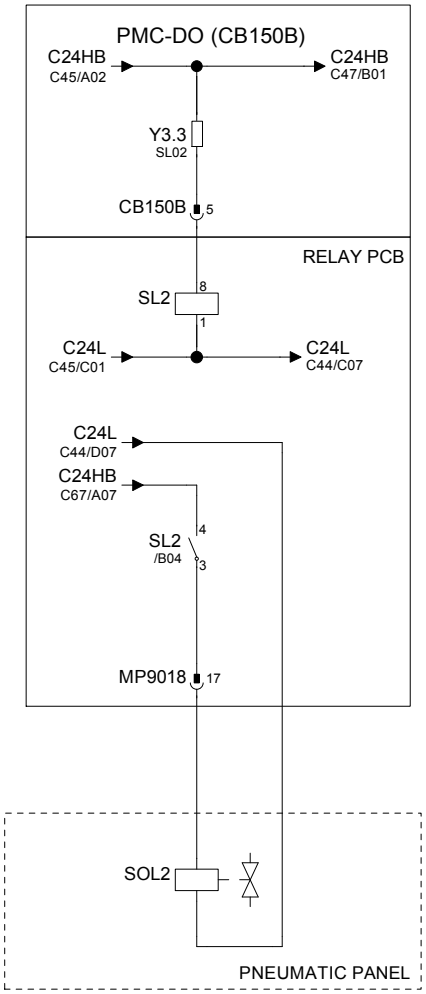
MACHINE LIGHT RELAY

POWER-MACHINE LIGHT

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
MACHINE LIGHT	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C45



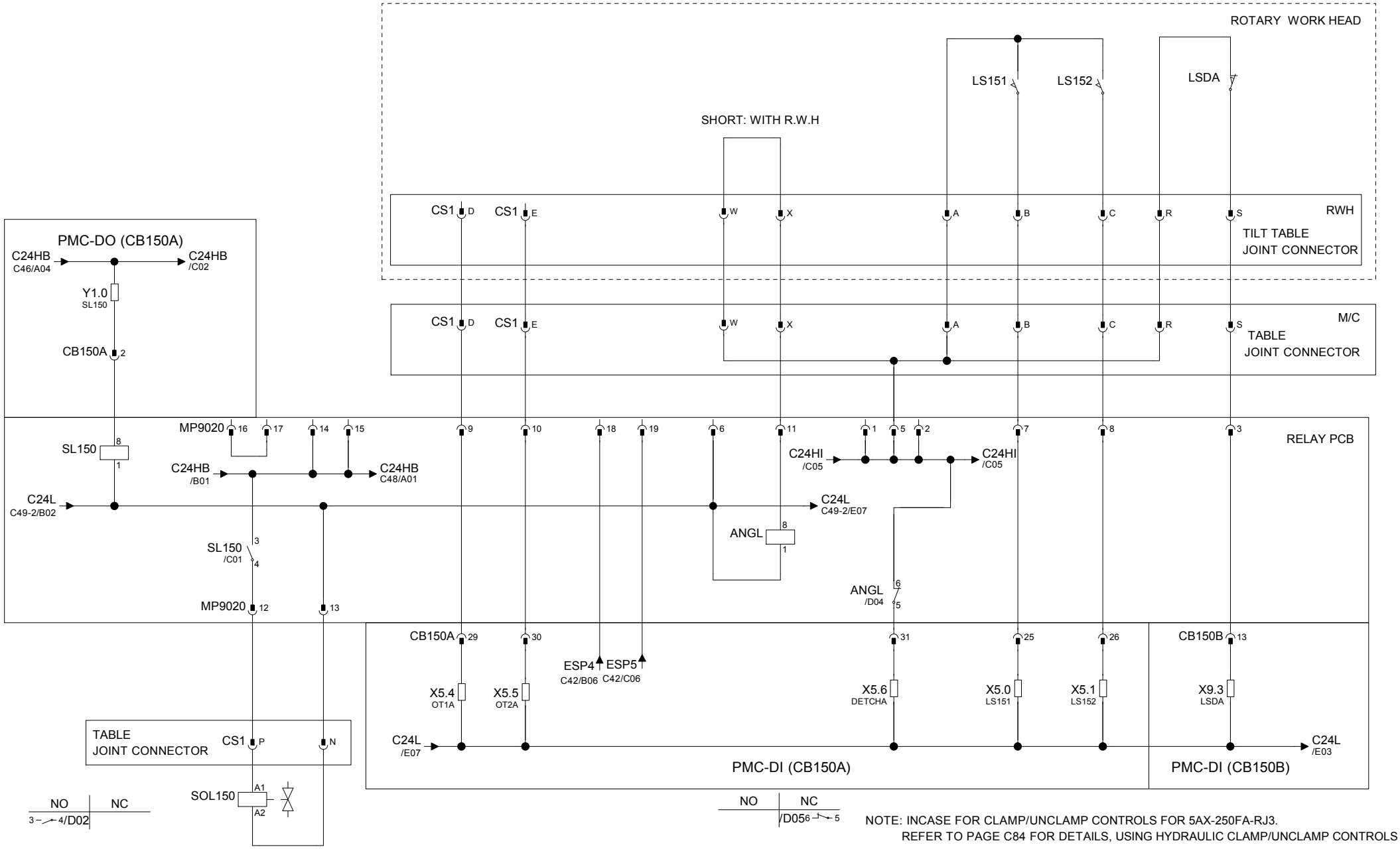
NO	NC
4 – 3/C04	

NOZZLE COOLANT SOLENOID

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
NOZZLE COOLANT	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C46

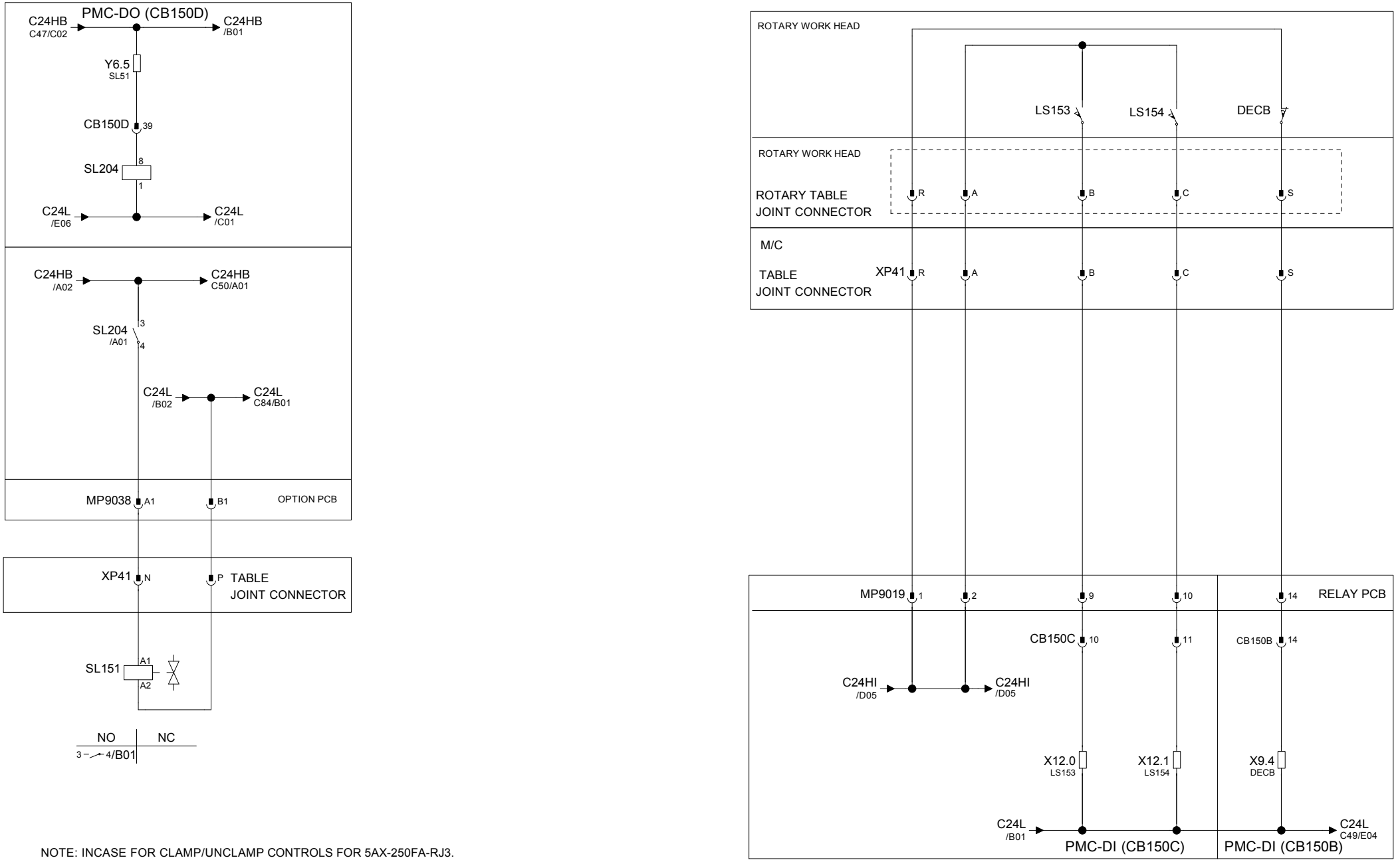


A AXIS UNCLAMP	A AXIS UNCLAMP SOLENOID	+OT ALARM ON	-OT ALARM ON	EMERGENCY STOP	A AXIS NEGLECT	A AXIS CLAMP	A AXIS UNCLAMP	A AXIS REFERENCE RETURN SLOWDOWN

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
ADDITIONAL A AXIS (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C47



NOTE: INCASE FOR CLAMP/UNCLAMP CONTROLS FOR 5AX-250FA-RJ3.
REFER TO PAGE C84 FOR DETAILS, USING HYDRAULIC CLAMP/UNCLAMP CONTROLS

B AXIS UNCLAMP	B AXIS UNCLAMP SOLENOID		B AXIS CLAMP	B AXIS UNCLAMP	B AXIS REFERENCE RETURN SLOWDOWN

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
ADDITIONAL B AXIS (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C48

NOTE: PRAGATI INDEX TABLE OPTION FOR STANDARD M/C

MTC POWER PANEL

PMC-DO (CB150D)

C24H /C01 → C24H /C02

Y6.0 M770 — Y6.1 M771

CB150D 34 35

M770 8 1 — M771 8 1

C24L C45/C02 → C24L C49-1/B02

NO	NC	NO	NC
3–4 C71/C02		3–4 C71/C02	
3–4 /D01		3–4 /D02	
3–4 C49-1/D01		3–4 C49-1/D02	

C03/D03

R2A C03/D02 — S2A — T2A C03/D04

MS20F /D01 (1 3 5 / 2 5 6) — MS20R /D02 (1 3 5 / 2 4 6)

TH20 3.2~5A (1 3 5 / 2 4 6)

13–14 C49-1/C07

95–96 /C07

SK20M C49-1/B03

TH20 SETTING VALUE	
200V/50HZ	3.9A
220V/50HZ	3.9A

C24HI /B07 → C24HI /B06

MP9031 A1 OPTION PCB

TH20 /B03 3.2~5A 95 96

C24H /A01 → C24H /A02

TB9012 (1~3)

OPTION PCB

MP9031 B1

M770 C71/B02 3 4 — M771 C71/B02 3 4

MP9031 B2 — MP9031 B3

MS20F A1 A2 — MS20R A1 A2

C24L C50/B01

NO	NC	NO	NC
1–2 /B03		1–2 /B03	
1–2 C49-1/B03		1–2 C49-1/B03	
3–5 /B03		3–4 /B04	
3–5 C49-1/B03		3–4 C49-1/B04	
5–6 /B03		5–6 /B04	
5–6 C49-1/B03		5–6 C49-1/B04	

INDXM U V W E

INDXM U V W — M 3 ~ — PE

ENC

BIT 1
BIT 2
BIT 3
BIT 4
STROBE
PARITY
0V
+24VDC
CLAMP

+24VDC
THERMAL FEEDBACK

VMP12 32 — 18 1 19 31 30 29 28 27 26 25 — MP9031 A2 OPTION PCB

CB150D 32 31 30 29 28 27 26 25 42

X14.7 UI08 — X14.6 UI07 — X14.5 UI06 — X14.4 UI05 — X14.3 UI04 — X14.2 UI03 — X14.1 UI02 — X14.0 UI01 — X13.0 TAFIN1

C24HI C91/B04

C24L C48/E07 → C24L C49-1/E07

PMC-DI (CB150D)

TPC JR. A START	TPC JR. A REF	PRAGATI INDEX MOTOR	INDEX TABLE MOTOR OVER HEAT	INDEX TABLE CLAMP PRX	PARITY	STROBE	BIT 4	BIT 3	BIT 2	BIT 1	THERMAL TRIP

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
PRAGATI INDEX TABLE (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C49

NOTE: PRAGATI INDEX TABLE OPTION FOR APC M/C (PAL 1)

MTC POWER PANEL

PMC-DO (CB150D)

C24H /C01 → C24H /C02

Y6.0 M770 — Y6.1 M771

CB150D 34 — 35

M770 8 1 — M771 8 1

C24L /D01 → C24L C49/B02

NO	NC	NO	NC
3–4 C71/C02		3–4 C71/C02	
3–4 C49/D01		3–4 C49/D02	
3–4 /D01		3–4 /D02	

C03/D03 C03/D04

R2A S2A T2A ← C03/D02

TH20-1 C49-2/A03

TH20-3 C49-2/A03

TH20-5 C49-2/A03

TH20 3.2~5A

13–14 /C07

95–96 C49/C07

MS20F C49/D01 — MS20R C49/D02

SK20M

TH20 SETTING VALUE	
200V/50HZ	3.9A
220V/50HZ	3.9A

C24HI /B07 → C24HI /B06

MP9031 A1 OPTION PCB

TH20-13 C49-2/B07

TH20 C49/B03 3.2~5A 13–14

C24H /A01 → C24H /A02

TB9012 (1~3)

OPTION PCB

MP9031 B1

M770 C71/B02 3–4 — M771 C71/B02 3–4

MP9031 B2 — MP9031 B3

MS20F A1 A2 — MS20R A1 A2

C24L /B01

NO	NC	NO	NC
1–2 C49/B03		1–2 C49/B03	
1–2 /B03		1–2 /B03	
3–5 C49/B03		3–4 C49/B04	
3–5 /B03		3–4 /B04	
5–6 C49/B03		5–6 C49/B04	
5–6 /B03		5–6 /B04	

INDXM R Y B E — U V W

INDXM M 3 ~ PE

ENC1

BIT 1
BIT 2
BIT 3
BIT 4
STROBE
PARITY
0V
+24VDC
CLMAP
+24VDC
THERMAL FEEDBACK

CS1 J K L M

CS1 B C S E P W N A D R T

WMP12 32 — 18 1 19 31 30 29 28 27 26 25 MP9031 A2 OPTION PCB

CB150D 32 — 31 30 29 28 27 26 25 42

X14.7 UI08 — X14.6 UI07 — X14.5 UI06 — X14.4 UI05 — X14.3 UI04 — X14.2 UI03 — X14.1 UI02 — X14.0 UI01 — X13.0 TAFIN1

C24HI C49-2/E05

C24L C49-2/E04 → C24L C49/E07

PMC-DI (CB150D)

TPC JR. A START	TPC JR. A REF	PRAGATI INDEX MOTOR	INDEX TABLE MOTOR OVER HEAT	INDEX TABLE CLAMP PRX	PARITY	STROBE	BIT 4	BIT 3	BIT 2	BIT 1	THERMAL TRIP

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
PRAGATI INDEX TBL - PAL 1(OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C49-1

NOTE: PRAGATI INDEX TABLE OPTION FOR APC M/C (PAL 2)

MTC POWER PANEL

PMC-DO (CB150D)

C24H /C01 → C24H /C02

Y6.2 M772 — Y6.3 M773

CB150D 36 — 37

M772 8 / 1 — M773 8 / 1

C24L /D01 → C24L C47/C01

NO	NC	NO	NC
3–4 C71/C02		3–4 C71/C03	
3–4 /D01		3–4 /D02	

C24H /A01 → C24H /A02

TB9012 (1~3)

OPTION PCB

MP9031 B1

M772 C71/B03 (3–4) — M773 C71/B03 (3–4)

MP9031 B4 — MP9031 B5

MS21F A1/A2 — MS21R A1/A2

C24L /B01

NO	NC
1–2/B03	
3–5/B03	
5–6/B03	

NO	NC
1–2/B03	
3–4/B03	
5–6/B04	

C49-1/A03

TH20-1 C49-1/A03 — TH20-3 — TH20-5 C49-1/A03

TH21 3.2~5A (1 3 5 / 2 4 6) — 13–14/C07

MS21F /D01 (1 3 5 / 2 5 6) — MS21R /D02 (1 3 5 / 2 4 6)

SK21M

TH21 SETTING VALUE	
200V/50HZ	3.9A
220V/50HZ	3.9A

CS2 J K L M

INDXM1 R Y B E / U V W — M 3~ — PE

ENC2 → CS2

BIT 1 → B
BIT 2 → C
BIT 3 → S
BIT 4 → E
STROBE → P
PARITY → W
0V → N
+24VDC → A
CLMAP → D
+24VDC → R
THERMAL FEEDBACK → T

TH20-13 C49-1/B07

TH21 /A03 3.2~5A (13 / 14)

MP9039 B3 B1 — MP9031 A4 — A6 — A8 A7 A6 A5 A4 A3 A1 — MP9031 A3 — OPTION PCB

CB150D 44 — 17 16 15 14 13 12 10 — 43

X13.2 TJALM1 — X15.7 UI15 — X15.6 UI14 — X15.5 UI13 — X15.4 UI12 — X15.3 UI11 — X15.2 UI10 — X15.0 UI09 — X13.1 TAFIN2

C24HI C49-1/E05

C24L C49-1/E04 → C24L C47/C05

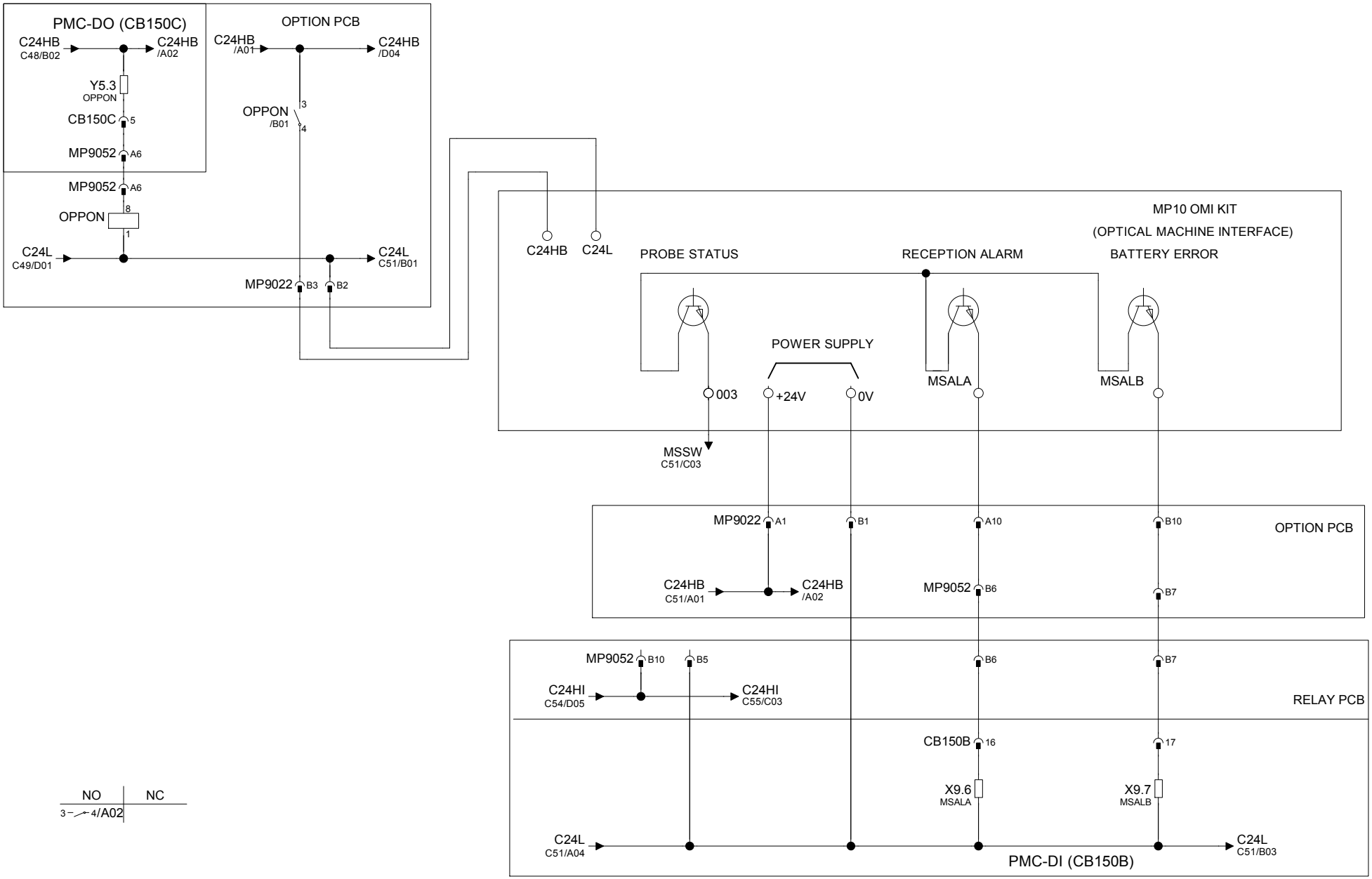
PMC-DI (CB150D)

TPC JR. A START	TPC JR. A REF	PRAGATI INDEX MOTOR	INDEX TABLE MOTOR OVER HEAT	INDEX TABLE CLAMP PRX	PARITY	STROBE	BIT 4	BIT 3	BIT 2	BIT 1	THERMAL TRIP

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
PRAGATI INDEX TBL - PAL 2(OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C49-2

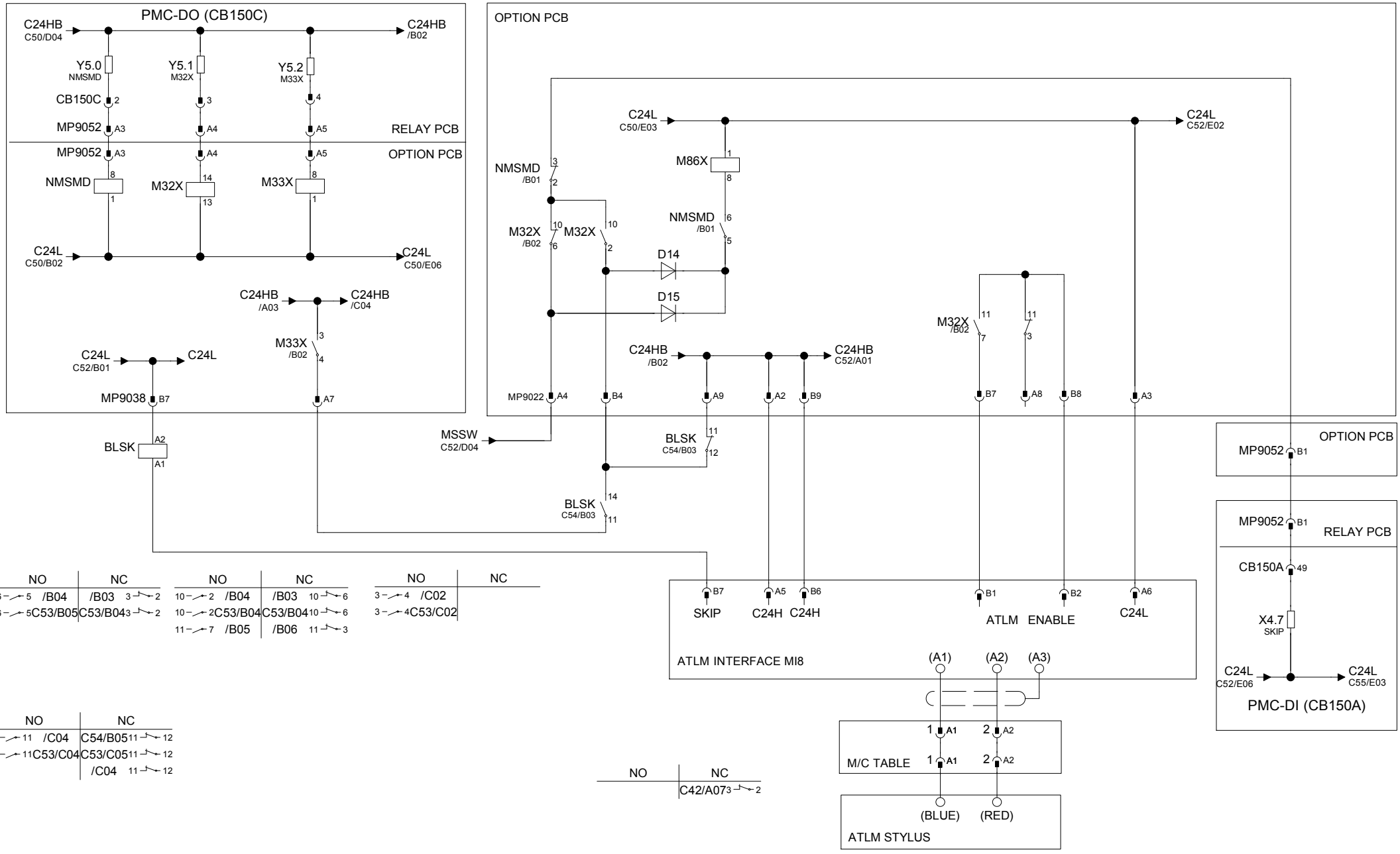


PROBE POWER ON	PROBE STATUS	ERROR RECEPTION ALARM	BATTERY ERROR

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
AWM-MARPOSS (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C50

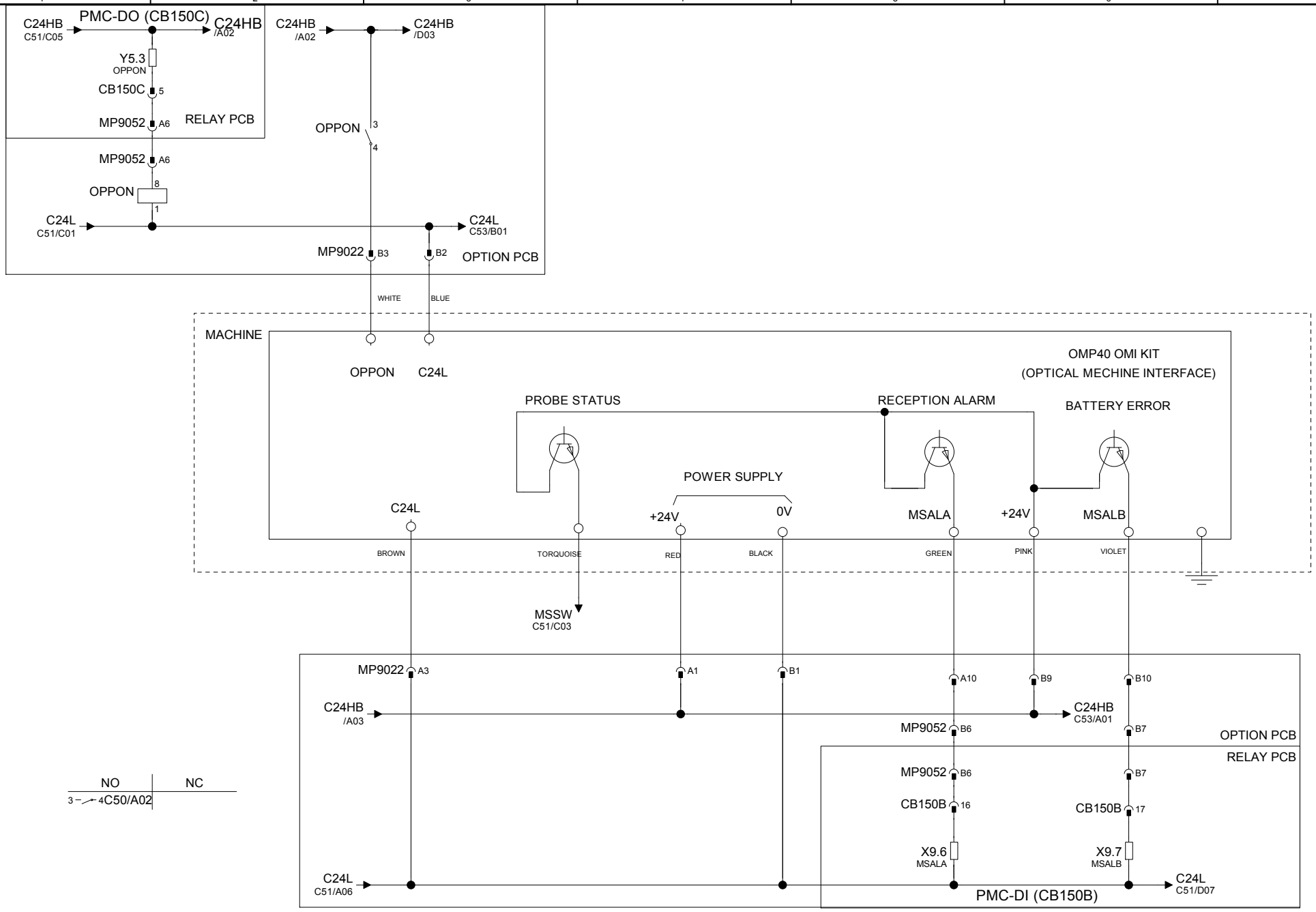


NOT MEASURING MODE	MEASURING ON	CIRCUIT CHECK	ATLM CONTACT TYPE	ATLM CONTROLS	SKIP

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
ATLM-RENISHAW TOUCH PROBE (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C51

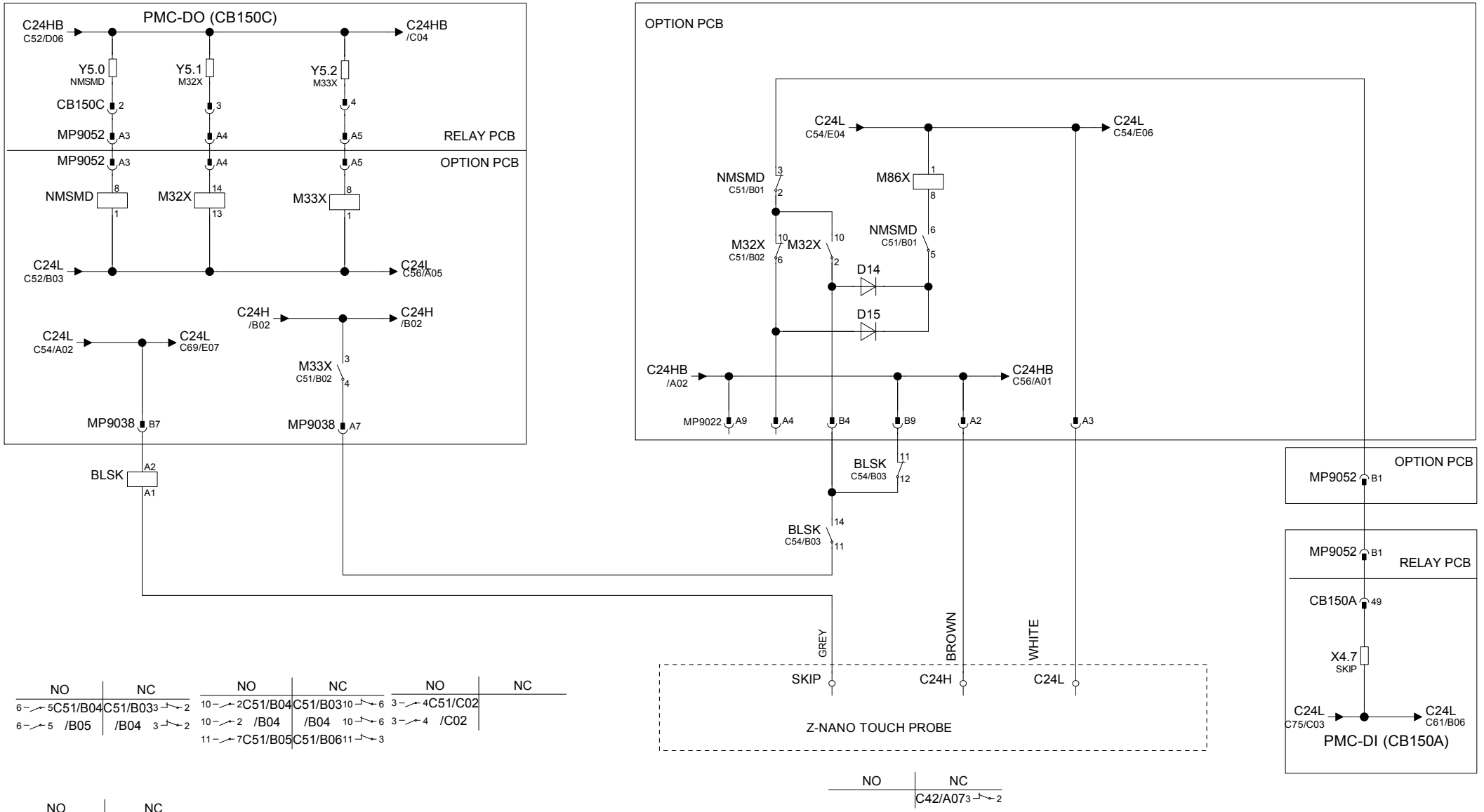


PROBE POWER ON	PROBE STATUS	RECEPTION ALARM	BATTERY ERROR

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
AWM-RENISHAW MP10 (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C52



NO	NC
6 – 5 C51/B04	C51/B03 3 – 2
6 – 5 /B05	/B04 3 – 2

NO	NC
10 – 2 C51/B04	C51/B03 10 – 6
10 – 2 /B04	/B04 10 – 6
11 – 7 C51/B05	C51/B06 11 – 3

NO	NC
3 – 4 C51/C02	
3 – 4 /C02	

NO	NC
C42/A07 3 – 2	

NO	NC
14 – 11 C51/C04	C54/B05 11 – 12
14 – 11 /C04	/C05 11 – 12
	C51/C04 11 – 12

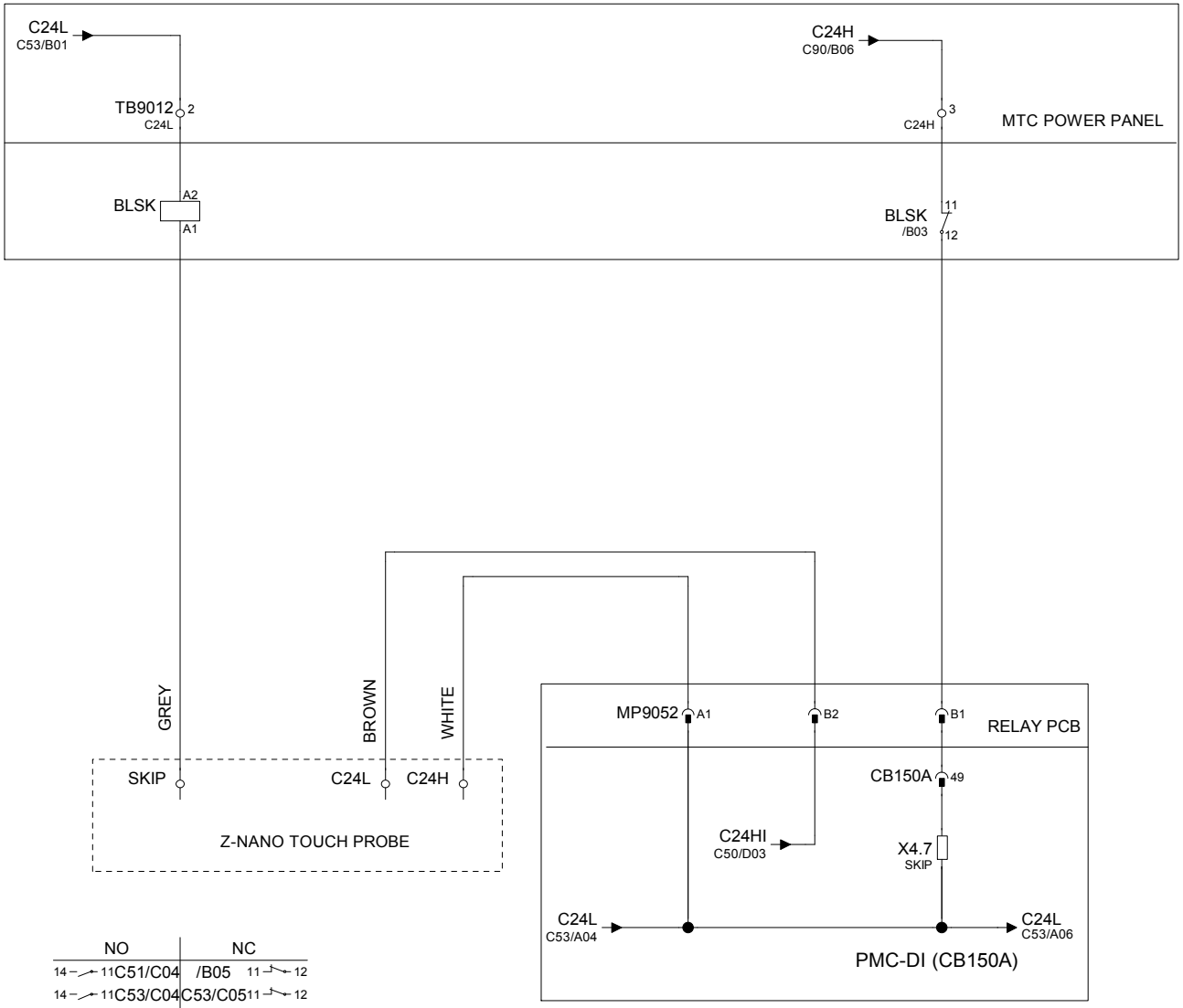
* APPLICABLE WHEN USING OPTION PCB - E6VS603A800-XX

NOT MEASURING MODE	MEASURING ON	CIRCUIT CHECK	CONTACT TYPE	SKIP

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
ATLM-BLUM Z-NANO TYPE (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C53



NO		NC	
14	11 C51/C04	/B05	11 12
14	11 C53/C04	C53/C05	11 12
		C51/C04	11 12

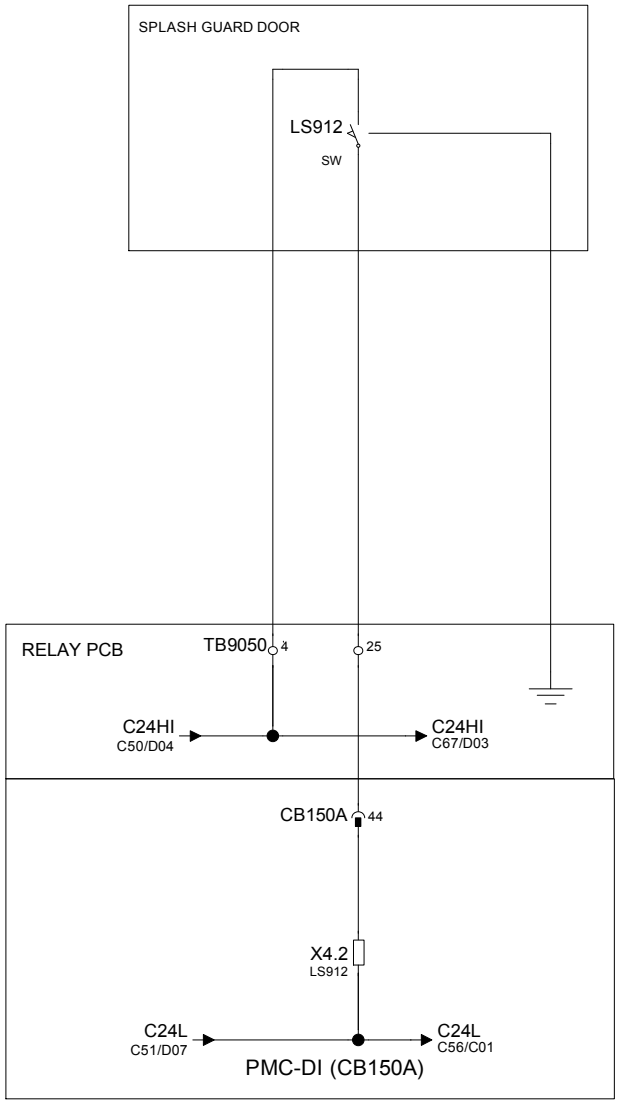
* APPLICABLE WHEN NOT USING OPTION PCB - E6VS603A800-XX

CONTACT TYPE	24VDC INPUT VOLTAGE	SKIP

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
ATLM-BLUM Z-NANO TYPE (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C54

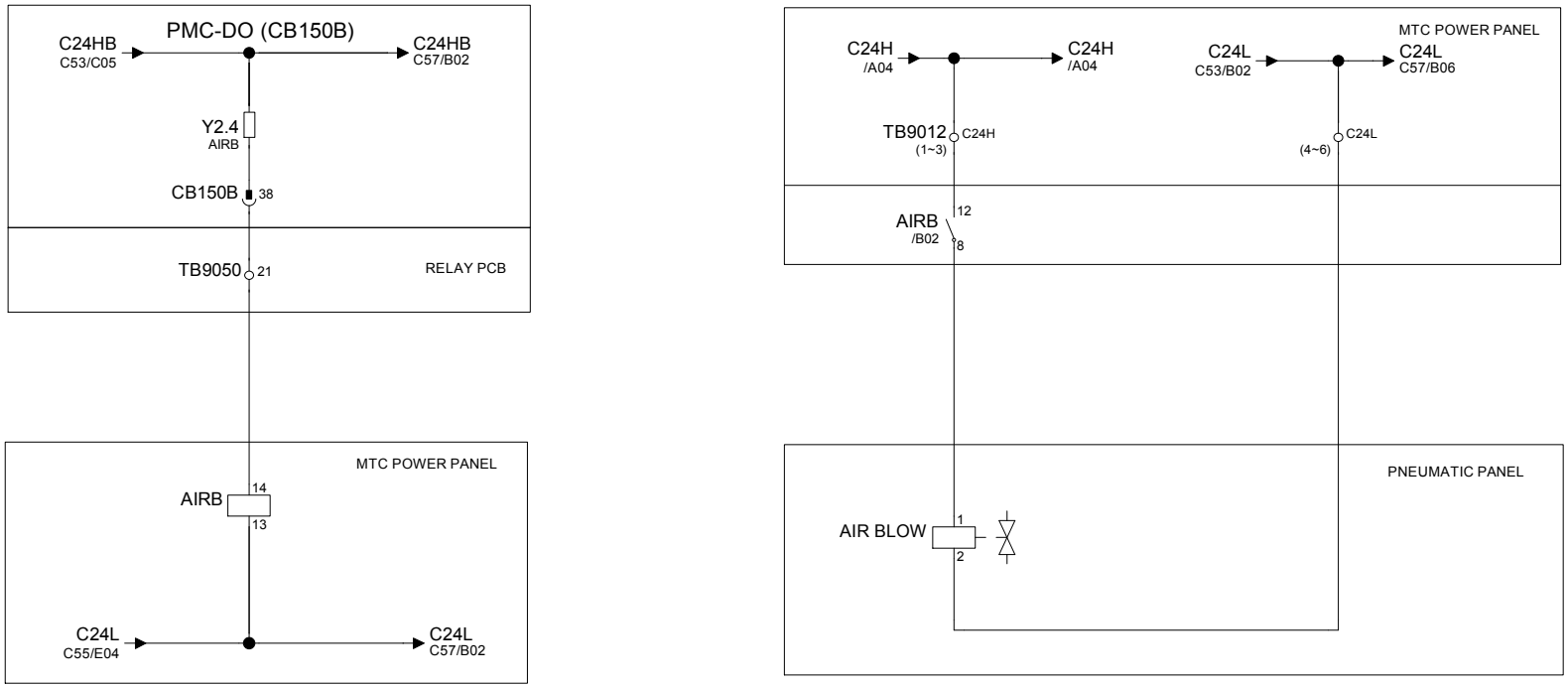


S/G FRONT DOOR INTERLOCK

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
S/G DOOR INTERLOCK	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C55



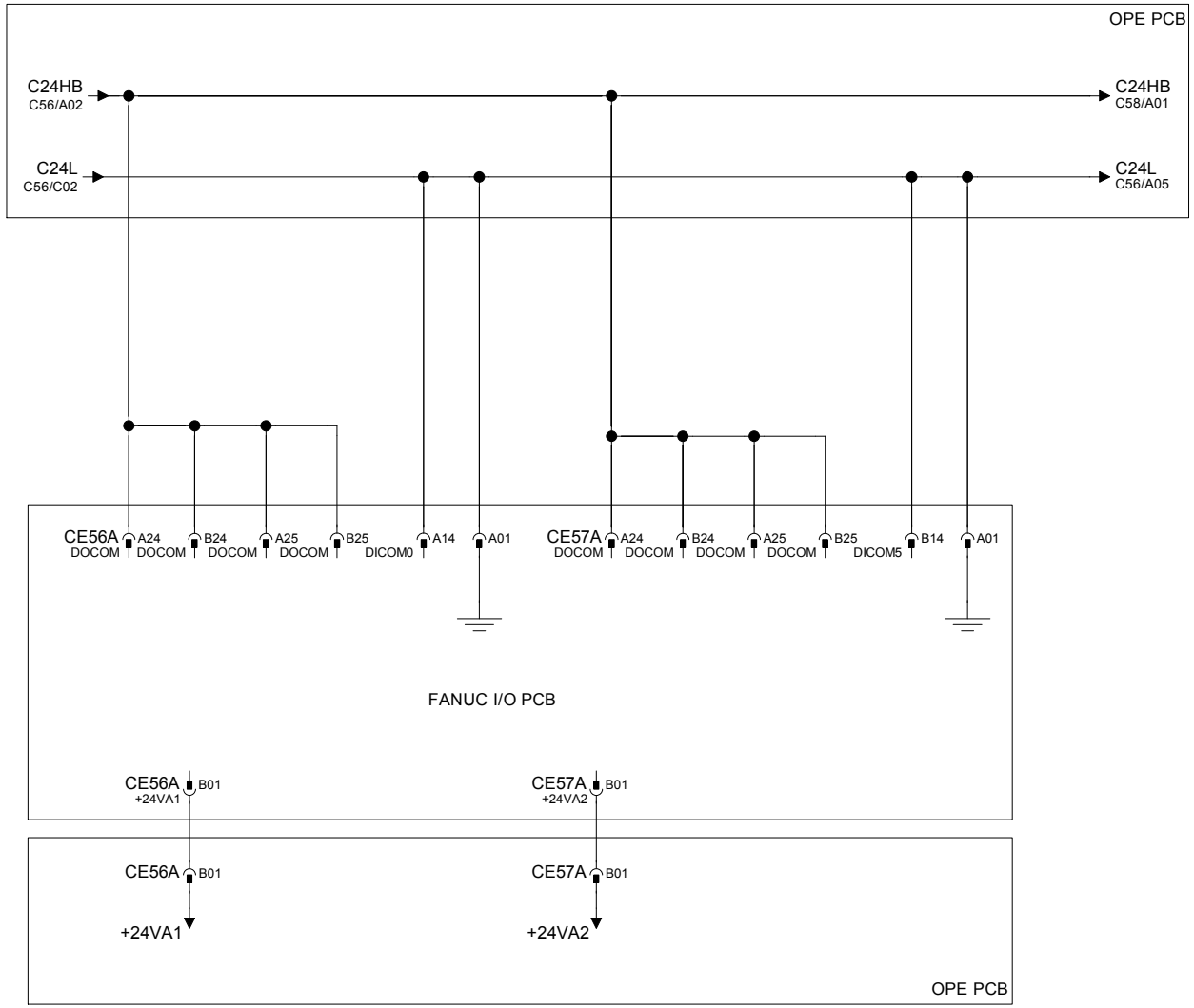
NO	NC
12 – 8/A04	

AIR BLOW		AIR BLOW SOLENOID	

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
AIR BLOW (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C56

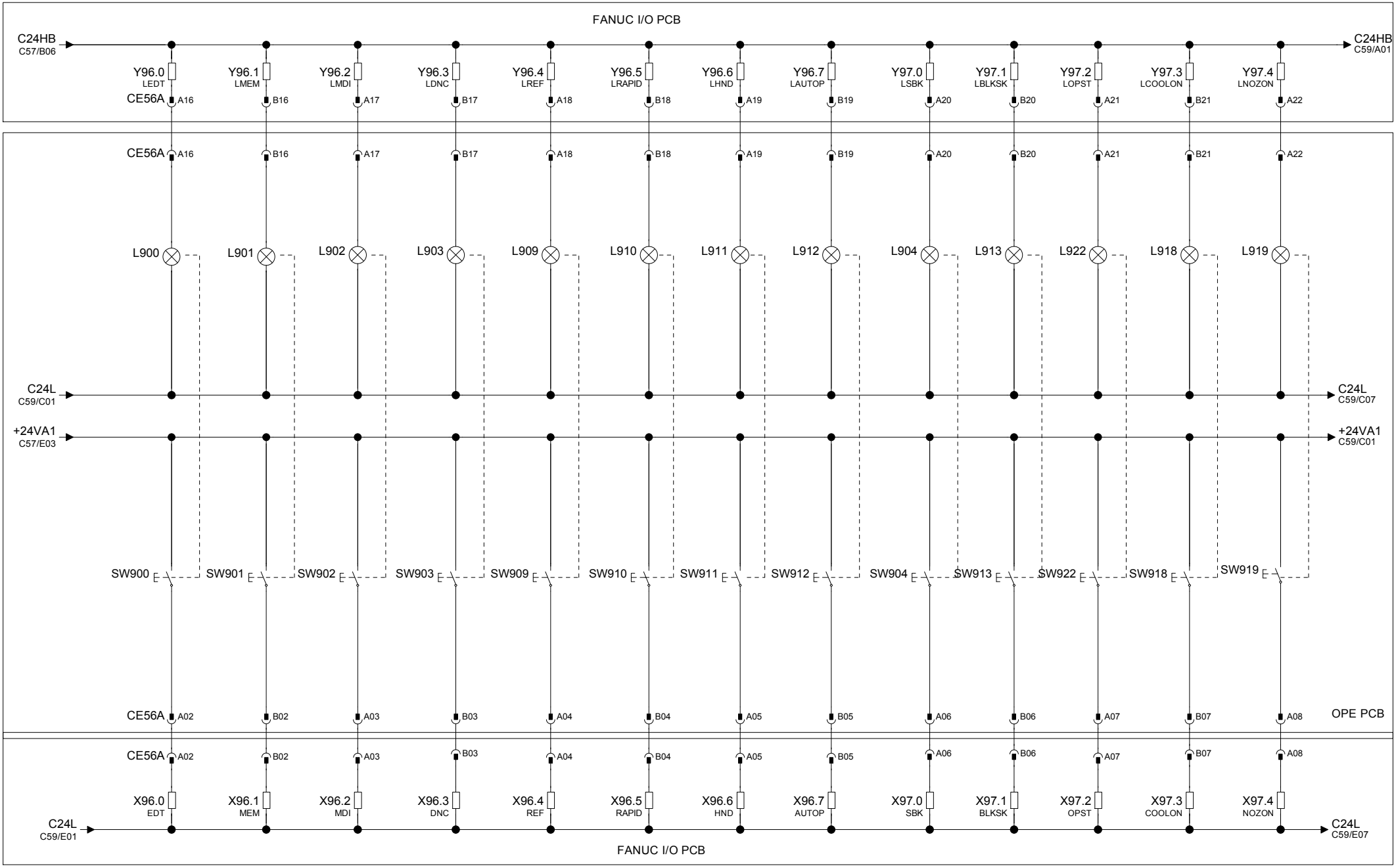


FANUC.I/O MODULE POWER

Revisions			Date	Name
	4			
	3	APPROVED	25-NOV-2010	GOH WILLIAM
	2	CHECKED	25-NOV-2010	S MAHESH
	1	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
MAIN OPERATION PANEL(24V FOR I/O MODULE)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C57

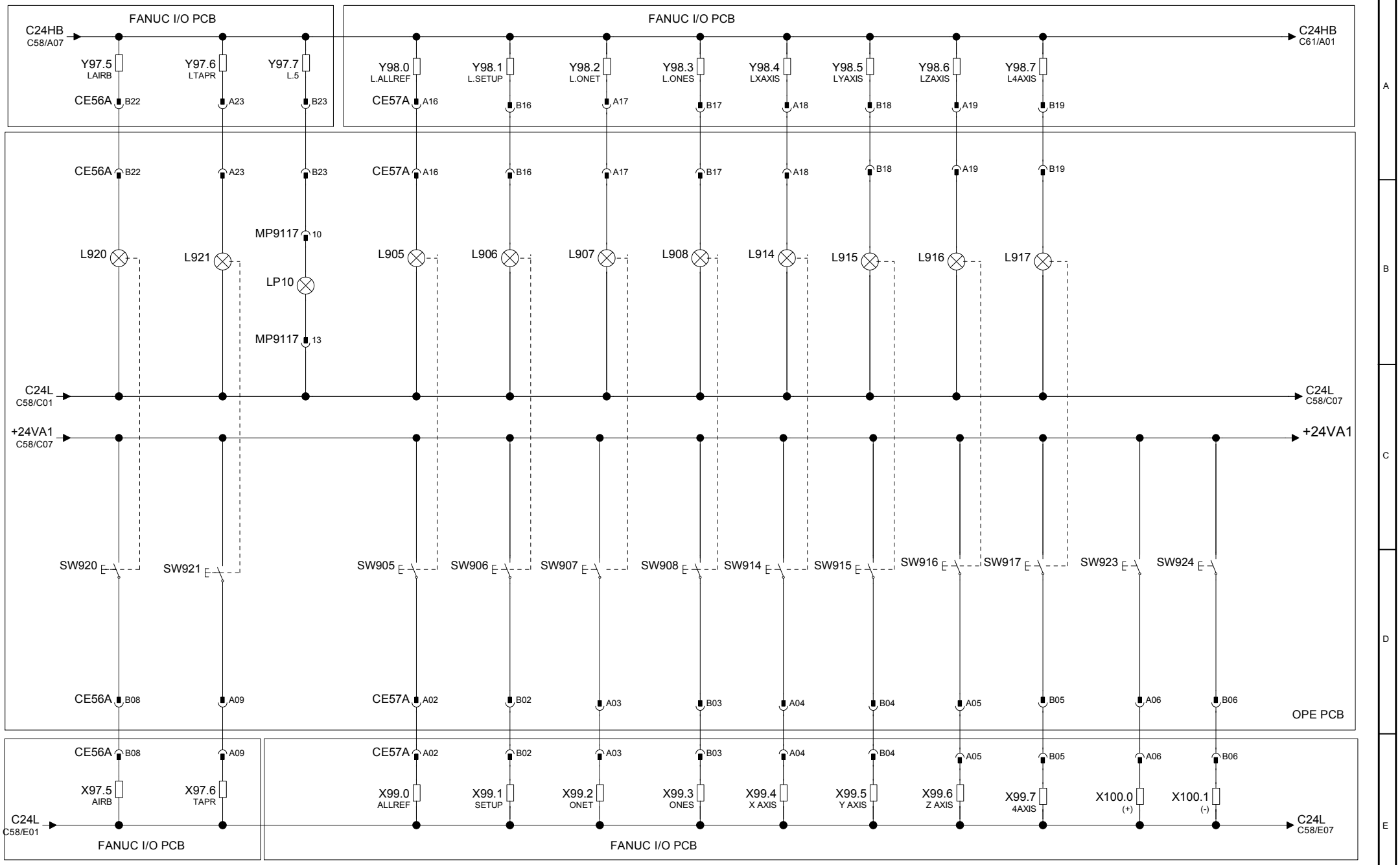


EDITOR	MEMORY	MDI	DNC	REFERENCE	RAPID	HANDLE	AUTO POWER OFF	SINGLE BLOCK	BLOCK SKIP	OPTIONAL STOP	COOLANT ON/OFF	NOZZLE ON/OFF

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
MODE SELECTION, FLEXIBLE SWITCHES	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C58



	AIR BLOW	TAP RETRACT	LAMP 5th AXIS	ALL AXES REFERENCE	SETUP	ONE TOUCH START	ONE TOUCH STOP	X AXIS	Y AXIS	Z AXIS	4TH AXIS	(+)	(-)

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE
MODE SELECTION, FLEXIBLE SWITCHES

MACHINE TYPE	SLIM3
NC TYPE	FS0i-MC

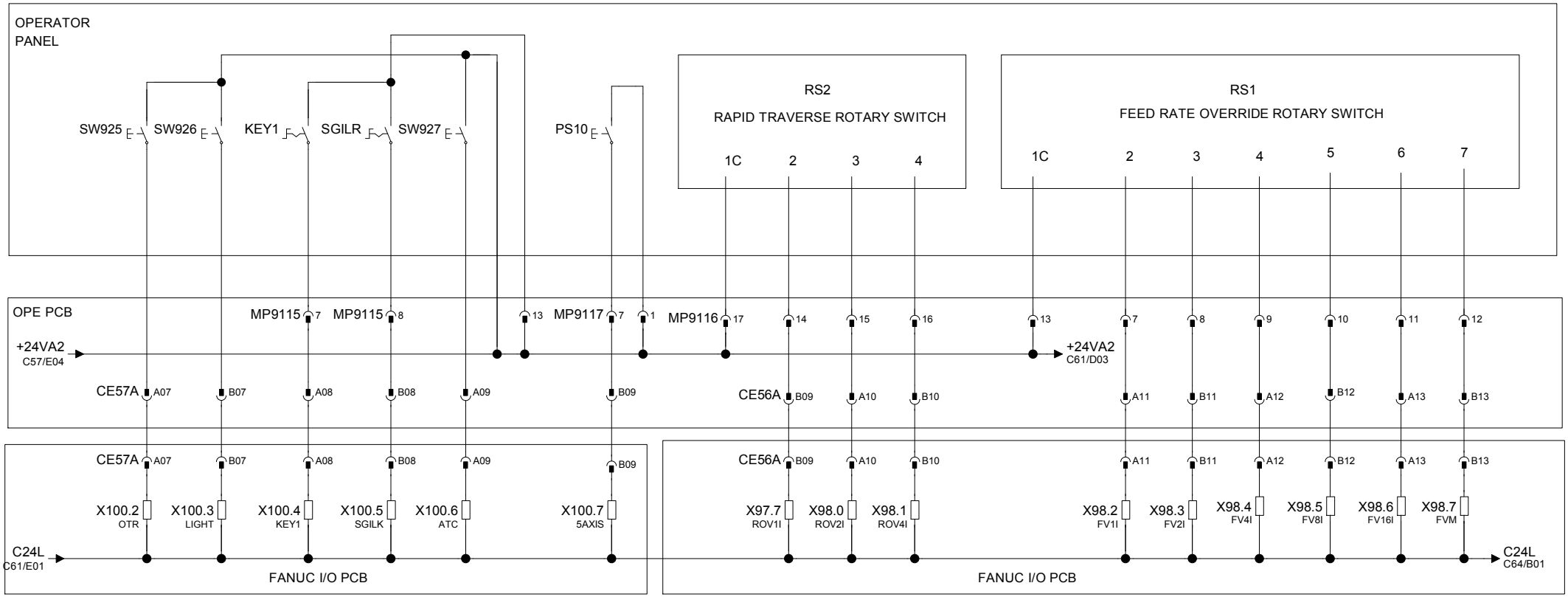
DRAWING No.
V090i-C1E-X03

GROUP
=

SHEET No.
C59

NOTE: SW927 IS LOCATED AT THE BACK SIDE OF THE PCB.
TO ACCESS THIS SW927, THE OPERATOR PANEL HAS TO
BE OPEN INORDER TO GAIN ACCESS

NOTE: PLEASE REFER TO OPERATOR PCB
PAGE H06 FOR MORE DETAILS.

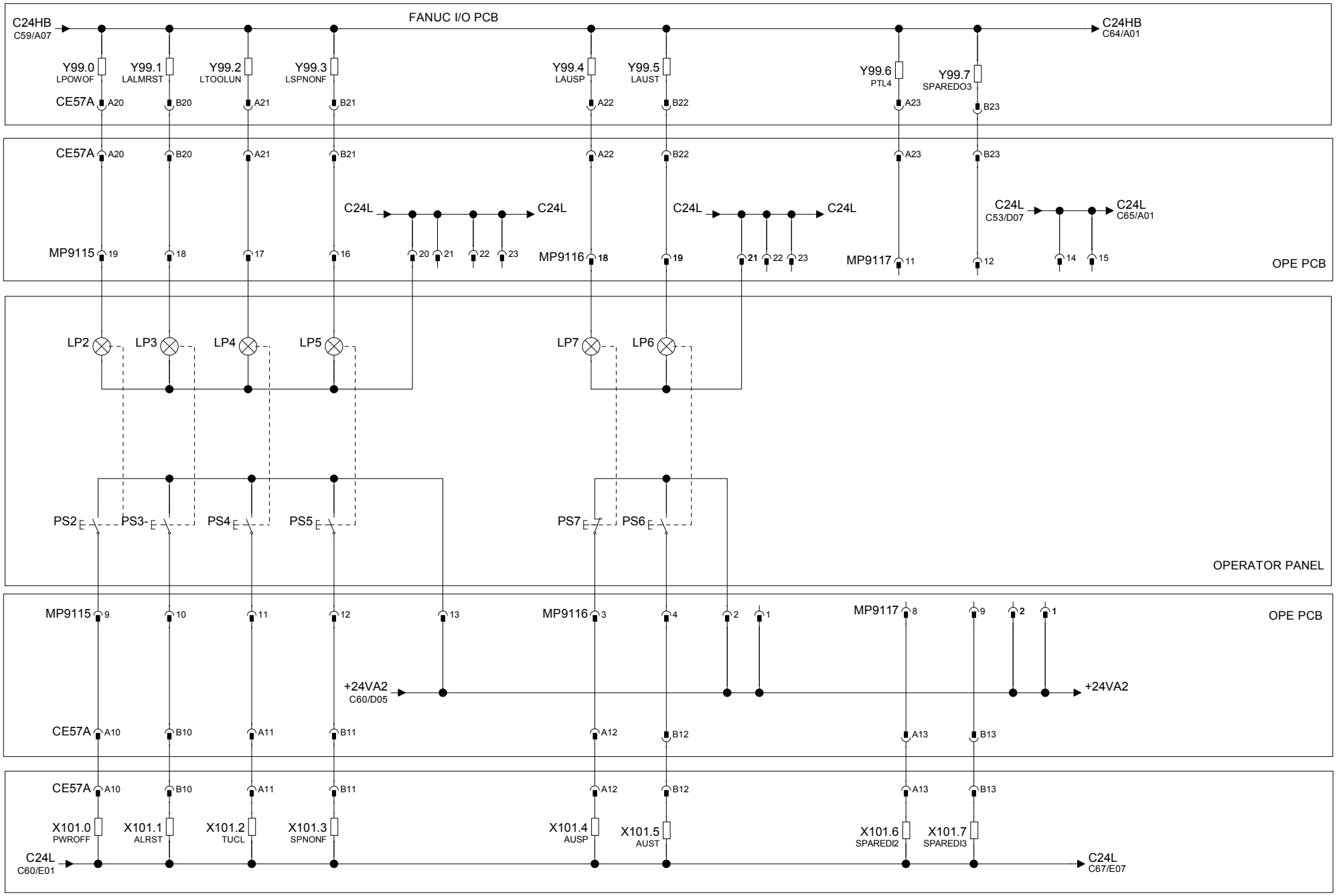


	OT RELEASE	LIGHT ON	MEMORY KEY	S/G DOOR ILK RELEASE	ATC AXIS SELECTION	5th AXIS SELECT	RAPID TRAVERSE OVERRIDE	FEED RATE OVERRIDE

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
MODE SELECTION, FLEXIBLE SWITCHES	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C60



Revisions		Date	Name
4			
3	APPROVED	25-NOV-2010	GOH WILLIAM
2	CHECKED	25-NOV-2010	S MAHESH
1	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

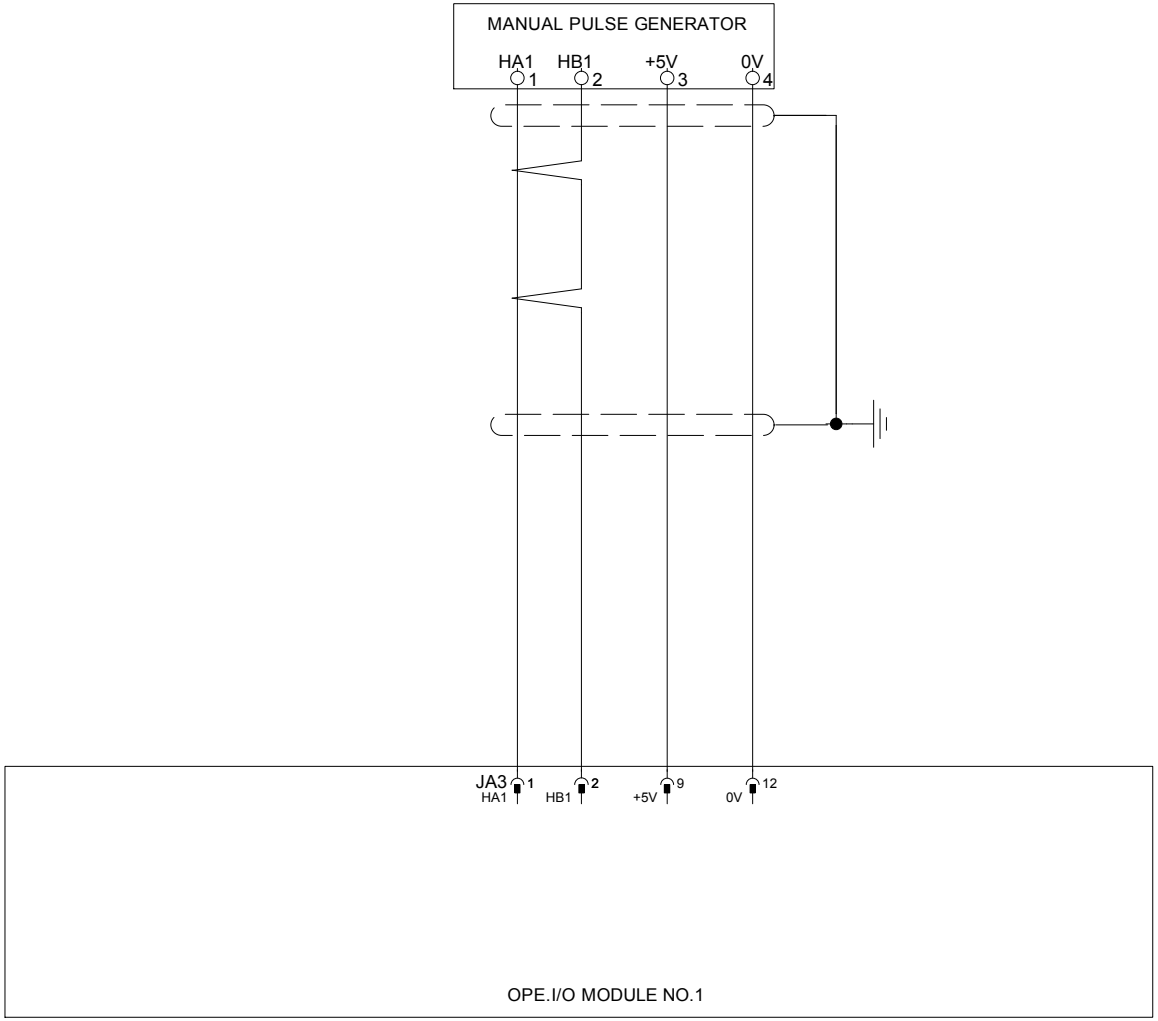
TITLE
MODE SELECTION, FLEXIBLE SWITCHES

MACHINE TYPE	SLIM3
NC TYPE	FS0i-MC

DRAWING No.
V090i-C1E-X03

GROUP
=

SHEET No.
C61

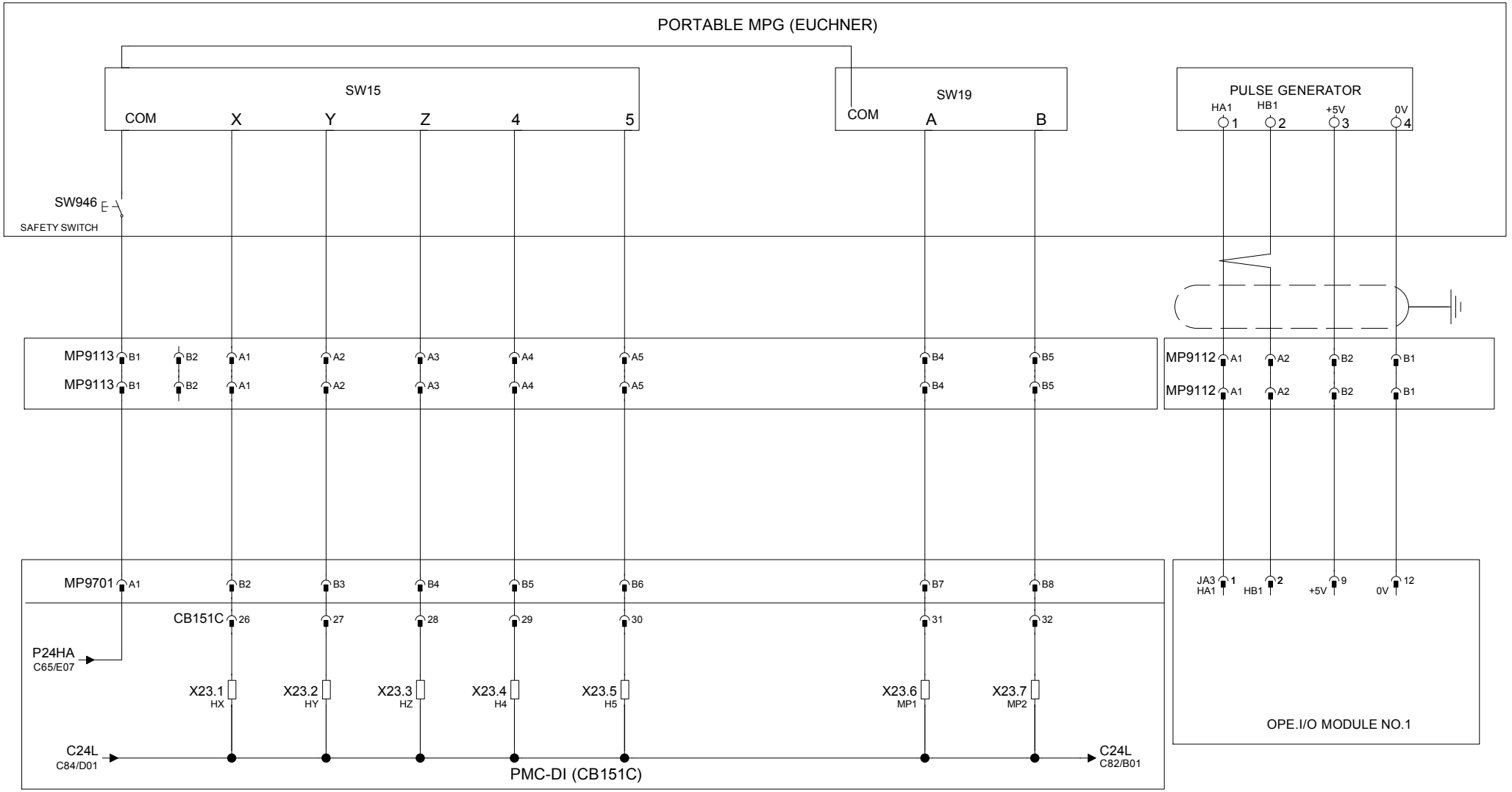


5V PULSE GENERATOR

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
MANUAL PULSE GENERATOR	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C62

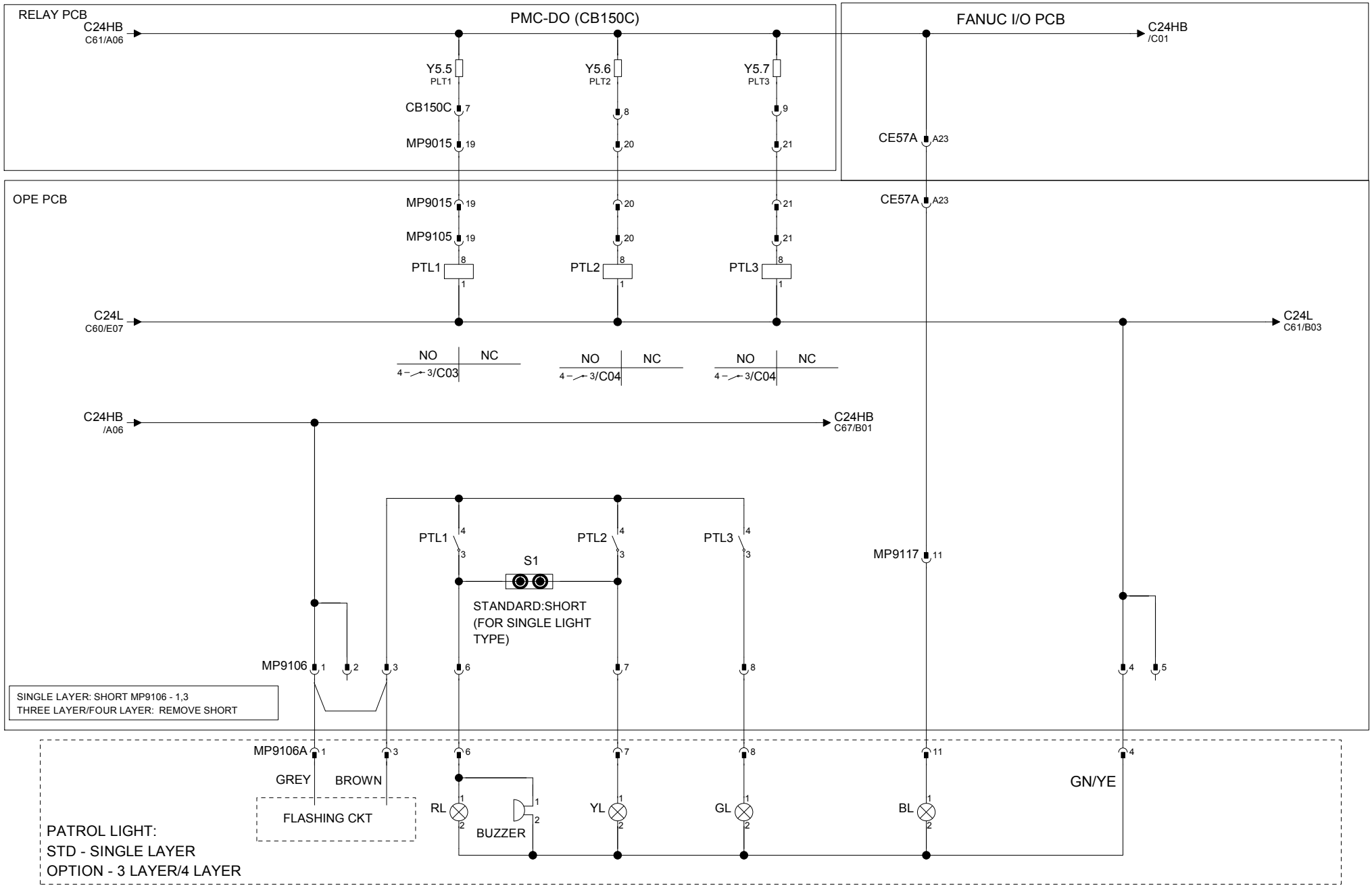


SAFETY SWITCH	X-AXIS SELECT	Y-AXIS SELECT	Z-AXIS SELECT	A-AXIS SELECT	B-AXIS SELECT		MULTIPLY X 10	MULTIPLY X 100	5V PULSE GENERATOR

Revisions			Date	Name
	4			
	3	APPROVED	25-NOV-2010	GOH WILLIAM
	2	CHECKED	25-NOV-2010	S MAHESH
	1	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
PORTABLE MANUAL PULSE GENERATOR (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C63

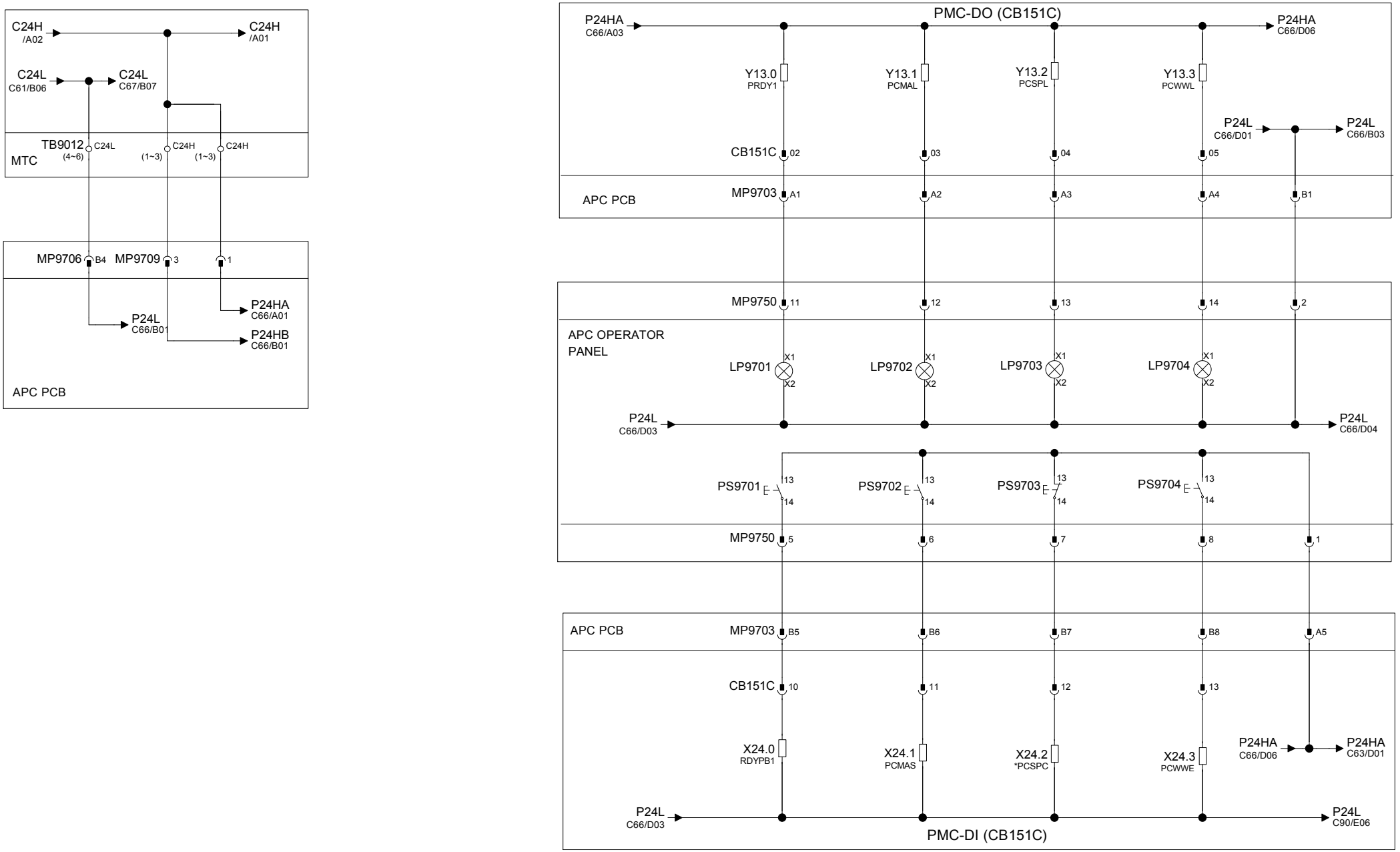


	SIGNAL LIGHT1 BUZZER	SIGNAL LIGHT2	SIGNAL LIGHT3	SIGNAL LIGHT4

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
SIGNAL LIGHT/PATROL LIGHT	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C64



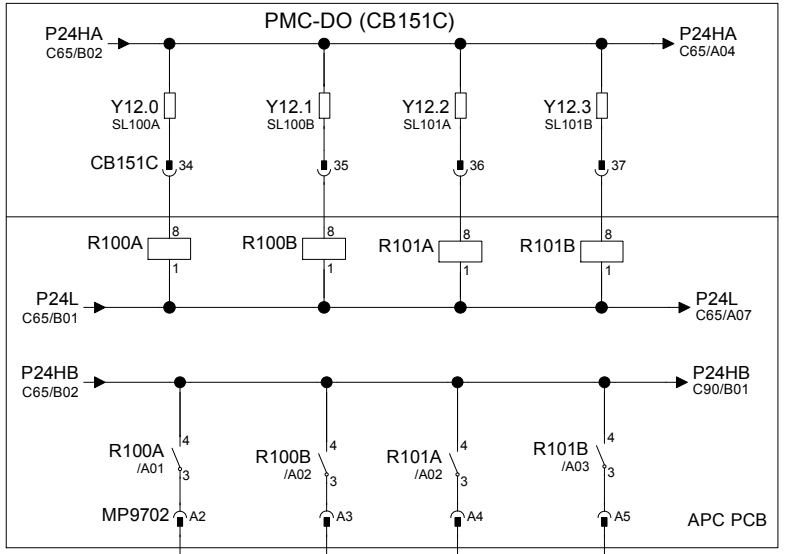
*NOTE: THIS OPTION IS NOT AVAILABLE ON STANDARD MACHINE.

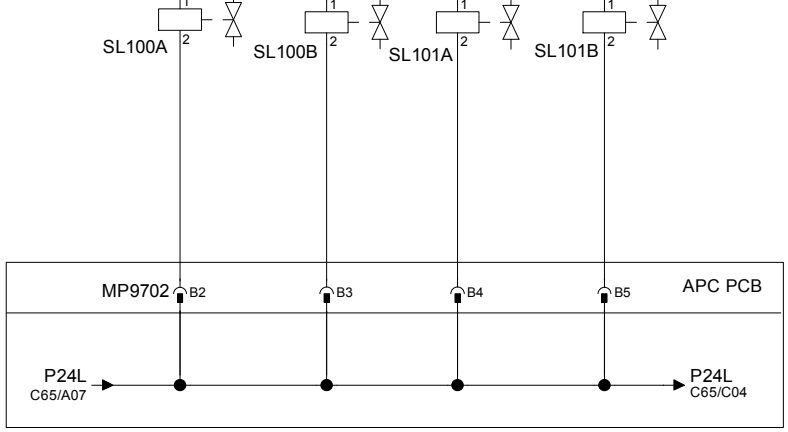
	INPUT VOLTAGE TO APC PCB		APC READY	APC MANUAL SWITCH	APC SIDE FEEDHOLD	WORK WASH ENABLE	

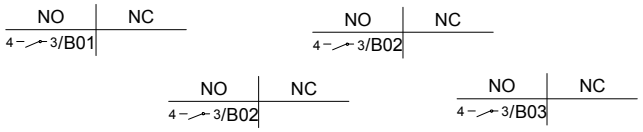
Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

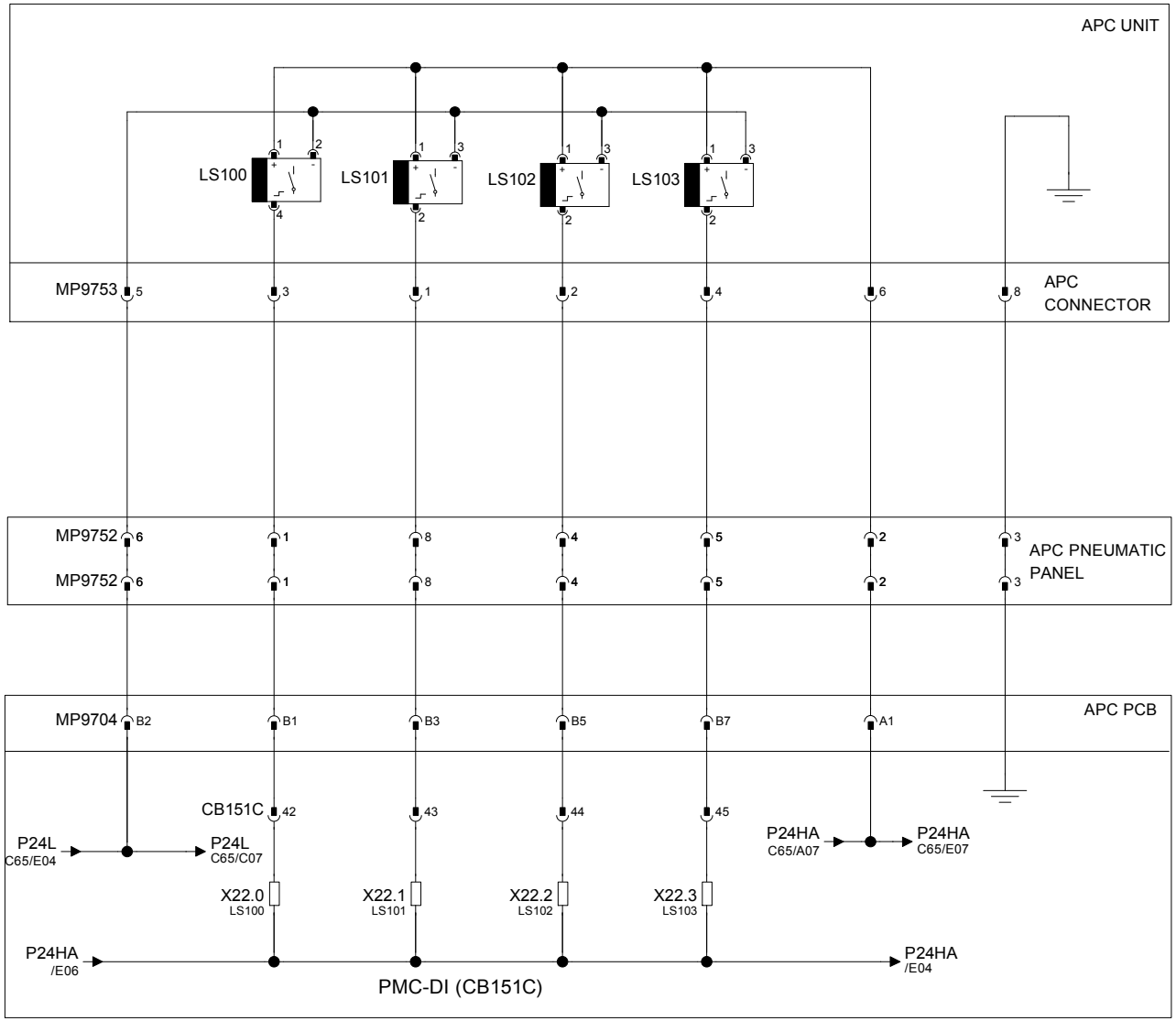
MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
APC PALLET OPERATOR PANEL (APC M/C)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C65









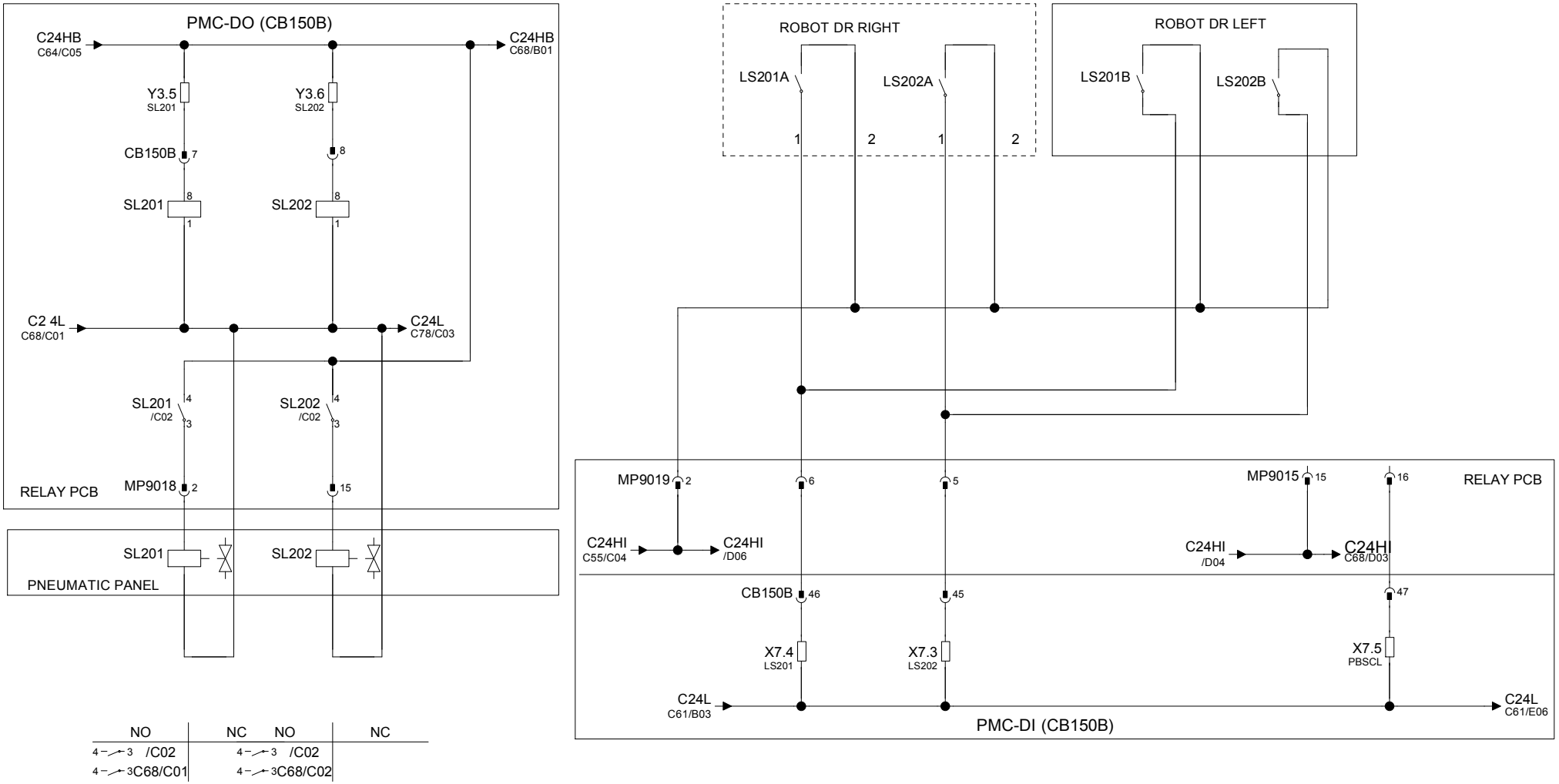
*NOTE: THIS OPTION IS NOT AVAILABLE ON STANDARD MACHINE.

APC ARM DOWN	APC ARM UP	APCARM CCW	APC ARM CW		APC ARM DOWN	APC ARM LIFT	APC ARM CCW	APC ARM CW

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
APC PALLET CONTROL (APC M/C)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C66

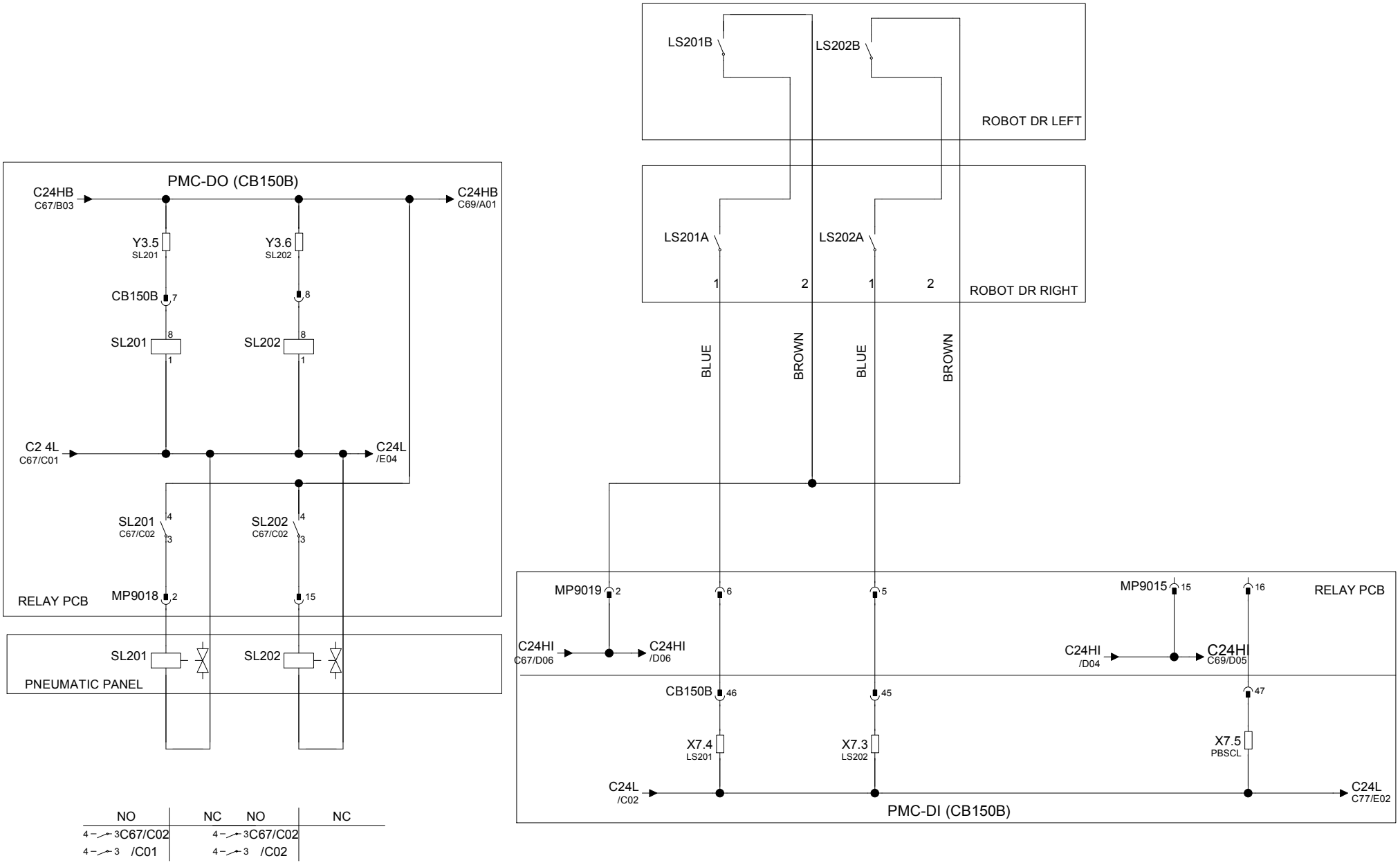


ROBOT DR CLOSE SOLENOID	ROBOT DR OPEN SOLENOID	ROBOT DR CLOSE CONFIRM	ROBOT DR OPEN CONFIRM	PUSHBUTTON ROBOT DR OPEN/CLOSE

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
ROBOT DR LEFT OR RIGHT (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C67

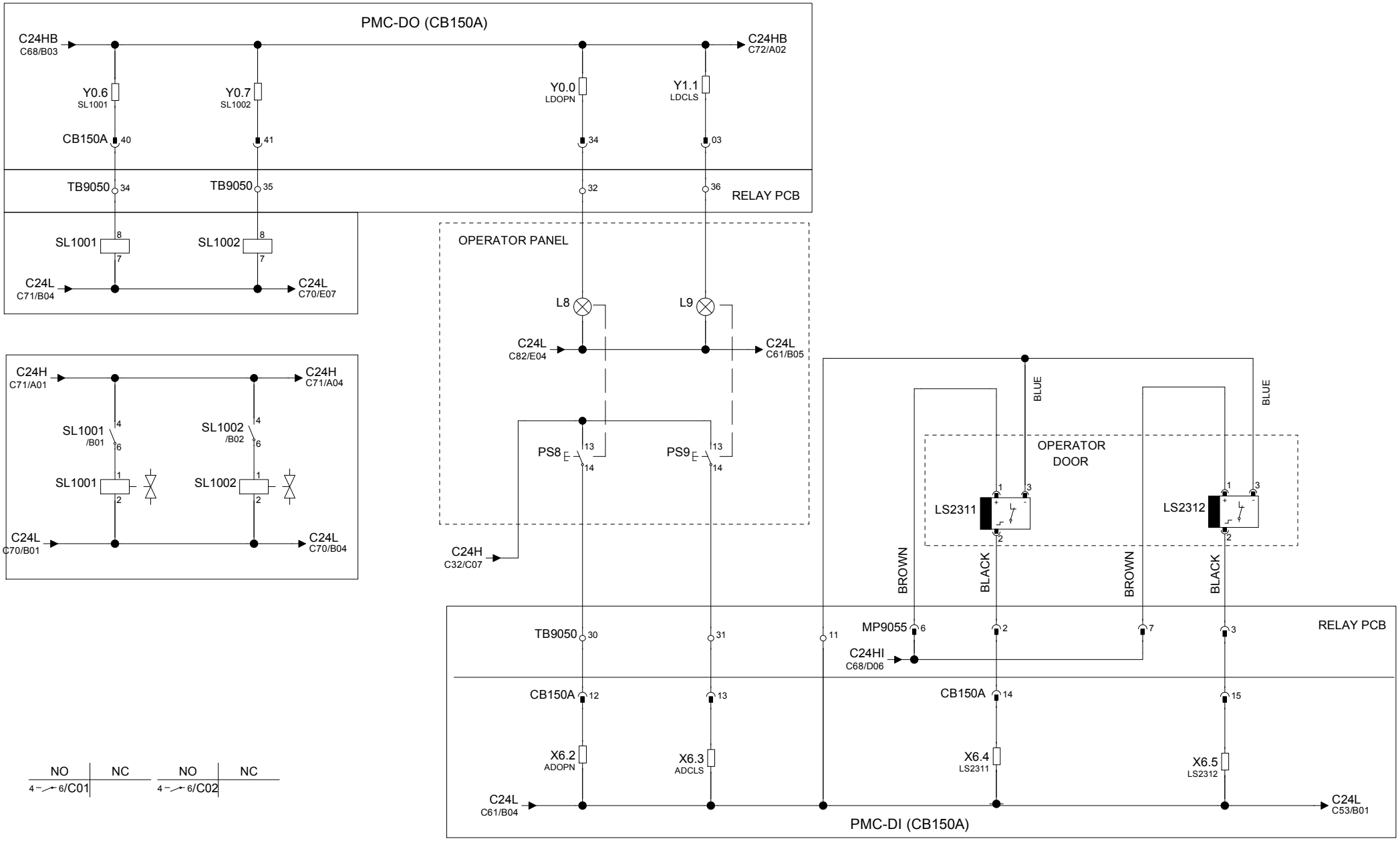


ROBOT DR CLOSE SOLENOID	ROBOT DR OPEN SOLENOID	ROBOT DR CLOSE CONFIRM	ROBOT DR OPEN CONFIRM	PUSHBUTTON ROBOT DR OPEN/CLOSE

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
ROBOT DR LEFT & RIGHT (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C68



*VALID FOR ALL MACHINES WITH AUTO DOOR OPTION

AUTO DOOR OPEN SOLENOID	AUTO DOOR CLOSE SOLENOID		AUTO DOOR OPEN PUSHBUTTON	AUTO DOOR CLOSE PUSHBUTTON		AUTO DOOR CLOSE	AUTO DOOR OPEN

Revisions			Date	Name
	4			
	3	APPROVED	25-NOV-2010	GOH WILLIAM
	2	CHECKED	25-NOV-2010	S MAHESH
	1	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

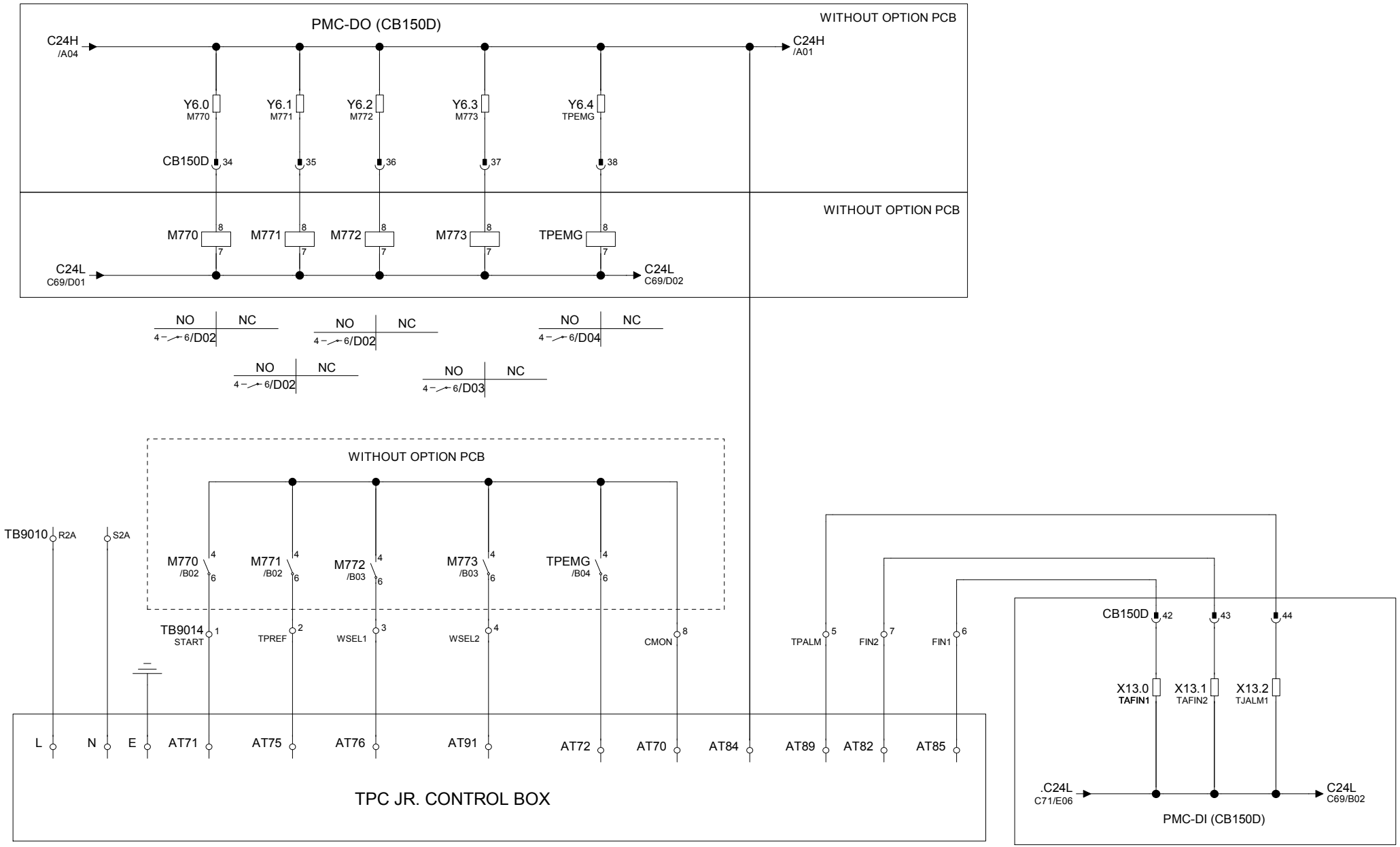
TITLE
AUTO DOOR WITH PUSHBUTTON (OPT)

MACHINE TYPE	SLIM3
NC TYPE	FS0i-MC

DRAWING No.
V090i-C1E-X03

GROUP
=

SHEET No.
C69

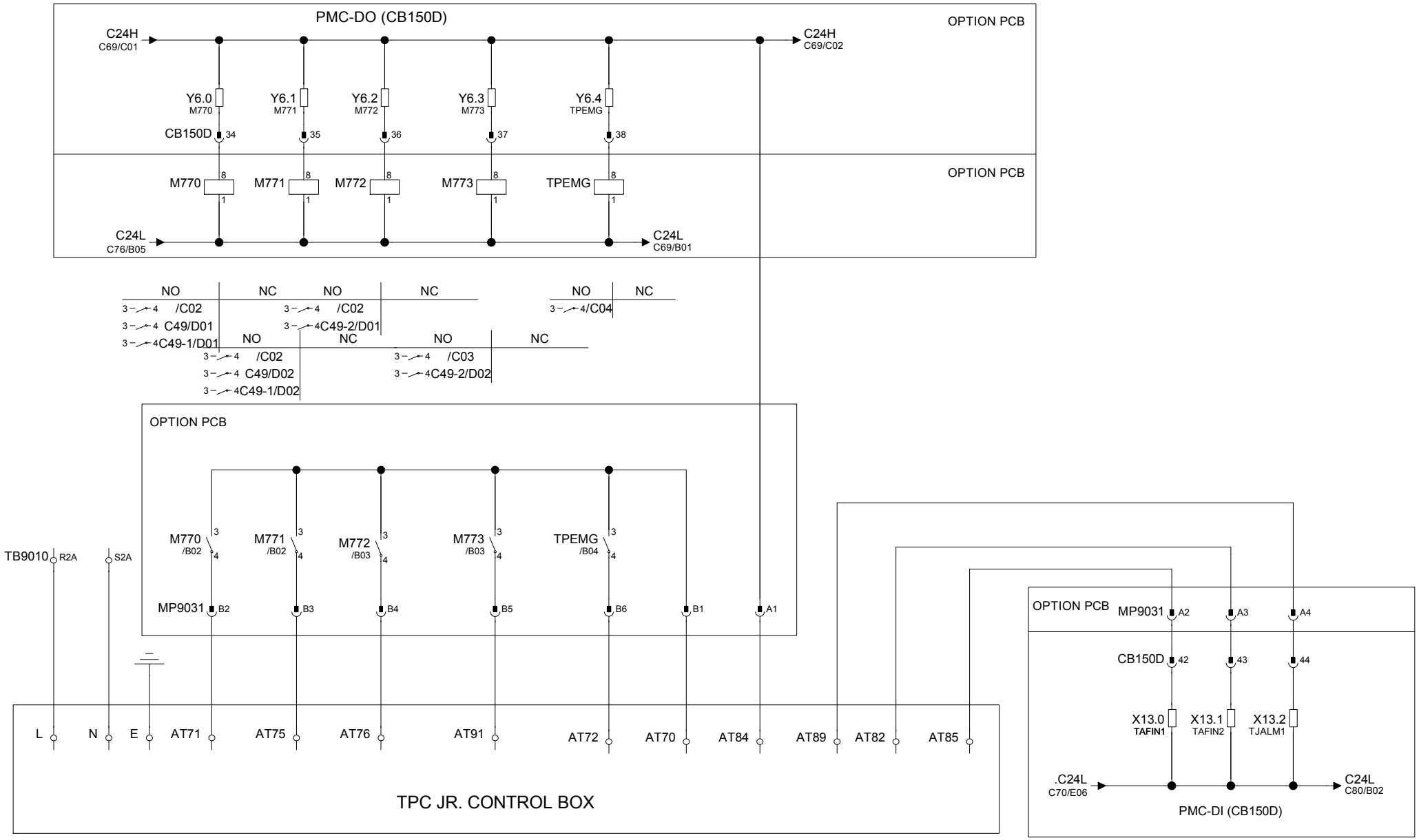


	TPC JR. INPUT POWER	TPC JR. A START	TPC JR. A REF	TPC JR. WORK SELECT 1	TPC JR. WORK SELECT 2	TPC JR. EMERGENCY	TPC JR. COMON I/P	TPC JR. COMON O/P	TPC JR. ALARM	TPC JR. FINISH 2	TPC JR. FINISH 1		TPC JR. FINISH 1	TPC JR. FINISH 2	TPC JR. ALARM

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
TPC JR (WITHOUT OPT PCB)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C70

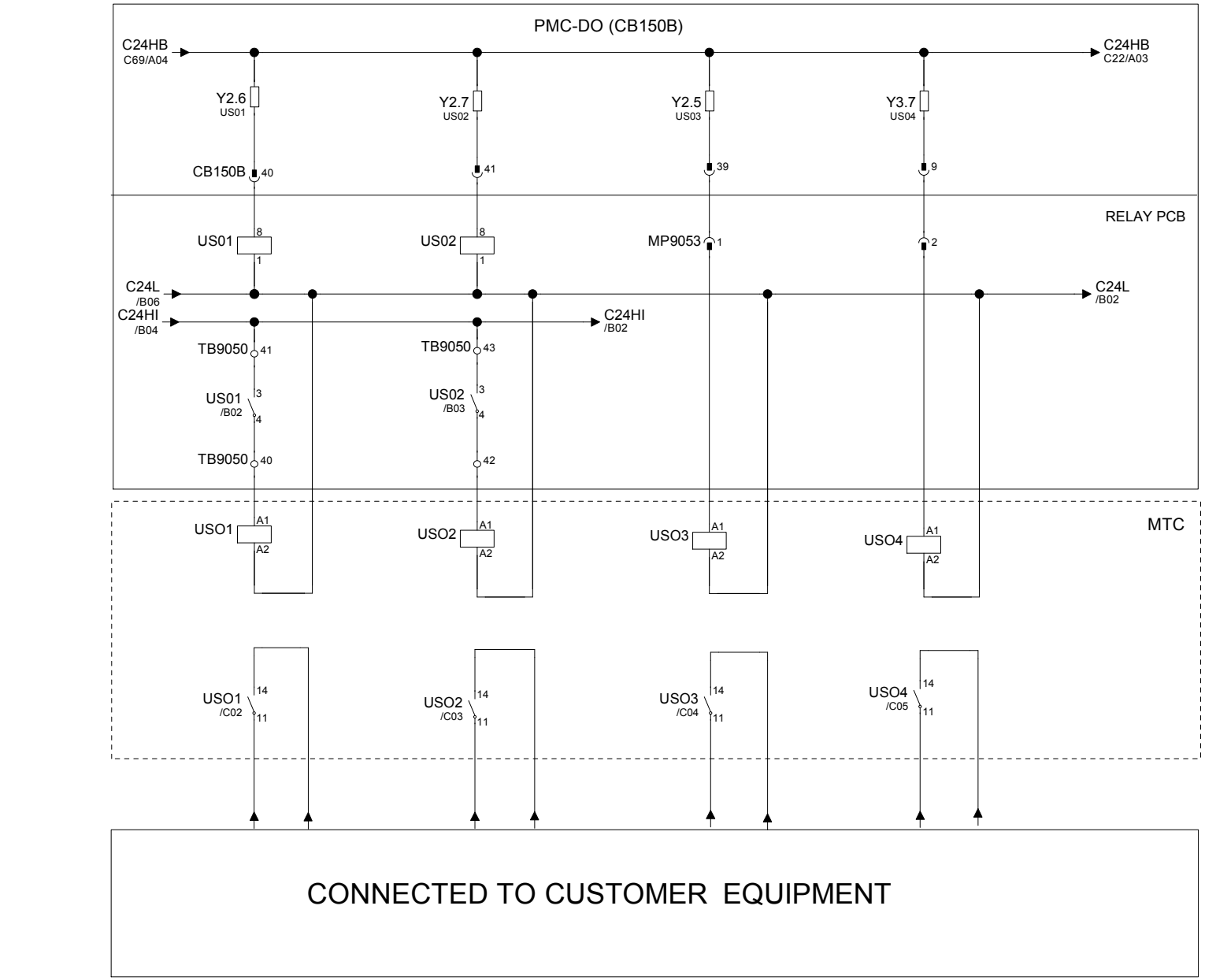


TPC JR. INPUT POWER	TPC JR. A START	TPC JR. A REF	TPC JR. WORK SELECT 1	TPC JR. WORK SELECT 2	TPC JR. EMERGENCY	TPC JR. COMON I/P	TPC JR. COMON O/P	TPC JR. ALARM	TPC JR. FINISH 2	TPC JR. FINISH 1		TPC JR. FINISH 1	TPC JR. FINISH 2	TPC JR. ALARM

Revisions			Date	Name
	4			
	3	APPROVED	25-NOV-2010	GOH WILLIAM
	2	CHECKED	25-NOV-2010	S MAHESH
	1	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
TPC JR (WITH OPT PCB)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C71

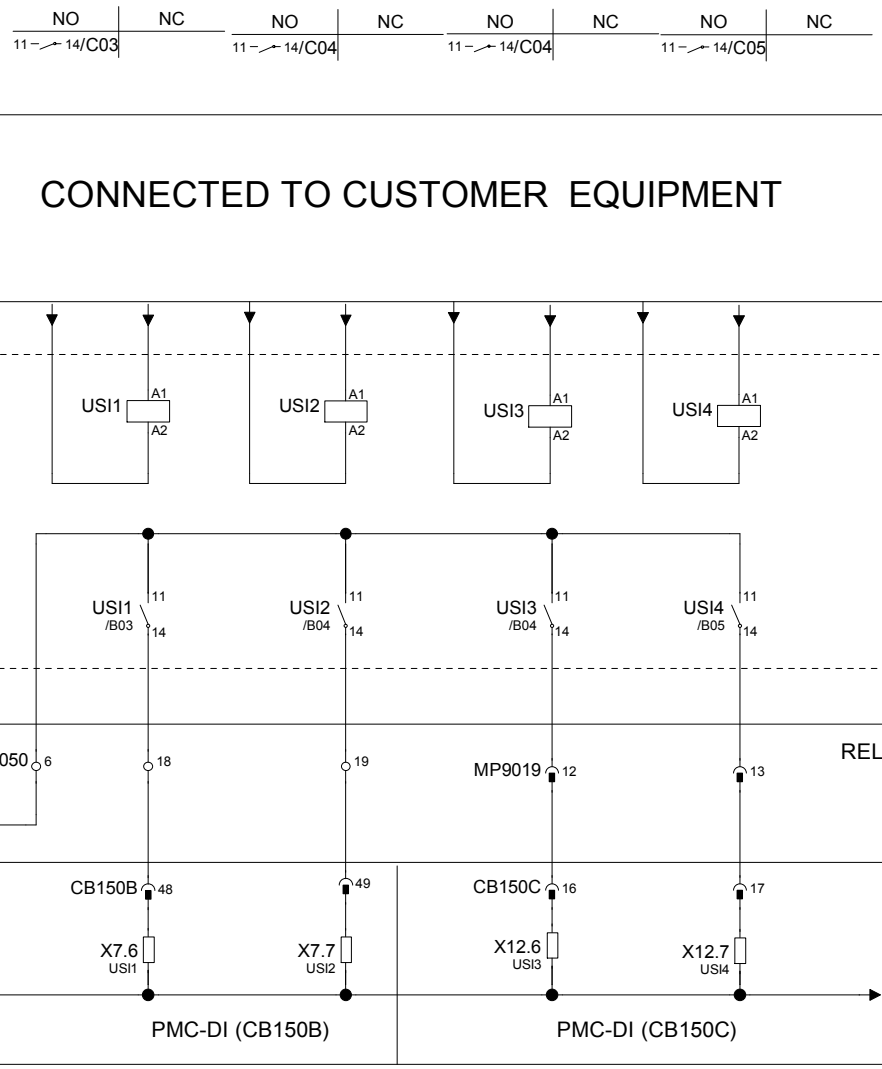


CUSTOMER MISCELLANEOUS CONTROL - M CODES OUTPUTS

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
USER MISCELLANEOUS CONTROL (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C72



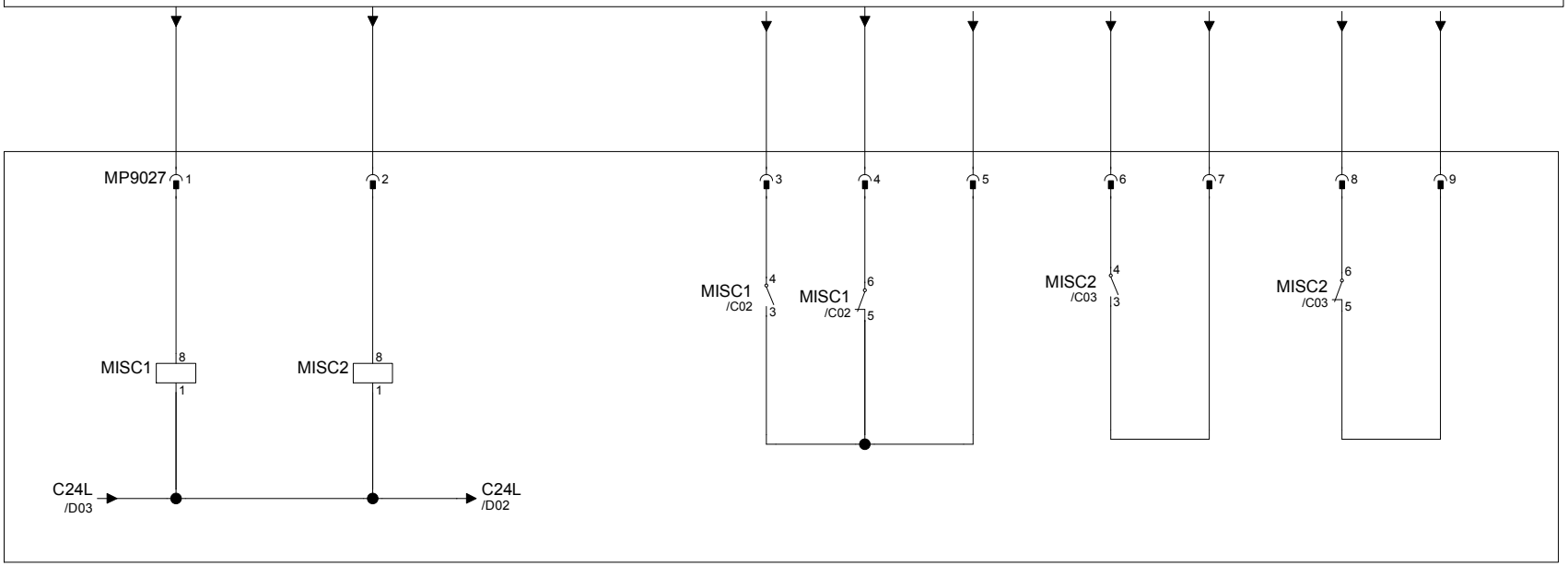
CUSTOMER MISCELLANEOUS CONTROL - M CODE INPUTS

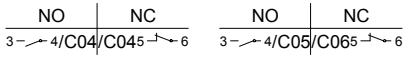
Revisions			Date	Name
	4			
	3	APPROVED	25-NOV-2010	GOH WILLIAM
	2	CHECKED	25-NOV-2010	S MAHESH
	1	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
USER MISCELLANEOUS CONTROL (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C73

CONNECTED TO CUSTOMER EQUIPMENT



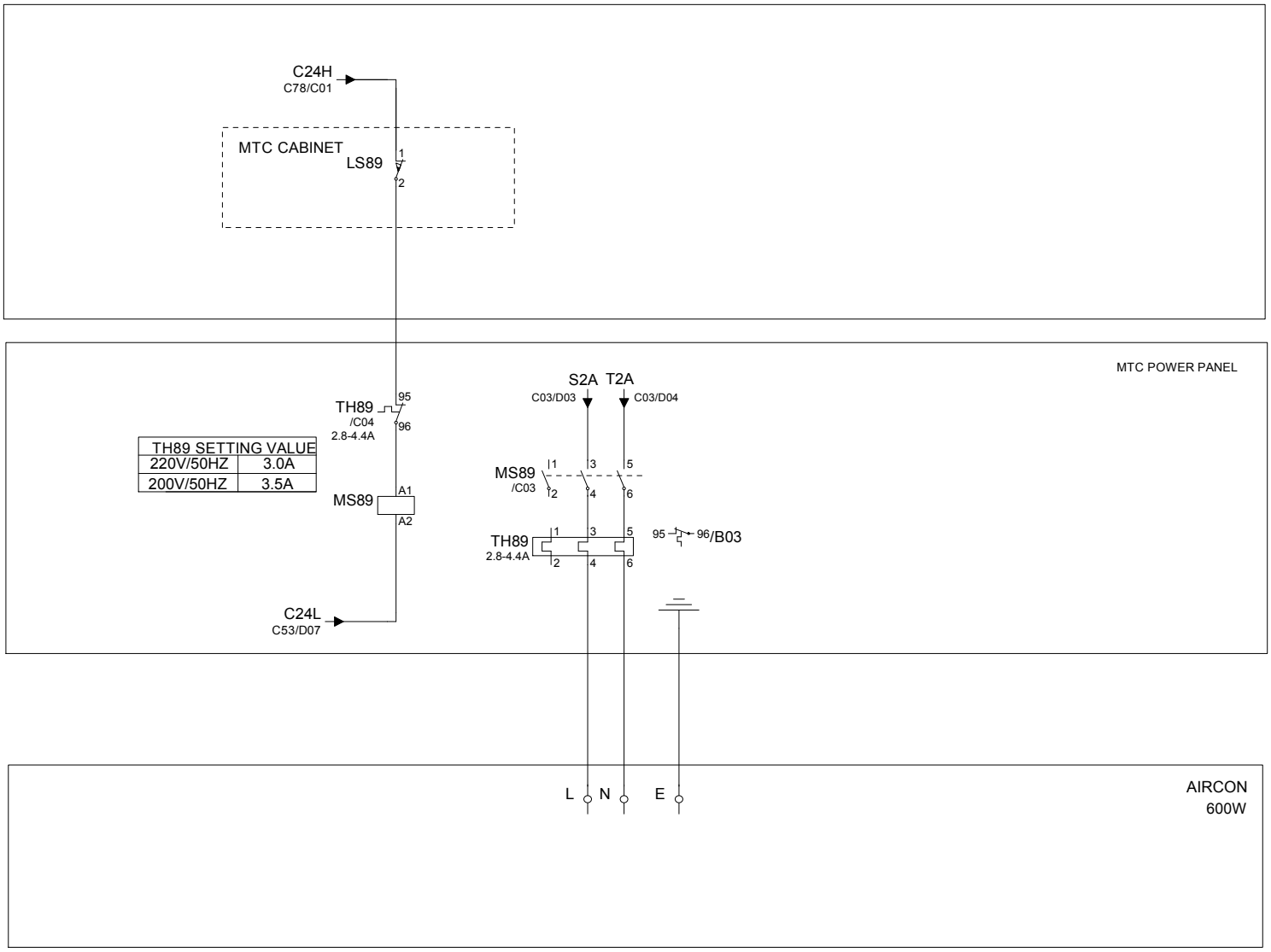


CUSTOMISED CONTROL

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CUSTOMISED CONTROL (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C74

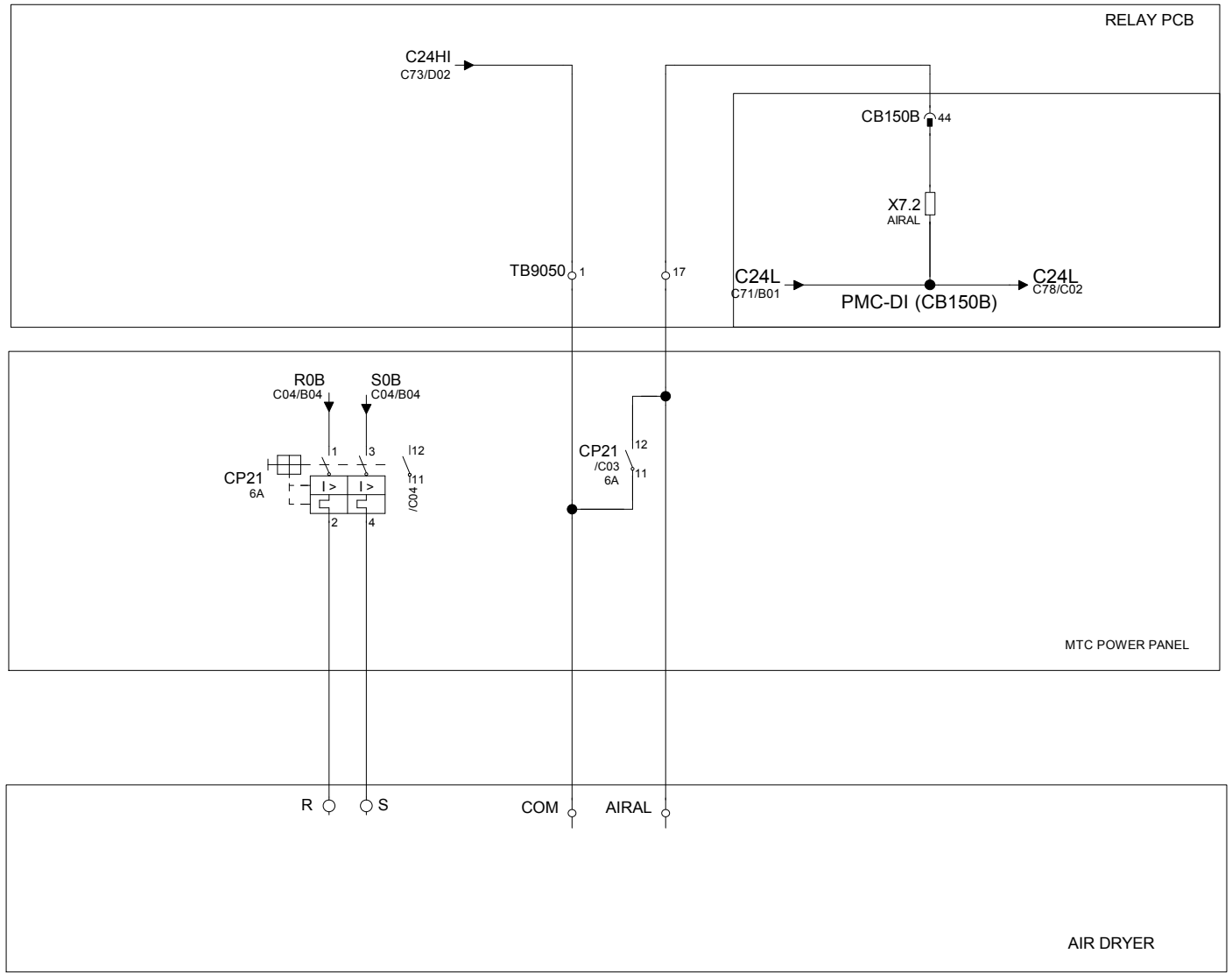


AIRCON CONTROL | INPUT POWER

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
AIRCON (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C75

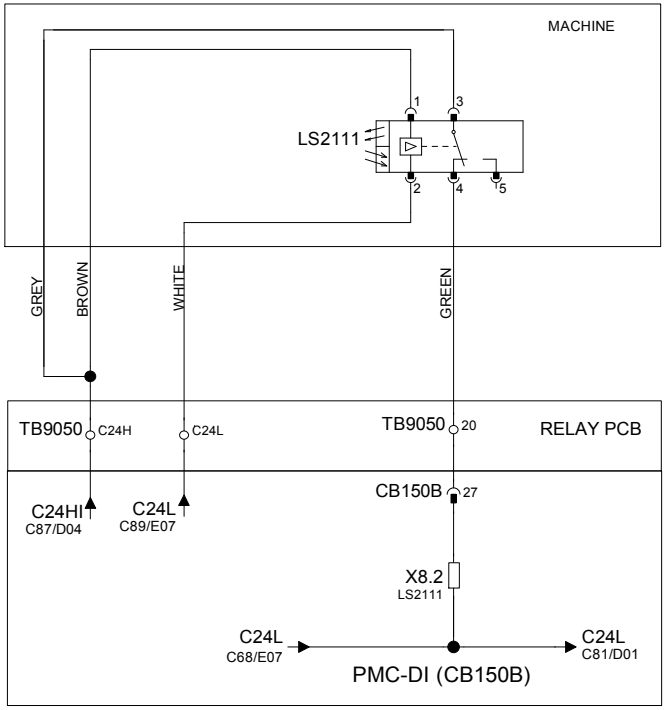


INPUT POWER		AIR DRYER ALARM	

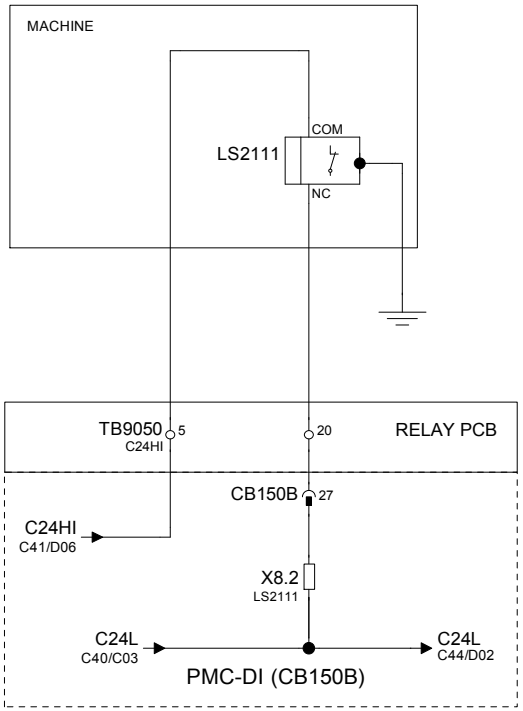
Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
AIR DRYER (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C76



BROKEN TOOL SENSOR - RENISHAW TRS2

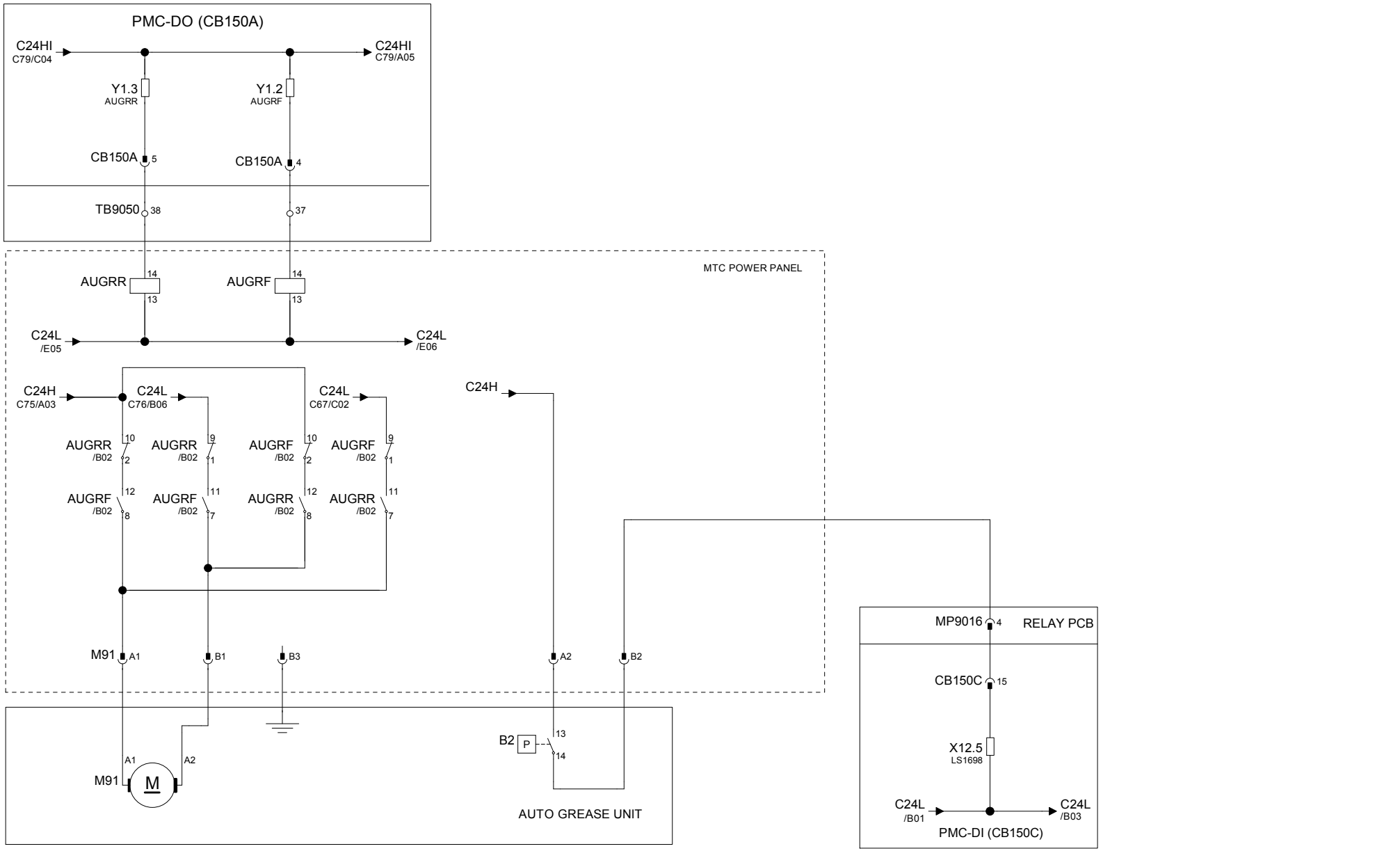


BROKEN TOOL SENSOR - OMRON

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
TOOL BROKEN SENSOR (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C77



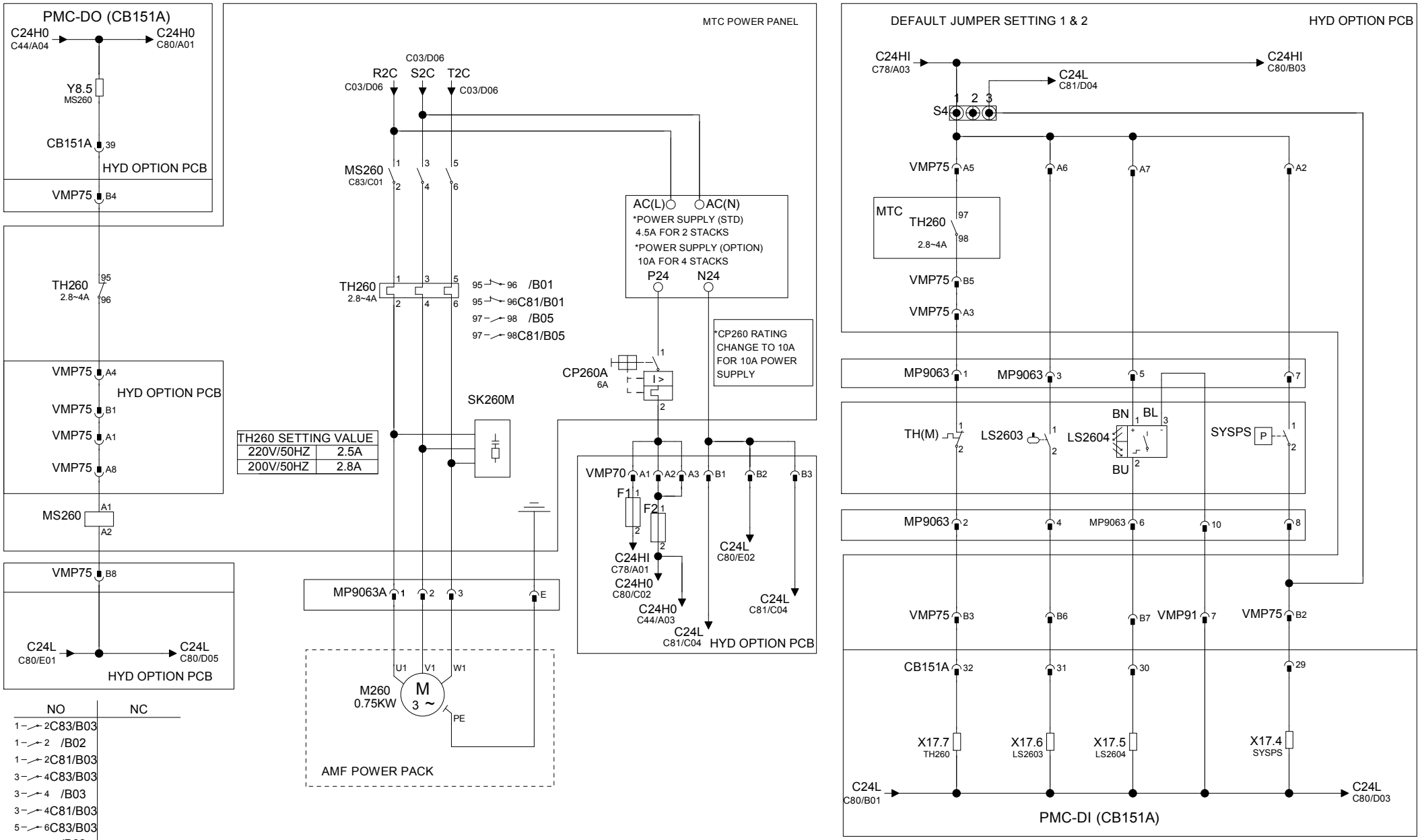
NOTE: WHEN AUTO GREASE IS INSTALLED, AN ADDITIONAL 10A POWER SUPPLY IS REQUIRED.
PLEASE REFER TO PAGE C32 FOR DETAILS OF ADDITIONAL 10A POWER SUPPLY.

AUTO GREASE MOTOR	PRESSURE SWITCH		PRESSURE SW SIGNAL

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
AUTO GREASE (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C78

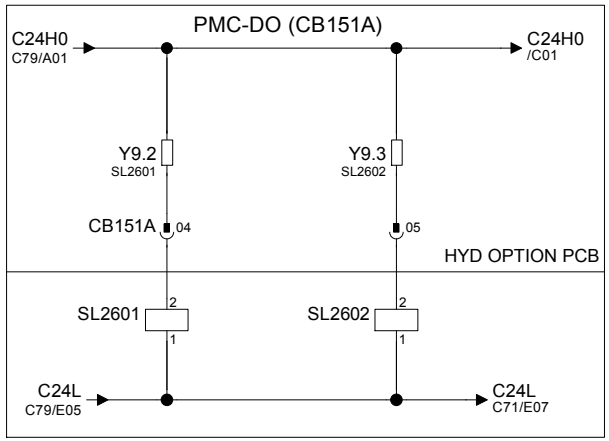


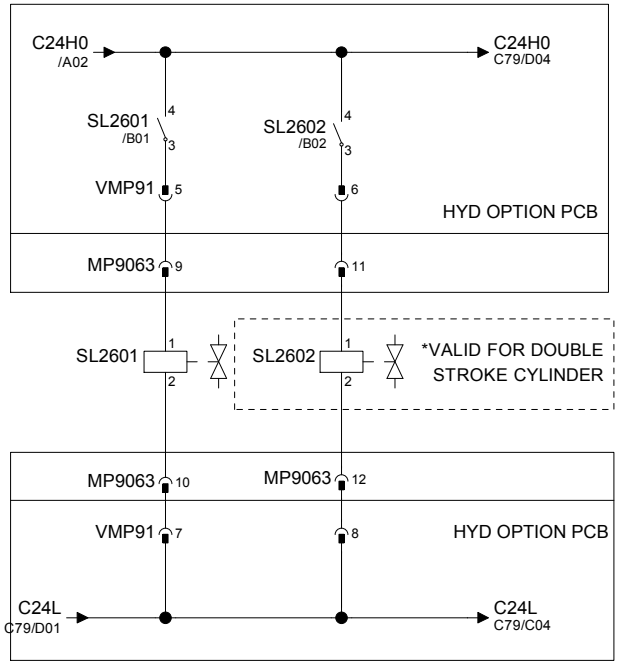
HYD POWERPACK ON/OFF CONTROL		HYD POWERPACK		HYD POWERPACK O/L THERMAL ALM	HYD OIL LEVEL LOW SWITCH	HYD OIL TEMP HIGH SENSOR	HYD POWRPACK SYSTEM PRESSURE	

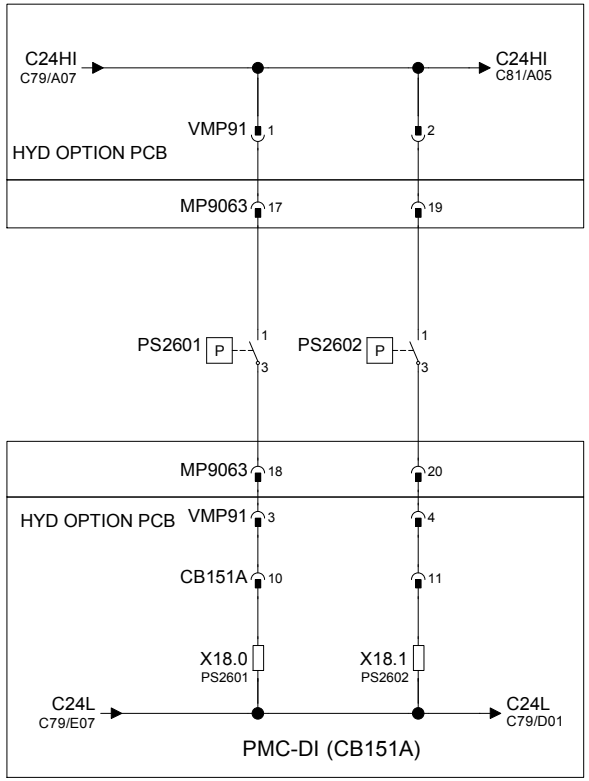
Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE				GROUP
HYDRAULIC OPT (AMF)	MACHINE TYPE	SLIM3	DRAWING No.	=
	NC TYPE	FS0i-MC	V090i-C1E-X03	SHEET No. C79







NO	NC
4–3 /C01	
4–3 C82/C01	
4–3 C84/C01	

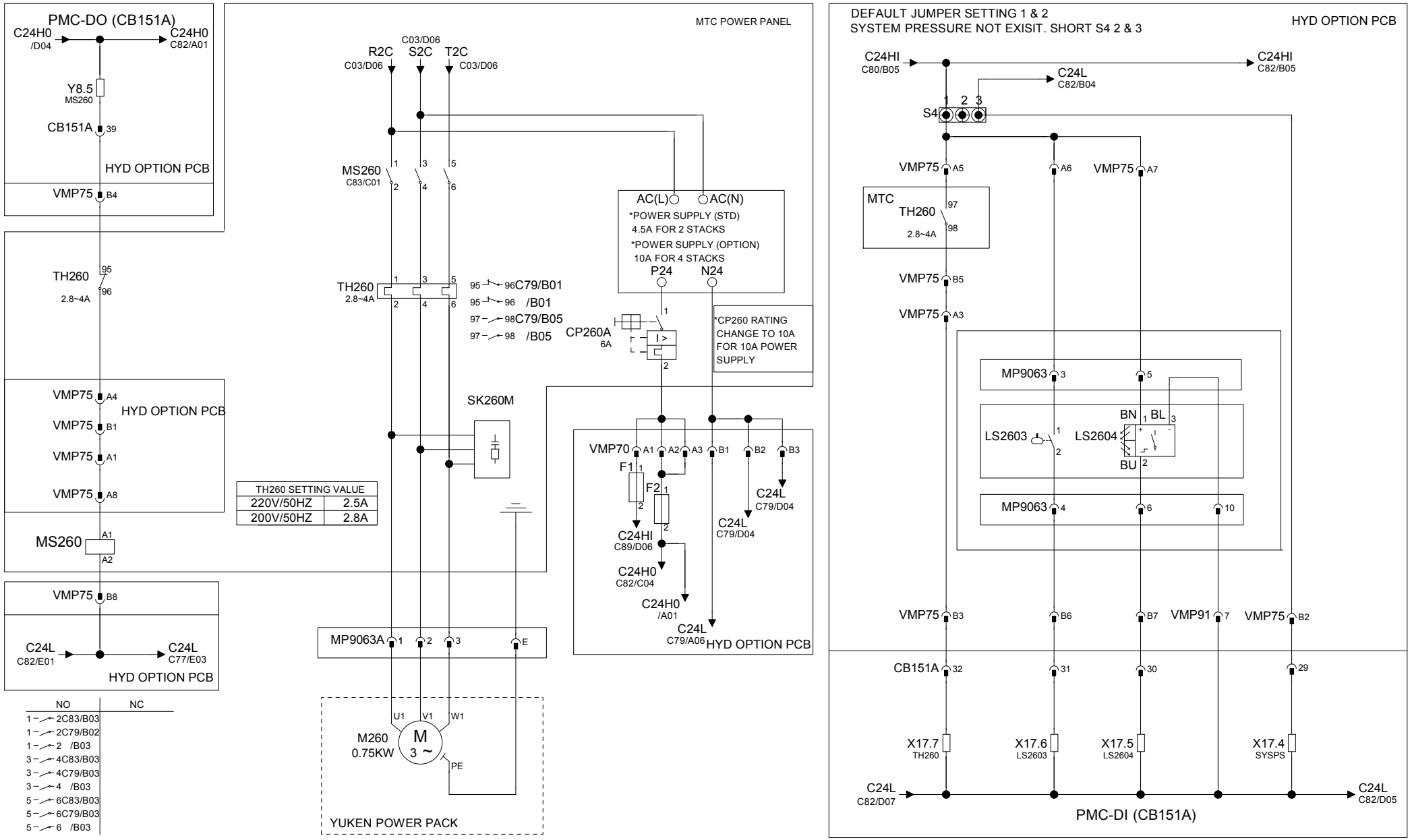
NO	NC
4–3 /C02	
4–3 C82/C02	
4–3 C84/C02	

SOL FIXTURE UN/CLAMP LINE1	SOL FIXTURE UN/CLAMP LINE2		PRESSURE SWITCH LINE1	PRESSURE SWITCH LINE2

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
HYDRAULIC PRESSURE SW SOL (AMF)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C80



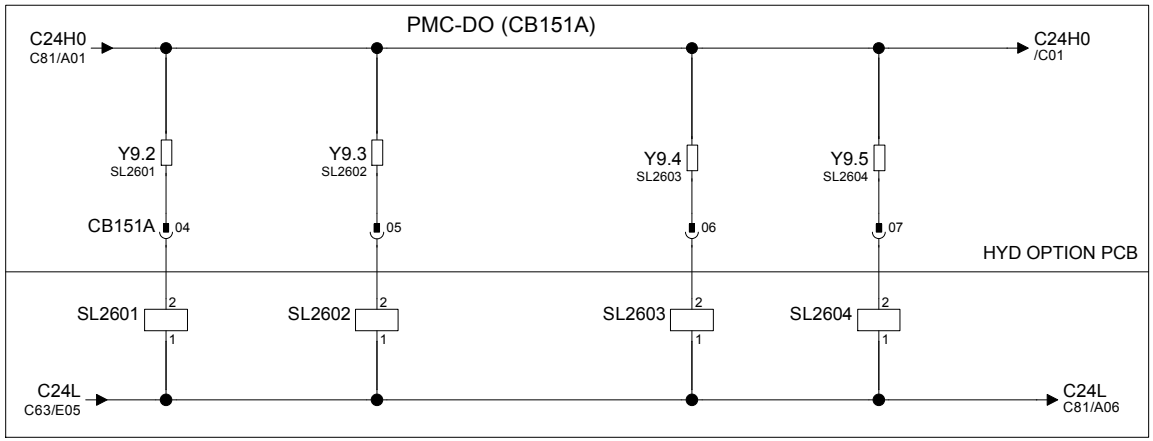
HYD POWRPACK ON/OFF CONTROL		HYD POWRPACK		HYD POWERPACK O/L THERMAL ALM	HYD OIL LEVEL LOW SWITCH	HYD OIL TEMP HIGH SENSOR	HYD POWERPACK SYSTEM PRESSURE

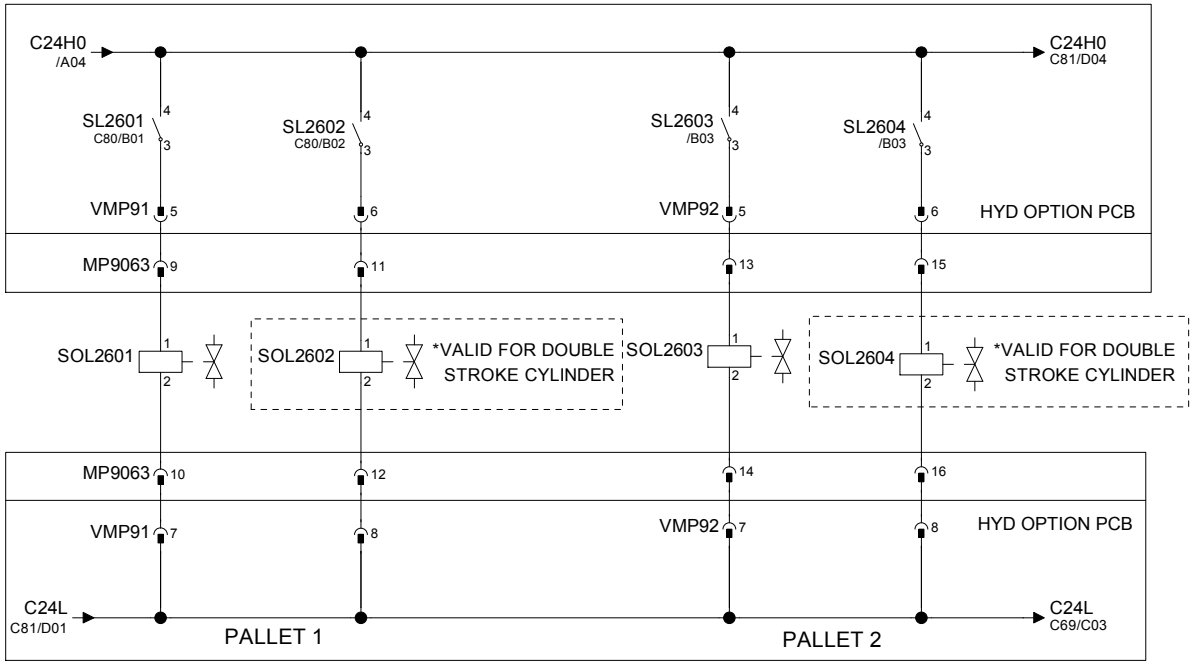
Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

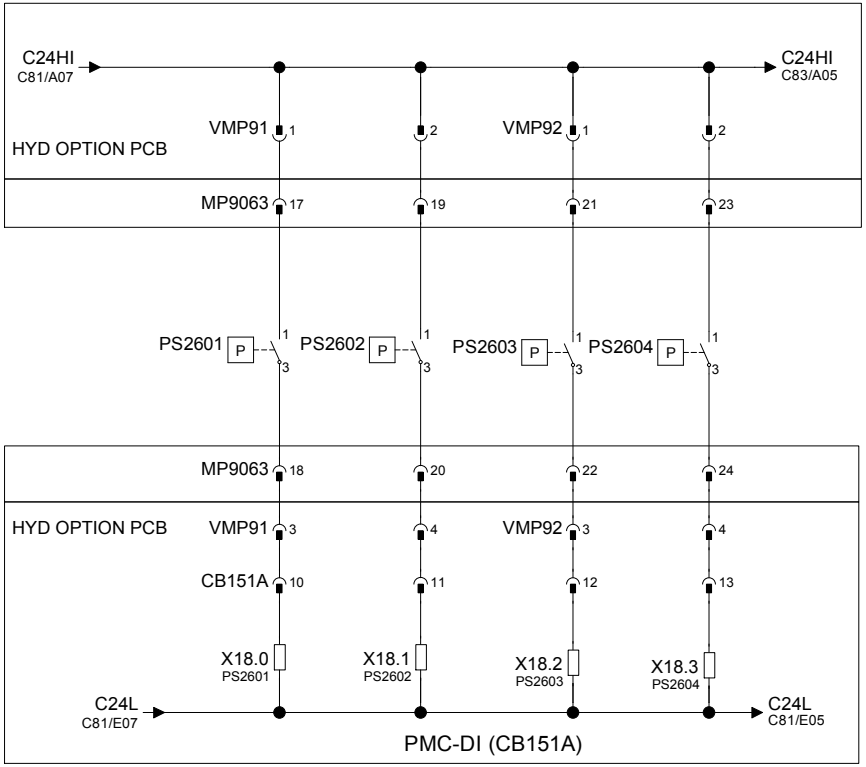
MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
HYDRAULIC OPT (YUKEN)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C81

NOTE:- INCASE OF APC MACHINE SL2601 & SL2602 IS USED FOR PALLET 1. SL2603 & SL2604 IS USED FOR PALLET 2.







NO	NC	NO	NC
4–3 C80/C01		4–3 C80/C02	
4–3 /C01		4–3 /C02	
4–3 C84/C01		4–3 C84/C02	

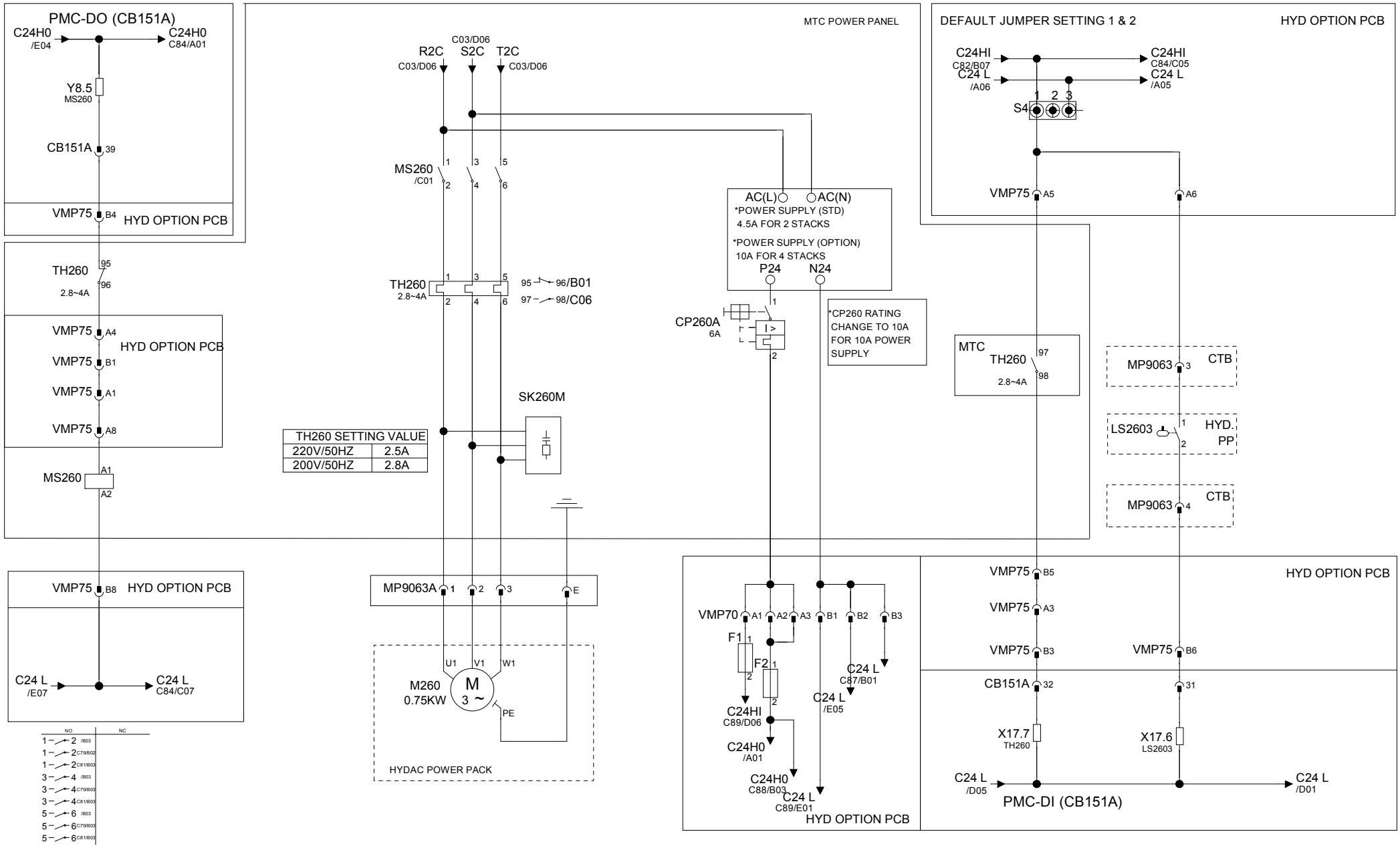
NO	NC	NO	NC
4–3 /C03		4–3 /C04	
4–3 C84/C03		4–3 C84/C04	

SOL FIXTURE UN/CLAMP LINE1	SOL FIXTURE UN/CLAMP LINE2		SOL FIXTURE UN/CLAMP LINE3	SOL FIXTURE UN/CLAMP LINE4		PRESSURE SWITCH LINE1	PRESSURE SWITCH LINE2	PRESSURE SWITCH LINE3	PRESSURE SWITCH LINE4	

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
HYDRAULIC PRESSURE SW, SOL(YUKEN)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C82



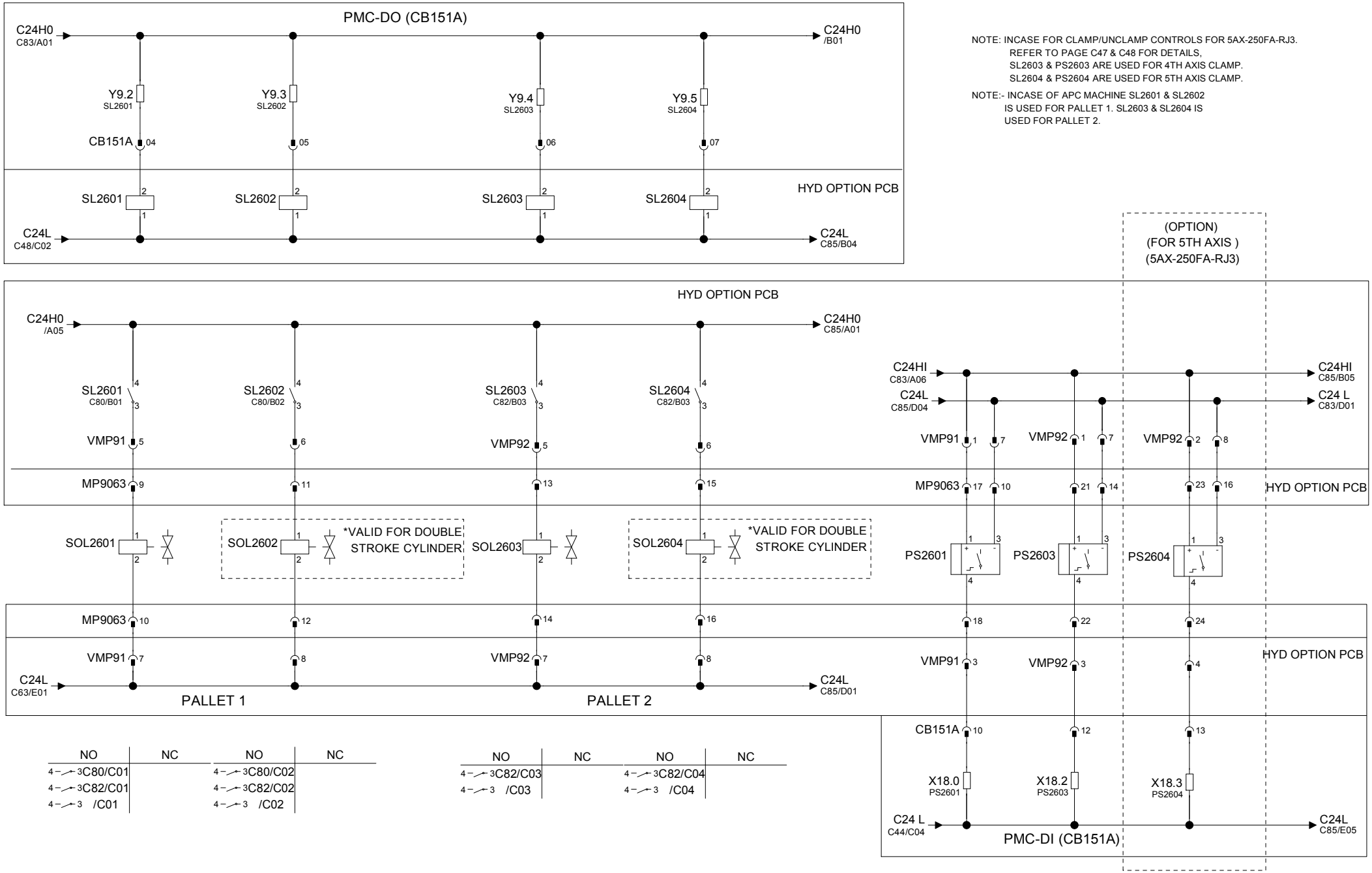
HYD POWRPACK ON/OFF CONTROL		HYD POWERPACK	
		HYD POWERPACK O/L THERMAL ALM	HYD OIL LEVEL LOW SWITCH

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.
HYDRAULIC OPT (HYDAC)	NC TYPE	FS0i-MC	V090i-C1E-X03

GROUP =
SHEET No. C83

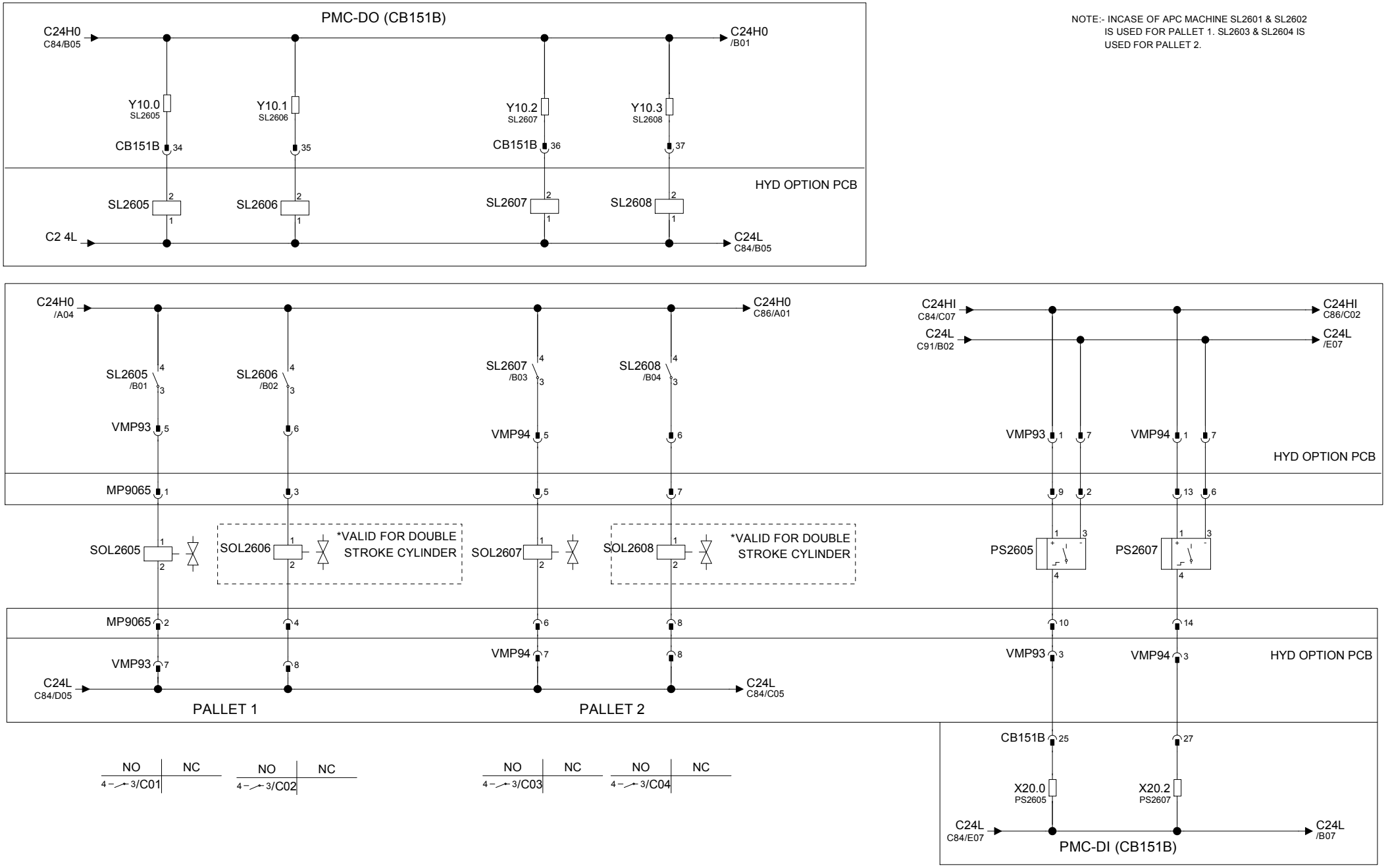


SOL FIXTURE UN/CLAMP LINE1	SOL FIXTURE UN/CLAMP LINE2		SOL FIXTURE UN/CLAMP LINE3	SOL FIXTURE UN/CLAMP LINE4		PRESSURE SWITCH LINE1	PRESSURE SWITCH LINE2	PRESSURE SWITCH LINE3

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
HYDRAULIC PR SW, SOL(1-2) STK (HYDAC)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C84

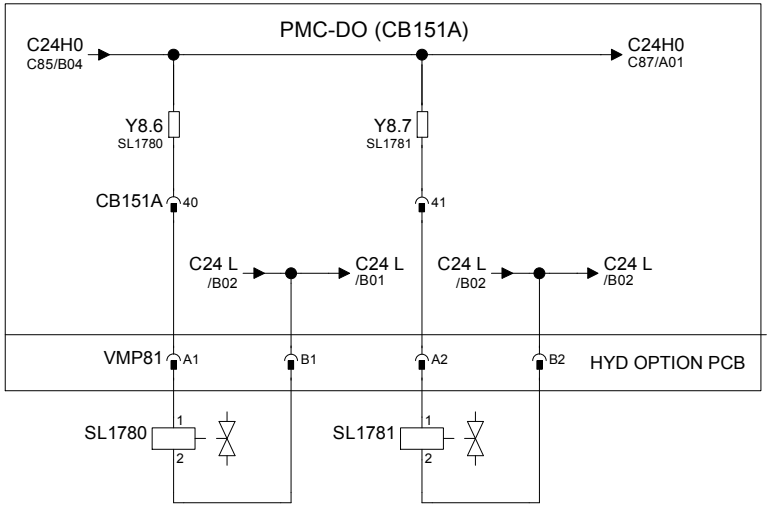


SOL FIXTURE UN/CLAMP LINE5	SOL FIXTURE UN/CLAMP LINE6		SOL FIXTURE UN/CLAMP LINE7	SOL FIXTURE UN/CLAMP LINE8		PRESSURE SWITCH LINE3	PRESSURE SWITCH LINE4

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

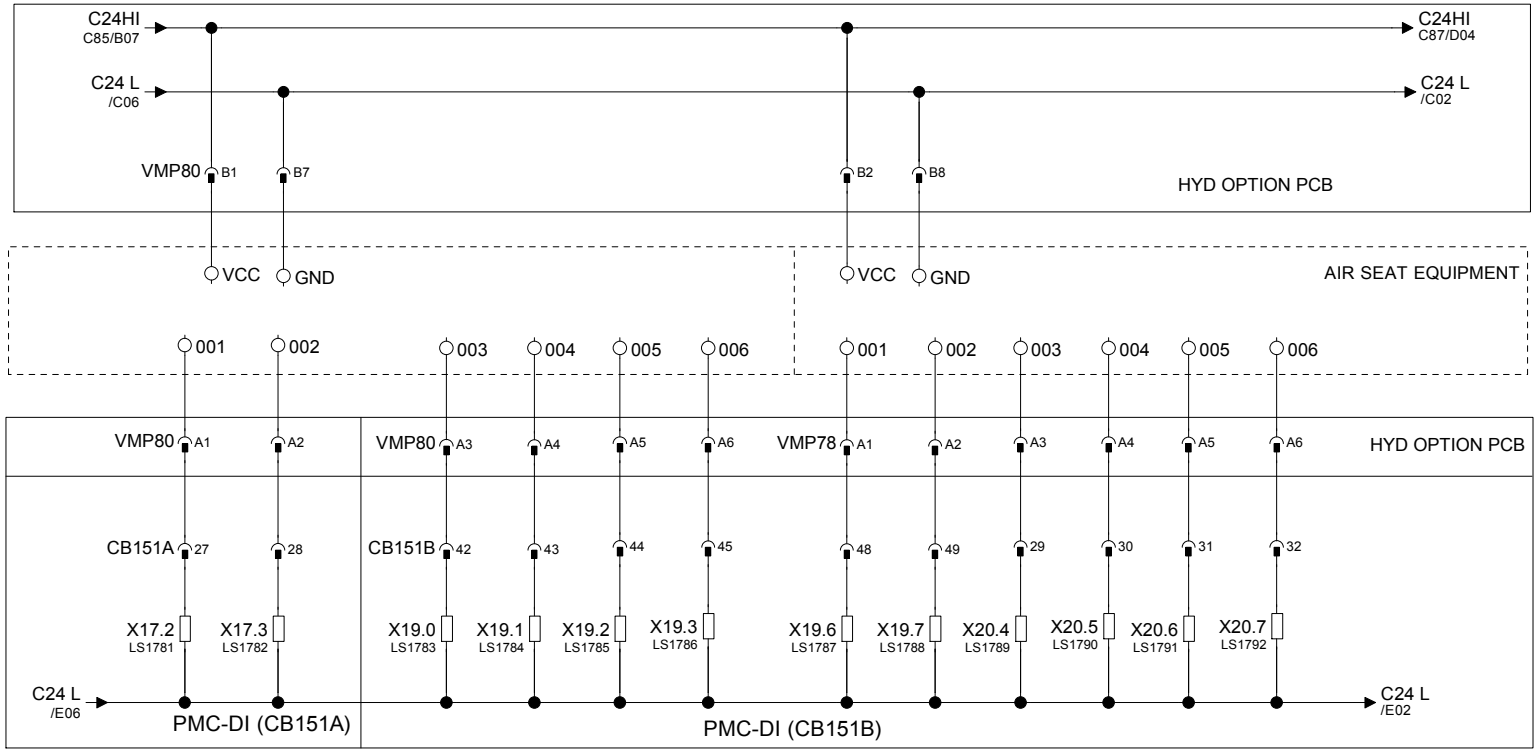
MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
HYDRAULIC PR SW, SOL(3-4) STK (HYDAC)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C85



NOTE:
IN CASE OF STANDARD MACHINE, SL1780 IS STANDARD.
AIRSEAT UP TO 12 UNITS CAN BE INSTALLED

NOTE:
IN CASE OF APC MACHINE, AIRSEAT UP TO 12 UNITS CAN BE INSTALLED.
SL1781 IS A MUST.
PALLET 1, AIRSEATS UP TO 6 UNITS CAN BE INSTALL (AIRSEAT 1, 3, 5, 7, 9 & 11)
PALLET 2, AIRSEATS UP TO 6 UNITS CAN BE INSTALL (AIRSEAT 2, 4, 6, 8, 10 & 12)

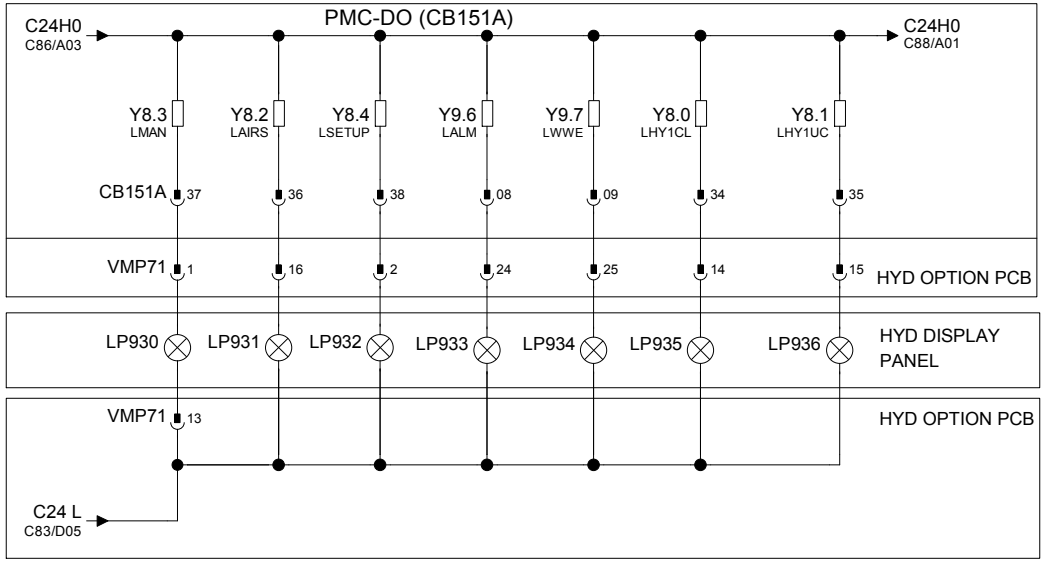


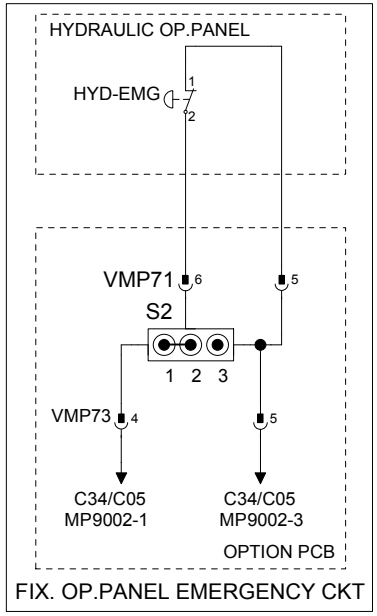
AIRSEAT SOLENOID 1	AIRSEAT SOLENOID 1 FOR APC MACHINE	AIRSEAT 1	AIRSEAT 2	AIRSEAT 3	AIRSEAT 4	AIRSEAT 5	AIRSEAT 6	AIRSEAT 7	AIRSEAT 8	AIRSEAT 9	AIRSEAT 10	AIRSEAT 11	AIRSEAT 12

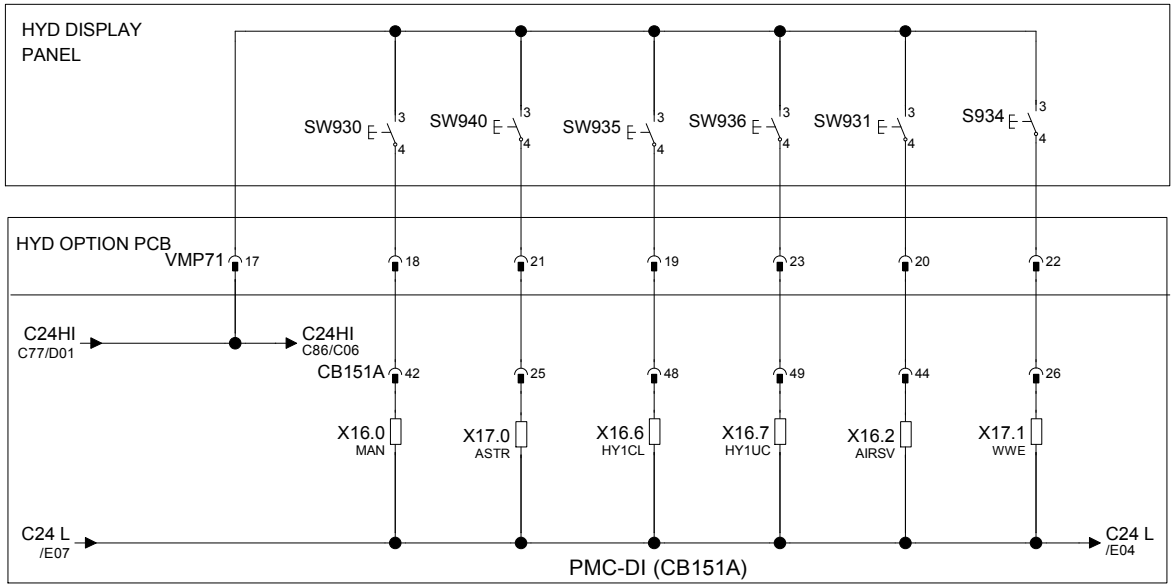
Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
AIR SEAT CHECK SENSOR (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C86





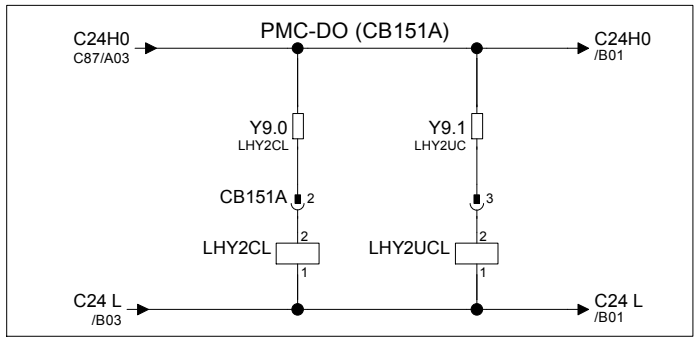


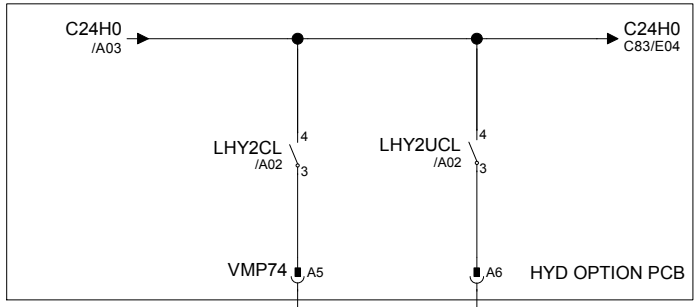
MANUAL MODE LAMP	SEAT CHECK LAMP	SETUP FINISH LAMP	ALARM LAMP	WORK WASH ENABLE LAMP	HYD CLAMP1 LAMP	HYD UNCLAMP1 LAMP		MANUAL MODE SW	AUTO START PB	CLAMP1 PB	UNCLAMP1 PB	AIRSEAT KEY SW	WORK WASH ENABLE PB

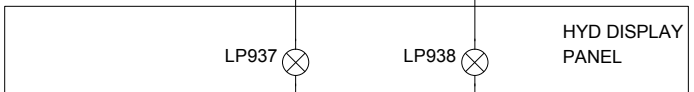
Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

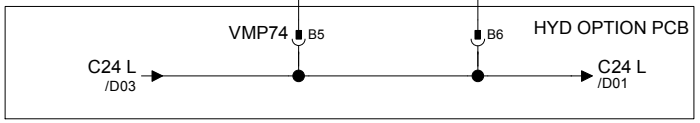
MAKINO
MAKINO ASIA PTE LTD

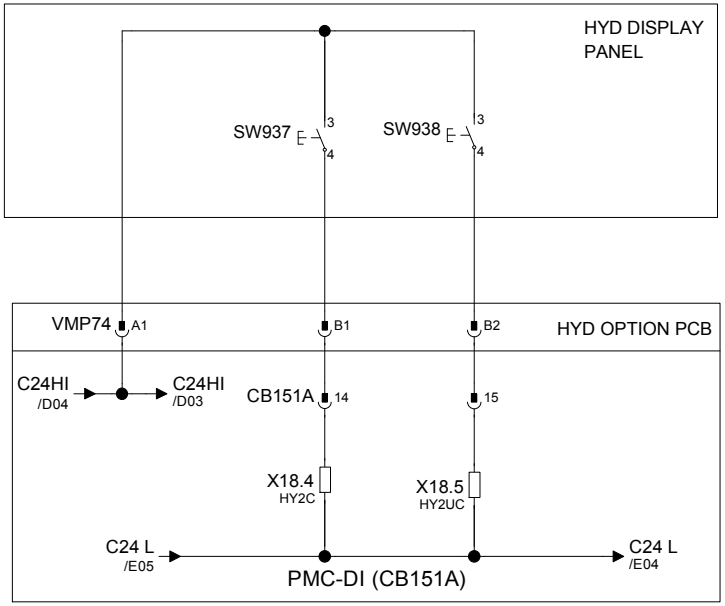
TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
HYDRAULIC DISPLAY (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C87

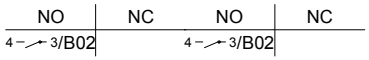












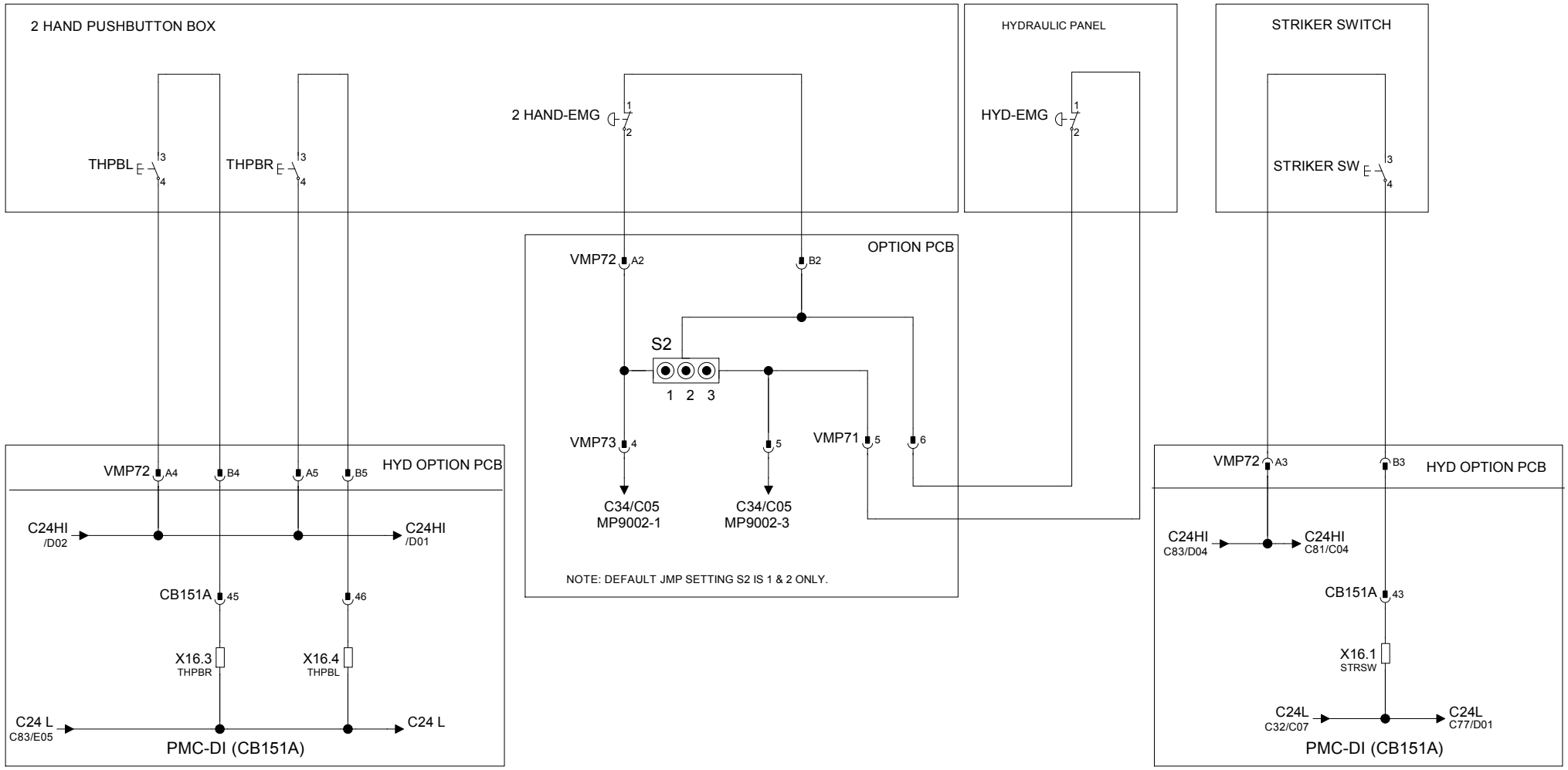
HYD CLAMP2 LAMP	HYD UNCLAMP2 LAMP	CLAMP2 PB	UNCLAMP2 PB

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
HYDRAULIC DISPLAY-2ND SET UN/CLAMP (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. C88

FOR TWO HAND PUSHBUTTON EMG CNTRL
- REMOVE S2 JUMPER FROM 1&2 ON HYDRAULIC PCB.
- REMOVE S10 SHORT FROM RELAY PCB.



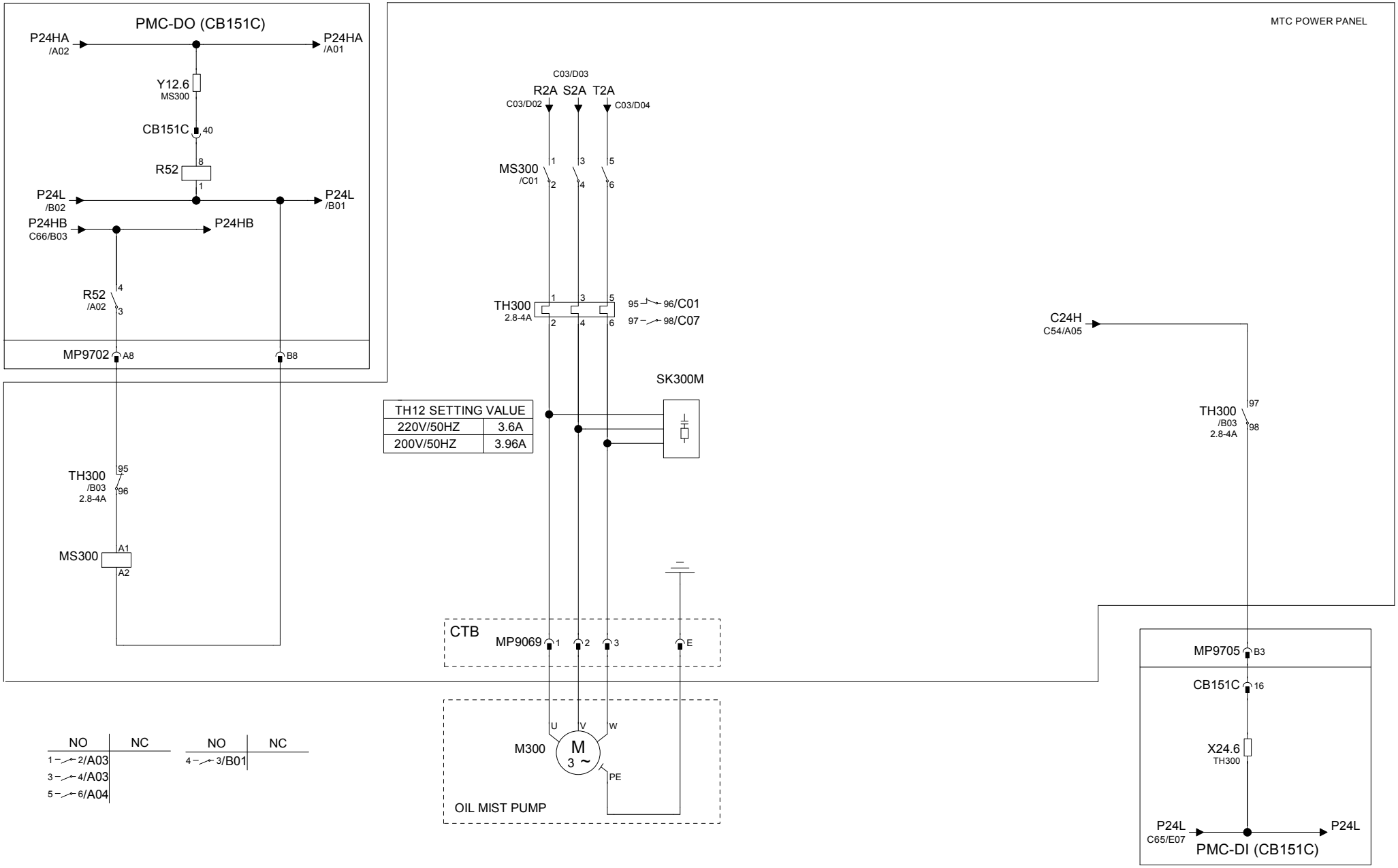
NOTE: ONLY VALID WITH HYDRAULIC FIXTURE CONTROL

	TWO HAND PB RIGHT	TWO HAND PB LEFT		EMERGENCY PUSHBUTTON CONTROL	STRIKER SWITCH

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
TWO HAND PUSHBUTTON (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C89

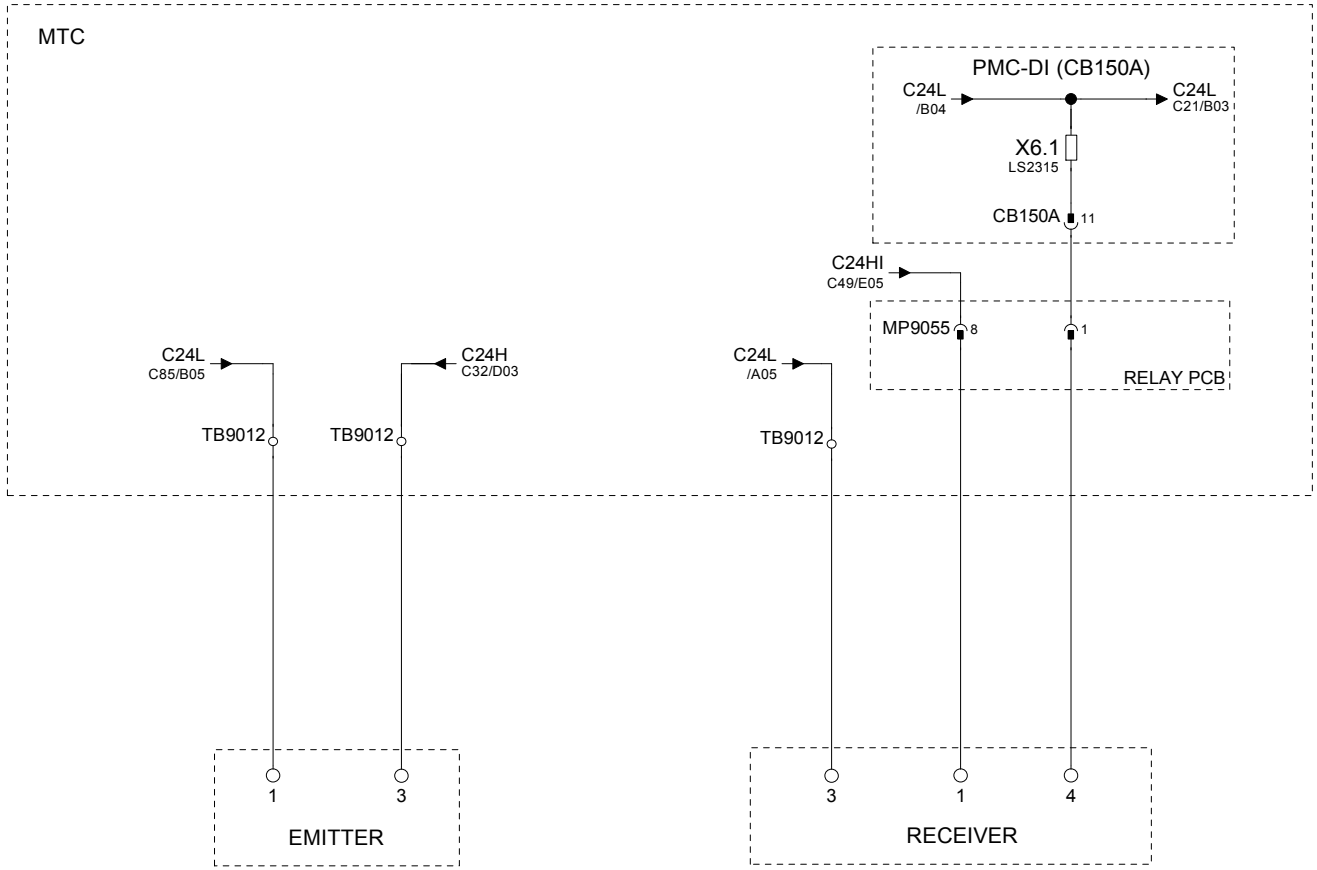


OIL MIST MOTOR ON	OIL MIST MOTOR	MIST COLLECTOR THERMAL

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
OIL MIST COLLECTOR (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C90



LIGHT CURTAIN SENSOR

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
SAFETY LIGHT CURTAIN (OPT)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. C91

NO. TB9001

CB0/ EXT. TRANS(OPTION)

1	R0/R0B
2	S0/S0B
3	T0/T0B
PE	E

NO. TB9012

1	C24H		1	C24H	1	C24H
2	C24H		2	C24H	2	C24H
3	C24H		3	C24H	3	C24H
4	C24L		4	C24L	4	C24L
5	C24L		5	C24L	5	C24L
6	C24L		6	C24L	6	C24L
7	LS25		6			

NO. TB9014

1	START		1	START
2	TPREF		2	TPREF
3	WSEL1		3	WSEL1
4	WSEL2		4	WSEL2
5	TPALM		5	TPALM
6	FIN1		6	FIN1
7	FIN2		7	FIN2
8	CMON		8	CMON
9	C24H		9	C24H
10	C24L		10	C24L
11	LS204		11	LS204
12	LS205		12	LS205

NO. TB9013

1	SW296-1
2	SW296-2
3	LS-COM
4	A1
5	A2
6	A3
7	A4
8	US03
9	US04

NO. TB9050

1	C24HI	26	LS196
2	C24HI	27	LS195
3	C24HI	28	LS191
4	C24HI	29	LS1340
5	C24HI	30	ADOPN
6	C24HI	31	ADCLS
7		32	LDOPN
8	SL1309	33	MS19
9	C24L	34	SL1001
10	C24HB	35	SL1002
11	C24L	36	LDCLS
12	LS04	37	AUGRF
13	CCHAL	38	AUGRR
14	LS03	39	RCHP
15	LS199	40	US01-4
16	SL1306B	41	US01-3
17	AIRAL	42	US02-4
18	USi1	43	US02-3
19	USi2		
20	LS2111		
21	AIRB		
22	SL1306A		
23	TH19		
24	LS190		
25	LS912		

NO. TB9010

PFM OPTION

1	R2A		1	R2A	1	R2A
2	R2A		2	R2A	2	R2A
3	R2A		3	R2A	3	R2A
4	S2A		4	S2A	4	S2A
5	S2A		5	S2A	5	S2A
6	S2A		6	S2A	6	S2A
7	T2A		7	T2A	7	T2A
8	T2A		8	T2A	8	T2A
9	T2A		9	T2A	9	T2A
10	R4		10	R4	10	R4
11	R4		11	R4	11	R4
12	S4		12	S4	12	S4
13	S4		13	S4	13	S4

NO. TB9011

PFM OPTION

1	100H		1	100H	1	100H
2	100H		2	100H	2	100H
3	100L		3	100L	3	100L
4	100L		4	100L	4	100L
5	CLK1		5	CLK1	5	CLK1
6	CLK2		6	CLK2	6	CLK2

AC CONTROL TERMINALS	DC CONTROL TERMINALS	TPC JR. TERMINALS	MTC EXTERNAL TERMINALS	RELAY PCB TERMINAL BLOCK

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
TERMINAL BLOCK - MTC	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. D01

No.	MP9002	Use	EMG-HYD
Type	1-178313-5		

1	S10	2		3	S10

No.	MP9003	Use	EMG-EXT
Type	1-178313-5		

1	S11	2		3	S11

No.	MP9004	Use	EMG-APC
Type	0-178313-5		

1	S8	2		3	S8

No.	MP9005	Use	X-AXIS LS
Type	DMS-9-P-1-M-G E-2-6-F		

1	C24HI	2	C24HI	3	ESP X
4	LSDX	5	OT2X	6	
7		8		9	

No.	MP9006	Use	Y-AXIS LS
Type	DMS-9-P-1-M-G E-2-6-F		

1	C24HI	2	ESP X	3	ESP Y
4	LSDY	5	OT2Y	6	
7		8		9	

No.	MP9007	Use	Z-AXIS LS
Type	DMS-9-P-1-M-G E-2-6-F		

1	C24HI	2	ESP Y	3	ESP Z
4	LSDZ	5	OT2Z	6	
7		8		9	

No.	MP9008	Use	Z-AX BRK
Type	0-178323-2		

B1		B2	C24L	B3	
A1		A2	EM1-5	A3	ZBRK-3

No.	MP9009	Use	HYD. PRESSURE SW
Type	1-178323-2		

B1	C24L	B2	C24L	B3	PS1201
A1	C24HI	A2	SL00-4	A3	

No.	MP9010	Use	OILMATIC CONTROL
Type	DMS-9-S-1-M-G E-2-6-F		

1	C24HI	2	C24HI	3	LS170
4	OMAL	5	C24HB	6	C24L
7	C24HB	8	OMON-4	9	OMON-3

No.	MP9011	Use	PFM INTERFACE.
Type	DMS-9-S-1-M-G E-2-6-F		

1	ZBRK-4	2	EM1-9	3	C24HI
4	TOOLR	5	S1	6	S1
7	S7	8	S7	9	

No.	MP9012	Use	AIR PRESSURE SW
Type	0-178313-5		

1	C24HI	2	C24L	3	LS905

No.	MP9015	Use	OPERATOR PANEL
Type	DMS-25-P-1-M-G E-2-6-F		

1	C24HB	2	C24HB	3	NCON-3	4	C24HB	5	NCON-4
6	EM2-4	7	C24L	8	EM2-4	9	LP1-1	10	C24L
11	C24L	12	EM2-4	13	S6	14	LP1-2	15	C24HI
16	PBSCL	17	C24L	18	PRTC	19	PTL1	20	PTL2
21	PTL3	22		23		24		25	

No.	MP9016	Use	ATC OP.PANEL
Type	DMS-9-S-1-M-G E-2-6-F		

1	C24HI	2	MCSP1	3	MCSP2
4	AUGRPB	5		6	
7		8	S6	9	S6

No.	MP9017	Use	SPINDLE.TL UNCLAMP
Type	DMS-9-S-1-M-G E-2-6-F		

1	LS00	2	LS01	3	LS602
4	LS601	5	C24HI	6	C24HI
7	C24HI	8	C24HI	9	C24L

No.	MP9018	Use	PNEUMATIC SOL
Type	DMS-25-P-1-M-G E-2-6-F		

1	SL2310	2	SL201	3	SL01	4	SL181	5	SL2111
6	SL1306	7	SL1306	8	SL1308	9		10	
11		12		13	C24L	14	SL2311	15	SL202
16	SL6B	17	SL02	18	SL192	19	----	20	
21		22		23		24		25	

No.	MP9019	Use	ATC LS
Type	DMS-25-S-1-M-G E-2-6-F		

1	C24HI	2	C24HI	3	C24L	4	C24L	5	LS202
6	LS201	7		8		9	LS154	10	LS155
11	LS25	12	USDi3	13	USDi4	14	LSDA	15	
16		17		18		19		20	
21		22		23		24		25	

No.	MP9020	Use	4th/5th AXIS
Type	DMS-25-S-1-M-G E-2-6-F		

1	C24HI	2	C24HI	3	LSDA	4	LSDB	5	C24HI
6	C24L	7	LS151	8	LS152	9	OT1A	10	OT2A
11	ANGL-8	12	SL150-4	13	C24L	14	C24HB	15	C24HB
16	MP9020 -17	17	MP9020 -16	18	ESP4	19	ESP5	20	
21		22		23		24		25	

RELAY PCB CONNECTORS

Revisions			Date	Name
	4			
	3	APPROVED	25-NOV-2010	GOH WILLIAM
	2	CHECKED	25-NOV-2010	S MAHESH
	1	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - RELAY PCB	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E01

No.	MP9021	Use	LUCC
Type	DMS-25-S-1-M-G E-2-6-F		

1	C24HI	2	C24L	3	TH16	4	LCMAS	5	LCFWD
6	LCREV	7	LSLIP	8		9	MS16F-4	10	MS16F-3
11		12	MS16R-4	13	MS16R-3	14		15	S9
16	S9	17		18	LCMAL	19		20	
21		22		23		24		25	

No.	MP9027	Use	MISC. CTRL
Type	DMS-9-P-1-M-G E-2-6-F		

1	MISC1-8	2	MISC2-8	3	MISC1-4
4	MISC1-6	5	MISC2-5	6	MISC2-4
7	MISC2-3	8	MISC2-6	9	MISC2-5

No.	MP9028	Use	OPTION SOL
Type	DMS-9-S-1-M-G E-2-6-F		

1	SL1306A	2	SL1306-3	3	SL1308-1
4		5		6	----
7	----	8	----	9	

No.	MP9050	Use	C24H/C24L
Type	1-0178323-2		

B1	C24HB	B2	C24L	B3	C24L
A1	C24HA	A2	C24HA	A3	C24HB

No.	MP9051	Use	100H/100L
Type	1-0178323-2		

B1	100L	B2	100L	B3	100H
A1	100L	A2	POUT-4	A3	LGT-4

No.	MP9052	Use	ATLM/AWM
Type	HIF3CB-20PA-2.54D5A		

B1	SKIP	B2	C24L	B3	EM1-8	B4	OTRL	B5	C24L	B6	MSALA	B7	MSALB	B8	C24HB	B9		B10	C24HB
A1	C24HI	A2		A3	NMSMD	A4	M32X	A5	M33X	A6	OPPON	A7	M34X	A8		A9	C24HB	A10	

No.	MP9053	Use	MTC-GENERAL
Type	MDS-9-P-1-M-G E-2-6-F		

1	US03	2	US04	3	CPTP
4	CB2	5	SPARE I/O	6	
7		8	EM1-6	9	EM1-10

No.	MP9054	Use	MTC-GENERAL
Type	MDS-9-P-1-M-G E-2-6-F		

1	C24HI	2	C24L	3	SPARE I/O
4	TH2	5	TH12	6	
7	MS2	8	SL1308	9	PPMON

No.	MP9055	Use	AUTO DOOR
Type	MDS-9-S-1-M-G E-2-6-F		

1	LS2315	2	LS2311	3	LS2312
4	LS2313	5	LS2314	6	C24HI
7	C24HI	8	C24HI	9	C24HI

No.	MP9056	Use	MTC-GENERAL
Type	1-178313-5		

1	NCON-3	2	C24L	3	

RELAY PCB CONNECTORS

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - RELAY PCB	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E02

No.	MP9052	Use	ATLM/AWM PCB
Type	HIF3CB-20PA-2.54D5A		

B1	SKIP	B2	C24L	B3	M86X-2	B4	M86X-3	B5	C24L	B6	MSALA	B7	MSALB	B8	C24HB	B9		B10	C24HB
A1	C24HI	A2		A3	NMSMD-8	A4	M32X-14	A5	M33X-8	A6	OPPON-8	A7	M34X-8	A8		A9	C24HB	A10	

No.	MP9023	Use	ATLM/AWM PCB
Type	178327-5		

B1	C24L	B2	C24L	B3	C24L	B4	S1	B5	M32X-8	B6	M32L-2	B7	M34X-4	B8	M34X-3
A1	C24HB	A2	C24HB	A3	C24HB	A4	S1	A5	M34X-5	A6	M32L-4	A7	M32L-3	A8	M34X-2

No.	MP9022	Use	ATLM/AWM PCB
Type	178328-5		

B1	C24L	B2	C24L	B3	OPPON-4	B4	M32X-2	B5	M32X-9	B6	M86X-7	B7	M32X-7	B8	M32X-11	B9	C24HB	B10	MSALB
A1	C24HB	A2	C24HB	A3	C24L	A4	M32X-6	A5	M32X-1	A6	M86X-5	A7	M86X-6	A8	M32X-3	A9	C24HB	A10	MSALA

No.	MP9031	Use	TPC JR
Type	172327-5		

B1	M770-3	B2	M770	B3	M771	B4	M772	B5	M773	B6	TPEMG	B7		B8	
A1	C24H	A2	TAFIN1	A3	TAFIN2	A4	TJALM1	A5	SPARE I/O	A6		A7		A8	C24L

No.	MP9038	Use	24 ATC
Type	316517-5		

B1	C24L	B2	C24L	B3	C24L	B4	C24L	B5	C24L	B6	C24L	B7	C24L	B8		B9	SL207-8	B10	SL2311-8
A1	SL51	A2	MS65	A3	SL206-4	A4	SL207-4	A5	SL2310-4	A6	SL2311-4	A7	M33X-4	A8	SL206-8	A9		A10	SL2310-8

No.	MP9039	Use	24 ATC
Type	178327-5		

B1	C24HI	B2	C24HI	B3	C24HI	B4	C24HI	B5	C24HI	B6	C24HI	B7	C24HI	B8	C24HI	B9	C24HI	B10	C24HI
A1	SPARE I/O	A2	TH65	A3	SPARE I/O	A4	SPARE I/O	A5	SPARE I/O	A6	SPARE I/O	A7	SPARE I/O	A8	SPARE I/O	A9	SPARE I/O	A10	SPARE I/O

No.	VMP12	Use	BLUM INTERFACE
Type	MR-50RFD2		

33	C24HI			1	C24HI
34				2	UO01
35				3	UO02
36	LSRCL	19	C24L	4	UO03
37		20	C24L	5	UO04
38		21	C24L	6	UO05
39		22	C24L	7	UO06
40		23	C24L	8	UO07
41		24	C24L	9	UO08
42		25	UI01	10	
43		26	UI02	11	
44		27	UI03	12	
45		28	UI04	13	
46	UI01	29	UI05	14	
47	UI02	30	UI06	15	
48	UI03	31	UI07	16	
49	UI04	32	UI08	17	
50	C24HI			18	C24HI

OPTION PCB CONNECTORS -E6VS603A800-XX

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - OPT PCB	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E03

No.	VMP70	Use	INPUT 24VDC
Type	177555-5		

A1	C24HI	A2	C24HO	A3	C24HO
B1	C24L	B2	C24L	B3	C24L

No.	VMP71	Use	OPE
Type	MR-25RMD2		

1	C24HI	2	AUT	3	AIRSV	4	HY1CL	5	DOPN
6	DCLS	7	HY1UC	8	LAUTO	9	LMAN	10	SYSPS
11	C24HI	12	C24HI	13	C24L	14	LHY1CL	15	LHY1UCL
16	LAIRS	17	LDOPN	18	LDCLS	19	OP EMG1	20	OP EMG1
21	OP EMG2	22	OP EMG2	23	C24L	24	C24L	25	C24L

No.	VMP72	Use	2HND PB
Type	178327-5		

B1	S1-1	B2	S2-2	B3	STRSW	B4	THPBR	B5	THPBL	B6	AREAS	B7	LMAN	B8	A8
A1	S1-2	A2	S2-1	A3	C24HI	A4	C24HI	A5	C24HI	A6	C24HI	A7	LAUTO	A8	B8

No.	VMP73	Use	EMG
Type	1-178315-5		

1	S1-1	2	S1-3	3	---	4	S2-1	5	S2-3

No.	VMP74	Use	AUT DR
Type	178326-5		

B1	LS2311	B2	LS2312	B3	LS2313	B4	LS2314	B5	C24L	B6	C24L
A1	C24HI	A2	C24HI	A3	S7	A4	S8	A5	SL2310-3	A6	SL2311-3

No.	VMP75	Use	HYD MOT
Type	178328-5		

B1	A4	B2	SYSPS	B3	TH260	B4	MS260	B5	A3	B6	LS2603	B7	LS2604	B8	C24L	B9	A9	B10	LMAN
A1	A8	A2	C24HI	A3	B5	A4	B1	A5	C24HI	A6	C24HI	A7	C24HI	A8	A1	A9	B9	A10	LAUTO

No.	VMP78	Use	SP DI
Type	1-0178317-2		

B1	C24HI	B2	C24HI	B3	C24HI	B4	C24HI	B5	C24HI	B6	C24HI	B7	C24HI	B8	C24HI	B9	C24HI	B10	C24HI
A1	LS1787	A2	LS1788	A3	LS1789	A4	LS1790	A5	LS1791	A6	LS1792	A7	SPDI7	A8	SPDI8	A9	SPDI9	A10	SPDI10

No.	VMP79	Use	SPARE DO
Type	1-0178317-2		

1	RL40-4	2	RL41-4	3	RL40-3	4	RL41-3	5	6	6	5	7	C24L	8	C24L

No.	VMP80	Use	AIRST
Type	178327-5		

B1	C24HI	B2	C24HI	B3	C24HI	B4	C24HI	B5	C24L	B6	C24L	B7	C24L	B8	C24L
A1	LS1781	A2	LS1782	A3	LS1783	A4	LS1784	A5	LS1785	A6	LS1786	A7	LS1787	A8	LS1788

No.	VMP81	Use	AIRST SOL
Type	178326-5		

B1	C24L	B2	C24L	B3	C24L	B4	C24L	B5	C24L	B6	C24L
A1	SL1780-3	A2	SL1781-3	A3	SL1782-3	A4	SL1783-3	A5	RL06-3	A6	RL07-3

OPTION PCB CONNECTORS -E7VS140A101-XX

Revisions			Date	Name
	④			
	③	APPROVED	25-NOV-2010	GOH WILLIAM
	②	CHECKED	25-NOV-2010	S MAHESH
	①	DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - OPT PCB	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E04

No.	VMP91	Use	AIR SEAT I/P
Type	1-0178317-2		

1	C24HI	2	C24HI	3	PS2601	4	PS2602	5	SL2601-3	6	SL2602-3	7	C24L	8	C24L

No.	VMP92	Use	AIR SEAT I/P
Type	1-0178317-2		

1	C24HI	2	C24HI	3	PS2603	4	PS2604	5	SL2603-3	6	SL2604-3	7	C24L	8	C24L

No.	VMP93	Use	AIR SEAT I/P
Type	1-0178317-2		

1	C24HI	2	C24HI	3	PS2605	4	PS2606	5	SL2605-3	6	SL2606-3	7	C24L	8	C24L

No.	VMP94	Use	AIR SEAT I/P
Type	1-0178317-2		

1	C24HI	2	C24HI	3	PS2607	4	PS2608	5	SL2607-3	6	SL2608-3	7	C24L	8	C24L

No.	VMP95	Use	AIR SEAT I/P
Type	1-0178317-2		

1	C24HI	2	C24HI	3	PS2701	4	PS2702	5	SL2701-3	6	SL2702-3	7	C24L	8	C24L

No.	VMP96	Use	AIR SEAT I/P
Type	1-0178317-2		

1	C24HI	2	C24HI	3	PS2703	4	PS2704	5	SL2703-3	6	SL2704-3	7	C24L	8	C24L

NO.	TB1
1	C24HI
2	C24HI
3	C24HI
4	C24L
5	C24L
6	C24L
7	RL08-3
8	RL09-3
9	LS912

OPTION PCB CONNECTORS -E7VS140A101-XX

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - OPT PCB	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E05

No.	CB150A	Use	I/O MODULE
Type	MR-50RFD2		

33	C24HB			1	C24HB
34	LDOPN			2	S150
35	MS19			3	LDCLS
36	SL192	19	C24L	4	AUDRF
37	SL1309	20	C24L	5	AUGRR
38	SL2310	21	C24L	6	LCMAL
39	SL2311	22	C24L	7	MS16F
40	SL1001	23	C24L	8	MS16R
41	SL1002	24	C24L	9	RCHP
42	TH19	25	LS151	10	LS1340
43	LS190	26	LS152	11	LS2315
44	LS912	27	LS196	12	ADPON
45	TH16	28	LS195	13	ADCLS
46	LCMAS	29	OT1A	14	LS2311
47	LCFWD	30	OT2A	15	LS2312
48	LCREV	31	DETCHA	16	LS2313
49	SKIP	32	LS191	17	LS2314
50	C24HB			18	C24HB

No.	CB150B	Use	I/O MODULE
Type	MR-50RFD2		

33	C24HB			1	C24HB
34	SL6B			2	SL1306A
35	LGT			3	MS2
36	SL181	19	C24L	4	SL1308
37	SL2111	20	C24L	5	SL02
38	AIRB	21	C24L	6	PRTC
39	US03	22	C24L	7	SL201
40	US01	23	C24L	8	SL202
41	US02	24	C24L	9	US04
42	CPTP	25	TOOLR	10	LSDX
43	CB2	26	SPARE	11	LSDY
44	AIRAL	27	LS2111	12	LSDZ
45	LS202	28	EMI	13	LSDA
46	LS201	29	ESP	14	LS27
47	PBSCL	30	OT2X	15	LSDB
48	USI1	31	OT2Y	16	MSALA
49	USI2	32	OT2Z	17	MSALB
50	C24HB			18	C24HB

No.	CB150C	Use	I/O MODULE
Type	MR-50RFD2		

33	C24HB			1	C24HB
34	POUT			2	NMSMD
35	ZBRK			3	M32X
36	OTRL	19	C24L	4	M33X
37	PPMON	20	C24L	5	OPPON
38	SL1306B	21	C24L	6	M34X
39	OMON	22	C24L	7	PTL1
40	SL01	23	C24L	8	PTL2
41	SL00	24	C24L	9	PTL3
42	LS00	25	SPARE	10	LS154
43	LS01	26	TH2	11	LS155
44	LS602	27	SPARE	12	LS25
45	LS601	28	CCHAL	13	MCSP1
46	LS170	29	LS03	14	MCSP2
47	OMAL	30	LS199	15	AUGRPB
48	LS905	31	TH12	16	USI3
49	LSLIP	32	PS1201	17	USI4
50	C24HB			18	C24HB

RELAY PCB - I/O MODULE CONNECTORS

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - MTC	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E06

No.	CB150D	Use	I/O MODULE
Type	MR-50RFD2		

33	C24HB			1	C24HB
34	M770			2	UO01
35	M771	19	C24L	3	UO02
36	M772	20	C24L	4	UO03
37	M773	21	C24L	5	UO04
38	TPEMG	22	C24L	6	UO05
39	SL51	23	C24L	7	UO06
40	MS65	24	C24L	8	UO07
41	LSRCL	25	UI01	9	UO08
42	TAFIN1	26	UI02	10	----
43	TAFIN2	27	UI03	11	TH65
44	TJALM1	28	UI04	12	----
45	SPARE	29	UI05	13	----
46	UI01	30	UI06	14	----
47	UI02	31	UI07	15	----
48	UI03	32	UI08	16	----
49	UI04			17	----
50	C24HB			18	C24HB

No.	CB151A	Use	I/O MODULE
Type	MR-50RFD2		

33	C24HB			1	C24HB
34	LHY1CL			2	LHY2CL
35	LHY1UC	19	C24L	3	LHY2UC
36	LAIRS	20	C24L	4	SL2601
37	LMAN	21	C24L	5	SL2602
38	LSETUP	22	C24L	6	SL2603
39	MS260	23	C24L	7	SL2604
40	SL1780	24	C24L	8	LALM
41	SL1781	25	ASTR	9	LWWE
42	MAN	26	WWE	10	PS2601
43	STRSW	27	LS1780	11	PS2602
44	AIRSV	28	LS1781	12	PS2603
45	THPBR	29	SYSPS	13	PS2604
46	THPBL	30	LS2604	14	HY2CL
47	AREAS	31	LS2603	15	HY2UC
48	HY1CL	32	TH260	16	SPARE
49	HY1UC			17	SPARE
50	C24HB			18	C24HB

No.	CB151C	Use	I/O MODULE
Type	MR-50RFD2		

33	P24HA			1	P24HA
34	SL100A			2	PRDY1
35	SL100B	19	C24L	3	PCMAL
36	SL101A	20	C24L	4	PCSPL
37	SL101B	21	C24L	5	PCSCL
38	SPARE	22	C24L	6	SPARE
39	SPARE	23	C24L	7	SPARE
40	SPARE	24	C24L	8	SPARE
41	SPARE	25	PS2605	9	SPARE
42	LS100	26	PS2606	10	RDYPB1
43	LS101	27	PS2607	11	PCMAS
44	LS102	28	PS2608	12	*PCSPC
45	LS103	29	SPARE	13	PCSCS
46	SPARE	30	SPARE	14	SPARE
47	SPARE	31	SPARE	15	SPARE
48	SPARE	32	SPARE	16	SPARE
49	SPARE			17	SPARE
50	P24HA			18	P24HA

No.	CB151B	Use	I/O MODULE
Type	MR-50RFD2		

33	C24HB			1	C24HB
34	SL2605			2	SL2701
35	SL2606	19	C24L	3	SL2702
36	SL2607	20	C24L	4	SL2703
37	SL2608	21	C24L	5	SL2704
38	SL1782	22	C24L	6	SPARE
39	SL1783	23	C24L	7	SPARE
40	SPARE	24	C24L	8	SPARE
41	SPARE	25	PS2605	9	SPARE
42	LS1783	26	PS2606	10	PS2701
43	LS1784	27	PS2607	11	PS2702
44	LS1785	28	PS2608	12	PS2703
45	LS1786	29	SPARE	13	PS2704
46	LS1787	30	SPARE	14	SPARE
47	LS1788	31	SPARE	15	SPARE
48	SPARE	32	SPARE	16	SPARE
49	SPARE			17	SPARE
50	C24HB			18	C24HB

OPTION PCB - I/O MODULE CONNECTORS

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - MTC	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E07

No.	MP9101	Use	IO POWER
Type	1-178313-5		

1	C24HB	2	C24L	3	

No.	MP9102	Use	LCD POWER
Type	1-178313-5		

1	NCON	2	C24L	3	

No.	MP9105	Use	24V & MTC
Type	MDS-25-P-1-M-G E-2-6-F		

1	C24HB	2	C24H	3	NCON	4	C24HB	5	NCON
6	ENON	7	C24L	8		9	PONL	10	C24L
11	C24L	12	EMG1	13	EMG2	14	PONL	15	SPDI1
16	SPDI1C	17	SPDO1	18	SPDO1C	19	PTL1	20	PTL2
21	PTL3	22		23		24		25	

No.	MP9106	Use	PATROL LGT
Type	MDS-9-P-1-M-G E-2-6-F		

1	C24HB	2	C24HB	3	PLT1-4	4	C24L	5	C24L
6	PTL1-3	7	PTL2-3	8	PTL3-3	9	PTL1-5		

No.	MP9115	Use	TOP PANEL
Type	DMS-25-S-1-M-G E-2-6-F		

1	C24HB	2	C24HB	3	NCON	4	EM2-4	5	EM2
6	LP2	7	KEY1	8	SGIR	9	PWROFF	10	ALRST
11	TUCL	12	SPNONF	13	+24VA2	14	EMG1	15	EMG2
16	LSPSP	17	LTUCL	18	LALRST	19	LPOFF	20	C24L
21	C24L	22	C24L	23	C24L	24		25	

No.	MP9116	Use	BOTTOM PANEL
Type	DMS-25-S-1-M-G E-2-6-F		

1	+24VA2	2	+24VA2	3	AUSP	4	AUST	5	
6		7	FV25	8	FV50	9	FV75	10	FV100
11	FV125	12	FV150	13	+24VA2	14	ROV25	15	ROV50
16	ROV100	17	+24VA2	18	LAUSP	19	LAUST	20	
21		22		23		24		25	

No.	MP9117	Use	SPARE I/O
Type	MDS-15-P-1-M-G E-2-6-F		

1	+24VA2	2	+24VA2	3	SPDI1	4	SPDI1C	5	SPDO1
6	SPDO1C	7	SPARE DI1	8	SPARE DI2	9	SPARE DI3	10	SPARE DO1
11	SPARE D02	12	SPARE D03	13		14	C24L	15	C24L

OPERATOR PANEL PCB CONNECTORS

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - OPR. PANEL	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E08

No.	CE56A	Use	I/O MODULE1
type	HIFBB-50D-2.54R		

	A		B
1	C24L	1	+24VA1
2	EDIT	2	MEM
3	MDI	3	DNC
4	REF	4	RAPID
5	HND	5	AUTOP
6	SBK	6	BLKSK
7	OPS1	7	COOLON
8	NOZON	8	AIRB
9	TAPR	9	ROV1I
10	ROV2I	10	ROV4I
11	FV1I	11	FV2I
12	FV4I	12	FV8I
13	FV16I	13	FVM
14	DICOM0	14	
15		15	
16	LEDT	16	LMEM
17	LMDI	17	LDNC
18	LREF	18	LRAPID
19	LHND	19	LAUTOP
20	LSBK	20	LBLKSK
21	LOPST	21	LCOOLON
22	LNOZON	22	LAIRB
23	LTAPR	23	SPAREDI1
24	C24HB	24	C24HB
25	C24HB	25	C24HB

No.	CE57A	Use	I/O MODULE1
type	HIFBB-50D-2.54R		

	A		B
1	C24L	1	+24VA2
2	ALLREF	2	SETUP
3	ONET	3	ONETS
4	X AXIS	4	Y AXIS
5	Z AXIS	5	4TH AXIS
6	'+'	6	'-'
7	OTR	7	LIGHT
8	KEY1	8	SGILK
9	ATC	9	SPAREDI1
10	PWROFF	10	ALRST
11	TUCL	11	SPNONF
12	AUSP	12	AUST
13	SPAREDI2	13	SPAREDI3
14		14	DICOM5
15		15	
16	LALLREF	16	LSETUP
17	LONET	17	LONES
18	LXAXIS	18	LYAXIS
19	LZAXIS	19	L4AXIS
20	LPOWOF	20	LALMRST
21	LTOOLUN	21	LSPNONF
22	LAUSP	22	LAUST
23	SPAREDO2	23	SPAREDO3
24	C24HB	24	C24HB
25	C24HB	25	C24HB

CONNECTOR TABLE - OP.PANEL

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - OPR. PANEL	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E09

No.	MP9060	Use	NOZZLE COOLANT
Type	09200030320 09200032711		

3	T2A	1	R2A
4	E	2	S2A

No.	MP9062	Use	TSC
Type	09300100302 09330102701		

6	C24H	1	C24H
7	LS190	2	LS196
8	R2A	3	C24L
9	S2A	4	C24H
10	T2A	5	LS195

No.	MP9063	Use	HYDRAULIC
Type	09300240302 09330242701		

13	SL2603-1	1	TH(M)-1
14	SL2601-2	2	TH(M)-2
15	SL2604-1	3	LS2603-1
16	SL2604-2	4	LS2603-2
17	PS2601-1	5	LS2604-1
18	PS2601-2	6	LS2604-2
19	PS2602-1	7	SYSPS-1
20	PS2602-2	8	SYSPS-2
21	PS2603-1	9	SL2601-1
22	PS2603-2	10	SL2601-2
23	PS2604-1	11	SL2602-1
24	PS2604-2	12	SL2602-2

No.	MP9064	Use	LUCC
Type	09330162701 09300160302		

9	LCMAL	1	R2A
10	C24L	2	S2A
11	LCMAS	3	T2A
12	LCFWD	4	M73R-21
13	LCREV	5	M73F-21
14	SW6-1	6	C24L
15	SW6-2	7	C24H
16	TH16	8	LSLIP

No.	MP9063A	Use	HYDRAULIC MOTOR
Type	09200030320 09200032711		

3	W260	1	U260
4	E	2	V260

CONNECTOR TABLE - COOLANT TERMINAL BOX

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - COOLANT TERMINAL BOX	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E10

No.	MP9700	Use	T'FORMER
Type	1-178140-5		

B1	0VAC	B2		B3	0VAC
A1	18VAC	A2		A3	18VAC

No.	MP9701	Use	SPARE I/O
Type	178327-5		

B1	SPARE I/O	B2	SPARE I/O	B3	SPARE I/O	B4	SPARE I/O	B5	SPARE I/O	B6	SPARE I/O	B7	SPARE I/O	B8	SPARE I/O	
A1	P24HA	A2	P24HA	A3	P24HA	A4	P24HA	A5	P24HA	A6	P24HA	A7	P24HA	A8	P24HA	

No.	MP9702	Use	RELAY NO
Type	316517-5		

B1	P24L	B2	P24L	B3	P24L	B4	P24L	B5	P24L	B6	P24L	B7	P24L	B8	P24L	B9		B10		
A1	R51-3	A2	R100A-3	A3	R100B-3	A4	R101A-3	A5	R101B-3	A6	R915-3	A7	R930-3	A8	R52-3	A9		A10		

No.	MP9703	Use	LAMP
Type	178327-5		

B1	P24L	B2	P24L	B3	P24L	B4	P24L	B5	RDYPB1	B6	PCMAS	B7	PCSPC	B8	PCSCS	
A1	PRDY1	A2	PCMAL	A3	PCSPL	A4	PCSCL	A5	P24HA	A6	P24HA	A7	P24HA	A8	P24HA	

No.	MP9704	Use	LS INPUT
Type	316517-5		

B1	LS100	B2	P24L	B3	LS101	B4	P24L	B5	LS102	B6	P24L	B7	LS103	B8	P24L	B9	LS110	B10	P24L	
A1	P24HA	A2		A3	P24HA	A4		A5	P24HA	A6		A7	P24HA	A8		A9	P24HA	A10		

No.	MP9705	Use	I/O SPARE
Type	178327-5		

B1	SPARE I/O	B2	SPARE I/O	B3	SPARE I/O	B4	SPARE I/O	B5	P24L	B6	P24L	B7	P24L	B8	P24L	
A1	P24HA	A2	P24HA	A3	P24HA	A4	P24HA	A5	D1	A6	D2	A7	D3	A8	HPON	

No.	MP9706	Use	LS,PS INPUT
Type	316517-5		

B1	LS55	B2	P24L	B3	LS56	B4	P24L	B5	MP9706 -B6	B6	MP9706 -B5	B7	MP9706 -B6	B8	P24L	B9	MP9706 -B6	B10	PS100	
A1	P24HA	A2	P24HA	A3	P24HA	A4	MP9706 -A5	A5	MP9706 -A4	A6	MP9706 -A5	A7	P24HA	A8	P24HB	A9	MP9706 -A5	A10	P24HA	

No.	MP9707	Use	RLY CHG OVER
Type	316517-5		

B1	R971-4	B2	R972-3	B3	R972-2	B4	R973-4	B5	MP9707 -B6	B6	MP9707 -B5	B7	MP9707 -B6	B8	MP9707 -B10	B9	MP9707 -B9	B10	P24L	
A1	R971-3	A2	R971-2	A3	R972-4	A4	R973-3	A5	R973-2	A6	MP9707 -A7	A7	MP9707 -A6	A8	MP9707 -A7	A9	P24HA	A10	P24HB	

No.	MP9708	Use	TO CP
Type	1-178136-5		

1	C24HA	2		3	C24HB

No.	MP9709	Use	FROM TB
Type	1-178136-5		

1	P24HA	2		3	P24HB

APC PCB CONNECTORS

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - APC PCB	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E11

No.	MP9750	Use	APC OPR PNL
Type	MR-20M		

1	PS9701-13	2	LP9701-X2	3		4		5	PS9701-14
6	PS9702-14	7	PS9703-22	8	PS9704-14	9	EMG-1	10	EMG-2
11	LP9701-X1	12	LP9702-X1	13	LP9703-X1	14	LP9704-X1	15	

No.	MP9753	Use	APC TABLE
Type	BS12, P/N: 002763, EUCHNER		

1	LS101	2	LS102	3	LS100	4	LS103	5	P24L
6	P24HA	7		8	E	9		10	
11		E							

No.	MP9752	Use	APC INTERFACE
Type	M12, 8 POLES CONNECTOR		

1	MP9753-3	2	MP9753-6	3	E	4	MP9753-2	5	MP9753-4
6	MP9753-5	7	UNUSED	8	MP9753-1				

No.	MP9751	Use	APC EMG
Type	178803-3		

B1	EMG-1	B2		B3	
A1	EMG-2	A2		A3	

No.	MP101	Use	TOOL STATUS
Type	178803-3		

B1	LS00	B2	LS01	B3	C24L
A1	C24H	A2	C24H	A3	

No.	MP101A	Use	SPDL SEAT CHECK
Type	178289-3		

B1	LS602	B2	LS601	B3	
A1	C24H	A2	C24H	A3	

No.	MP9013	Use	CLAMP/UNCLAMP
Type	178289-3		

B1	C24L	B2		B3	
A1	SL00A-5	A2		A3	

No.	MP9013A	Use	CLAMP/UNCLAMP
Type	178803-3		

B1	C24L	B2		B3	
A1	SL00A-5	A2		A3	

APC INTERFACE CONNECTORS

SPINDLE SEAT CHECK CONNECTORS

TOOL CLAMP CONNECTORS

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

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MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
CONNECTOR TABLE - INTERFACE CONNECTORS	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. E12

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150A		PAGE
X004.7	SKIP	SKIP SIGNAL	49	COM	C51
X004.6	LCREV	LUCC REVERSE	48	C24HI	C31
X004.5	LCFWD	LUCC FORWARD	47	(1) (18) (33) (50)	C31
X004.4	LCMAS	LUCC MANUAL	46		C31
X004.3	TH16	LUCC O/L	45	C24L	C31
X004.2	LS912	FRONT/RIGHT DOOR ILK	44	(19) (20) (21) (22) (23) (24)	C55
X004.1	LS190	TSC FILTER CLOGGED	43		C30
X004.0	TH19	O/L THRU' SPDL COOLANT	42		C24

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150A		PAGE
X005.7	LS191 LS197	SECONDARY BAG FILTER CLOG TSC COOLANT LEVEL HIGH	32	COM	C30
X005.6	DETCHA	A AXIS - OVER TRAVEL	31	C24HI	C47
X005.5	OT2A	A AXIS OT	30	(1) (18) (33) (50)	C47
X005.4	OT1A	TILT AXIS OT	29		C47
X005.3	LS195	TSC FLOW SWITCH	28	C24L	C25
X005.2	LS196	TSC COOLANT LEVEL LOW	27	(19) (20) (21) (22) (23) (24)	C25
X005.1	LS152	A AXIS UNCLAMP	26		C47
X005.0	LS151	A AXIS CLAMP	25		C47

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150A		PAGE
X006.7	LS2314	LEFT DOOR OPEN	17	COM	--
X006.6	LS2313	LEFT DOOR CLOSE	16	C24HI	--
X006.5	LS2312	RIGHT DOOR OPEN	15	(1) (18) (33) (50)	C69
X006.4	LS2311	RIGHT DOOR CLOSE	14		C69
X006.3	ADCLS	AUTO DOOR CLOSE PUSHBUTTON	13	C24L	C69
X006.2	ADOPN	AUTO DOOR OPEN PUSHBUTTON	12	(19) (20) (21) (22) (23) (24)	C69
X006.1	LS2315	LIGHT CURTAIN SENSOR	11		C91
X006.0	LS1340	ROTARY UNION WEAR & TEAR LS	10		C25

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150A		PAGE
Y000.7	SL1002	AUTO DOOR CLOSE	41	COM	C69
Y000.6	SL1001	AUTO DOOR OPEN	40	C24HI	C69
Y000.5	SL2311	NOZZLE COOLANT AIR PURGE	39	(1) (18) (33) (50)	C44
Y000.4	SL2310	SPINDLE AIRSEAL	38		C44
Y000.3	SL1309	ROTARY UNION DRAIN	37	C24L	C25
Y000.2	SL192	R THRU' SPDL AIR/TOOL THRU AIR	36	(19) (20) (21) (22) (23) (24)	C25
Y000.1	MS19	R THRU' SPDL CLNT/TOOL THRU	35		C24
Y000.0	LDOPN	L AUTO DOOR OPEN	34		C69

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150A		PAGE
Y001.7	RCHP	SPINDLE SPEED CONTROL	09	COM	C36
Y001.6	MS16R	LUCC, REVERSE	08	C24HI	C31
Y001.5	MS16F	LUCC, FORWARD	07	(1) (18) (33) (50)	C31
Y001.4	LCMAL	LUCC MANUAL LAMP	06		C31
Y001.3	AUGRR	AUTO GREASE REVERSE	05	C24L	C78
Y001.2	AUGRF	AUTO GREASE FORWARD	04	(19) (20) (21) (22) (23) (24)	C78
Y001.1	LDCLS	L AUTO DOOR CLOSE	03		C69
Y001.0	SL150	A AXIS UNCLAMP	02		C47

I/O ADDRESS TABLE - CB150A

Revisions			Date	Name
	4			
	3	APPROVED	25-NOV-2010	GOH WILLIAM
	2	CHECKED	25-NOV-2010	S MAHESH
	1	DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
DI/DO ADDRESS TABLE (MACHINE I/O)	NC TYPE	FS0i-MC	V090i-C1E-X03	= SHEET No. F01

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150B		PAGE
X007.7	USI2	USER DI2	49	COM	C73
X007.6	USI1	USER DI1	48	C24HI	C73
X007.5	PBSCL	PUSHBUTTON ROBOT DR OPEN/CLOSE	47	(1)	C67
X007.4	LS201	ROBOT DR CLOSE CONFIRM	46	(18) (33) (50)	C67
X007.3	LS202	ROBOT DR OPEN CONFIRM	45	C24L	C67
X007.2	AIRAL	AIR DRYER ALARM	44	(19) (20)	C76
X007.1	CB2	ASSIST POWER LINE BRK	43	(21) (22)	C35
X007.0	CPTP	CABINET PROTECTOR TRIP	42	(23) (24)	C35

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150B		PAGE
X008.7	OT2Z	LS OVER TRAVEL Z	32	COM	C42
X008.6	OT2Y	LS OVER TRAVEL Y	31	C24HI	C42
X008.5	OT2X	LS OVER TRAVEL X	30	(1)	C42
X008.4	ESP	EMERGENCY STOP	29	(18) (33) (50)	C42
X008.3	EMI	EMG STOP SIGNAL	28	C24L	C35
X008.2	LS2111	TOOL BROKEN SENSOR	27	(19) (20)	C77
X008.1	LS1678	TOOL CL/UC PP TEMP. ALARM	26	(21) (22)	C41
X008.0	TOOLR	TOOL RETRACT	25	(23) (24)	C19

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150B		PAGE
X009.7	MSALB	BATTERY LOW ALARM	17	COM	C50
X009.6	MSALA	COMMUNICATION ALARM	16	C24HI	C50
X009.5	DECC	LS DECELERATION C	15	(1)	--
X009.4	DECB	LS DECELERATION B	14	(18) (33) (50)	C48
X009.3	LSDA	LS DECELERATION A	13	C24L	C47
X009.2	LSDZ	LS DECELERATION Z	12	(19) (20)	C42
X009.1	LSDY	LS DECELERATION Y	11	(21) (22)	C42
X009.0	LSDX	LS DECELERATION X	10	(23) (24)	C42

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150B		PAGE
Y002.7	USO2	USER DO2	41	COM	C72
Y002.6	USO1	USER DO1	40	C24HI	C72
Y002.5	USO3	USER DO3	39	(1)	C72
Y002.4	AIRB	AIR BLOW	38	(18) (33) (50)	C56
Y002.3	SL2111	AIR TOOL BREAK SENSOR	37	C24L	--
Y002.2	SL181	SHOWER COOLANT	36	(19) (20)	C44
Y002.1	LGT	R MACHINE LIGHT	35	(21) (22)	C45
Y002.0	SL6B	ATC AIR	34	(23) (24)	C43

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150B		PAGE
Y003.7	USO4	USER DO4	09	COM	C72
Y003.6	SL202	ROBOT DR OPEN SOLENOID	08	C24HI	C67
Y003.5	SL201	ROBOT DR CLOSE SOLENOID	07	(1)	C67
Y003.4	PRTC	PART COUNTER	06	(18) (33) (50)	--
Y003.3	SL02	NOZZLE COOLANT SOL	05	C24L	C46
Y003.2	SL1308	ROTARY UNION BACKUP	04	(19) (20)	C26
Y003.1	MS2	R NOZZLE COOLANT	03	(21) (22)	C23
Y003.0	SL1306A	COOLANT DRAWBACK ON	02	(23) (24)	C26

I/O ADDRESS TABLE - CB150B

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
DI/DO ADDRESS TABLE (MACHINE I/O)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. F02

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150C		PAGE
X010.7	LSLIP	LUCC SLIP	49	COM	C31
X010.6	LS905	MAIN AIR PRESSURE SWITCH	48	C24HI	C38
X010.5	OMAL	SPINDLE TEMP CONTROL ALARM	47	(1)	C37
X010.4	LS170	SPINDLE OIL FLOW SW	46	(18) (33) (50)	C37
X010.3	LS601	SPINDLE TOOL SEATING CHECK	45	C24L	C38
X010.2	LS602	SPINDLE TOOL CONFIRM	44	(19) (20)	C38
X010.1	LS01	TOOL UNCLAMP	43	(21) (22)	C38
X010.0	LS00	TOOL CLAMP	42	(23) (24)	C38

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150C		PAGE
X011.7	PS1201	HYD. PRESSURE SWITCH	32	COM	C41
X011.6	TH12	HYD. POWER PACK ALARM	31	C24HI	C41
X011.5	---	---	30	(1)	--
X011.4	LS03	COOLANT LEVEL SENSOR - LOW	29	(18) (33) (50)	C27
X011.3	CCHAL	COOLANT CHILLER ALARM	28	C24L	C29
X011.2	LS04	COOLANT LEVEL SENSOR - HIGH	27	(19) (20)	C27
X011.1	TH2	O/L NOZZLE COOLANT MOTOR	26	(21) (22)	C23
X011.0	LS1679	TOOL CL/UC PP OIL LEVEL LOW	25	(23) (24)	C41

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150C		PAGE
X012.7	USI4	USER DI4	17	COM	C73
X012.6	USI3	USER DI3	16	C24HI	C73
X012.5	LS1698	AUTOGREASE PR.SWITCH	15	(1)	C78
X012.4	MCSP2	HIGH SPINDLE SPEED	14	(18) (33) (50)	C36
X012.3	MCSP1	LOW SPINDLE SPEED	13	C24L	C36
X012.2	---	---	12	(19) (20)	--
X012.1	LS154	B AXIS UNCLAMP	11	(21) (22)	C48
X012.0	LS153	B AXIS CLAMP	10	(23) (24)	C48

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150C		PAGE
Y004.7	SL00	R TOOL UNCLAMP	41	COM	C39
Y004.6	SL01	SP. TAPER CLEANING AIR	40	C24HI	C38
Y004.5	OMON	R SP. TEMP CONTROL ON	39	(1)	C37
Y004.4	SL1306B	COOLANT DRAWBACK OFF	38	(18) (33) (50)	C26
Y004.3	PPMON	TCLAMP P PACK MOTOR ON	37	C24L	C41
Y004.2	OTRL	OVER TRAVEL RELEASE	36	(19) (20)	C42
Y004.1	ZBRK	SERVO READY(Z BRAKE)	35	(21) (22)	C19
Y004.0	POUT	R POWER OUT	34	(23) (24)	C33

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150C		PAGE
Y005.7	PTL3	PATROL LIGHT 3	09	COM	C64
Y005.6	PTL2	PATROL LIGHT 2	08	C24HI	C64
Y005.5	PTL1	PATROL LIGHT 1	07	(1)	C64
Y005.4	M34X	R TOOL LENGTH MEASURE	06	(18) (33) (50)	--
Y005.3	OPPON	PROBE POWER ON	05	C24L	C50
Y005.2	M33X	MEASURING CIRCUIT CHECK	04	(19) (20)	C51
Y005.1	M32X	R TOOL LENGTH MEASURE	03	(21) (22)	C51
Y005.0	NMSMD	NOT MEASURING MODE	02	(23) (24)	C51

I/O ADDRESS TABLE - CB150C

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
DI/DO ADDRESS TABLE (MACHINE I/O)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. F03

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150D		PAGE
X0013.7	UI03	BLUM PCB INTERFACE	49	COM	---
X0013.6	UI02	BLUM PCB INTERFACE	48	C24HI	---
X0013.5	UI01	BLUM PCB INTERFACE	47	(1) (18) (33) (50)	---
X0013.4	LSROK	LASER OK	46		---
X0013.3	SPARE	---	45	C24L	---
X0013.2	TJALM1	TPC JR. ALARM/PRAGATI MTR OH	44	(19) (20) (21) (22) (23) (24)	C49-2
X0013.1	TAFIN2	TPC JR. FIN2/THERMAL TRIP	43		C49-2
X0013.0	TAFIN1	TPC JR. FIN1/THERMAL TRIP	42		C49

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150D		PAGE
Y0006.7	LSRCL	LASER ATLM	41	COM	---
Y0006.6	MS65	TRANSFER PUMP ON	40	C24HI	C29
Y0006.5	SL51	5TH AXIS UNCLAMP	39	(1) (18) (33) (50)	C48
Y0006.4	TPEMG	TPC JR. EMERGENCY	38		C70
Y0006.3	M773	TPC JR. WSEL2	37	C24L	C49-2
Y0006.2	M772	TPC JR. WSEL1	36	(19) (20) (21) (22) (23) (24)	C49-2
Y0006.1	M771	TPC JR. A REF	35		C49
Y0006.0	M770	TPC JR. A START	34		C49

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150D		PAGE
X0014.7	UI08	INDEX TABLE MOTOR OVERHEAT	32	COM	C49
X0014.6	UI07	INDEX TABLE CLAMP PRX	31	C24HI	C49
X0014.5	UI06	PARITY	30	(1) (18) (33) (50)	C49
X0014.4	UI05	STROBE	29		C49
X0014.3	UI04	BIT 4	28	C24L	C49
X0014.2	UI03	BIT 3	27	(19) (20) (21) (22) (23) (24)	C49
X0014.1	UI02	BIT 2	26		C49
X0014.0	UI01	BIT 1	25		C49

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150D		PAGE
Y0007.7	UO08	MACRO OUTPUT INTERFACE	09	COM	---
Y0007.6	UO07	MACRO OUTPUT INTERFACE	08	C24HI	---
Y0007.5	UO06	MACRO OUTPUT INTERFACE	07	(1) (18) (33) (50)	---
Y0007.4	UO05	MACRO OUTPUT INTERFACE	06		---
Y0007.3	UO04	MACRO OUTPUT INTERFACE	05	C24L	---
Y0007.2	UO03	MACRO OUTPUT INTERFACE	04	(19) (20) (21) (22) (23) (24)	---
Y0007.1	UO02	MACRO OUTPUT INTERFACE	03		---
Y0007.0	UO01	MACRO OUTPUT INTERFACE	02		---

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB150D		PAGE
X0015.7	UI15	INDEX TABLE CLAMP PRX	17	COM	C49-2
X0015.6	UI14	PARITY	16	C24HI	C49-2
X0015.5	UI13	STROBE	15	(1) (18) (33) (50)	C49-2
X0015.4	UI12	BIT 4	14		C49-2
X0015.3	UI11	BIT 3	13	C24L	C49-2
X0015.2	UI10	BIT 2	12	(19) (20) (21) (22) (23) (24)	C49-2
X0015.1	TH65	TRANSFER PUMP TRIP	11		C29
X0015.0	UI09	BIT 1	10		C49-2

I/O ADDRESS TABLE - CB150D

Revisions			Date	Name
	④			
	③	APPROVED	25-NOV-2010	GOH WILLIAM
	②	CHECKED	25-NOV-2010	S MAHESH
	①	DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
DI/DO ADDRESS TABLE (MACHINE I/O)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. F04

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151A		PAGE
X0016.7	HY1UC	HYD UNCLAMP PB	49	COM	C87
X0016.6	HY1CL	HYD UNCLAMP PB	48	C24HI	C87
X0016.5	---	---	47	(1) (18) (33) (50)	----
X0016.4	THPBL	TWO HAND PB L	46		C89
X0016.3	THPBR	TWO HAND PB R	45	C24L	C89
X0016.2	AIRSV	AIR SEAT CHECK SW	44	(19) (20) (21) (22) (23) (24)	C87
X0016.1	STRSW	STRIKER SW	43		C89
X0016.0	MAN	MANUAL MODE SEL SWITCH	42		C87

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151A		PAGE
Y0008.7	SL1781	AIR SEAT CHECK SOLENOID 2	41	COM	C86
Y0008.6	SL1780	AIR SEAT CHECK SOLENOID 1	40	C24HI	C86
Y0008.5	MS260	HYDRAULIC MOTOR ON	39	(1) (18) (33) (50)	C79
Y0008.4	LSETUP	LAMP SETUP FINISH	38		C87
Y0008.3	LMAN	LAMP MANUAL MODE	37	C24L	C87
Y0008.2	LAIRS	LAMP AIR SEAT CHECK VALID	36	(19) (20) (21) (22) (23) (24)	C87
Y0008.1	LHY1UC	LAMP HYD UNCLAMP1	35		C87
Y0008.0	LHY1CL	LAMP HYD CLAMP1	34		C87

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151A		PAGE
X0017.7	TH260	HYD MOTOR OVERLOAD	32	COM	C79
X0017.6	LS2603	HYD OIL LEVEL FLOAT SWITCH	31	C24HI	C79
X0017.5	LS2604	HYD OIL TEMPERATURE SENSOR	30	(1) (18) (33) (50)	C79
X0017.4	SYSPS	SYSTEM PRESSURE SWITCH	29		C79
X0017.3	LS1782	AIR SEAT SENSOR 2	28	C24L	C86
X0017.2	LS1781	AIR SEAT SENSOR 1	27	(19) (20) (21) (22) (23) (24)	C86
X0017.1	WWE	WORK WASH ENABLE PUSHBUTTON	26		C87
X0017.0	ASTR	AUTO START PUSHBUTTON	25		C87

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151A		PAGE
Y0009.7	LWWE	LAMP WORK WASH ENABLE	09	COM	C87
Y0009.6	LALM	LAMP ALARM	08	C24HI	C87
Y0009.5	SL2604	HYD SOLENOID 4	07	(1) (18) (33) (50)	C82
Y0009.4	SL2603	HYD SOLENOID 3	06		C82
Y0009.3	SL2602	HYD SOLENOID 2	05	C24L	C80
Y0009.2	SL2601	HYD SOLENOID 1	04	(19) (20) (21) (22) (23) (24)	C80
Y0009.1	LHY2UC	LAMP HYD UNCLAMP2	03		C88
Y0009.0	LHY2CL	LAMP HYD CLAMP2	02		C88

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151A		PAGE
X0018.7	LS1881	COMPONENT PRESENT SENSOR 1	17	COM	----
X0018.6	LS1882	COMPONENT PRESENT SENSOR 2	16	C24HI	----
X0018.5	HY2UC	HYD UNCLAMP2 PUSHBUTTON	15	(1) (18) (33) (50)	C88
X0018.4	HY2CL	HYD CLAMP2 PUSHBUTTON	14		C88
X0018.3	PS2604	HYD PRESSURE SWITCH 4	13	C24L	C82
X0018.2	PS2603	HYD PRESSURE SWITCH 3	12	(19) (20) (21) (22) (23) (24)	C82
X0018.1	PS2602	HYD PRESSURE SWITCH 2	11		C80
X0018.0	PS2601	HYD PRESSURE SWITCH 1	10		C80

I/O ADDRESS TABLE - CB151A

Revisions			Date	Name
	4			
	3	APPROVED	25-NOV-2010	GOH WILLIAM
	2	CHECKED	25-NOV-2010	S MAHESH
	1	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
DI/DO ADDRESS TABLE (MACHINE I/O)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. F05

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151B		PAGE
X0019.7	LS1788	AIR SEAT SENSOR 8	49	COM	C86
X0019.6	LS1787	AIR SEAT SENSOR 7	48	C24HI	C86
X0019.5	---	---	47	(1)	----
X0019.4	---	---	46	(18)	----
X0019.3	LS1786	AIR SEAT SENSOR 6	45	(33)	C86
X0019.2	LS1785	AIR SEAT SENSOR 5	44	(50)	C86
X0019.1	LS1784	AIR SEAT SENSOR 4	43	C24L	C86
X0019.0	LS1783	AIR SEAT SENSOR 3	42	(19)	C86
				(20)	
				(21)	
				(22)	
				(23)	
				(24)	

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151B		PAGE
Y0010.7	SPARE	----	41	COM	----
Y0010.6	SPARE	----	40	C24HI	----
Y0010.5	SL1783	AIRSEAT CHECK SOLENOID 4	39	(1)	----
Y0010.4	SL1782	AIRSEAT CHECK SOLENOID 3	38	(18)	----
Y0010.3	SL2608	HYD SOLENOID 8	37	(33)	C85
Y0010.2	SL2607	HYD SOLENOID 7	36	(50)	C85
Y0010.1	SL2606	HYD SOLENOID 6	35	C24L	C85
Y0010.0	SL2605	HYD SOLENOID 5	34	(19)	C85
				(20)	
				(21)	
				(22)	
				(23)	
				(24)	

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151B		PAGE
X0020.7	LS1792	AIR SEAT SENSOR 12	32	COM	C86
X0020.6	LS1791	AIR SEAT SENSOR 11	31	C24HI	C86
X0020.5	LS1790	AIR SEAT SENSOR 10	30	(1)	C86
X0020.4	LS1789	AIR SEAT SENSOR 9	29	(18)	C86
X0020.3	PS2608	HYD PRESSURE SWITCH 8	28	(33)	----
X0020.2	PS2607	HYD PRESSURE SWITCH 7	27	(50)	C85
X0020.1	PS2606	HYD PRESSURE SWITCH 6	26	C24L	----
X0020.0	PS2605	HYD PRESSURE SWITCH 5	25	(19)	C85
				(20)	
				(21)	
				(22)	
				(23)	
				(24)	

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151B		PAGE
Y0011.7	SPARE	----	09	COM	----
Y0011.6	SPARE	----	08	C24HI	----
Y0011.5	SPARE	----	07	(1)	----
Y0011.4	SL1801	FIXTURE COOLANT SOL	06	(18)	C44
Y0011.3	SL2704	HYD SOLENOID 2	05	(33)	----
Y0011.2	SL2703	HYD SOLENOID 1	04	(50)	----
Y0011.1	SL2702	AUTO DOOR CLOSE	03	C24L	----
Y0011.0	SL2701	AUTO DOOR OPEN	02	(19)	----
				(20)	
				(21)	
				(22)	
				(23)	
				(24)	

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151B		PAGE
X0021.7	SPARE	----	17	COM	----
X0021.6	SPARE	----	16	C24HI	----
X0021.5	SPARE	----	15	(1)	----
X0021.4	SPARE	----	14	(18)	----
X0021.3	SPARE	----	13	(33)	----
X0021.2	SPARE	----	12	(50)	----
X0021.1	SPARE	----	11	C24L	----
X0021.0	SPARE	----	10	(19)	----
				(20)	
				(21)	
				(22)	
				(23)	
				(24)	

I/O ADDRESS TABLE - CB151B

Revisions			Date	Name
	4			
	3	APPROVED	25-NOV-2010	GOH WILLIAM
	2	CHECKED	25-NOV-2010	S MAHESH
	1	DRAWN	25-NOV-2010	PRAVEEN B

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TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
DI/DO ADDRESS TABLE (MACHINE I/O)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. F06

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151C		PAGE
X0022.7	SPARE	----	49	COM	----
X0022.6	SPARE	----	48	P24HA	----
X0022.5	SPARE	----	47	(1) (18)	----
X0022.4	SPARE	----	46	(33) (50)	----
X0022.3	LS103	APC ARM CW	45	C24L	C66
X0022.2	LS102	APC ARM CCW	44	(19) (20)	C66
X0022.1	LS101	APC ARM LIFT	43	(21) (22)	C66
X0022.0	LS100	APC ARM DOWN	42	(23) (24)	C66

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151C		PAGE
Y0012.7	SPARE	----	41	COM	----
Y0012.6	MS300	MISTCOLLECTOR	40	P24HA	C90
Y0012.5	SPARE	----	39	(1) (18)	----
Y0012.4	SPARE	----	38	(33) (50)	----
Y0012.3	SL101B	APC ARM CW	37	C24L	C66
Y0012.2	SL101A	APC ARM CCW	36	(19) (20)	C66
Y0012.1	SL100B	APC ARM UP	35	(21) (22)	C66
Y0012.0	SL100A	APC ARM DOWN	34	(23) (24)	C66

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151C		PAGE
X0023.7	MP2	MULTIPLY X100	32	COM	C63
X0023.6	MP1	MULTIPLY X10	31	P24HA	C63
X0023.5	H5	B AXIS SELECT	30	(1) (18)	C63
X0023.4	H4	A AXIS SELECT	29	(33) (50)	C63
X0023.3	HZ	Z AXIS SELECT	28	C24L	C63
X0023.2	HY	Y AXIS SELECT	27	(19) (20)	C63
X0023.1	HX	X AXIS SELECT	26	(21) (22)	C63
X0023.0	SPARE	----	25	(23) (24)	----

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151C		PAGE
Y0013.7	SPARE	----	09	COM	----
Y0013.6	SPARE	----	08	P24HA	----
Y0013.5	SPARE	----	07	(1) (18)	----
Y0013.4	SPARE	----	06	(33) (50)	----
Y0013.3	PCWWL	WORK WASH GUN LAMP	05	C24L	C65
Y0013.2	PCSPL	APC APC SIDE FEEDHOLD LIGHT	04	(19) (20)	C65
Y0013.1	PCMAL	APC MANUAL SWITCH LIGHT	03	(21) (22)	C65
Y0013.0	PRDY1	APC READY LIGHT	02	(23) (24)	C65

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CB151C		PAGE
X0024.7	SPARE	----	17	COM	----
X0024.6	TH300	MISTCOLLECTOR THERMAL TRIP	16	P24HA	C90
X0024.5	SPARE	----	15	(1) (18)	----
X0024.4	SPARE	----	14	(33) (50)	----
X0024.3	PCWWE	WORK WASH GUN ENABLE	13	C24L	C65
X0024.2	PCSPC	APC APC SIDE FEEDHOLD	12	(19) (20)	C65
X0024.1	PCMAS	APC MANUAL SWITCH	11	(21) (22)	C65
X0024.0	RDYPB1	APC READY	10	(23) (24)	C65

I/O ADDRESS TABLE - CB151C

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
DI/DO ADDRESS TABLE (MACHINE I/O)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. F07

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CE56A		PAGE
X096.7	AUTOP	AUTO POWER OFF	B05	COM	C58
X096.6	HND	HANDLE MODE	A05		C58
X096.5	RAPID	RAPID MODE	B04	+24V (B01)	C58
X096.4	REF	REFERENCE MODE	A04		C58
X096.3	DNC	DNC MODE	B03	C24L (A01)	C58
X096.2	MDI	MDI MODE	A03		C58
X096.1	MEM	MEMORY MODE	B02	DICOM (A14)	C58
X096.0	EDT	EDIT MODE	A02		C58

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CE56A		PAGE
Y096.7	LAUTOP	AUTO POWER OFF LIGHT	B19	COM	C58
Y096.6	LHND	HANDLE LIGHT	A19		C58
Y096.5	LRAPID	RAPID LIGHT	B18	C24H (A24) (B24) (A25) (B25)	C58
Y096.4	LREF	REFERENCE LIGHT	A18		C58
Y096.3	LDNC	DNC LIGHT	B17		C58
Y096.2	LMDI	MDI MODE LIGHT	A17	C24L (A01)	C58
Y096.1	LMEM	MEMORY MODE LIGHT	B16		C58
Y096.0	LEDT	EDIT MODE LIGHT	A16		C58

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CE56A		PAGE
X097.7	ROV1I	RAPID TRAVERSE OVERRIDE SW.25	B09	COM	C60
X097.6	TAPR	TAP RETRACT	A09		C59
X097.5	AIRB	AIR BLOW	B08	+24V (B01)	C59
X097.4	NOZON	NOZZLE ON/OFF	A08		C58
X097.3	COOLON	COOLANT ON	B07	C24L (A01)	C58
X097.2	OPST	OPTIONAL STOP	A07		C58
X097.1	BLKSK	BLOCK SKIP	B06	DICOM (A14)	C58
X097.0	SBK	SINGLE BLOCK	A06		C58

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CE56A		PAGE
Y097.7	L.5	LAMP 5TH AXIS SELECT	B23	COM	C59
Y097.6	LTAPR	TAP RETRACT LIGHT	A23		C59
Y097.5	LAIRB	AIR BLOW LIGHT	B22	C24H (A24) (B24) (A25) (B25)	C59
Y097.4	LNOZON	NOZZLE ON/OFF LIGHT	A22		C58
Y097.3	LCOOLON	COOLANT ON/OFF LIGHT	B21		C58
Y097.2	LOPST	OPTIONAL STOP LIGHT	A21	C24L (A01)	C58
Y097.1	LBLKSK	BLOCK SKIP LIGHT	B20		C58
Y097.0	LSBK	SINGLE BLOCK LIGHT	A20		C58

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CE56A		PAGE
X098.7	FVM	FEED RATE OVERRIDE SWITCH (INHIBIT)	B13	COM	C60
X098.6	FV16I	FEED RATE OVERRIDE SWITCH (16)	A13		C60
X098.5	FV8I	FEED RATE OVERRIDE SWITCH (8)	B12	+24V (B01)	C60
X098.4	FV4I	FEED RATE OVERRIDE SWITCH (4)	A12		C60
X098.3	FV2I	FEED RATE OVERRIDE SWITCH (2)	B11	C24L (A01)	C60
X098.2	FV1I	FEED RATE OVERRIDE SWITCH (1)	A11		C60
X098.1	ROV4I	RAPID TRAVERSE OVERRIDE SW.100	B10	DICOM (A14)	C60
X098.0	ROV2I	RAPID TRAVERSE OVERRIDE SW.50	A10		C60

I/O ADDRESS TABLE - OPE.I/O MODULE NO.1

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
DI/DO ADDRESS TABLE (OPR. PANEL I/O)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. F08

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CE57A		PAGE
X099.7	4AXIS	4TH AXIS	B05	COM	C59
X099.6	Z AXIS	Z AXIS	A05		C59
X099.5	Y AXIS	Y AXIS	B04	+24V (B01)	C59
X099.4	X AXIS	X AXIS	A04		C59
X099.3	ONES	ONE TOUCH FUNCTION STOP	B03	C24L (A01)	C59
X099.2	ONET	ONE TOUCH FUNCTION START	A03		C59
X099.1	SETUP	SETUP	B02	DICOM (A14)	C59
X099.0	ALLREF	ALL AXES REFERENCE	A02		C59

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CE57A		PAGE
X100.7	5AXIS	5TH AXIS INPUT	B09	COM	C60
X100.6	ATC	ATC AXIS SELECT	A09		C60
X100.5	S/GILK	S/G ILK RELEASE	B08	+24V (B01)	C60
X100.4	KEY1	MEMORY KEY	A08		C60
X100.3	LIGHT	LIGHT ON	B07	C24L (A01)	C60
X100.2	OTR	OT RELEASE	A07		C60
X100.1	'-'	(-)MINUS DIRECTION	B06	DICOM (A14)	C59
X100.0	'+'	(+)PLUS DIRECTION	A06		C59

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CE57A		PAGE
X101.7	SPAREDI3	SPARE INPUT3	B13	COM	C61
X101.6	SPAREDI2	SPARE INPUT2	A13		C61
X101.5	AUST	CYCLE START	B12	+24V (B01)	C61
X101.4	AUSP	FEEDHOLD	A12		C61
X101.3	SPNONF	SPINDLE ON/OFF	B11	C24L (A01)	C61
X101.2	TUCL	TOOL UNCLAMP	A11		C61
X101.1	ALRST	ALARM RESET	B10	DICOM (A14)	C61
X101.0	PWROFF	POWER OFF	A10		C61

DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CE57A		PAGE
Y098.7	L4AXIS	4TH AXIS LIGHT	B19	COM	C59
Y098.6	LZAXIS	Z AXIS LIGHT	A19		C59
Y098.5	LYAXIS	Y AXIS LIGHT	B18	C24H (A24) (B24) (A25) (B25)	C59
Y098.4	LXAXIS	X AXIS LIGHT	A18		C59
Y098.3	LONES	ONE TOUCH STOP LIGHT	B17		C59
Y098.2	LONET	ONE TOUCH START LIGHT	A17	C24L (A01)	C59
Y098.1	LSETUP	ONE TOUCH FUNCTION2 SELECT	B16		C59
Y098.0	LALLREF	ONE TOUCH FUNCTION1 SELECT	A16		C59

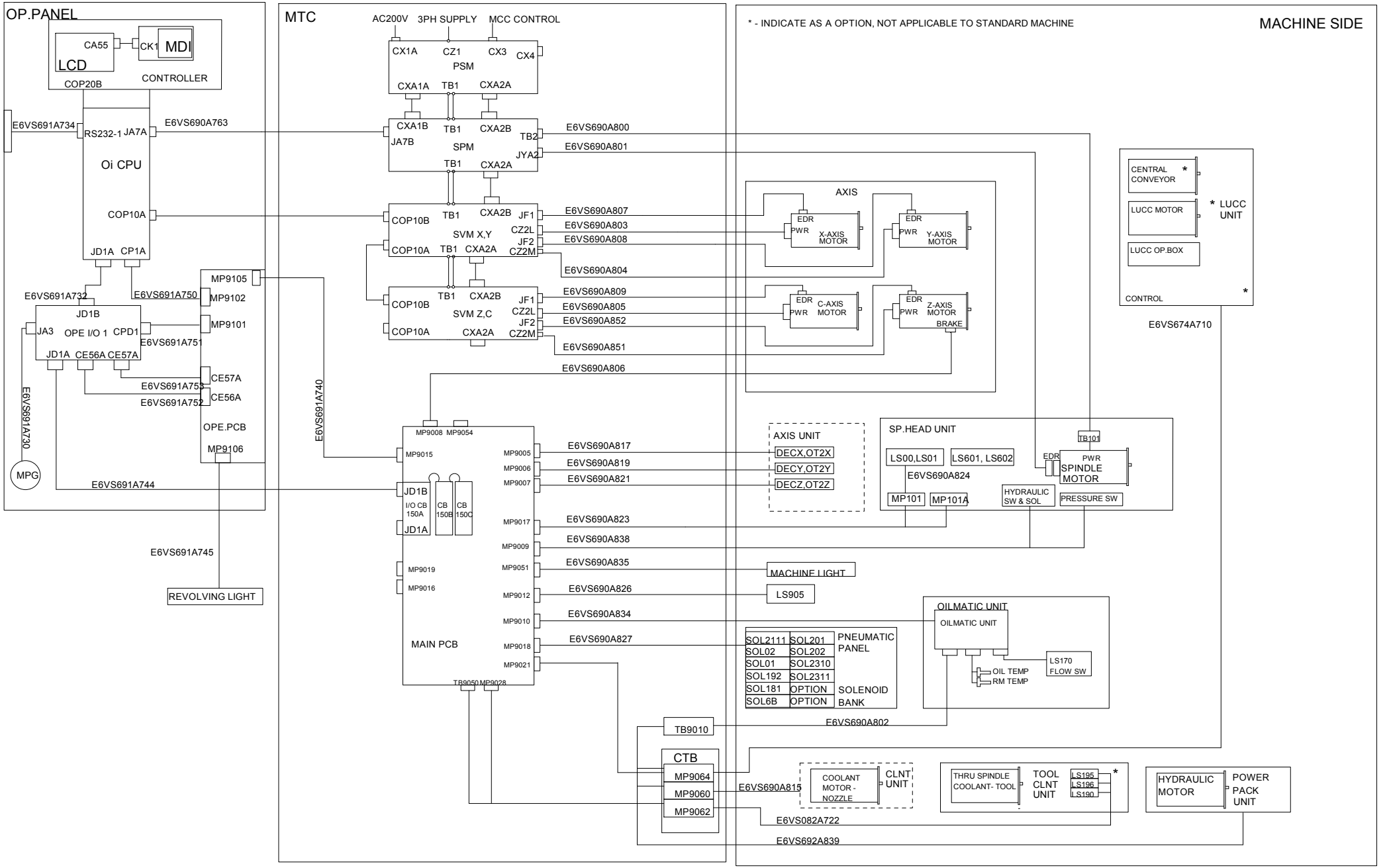
DGN.NO.	SYMBOL	SIGNAL NAME	CONNECTOR NO. CE57A		PAGE
Y099.7	SPARED03	SPARE OUTPUT3	B23	COM	C61
Y099.6	PTL4	SPARE OUTPUT2	A23		C61
Y099.5	LAUST	CYCLE START LIGHT	B22	C24H (A24) (B24) (A25) (B25)	C61
Y099.4	LAUSP	FEEDHOLD LIGHT	A22		C61
Y099.3	LSPNONF	SPINDLE STOP LIGHT	B21		C61
Y099.2	LTOOLUN	TOOL UNCLAMP LIGHT	A21	C24L (A01)	C61
Y099.1	LALMRST	ALARM RESET LIGHT	B20		C61
Y099.0	LPOWOF	POWER OFF LIGHT	A20		C61

I/O ADDRESS TABLE - OPE.I/O MODULE NO.1

Revisions			Date	Name
	④			
	③	APPROVED	25-NOV-2010	GOH WILLIAM
	②	CHECKED	25-NOV-2010	S MAHESH
	①	DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
DI/DO ADDRESS TABLE (OPR. PANEL I/O)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. F09

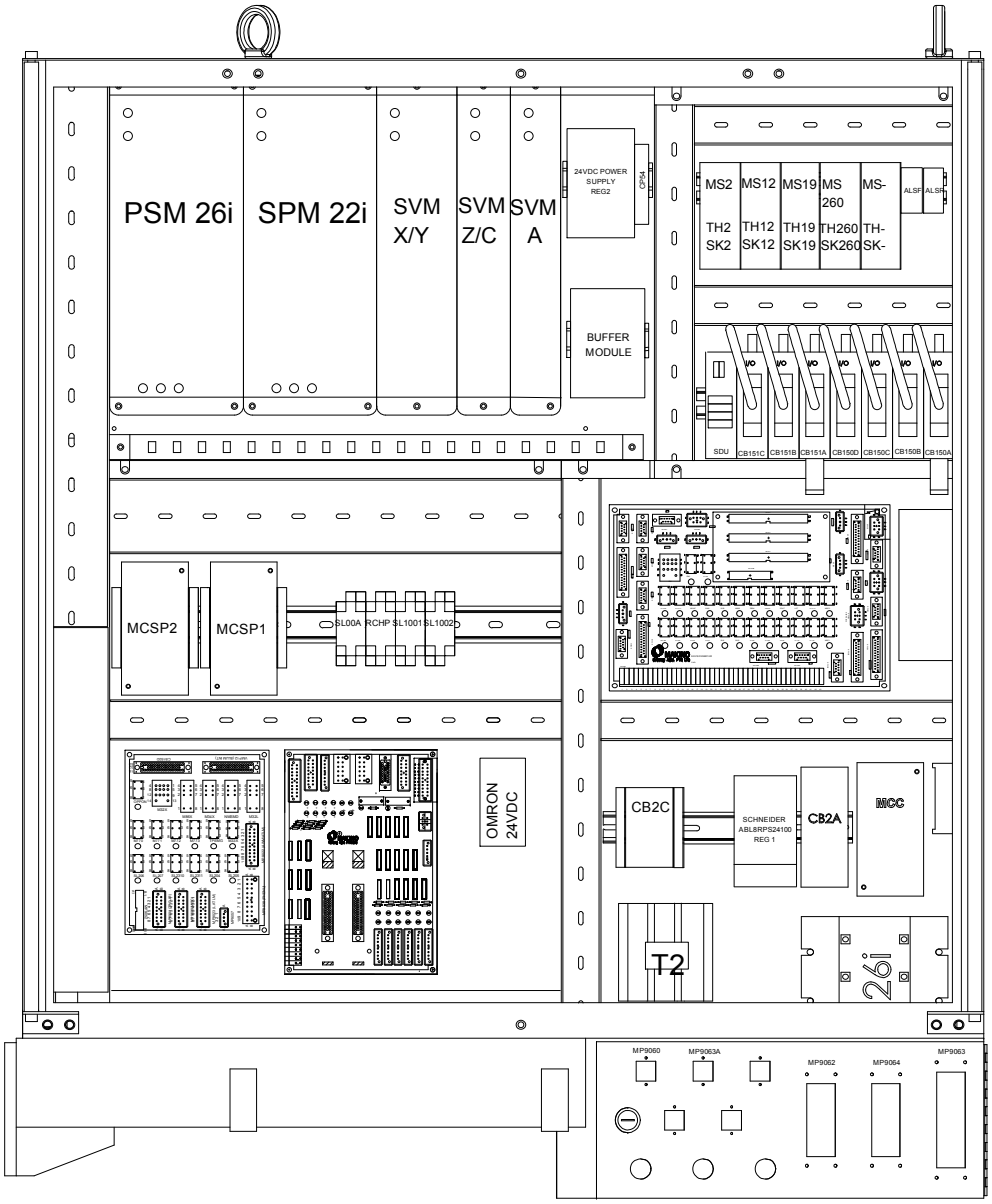


OPERATION PANEL	MTC	MACHINE SIDE

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
MACHINE INTERCONNECTION (STD & APC M/C)	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. G01



NO.	DESIGNATION	DESCRIPTION	MAKE	MFR PART NO.	MAKINO PART NO.
1	PSM	PSM-26i	FANUC	---	A06B-6140-H026
2	SPM	SPM-22i	FANUC	---	A06B-6141-H022#580
3	SVM X/Y	SVM2 X/Y	FANUC	---	A06B-6117-H210
4	SVM Z/C	SVM2 Z/C	FANUC	---	A06B-6117-H208
5	PFM	BUFFER MODULE	SCHNEIDER	ABL8BUF24400	L35A009
6	SVM A	SVM A	FANUC	---	A06B-6117-H104
7	MCSP1, MCSP2	MAG CONT AC100,120A COVER, MAGNETIC CONTACTOR	SIEMENS SIEMENS	3TF48220AF0 3TF46/47, 3TX7460-0E	L16A053 L16A063
8	MCC	MAG CONT AC100,120A COVER, MAGNETIC CONTACTOR	SIEMENS SIEMENS	3TF48220AF0 3TF46/47, 3TX7460-0E	L16A053 L16A063
9	24VDC REG	DC24V SUPPLY	SIEMENS	6EP1334-2AA01	L35A006
10	T2	TRANSFORMER 250VA,	SCHNEIDER	ABL6TS25G	L29A112
11	ACR	AC REACTOR PSM26i	FANUC	---	A81L-0001-0159
12	RELAY PCB	RELAY PCB	MAKINO	---	E6VS090A201-00
13	CB150A,CB151A	I/O MODULE BASIC 24/26	FANUC	---	A03B-0815-C001
14	CB150B , CB150C	I/O UNIT 24/26 POINTS	FANUC	---	A03B-0815-C003
15	CB2A CB2A AUX	CB, 3P,32A W/AUX SW1NO/1NC	SIEMENS	3VU1640-1MP00	L14A049
16	CB2C	CB, 3P,10A W/AUX SW1NO/1NC	SIEMENS	3VU1640-1ML00	L14A052
17	MS2 TH2 SK2	CONTACTOR 12A 24VDC SURGE SUPRESSOR OVERLOAD RELAY SPARK KILLER	SIEMENS SIEMENS SIEMENS MURR	3RT1017-1BB41 3RT1916-1JK00 3RU1116-1GB0 23004	L16A030 L13C012 L16B066 L13C011
18	MS12 TH12 SK12	CONTACTOR 7A 24VDC SURGE SUPRESSOR OVERLOAD RELAY SPARK KILLER	SIEMENS SIEMENS SIEMENS MURR	3RT1015-1BB41 3RT1916-1JK00 3RU1116-1DB0 23004	L16A031 L13C012 L16B065 L13C011
19	MS19 TH19 SK19	CONTACTOR 12A 24VDC SURGE SUPRESSOR OVERLOAD RELAY SPARK KILLER	SIEMENS SIEMENS SIEMENS MURR	3RT1017-1BB41 3RT1916-1JK00 3RU1116-1KB0 23004	L16A030 L13C012 L16B067 L13C011
20	MS260 TH260 SK260	CONTACTOR 12A OVERLOAD RELAY CONTACTOR 12A OVERLOAD RELAY 24VDC SURGE SUPRESSOR SPARK KILLER	SIEMENS SIEMENS SIEMENS SIEMENS SIEMENS MURR	3RT1017-1BB41 3RU1116-1HB0 3RT1017-1BB41 3RU1116-1DB0 3RT1916-1JK00 23004	L16A030 (YUKEN) L16B078 (YUKEN) L16A031 (AMF) L16B065 (AMF) L13C012 (YUKEN/AMF) L13C011 (YUKEN/AMF)
21	RCHP, SL00A, SL1001, SL1002	RELAY RELAY SOCKET CLIP	OMRON OMRON OMRON	MY2N-D2 24VDC PYF08A-E CLIP FOR MY	L16B081 L16B082 L16B059
22	POWER SUPPLY 24V 4.5A	POWER SUPPLY	OMRON	---	L35A002
23	POWER SUPPLY 24V 10A	POWER SUPPLY	SCHNEIDER	ABL8RPS24100	L35A008

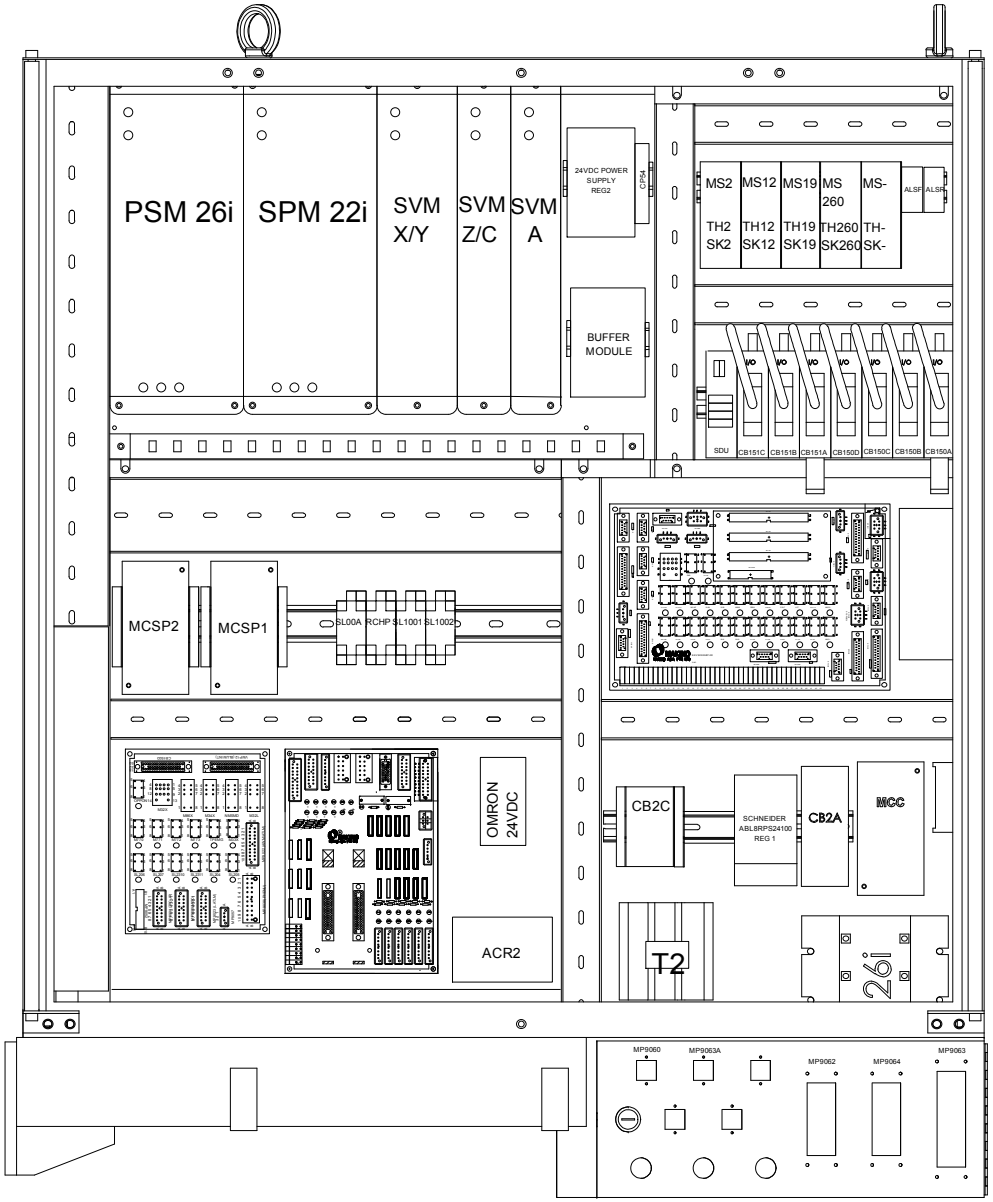
MTC FRONT VIEW

PARTS LIST

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
MTC LAYOUT, PARTS	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. H01



NO.	DESIGNATION	DESCRIPTION	MAKE	MFR PART NO.	MAKINO PART NO.
24	OPTION PCB	OPTION PCB	MAKINO	---	E6VS603800-00
25	MACRO PCB	MACRO PCB	MAKINO	---	---
26	HYDRAULIC PCB	HYDRAULIC PCB	MAKINO	---	E7VS140A101-00
27	24VDC P/S	OMRON POWER SUPPLY	OMRON	---	L35A002
28	ACR2	AC REACTOR BETA	FANUC	---	A81L-0001-0083#3C
29	SDU	SDU UNIT FOR SERIAL SCALE	FANUC	---	A02B-0236-C205

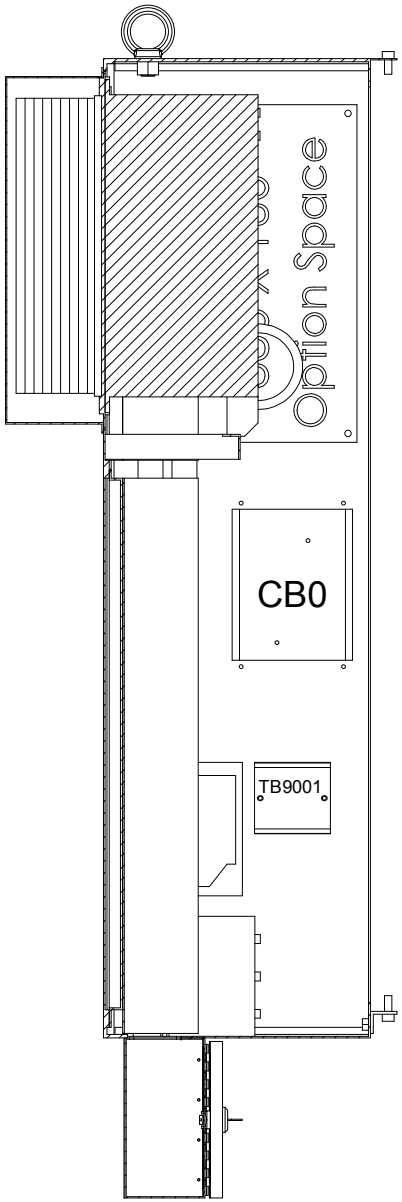
MTC FRONT VIEW

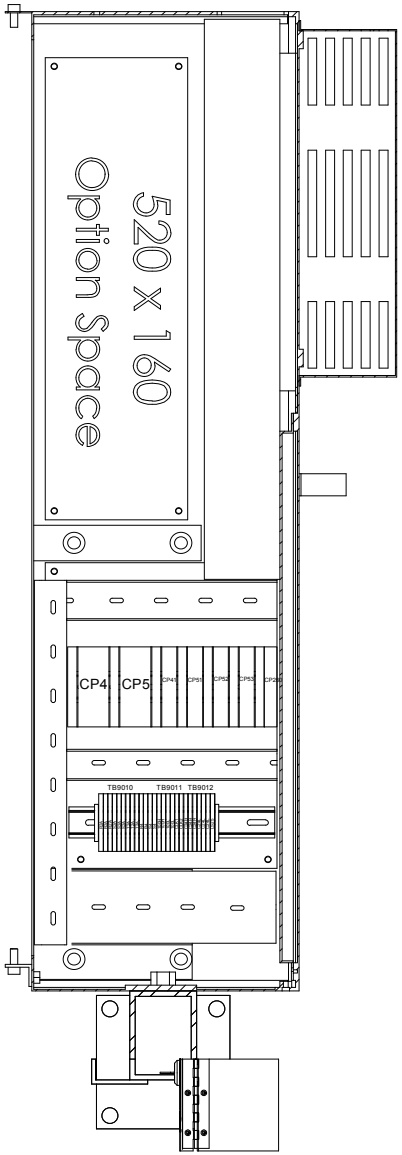
PARTS LIST

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
MTC LAYOUT, PARTS	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. H02





NO.	DESIGNATION	DESCRIPTION	MAKE	MFR PART NO.	MAKINO PART NO.
1	CP4	CP, 2P, 10A	SIEMENS	5SX4210-7RC	L14B037
	CP5	CP, 2P, 6A		5SX4206-7RC	L14B038
2	CP41,CP51 CP52,CP53	CP, 1P, 8A	SIEMENS		
		CP, 1P, 6A		5SX4106-7RC	L14B040 MAKINO INDIA SLIM3
3	CP4AUX,CP5AUX, CP41AUX,CP51AUX CP52AUX,CP53AUX	AUX CONTACT 1NO/1NC	SIEMENS	ST3020	
					L17B045 MAKINO INDIA SLIM3
4	CB0	CIRCUIT BREAKER 3P. AC415, 100A (WITH SHUNT)	SIEMENS		
				3VL1710-1DD33-8RA0	L14A035 MAKINO INDIA SLIM3
5	TB9001	TERMINAL BLOCKS	FUJI	ABYBN103-5,3P	L15C043
6	TB9010 TB9011 TB9012	TERMINAL BLOCKS	WAGO	280 SERIES	L15C070, L15C071, L15C051
7	CP260	CP, 1P, 6A	SIEMENS		
				5SX4106-7RC	L14B040 MAKINO INDIA SLIM3

MTC RIGHT VIEW

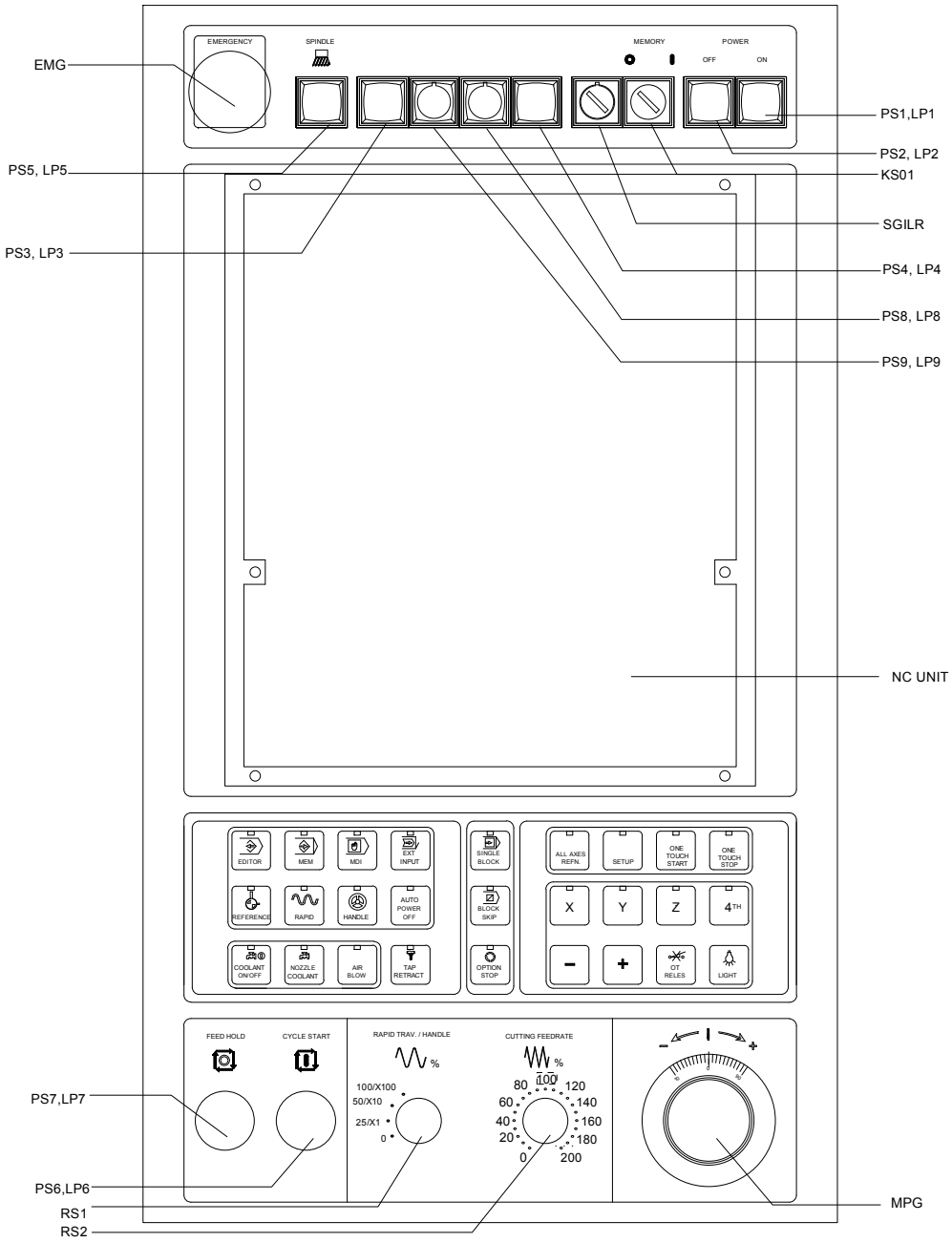
MTC LEFT VIEW

PARTS LIST

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
MTC LAYOUT, PARTS	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. H03



NO.	DESIGNATION	DESCRIPTION	MAKE	MFR PART NO.	MAKINO PART NO.
1	EMG	PB EMERGENCY	TEKNIC	2PSML401	L11B109
2	KS01 SGILR	KEY SWITCH	SCHLEGEL	QXSSA17E AT	L11C045 L11B008
3	RS1 RS2	ROTARY SWITCH ROTARY SWITCH SWITCH KNOB	TOSOKU TOSOKU TOSOKU	RS400P-1-1-4BP DPP01020J16R BC60	L11C038 L11C040 L11C058
4	PS2, LP2 PS3, LP3 PS5, LP5 PS4, LP4	PB ACTUATOR PB CONTACT BLOCK PB LED PB LENS PB COVER	SCHLEGEL	QXD BZLOI W2X4,6/24 T25DKL KDQN16	L11B046 L11B038 L11B030 L11B055 L11B063
5	PS7,LP7	PB & LED 1NC (RED)	FUJI	AR22G4L-01E3W	L11B057
6	PS6,LP6	PB & LED 1NO (GREEN)	FUJI	AR22G4L-10E3W	L11B056
7	PS1,LP1	PB ACTUATOR PB CONTACT BLOCK PB LED PB LENS	SCHLEGEL	QXD BZLII W2X4,6/24 T25DKL	L11B046 L11B106 L11B030 L11B055
8	MPG	HKD DIAL SILVER HANDWHEEL	EUCHNER	057 314 MPG,EUCHNER, HKD100S100G05C2136	L11C054 L11C053

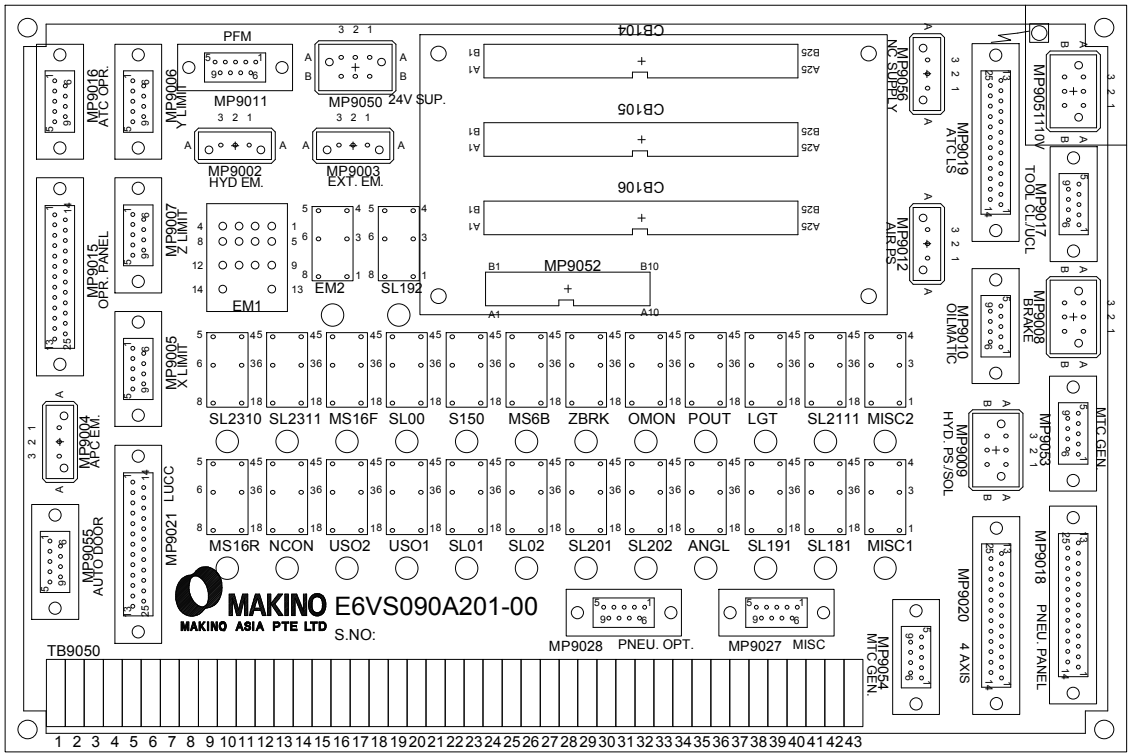
OPERATOR PANEL FRONT VIEW - E6VS091A403-XX

PARTS LIST

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
OPERATOR PANEL LAYOUT, PARTS	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. H04



NO.	DESIGNATION	DESCRIPTION	MAKE	MFR PART NO.	MAKINO NO.
01	MP9002,MP9003 MP9004,MP9012 MP9056	CONN.AMP 3WAY, X TYPE	AMP	1-178313-5	L15A087
02	MP9050,MP9051 MP9008,MP9009	CONN.AMP 6WAY, X TYPE	AMP	0-178323-2	L15A144
03	MP9055,MP9016 MP9011,MP9017 MP9010,MP9028	D-CONN. 9WAY, FEMALE	REPUTED	DMS-9-S-1-M-G E-2-6-F	L15A246
04	MP9005,MP9006 MP9007,MP9027 MP9054,MP9053	D-CONN. 9WAY, MALE	REPUTED	DMS-9-P-1-M-G E-2-6-F	L15A243
05	MP9015,MP9018	D-CON 25 WAY MALE	REPUTED	DMS-25-P-1-M-G E-2-6-F	L15A245
06	MP9020,MP9021 MP9019	D-CON 25 WAY FEMALE	REPUTED	DMS-25-S-1-M-G E-2-6-F	L15A248
07	CB104,CB105 CB106	CONN. TAP HRS 50 WAY	HIROSE	HIF3CB-50PA-2.54D5A	L15A114
08	MP9052	CONN. TAP HRS 20 WAY	HIROSE	HIF3CB-20PA-2.54D5A	L15A115
09	TB9050	PCB TERMINAL 5/5.08 SPACING	WAGO	255-401	L15A249
10	PE	EARTH STUD M4	---	---	---
11	POUT,ZBRK,SL01 SL00,LGT,SL181 SL02,SL201,SL202 EM2,ANGL,MISC1 MISC2	RELAY 1NO+1NC	OMRON	G6B-2114P-US	L16B055
12	SL2111,MS6B,US01 US02,SL192,SL191 SL2310,SL2311 SL150,MS16F MS16R,OMON	RELAY BASE 1NO	OMRON	G6B-04P	L16B061
13	NCON	RELAY 2NO	OMRON	G6B-2214P-US	L16B056
14	R	510Ohm, 5W CEMENT	VB&LEE	---	L13B006
15	C	0.22MicroF, 250V CERAMIC	REPUTED	---	L13A001
16	S1,S2,S3,S4	SHORTING LINK 2 WAY	REPUTED	---	---
17	S5	SHORTING LINK 5 WAY	REPUTED	---	---
18	D1~D27	DIODE 1N4007	REPUTED	---	---
19	OTRL,SL181,NMSMD UMD1,RCHP,SL01 SL00,UMD2,M34X SL21,SL20,SL150 MS16F,MS16R,OPON M86X,SGA,MS6B ANGL,M33X,SL913 SL912	RELAY SOCKET, PCB MOUNT	OMRON	P2R-08P	L16B012
20	EM1,EM2	RELAY 4PDT, 24VDC	FUJI	HH54P-FL 24VDC	L16B026
21	M32X,M32L	RELAY 4PDT, 24VDC	FUJI	HH54P-FL 24VDC	L16B026
22	EM1,EM2 M32X,M32L	RELAY SOCKET, PCB MOUNT	FUJI	TP514B	L16B038
23	NCN2,NCON,LGT ZBRK,POUT	RELAY 2PDT, 24VDC	FUJI	HH62P-FL 24VDC	L16B025
24	NCN2,NCON,LGT ZBRK,POUT	RELAY SOCKET, PCB MOUNT	FUJI	TP68B	L16B039

RELAY PCB FRONT VIEW

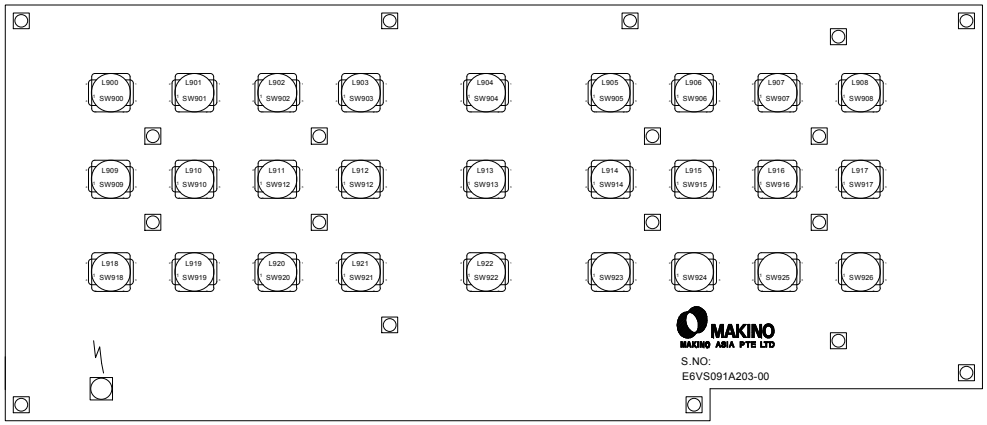
PARTS LIST

Revisions			Date	Name
	④			
	③	APPROVED	25-NOV-2010	GOH WILLIAM
	②	CHECKED	25-NOV-2010	S MAHESH
	①	DRAWN	25-NOV-2010	PRAVEEN B

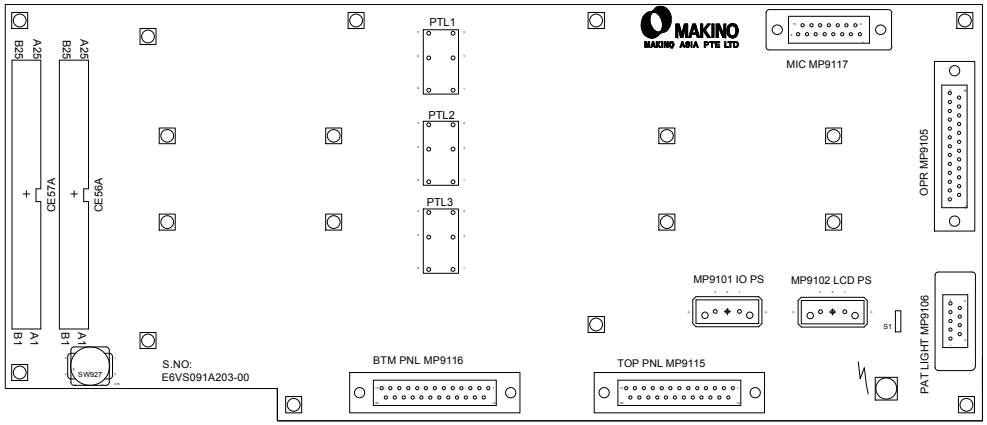
MAKINO
MAKINO ASIA PTE LTD

TITLE
RELAY PCB LAYOUT, PARTS

MACHINE TYPE	SLIM3	DRAWING No.	GROUP =
NC TYPE	FS0i-MC	V090i-C1E-X03	SHEET No. H05



(LFRONT VIEW FOR PANEL MOUNTING)



(LBACK VIEW(FOR CABLE CONNECTIONS)

NO.	DESIGNATION	DESCRIPTION	MAKE	MFR PART NO.	MAKINO PART NO.
1	SW900 ~ SW927	SW CAP JAE SW TACTILE FB JAE	JAE	S-N066-1065 61T-0101	L11C030 L11C026
2	S1	SHORTING LINK	REPUTED	--	
3	PTL1,PTL2,PTL3	OMRON G6B SERIES	OMRON	G6B-2214P-US	L16B056
4	L900~L932	GREEN LED (3MM)	REPUTED	--	L23A029
5	R900~R932	RESISTOR 2.2K, 1/2W	REPUTED	--	L17B011
6	D932~D934	DIODE 1N4007	REPUTED	--	
7	CE56A,CE57A	CONN. TAP HRS 50 WAY	HIROSE	HIF3CB-50PA-2.54D5A	L15A114
8	MP9117	D-CON 15 WAY MALE HEX CLINCHED NUT FRONT	REPUTED	DMS-15-P-1-M-G E-2-6-F	L15A244
9	MP9115,MP9116	D-CON 25 WAY FEMALE HEX CLINCHED NUT FRONT	REPUTED	DMS-25-S-1-M-G E-2-6-F	L15A248
10	MP9105	D-CON 25 WAY MALE HEX CLINCHED NUT FRONT	REPUTED	DMS-25-P-1-M-G E-2-6-F	L15A245
11	MP9106	D-CON 9 WAY MALE HEX CLINCHED NUT FRONT	REPUTED	DMS-9-P-1-M-G E-2-6-F	L15A243
12	MP9101,MP9102	CONN.AMP 3 WAY, X TYPE	AMP	1-178313-5	L15A087

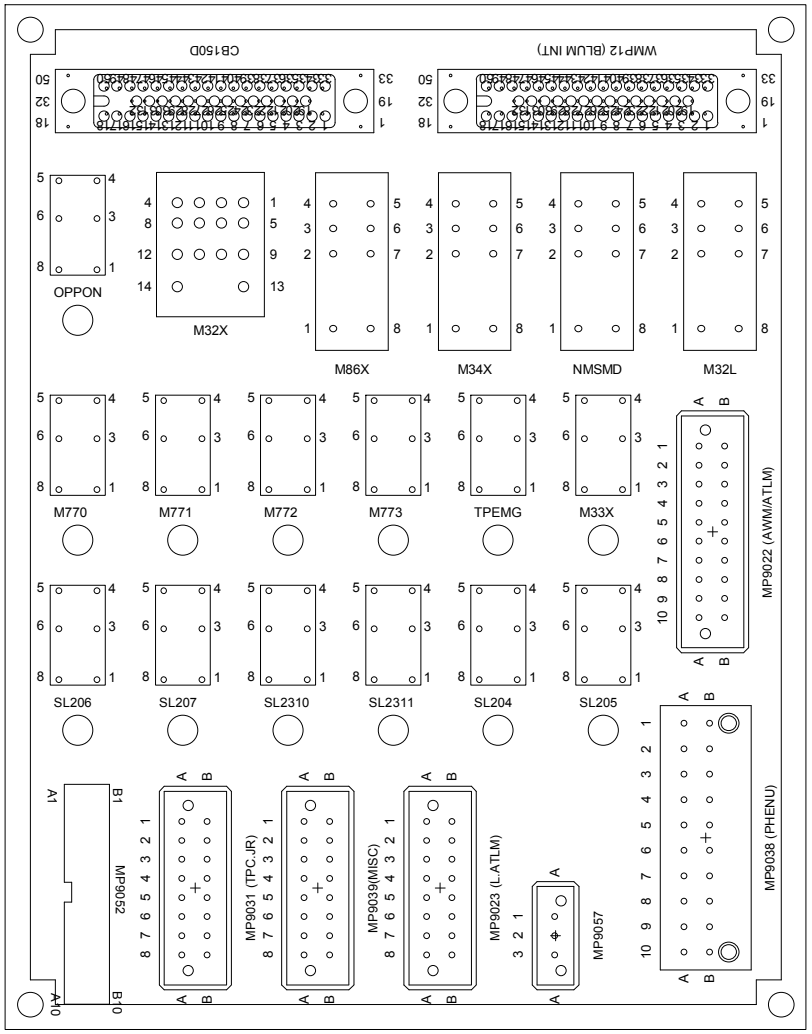
OPERATOR PCB-E6VS091A203-00

PARTS LIST

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
OPERATOR PCB LAYOUT, PARTS	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. H06



NO.	DESIGNATION	DESCRIPTION	MAKE	MFR PART NO.	MAKINO PART NO.
1	M86X, M34X, NMSMD M32L	RELAY SOCKET, PCB MOUNT	OMRON	P2R-08P	L16B012
2	OPPON, M770, M771, M772, M773, TPEMG, SL204, SL205, SL206, SL207, SL2310, SL2311, M33X	RELAY BASE	OMRON	G6B-04P	L16B061
3	M32X	RELAY SOCKET, PCB MOUNT	FUJI	TP514B	L16B038
4	MP9031, MP9039, MP9023	CONNECTOR, AMP, 16 WAY, X TYPE	AMP	178327-5	L15A085
5	MP9038	CONNECTOR, AMP 20 WAY	AMP	316517-5	L15A183
6	MP9057	CONNECTOR, AMP 3 WAY X TYPE	AMP	1-178313-5	L15A087
7	MP9022	CONNECTOR AMP, 20WAY	AMP	178328-5	L15A184
8	VMP12, CB150D	CONNECTOR HONDA, 50PIN, FEMALE	HONDA	MR-50RFD2	L15A187
9	MP9052	CONNECTOR TAP HRS 20WAY	HIROSE	HIF3CB-20PA-2.54D5A	L15A115
10	R1~R12	RESISTOR, 2.2K, 0.25W	REPUTED		L13B023
11	L1~L12	GREEN LED 3MM DIA	REPUTED		L23A029
12	D1~D15	DIODE 1N4007	REPUTED		

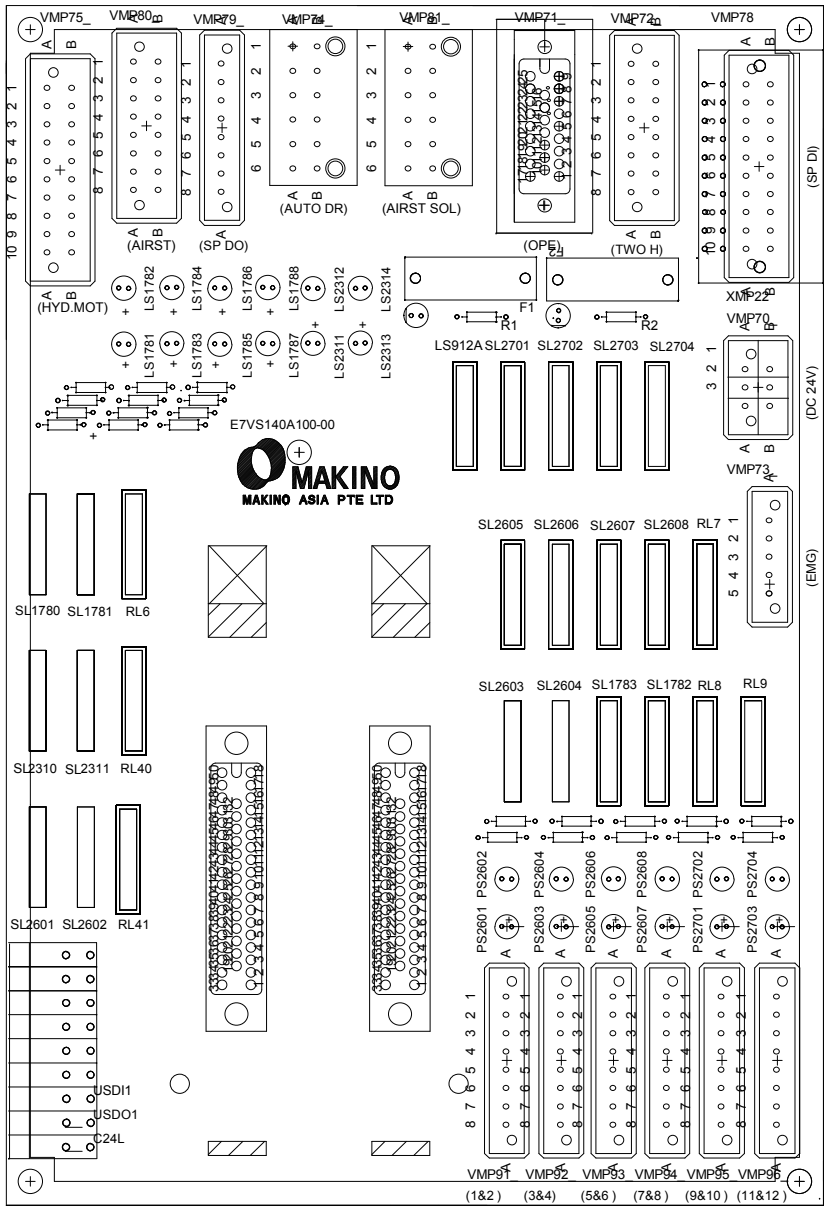
OPTION PCB-E6VS603A800-00

PARTS LIST

Revisions			Date	Name
④				
③		APPROVED	25-NOV-2010	GOH WILLIAM
②		CHECKED	25-NOV-2010	S MAHESH
①		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
OPTION PCB LAYOUT, PARTS	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. H07



NO.	DESIGNATION	DESCRIPTION	MAKE	MFR PART NO.	MAKINO PART NO.
1	VMP70	CONNECTOR AMP, 6WAY XY TYPE	AMP	3-178140-5	L15A088
2	VMP73	CONNECTOR AMP 5 WAY X TYPE	AMP	1-178315-5	L15A272
3	VMP79, VMP91, VMP92, VMP93, VMP94 VMP95, VMP96 (DUMMY)	CONNECTOR AMP, 8WAY X TYPE	AMP	1-178317-2	L15A143
4	VMP71	CONNECTOR HONDA, 25PIN	HONDA	MR-25RM	L15A309
5	F1	FUSE (3A)	REPUTED	--	--
6	F2	FUSE (5A)	REPUTED	--	--
7	CB15XA, CB15XB	CONNECTOR HONDA, 50PIN, FEMALE	HONDA	MR-50RFD2	L15A187
8	VMP78	CONNECTOR AMP, 20WAY	AMP	178308-5	L15A314
9	VMP72	CONNECTOR AMP, 16WAY	AMP	178327-5	L15A085
10	VMP74	CONNECTOR AMP, 12WAY	AMP	178326-5	L15A270
11	VMP75	CONNECTOR AMP, 20WAY	AMP	178328-5	L15A184
12	VMP80	CONNECTOR AMP, 16WAY	AMP	178327-5	L15A085
13	VMP81, VMP74	CONNECTOR AMP, 12WAY	AMP	178326-5	L15A270
14	LED	RED LED 5MM DIA	REPUTED	--	--
15	LED	GREEN LED 3MM DIA	REPUTED	--	--
16	TB1 ~ TB9	PCB TERMINAL S/5.08 SPACING	WAGO	255-401,255-100	L16B038
17	SL1780, SL1781 SL2601, SL2602, SL2603, SL2604, SL2311, SL1780, SL1781, SL2310, SL2311,	PHOTO MOS RELAY	PANASONIC	AQZ102	L16B073
18	LS912A, SL2701 SL2702, SL2703, SL2704, SL2705, SL2606, SL2607, SL2608, SL1783, SL1782,	PHOTO MOS RELAY SOCKET	PANASONIC	RL PA1a-PS MATSU	L16B074
19	RL6, RL7, RL8, RL9, RL40, RL41 (DUMMY)	PHOTO MOS RELAY	PANASONIC	AQZ102	L16B073

OPTION PCB-E7VS140A101-00

PARTS LIST

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

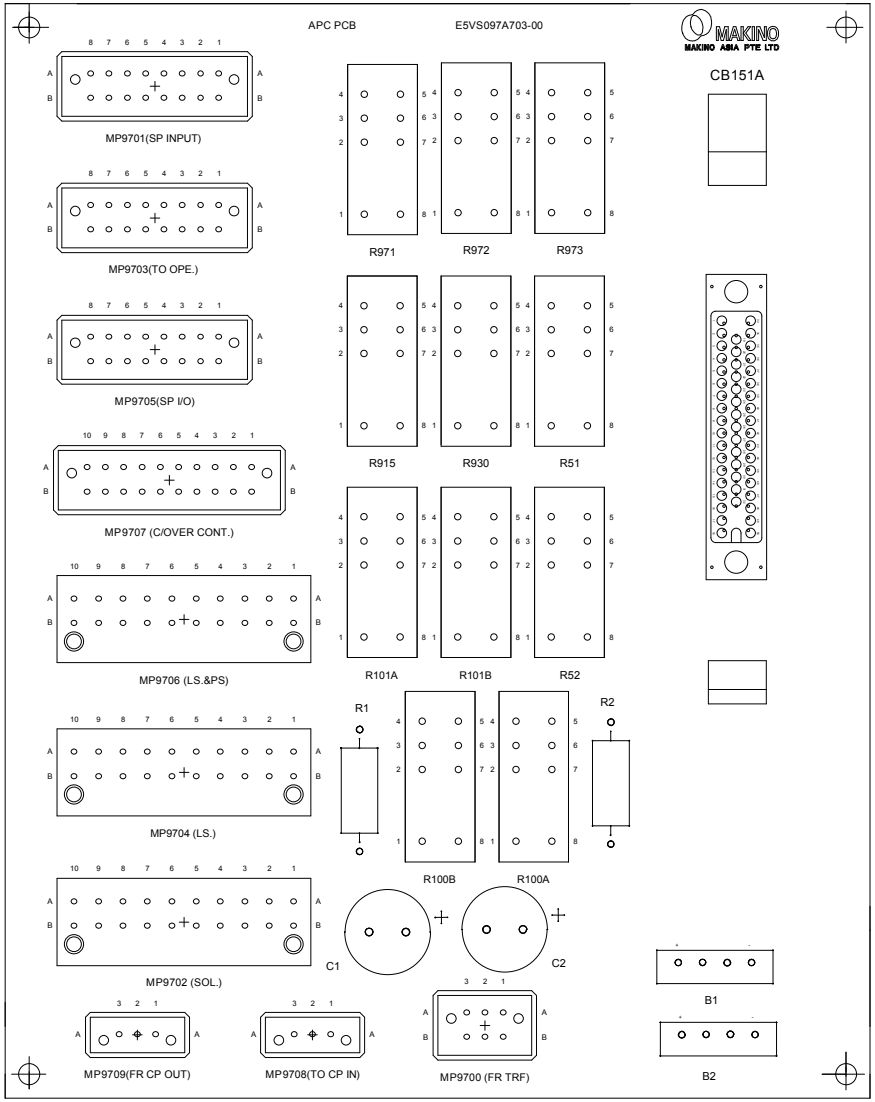
TITLE
HYDRAULIC PCB LAYOUT, PARTS

MACHINE TYPE	SLIM3
NC TYPE	FS0i-MC

DRAWING No.
V090i-C1E-X03

GROUP
=

SHEET No.
H08



NO.	DESIGNATION	DESCRIPTION	MAKE	MFR PART NO.	MAKINO PART NO.
1	MP9700	CONN TAP HDR 6W AMP	AMP	1-178140-5	L15A117
2	MP9701, MP9703, MP9705	CONN TAP HDR 16W AMP	AMP	178327-5	L15A085
3	MP9702, MP9704 MP9706	CONN AMP HDR20W	AMP	316517-5	L15A183
4	MP9707	CONN AMP HDR20W	AMP	178328-5	L15A184
5	MP9708, MP9709	CONN AMP	AMP	1-178136-5	L15A086
6	R51, R52, R100A, R100B, R101A, R101B, R925, R930, R971, R972, R973	RELAY OMRON G2R-2-SND DC24V	OMRON	G2R-2-SND	L16B024
7	R51, R52, R100A, R100B, R101A, R101B, R925, R930, R971, R972, R973	RELAY SOCKET OMRON P2R-08P	ORMON	P2R-08P	L16B012
8	CB151A	CONN HONDA 50 PIN PCB FEMALE	HONDA	MR-50RFD2	L15A187
9	R1, R2	RESISTOR 10K OHM,2W	REPUTED	---	---
10	C1, C2	CAPACITOR,2200 UF,50V,ELECT	REPUTED	---	---
11	B1, B2	RECTIFIER BRIDGE 4A QBL406	REPUTED	---	---
12	D1, D2, D3	DIODE 1N4007	REPUTED	----	---

OPTION PCB-EE5VS97A703-00

PARTS LIST

Revisions			Date	Name
4				
3		APPROVED	25-NOV-2010	GOH WILLIAM
2		CHECKED	25-NOV-2010	S MAHESH
1		DRAWN	25-NOV-2010	PRAVEEN B

MAKINO
MAKINO ASIA PTE LTD

TITLE	MACHINE TYPE	SLIM3	DRAWING No.	GROUP
APC PCB LAYOUT, PARTS	NC TYPE	FS0i-MC	V090i-C1E-X03	=
				SHEET No. H09